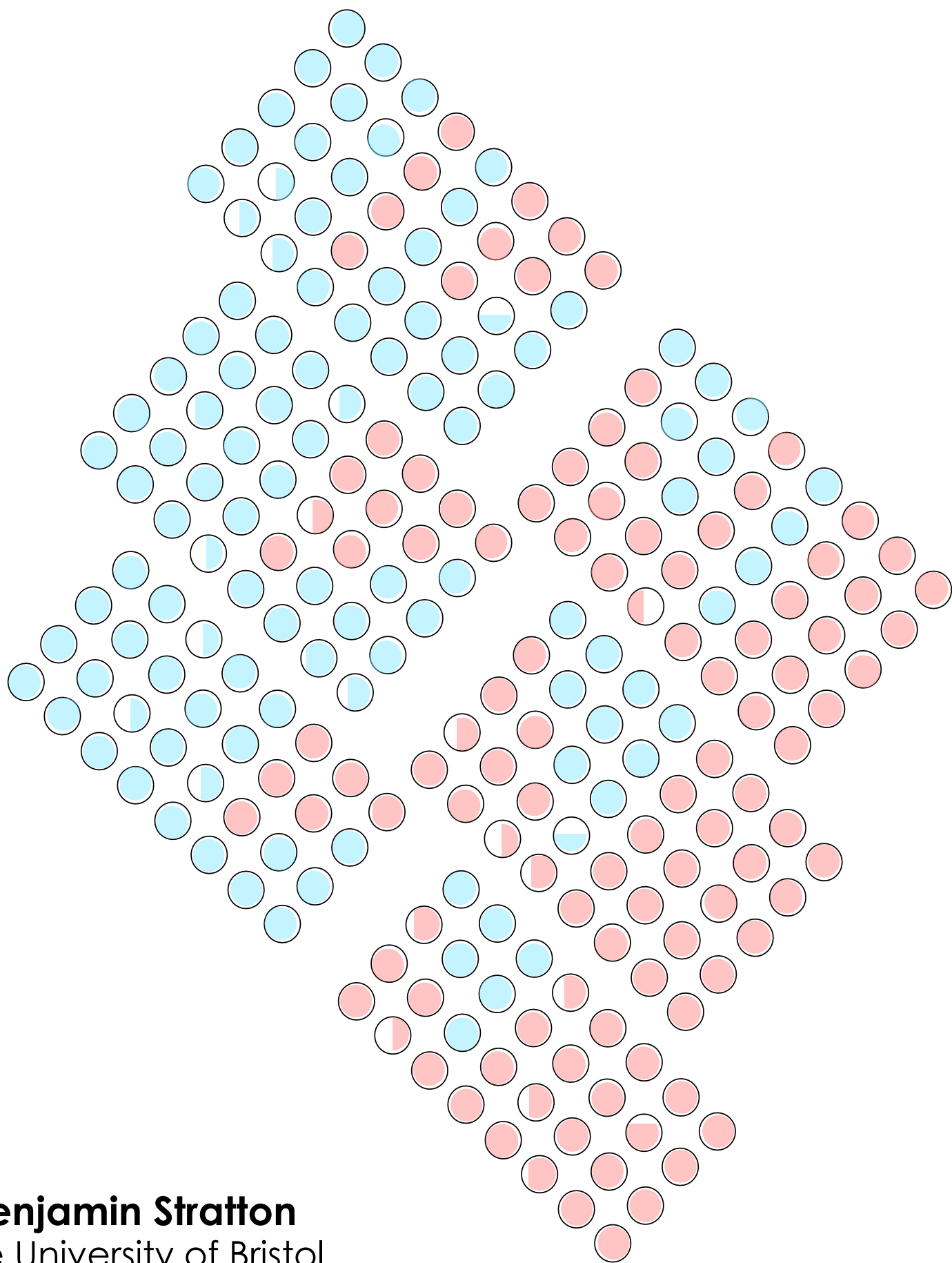


The More the Merrier:

Resource Preservation and Enhancement in
Quantum Resource Theories



Benjamin Stratton

The University of Bristol

The cover art of this thesis is an abstract graphical depiction of optimally cooling a thermal qubit using three other thermal qubits and a global unitary operation.

See Section 7.6 for full mathematical details.



University of
BRISTOL

Doctoral Thesis

**The More the Merrier: Resource Preservation
and Enhancement in Quantum Resource Theories**

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A dissertation submitted to **The University of Bristol** in accordance with the requirements for the award of the degree of **Doctor of Philosophy** in the Faculty of Science, School of Physics.

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This work was completed whilst a member of both the **Quantum Information Theory Group** in the School of Physics, and the 8th Cohort of the **Quantum Engineering Centre for Doctoral Training**.

Author's Declaration

I, **Benjamin Stratton**, declare that the work in this dissertation was carried out in accordance with the requirements of the University's Regulations and Code of Practice for Research Degree Programmes and that it has not been submitted for any other academic award.

Except where indicated by specific reference in the text, the work is my own work.

Work done in collaboration with, or with the assistance of, others, is indicated as such.

Any views expressed in the dissertation are those of the author.

SIGNED: Benjamin Stratton

DATE: 19th December 2025

Abstract

The More the Merrier: Resource Preservation and Enhancement in Quantum Resource Theories

Resource theories provide a rigorous mathematical framework for quantifying and studying different quantum phenomena; they achieve this by studying the limitations placed on a system by a set of physical constraints. Through this operational approach, they have successfully been used to understand a diverse range of properties, including those of quantum states, measurements and dynamics.

Within this framework, questions about resource preservation and resource enhancement are of both practical and fundamental interest. The first part of this thesis investigates the notion of resource preservation and enhancement within the resource theories of informational non-equilibrium, unspeakable coherence, and athermality. First, the framework of resource preservability theories is applied to the resource theory of informational non-equilibrium, yielding a characterisation of the ability of the allowed operations to preserve informational non-equilibrium (the resource). To achieve this, resource preservation itself is treated as a dynamical resource, leading to the definition of a dynamical resource theory over the set of allowed operations. Secondly, the framework of resource concentration is introduced to study the minimal feasible scenario of local resource enhancement under resource non-generating operations. This framework is applied to both the resource theory of informational non-equilibrium and the resource theory of unspeakable coherence, with insights gained from this minimal case then applied to gain an understanding of resource enhancement in more general cases. Finally, local resource enhancement via resource-generating operations is investigated within the resource theory of athermality, specifically in the context of cooling. The effect of the energetic structure of a machine composed of n thermal qubits on its ability to cool an additional thermal qubit via energy non-conserving unitary dynamics is analysed. A set of inequalities characterising the optimal achievable cooling, the protocols that attain it, and their associated complexity is found.

In the second part of this thesis, the Choi-state, an indispensable tool in the study and analysis of quantum channels, is investigated. Specifically, the Choi-rank is shown to provide a universal bound on how successfully two agents can perform in an entanglement-assisted exclusion task. To achieve this, our understanding of the task of conclusive k -state exclusion in general is developed.

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List of publications

Publications and preprints completed during PhD studies

The following list of publications and preprints, ordered chronologically, I have co-authored throughout my PhD and are discussed in this thesis.

- **Unspeakable Coherence Concentration**, B.Stratton, C.-Y.Hsieh, P.Skrzypczyk [arXiv:2512.04255 \(2025\)](#) [1] [Chapter 6].
- **Cooling a Qubit Using n Others**, J.Xuereb, B.Stratton, A.Rolandi, J.He, M.Huber, P.Bakhshinezhad, [PRX Quantum](#) (2025) [2] [Chapter 7].
- **Informational non-equilibrium concentration**, C.-Y.Hsieh, B.Stratton, H.-C.Weng, V.Scarani, [Phys. Rev. A](#) (2025) [3] [Chapter 6].
- **Operational Interpretation of the Choi Rank Through k-State Exclusion**, B.Stratton, C.-Y.Hsieh, P.Skrzypczyk, [Phys. Rev. A](#) (2024) [4] [Chapter 8].
- **Dynamical Resource Theory of Informational Non-equilibrium Preservability**, B.Stratton, C.-Y.Hsieh, P.Skrzypczyk, [Phys. Rev. Lett.](#) (2024) [5] [Chapter 5].

The following list of publications and preprints, whilst co-authored during my PhD, are not discussed in this thesis:

- **An Algorithm for Estimating α -Stabilizer Rényi Entropies via Purity**, B.Stratton, [arXiv:2507.02540](#) (2025) [6].
- **Dynamical resource theory of incompatibility preservability**, C.-Y.Hsieh, B.Stratton, C.-H.Wu, H.-Yu.Ku [Phys. Rev. A](#) (2025) [7].

Statement on the inclusion of published and collaborative work

My dissertation does include work that has been published, submitted or prepared for publication with me as an author.

My dissertation does include non-published collaborative work that contains data or contributions from others.

As there are publications with me as an author and/or any collaborative work that contains data or contributions from others, I have completed the following statement:

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Author contributions

I led this work, performing the proofs, writing the paper and dealing with the publication process. Although, I had frequent meetings with my supervisors Paul Skrzypczyk and Chung-Yun Hsieh throughout, which impacted both the conceptual direction of the project as well as certain technical details.

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Result 6.2.1 and Corollary 6.2.2 were proved by Chung-Yun Hsieh, but are included in this thesis for necessity and completeness. Also, the contrapositive of Result 6.2.5 was pointed out to me by Ludovico Lami at the 2025 Quantum Resource Theories Workshop. The remainder of the included results are conceptually developed and proved by me. In addition, I contributed to writing these sections within the manuscript [3].

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Chapter 8

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Author contributions

I led this work, performing the proofs, writing the paper and dealing with the publication process. Although, I had frequent meetings with my supervisors Paul Skrzypczyk and Chung-Yun Hsieh throughout, which impacted both the conceptual direction of the project as well as certain technical details.

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INTRODUCTION

1 Introduction

Classical information theory is the study of encoding, processing and decoding information in physical systems that obey the laws of classical physics [8]; quantum information theory is the study of encoding, processing and decoding information in physical systems that obey the laws of quantum physics [9, 10]. Given classical systems can be modelled by restricted quantum systems, classical information theory is a subtheory of quantum information theory. As a result, the possibilities for processing information within quantum systems is larger. This remarkable fact underpins the now intellectually and monetarily flourishing field of quantum technologies.

In this field, a deeper understanding of the behaviour of physical systems has been utilised to develop novel methods of information processing [9]. Interestingly, the reverse is now true. Despite initially being conceived as an abstract mathematical theory for studying communication over noisy channels — that was specifically independent of the underlying physical system encoding the information to be communicated — information theory has now become a tool for studying the physical systems themselves. By considering what information can be encoded, processed, and decoded in a system, the physical properties of that system can be understood and modelled. A clear example of this is the no-cloning theorem [11]. This is an information-theoretic statement about the impossibility of perfectly copying an unknown state, but it turns out to reflect deep physical constraints on the allowed dynamics of quantum systems. Moreover, it applies operational constraints too, preventing all quantum states from being perfectly distinguishable.

Since its inception, information theory has been operationally focused, with key quantities defined through the limitations of what can be accomplished in some physical task. This philosophy has been extended and formalised within quantum information theory through the notion of *resources*, where, colloquially, a property of a quantum object is considered a resource if it is able to provide an advantage in some operational task [12]. It has been shown that many different phenomena of quantum systems can be modelled as resources, with this perspective offering a systematic way to quantify and study the underlying phenomena.

Quantum resource theories [12, 13] provide a rigorous mathematical framework to formally study the notion of resources. They can be considered to provide ‘rulers’ that enable the quantification of abstract quantities of quantum systems, particularly those properties that do not have an associated physical observable, meaning the amount of the property cannot be directly measured. A plethora of quantum properties, such as, but not limited to, entanglement [14], coherence [15, 16], athermality [17–19], non locality [20], measurement informativeness [21], incompatibility [22], and decoherence [23], have been successfully quantified via quantum resource theories. As is clear from the list, they have been fruitfully applied to properties of quantum states, measurements and dynamics.

Within a quantum resource theory, a set of physical constraints is considered. This then places limitations on the operations that can be performed. Under this limited set of operations — the so-called *allowed operations* — only objects without the property of interest (i.e., not a resource) can be generated. One can then ask what it is, and what it is not, possible to achieve when unable to generate the resource. Separating the allowed operations in this way paves the way for the resource to be quantified by a variety of methods. If these physical constraints coincide with real limitations in the lab, then the resource theories can also provide a framework for determining what it is possible to achieve when subject to such constraints. The quintessential example of this is within the resource theory of entanglement [14] when the allowed operations are Local Operations and Classical Communication (LOCC). Such operations are not only entanglement non-generating, but are also an accurate description of the physical limitations of contemporary technology i.e., with our current technology it is difficult to generate entanglement between spatially separated laboratories (parties). Hence, the resource theory of entanglement has not only allowed entanglement to be operationally quantified, it has also allowed the limitations of what can be achieved when constrained to LOCC to be systematically explored.

The fact that resources cannot be freely generated under the physical constraints ascribes them with

value. This then creates a strong incentive to make sure that they are efficiently utilised. In addition, it ensures there always exists a desire to acquire more resources. And, lastly, it also means that one would like these resources to be sufficiently maintained when not being used. Questions concerning the efficient use, enhancement, and preservation of resources are of both practical and foundational importance, as they span from magic state generation [24, 25] to quantifying thermalisation [3, 5, 17, 19, 26]. A central aim of this thesis is to explore the processes of resource preservation and enhancement within quantum resource theories, testing and developing tools that allow us to better understand these crucial processes.

Below, an overview of what is included in each chapter of this thesis is given.

1.1 Thesis Overview

1.1.1 Preliminaries

In this Section, an overview of the background theory used throughout this thesis will be given. Initially, in Chapter 2, an overview of the theory of quantum information is given, including the mathematical underpinnings, along with a discussion of how quantum states, measurements and dynamics are modelled. Following this, several other key concepts used through this thesis will be detailed, namely, supermaps, norms, entropy functions, stochastic matrices and discrete Weyl operators.

In Chapter 3, three quantum information processing tasks of fundamental importance are discussed. An overview of the task of quantum state discrimination is first given, before the task of quantum state exclusion is discussed in detail. Finally, Superdense Coding is discussed from the prospective of the task of quantum state discrimination.

To finish this Section, in Chapter 4 an overview of quantum resource theories is given. Key concepts within the quantum resource theories framework are discussed, before the examples of the resource theory of informational non-equilibrium and the resource theory of unspeakable coherence are given.

1.1.2 Results Part 1 — Resource Preservation and Enhancement

This section focuses on results related to the preservation and enhancement of resources in the static resource theories of informational non-equilibrium [18], unspeakable coherence [16] and athermality [17, 19].

In Chapter 5, the framework of resource preservability theories is applied to the resource theory of informational non-equilibrium. These theories treat the ability of the allowed operations of a static resource theory to preserve the resource as a *dynamical resource*. Hence, a dynamical resource theory is developed that places a partial-order on the set of allowed operations of the resource theory of informational non-equilibrium based on their ability to *maintain* informational non-equilibrium. A characterisation of the allowed operations of this dynamical resource theory is given for qubit channels and the n -dimensional Weyl-covariant channels — a physically relevant subset of the general channels. Moreover, an operational interpretation of the dynamical resource is given through a state discrimination game with fixed Bell state measurements, before an explicit link between a channel's classical capacity and its ability to preserve informational non-equilibrium is made. This work demonstrates the viability of resource preservability theories [26] via a concrete example able to provide non-trivial insights. Moreover, it provides a method for quantifying how 'thermalising' dynamics are.

In Chapter 6, the framework of resource concentration is introduced as a method for studying the minimal feasible scenario of local resource enhancement under resource non-generating operations. The framework is then applied to both the resource theories of informational non-equilibrium [18] and unspeakable coherence [16]. Understanding resource enhancement in this minimal limiting case is not only practically relevant to contemporary quantum technologies — under the assumption that smaller equals more practically achievable — but also allows for a bottom up understanding of resource enhancement to be built. For both resources, what is learnt in this minimal scenario is then used to consider resource enhancement in more general cases.

In Chapter 7, local resource enhancement in the resource theory of athermality is considered. This

is done specifically within the context of *cooling* a thermal qubit i.e., increasing the probability of the qubit being in its ground state, when given access to n other thermal qubits — known as a machine — and energy non-conserving (resource-generating) unitary operations. Cooling is a task of fundamental importance in thermodynamics, and it has been considered in many different scenarios [27–38]. Here, we look to probe how the energetic structure of the machine impacts on both the ability to perform cooling, and the complexity of the necessary optimal cooling protocol. To achieve this, we consider a model where each of the n thermal qubits in the machine have local Hamiltonians with varying energy differences. We then show that there exists a set of inequalities related to this energetic structure of the n -qubit machine that determines the optimal cooling possible, the necessary protocol to implement it, and the complexity of this protocol. This work builds on the notion of algorithmic cooling, and establishes new fundamental bounds on cooling systems. Moreover, it introduces a new framework with which to consider cooling protocols, which we use to inform the development of a purity enhancement protocol.

1.1.3 Results Part 2 - Exclusion

In the final chapter of this thesis, Chapter 8, we investigate the Choi-state, an indispensable tool in the study and analysis of quantum channels. Considering a channel in terms of its associated Choi-state can greatly simplify problems. It also offers an alternative approach to the characterisation of quantum channels, with properties of the Choi-state providing novel insight into a channel's behaviour. The rank of a Choi-state, termed the *Choi-rank*, has proven to be an important characterising property, and here, its significance is further elucidated through an operational interpretation. Specifically, the Choi-rank is shown to provide a universal bound on how successfully two agents, Alice and Bob, can perform an entanglement-assisted exclusion task. The task can be considered an extension of super-dense coding, where Bob can only output information about Alice's encoded bit-string with certainty. Conclusive state exclusion, in place of state discrimination, is therefore considered at the culmination of the super-dense coding protocol. In order to prove this result, we develop our understanding of the task of conclusive k -state exclusion in general, introducing a novel necessary condition for when it is possible to perform the task on some arbitrary set of states. In the process, we introduce the notion of weak and strong quantum state exclusion.

1.1.4 Conclusion

To finish, we provide some outlook on the field of resource theories as a whole, and discuss possible future directions for each of the chapters of this thesis.

1.2 Formatting Details

Definitions and notable results are presented in titled boxes with the following colour code:

Definition 1.2.1. *A precise and organised meaning to a new term.*

Proposition 1.2.1. *A statement that is derived from what has been discussed.*

Observation 1.2.2. *A statement or a remark that is made based on informal analysis.*

Task 1.2.3. *A physical procedure to be performed.*

Lemma 1.2.4. *A minor, proven proposition.*

Result 1.2.5. *A larger result.*

Corollary 1.2.6. *A proposition that follows from a larger result.*

Theorem 1.2.7. *A result that does not form part of my own work.*

PRELIMINARIES

“The map is not the territory” — Alfred Korzybski

In this chapter we will cover the mathematical background necessary to understand the work in this thesis. The aim is to introduce the tools and notion used, and not to provide an extensive exploration of the underlying theory of quantum information. For this, the reader should refer to [39] or [10].

2.1 Mathematical Preliminaries

A Hilbert space $\mathcal{H}$ is a complex vector space with a complete inner product. We will denote the Hilbert space of different physical systems with a subscript, for example $\mathcal{H}_A$ for the system A . In this thesis we only consider finite dimensional Hilbert spaces and, where necessary, we will denote the specific dimension of the system with a superscript e.g., $\mathcal{H}^d$ for a d -dimensional Hilbert space. If considering two separate physical systems, A and B , we will consider the total Hilbert space to be the tensor product of their Hilbert spaces, $\mathcal{H}_A \otimes \mathcal{H}_B$.

Vectors in the Hilbert space will be denoted by a ket $|a\rangle \in \mathcal{H}$, where a is some label used to identify the vector. All vector spaces have a dual space, $\mathcal{H}^*$, that is the space of linear functionals on the original space. Elements of the dual space will be denoted by bra $\langle b| \in \mathcal{H}^*$, where $\langle b|$ is a functional that takes an element of $\mathcal{H}$ to a complex number. To each element $|a\rangle \in \mathcal{H}$, there is an associated element in the dual space $\langle a| \in \mathcal{H}^*$.

An inner product is a function that takes two vectors from $\mathcal{H}$ as its input and returns a scalar. It must also be positive semi-definite if both inputs are the same, have conjugate symmetry, be linear in the second argument, and additive. Here, we consider $\langle \cdot | \cdot \rangle : \mathcal{H} \times \mathcal{H} \rightarrow \mathbb{C}$, where $\mathbb{C}$ is the set of complex numbers. Two non-zero vectors $|a\rangle, |b\rangle \in \mathcal{H}$ are said to be orthogonal if $\langle a|b\rangle = 0$ i.e., their inner product is zero.

A basis of a Hilbert space is a set of vectors, $\{|i\rangle\}_i$ that are linearly independent and span the whole space. If the inner product of the basis vectors is $\langle i|j\rangle = \delta_{i,j}$, where $\delta_{i,j}$ is the Kronecker delta function, then the basis is said to be an orthonormal basis.

A linear operator is a map $X : \mathcal{H} \rightarrow \mathcal{V}$, where $\mathcal{V}$ is also a Hilbert space, such that

$$X(a|\psi\rangle + b|\phi\rangle) = aX(|\psi\rangle) + bX(|\phi\rangle) \quad \forall a, b \in \mathbb{C} \text{ and } |\psi\rangle, |\phi\rangle \in \mathcal{H}. \quad (2.1)$$

The set of all linear operators of the form $X : \mathcal{H} \rightarrow \mathcal{V}$ will be denoted $\mathcal{L}(\mathcal{H} \rightarrow \mathcal{V})$. If $\mathcal{H} = \mathcal{V}$, meaning X maps from the Hilbert space $\mathcal{H}$ to itself, then X is said to be a linear operator *on* $\mathcal{H}$. The set of linear operators on $\mathcal{H}$ will be denoted $\mathcal{L}(\mathcal{H})$. Given a linear operator $X \in \mathcal{L}(\mathcal{H})$ and a basis of $\mathcal{H}$, denoted by $\{|i\rangle\}_i$, the linear operator X can be decomposed into a square matrix with respect to this basis as $X = \sum_{i,j} x_{i,j} |i\rangle\langle j|$, where $x_{i,j} = \langle i|X|j\rangle$ are the matrix elements. Here, we have used the standard notation that if $|j\rangle \in \mathcal{H}$ and $|i\rangle \in \mathcal{V}$ then an operator that maps $|j\rangle \rightarrow |i\rangle$ is denoted $|i\rangle\langle j|$.

If there exists an $|a\rangle \in \mathcal{H}$ such that $X|a\rangle = x_a|a\rangle$, where X is an operator on $\mathcal{H}$ and $x_a \in \mathbb{C}$, then $|a\rangle$ is an eigenvector of the operator X with eigenvalue x_a . If one can find a basis of eigenvectors of X , termed an eigenbasis of X , then the matrix of X with respect to this basis is diagonal, such that $X = \sum_i a_i |a_i\rangle\langle a_i|$ if $\{|a_i\rangle\}_i$ is an eigenbasis of X . The set of all eigenvalues of X is called its spectrum, which will be denoted by $\text{spec}(X)$.

An operator X is said to be positive semi-definite, denoted $X \geq 0$, if $\langle a|X|a\rangle \geq 0 \quad \forall |a\rangle \in \mathcal{H}$. Equivalently, it means that all the eigenvalues of a positive semi-definite operator are greater than or equal to zero.

The adjoint of an operator is the operator $X^\dagger$ such that $\langle a|X^\dagger|b\rangle = \langle b|X|a\rangle^* \quad \forall |a\rangle, |b\rangle \in \mathcal{H}$, where $(\cdot)^*$ is the complex conjugate. An operator is then said to be Hermitian if $X^\dagger = X$. The eigenvectors of a Hermitian operators always form an orthonormal basis and all its eigenvalues are real.

The trace of a square matrix is defined to be the sum of the diagonal elements of the matrix. Given an operator $X \in \mathcal{L}(\mathcal{H})$, this can be seen to be given by $\text{tr}[X] = \sum_i \langle i | X | i \rangle$, where $\{|i\rangle\}_i$ is any basis of $\mathcal{H}$.

Let $X \in \mathcal{L}(\mathcal{H} \rightarrow \mathcal{V})$. There exists a basis $\{|a_i\rangle\}_i$ of $\mathcal{H}$ and a basis $\{|b_j\rangle\}_j$ of $\mathcal{V}$ such that $X = \sum_i \lambda_i |b_i\rangle \langle a_i|$. The set of all λ_i are called the singular values of X and all λ_i are real and non-negative. The set of singular values of X is equal to the spectrum of $\sqrt{X^\dagger X}$. For Hermitian square matrices, the singular values are the absolute value of the eigenvalues.

Lastly, another inner product considered in this thesis is the Hilbert-Schmidt inner product, which for two operators $X, Y \in \mathcal{L}(\mathcal{H})$ is defined as $\text{tr}[X^\dagger Y] : \mathcal{L}(\mathcal{H}) \times \mathcal{L}(\mathcal{H}) \rightarrow \mathbb{C}$.

2.2 States

A state of a quantum system is a complete description of the system at a given moment in time: it allows one to calculate the probabilities of all possible measurement outcomes. Mathematically, quantum states are modelled by the set of density operators on a Hilbert space $\mathcal{H}$,

$$\mathcal{D}(\mathcal{H}) := \{\rho : \rho \geq 0, \text{tr}[\rho] = 1\}. \quad (2.2)$$

These are the set of linear operators on $\mathcal{H}$ that are positive semi-definite and have unit trace, such that $\mathcal{D}(\mathcal{H}) \subset \mathcal{L}(\mathcal{H})$. Density operators will be denoted by Greek letters such as ρ, σ and ω throughout this thesis. Where necessary, if considering the state of a specific systems, a subscript will be used e.g, $\rho_A \in \mathcal{D}(\mathcal{H}_A)$.

All density operators are Hermitian, and therefore their eigenvectors can form a complete orthonormal basis and their eigenvalues are real. If ρ has only a single eigenvector $|\psi\rangle \in \mathcal{H}$ with non-zero eigenvalue, then it can be decomposed with respect to its eigenbasis as $\rho = |\psi\rangle \langle \psi|$. Such a state is termed a *pure state*, as it can be described by a single vector in the Hilbert space. These states can be physically generated by preparing the state $|\psi\rangle$ with probability one.

If ρ instead has multiple eigenvectors $\{|\phi_i\rangle\}_i$ with non-zero eigenvalues, then it can be decomposed with respect to its eigenbasis as $\rho = \sum_i p_i |\phi_i\rangle \langle \phi_i|$, where p_i is the eigenvalue associated to the eigenvector $|\phi_i\rangle$. As $\rho \geq 0$ and $\text{tr}[\rho] = 1$, meaning all the eigenvalues are positive and sum to one, they can be interpreted as probabilities. Hence, ρ can be considered to be a probabilistic mixture of the pure states $\{|\phi_i\rangle \langle \phi_i|\}_i$. Such states are therefore termed *mixed states*, and can be physically generated by a source that outputs a state drawn from the ensemble $(p_i, |\phi_i\rangle)$.

If all of the eigenvectors of ρ have equal eigenvalue, then the state decomposed with respect to its eigenbasis is $\rho = d^{-1} \sum_i |\psi_i\rangle \langle \psi_i|$, where $\{|\psi_i\rangle\}_i$ is a basis of the full Hilbert space, meaning it has d elements if $\rho \in \mathcal{D}(\mathcal{H}^d)$, and each eigenvalue equals d^{-1} . Therefore, $\rho = \mathbb{I}/d$, as the sum over an orthonormal basis is the identity operator, denoted $\mathbb{I}$. This is the most mixed state possible (has the least amount of purity) and is termed the *maximally mixed* state. Given that the sum of any orthonormal basis is the identity, a maximally mixed state is maximally mixed in all bases. It can therefore be physically generated by selecting any orthonormal basis of states $\{|\psi_i\rangle\}_i$, and then drawing a state from the ensemble $(d^{-1}, |\psi_i\rangle)$.

The purity of a state, from pure to maximally mixed, can be captured numerically via the following definition.

Definition 2.2.1 (Purity). Let $\rho \in \mathcal{D}(\mathcal{H}^d)$. The purity of ρ is defined as $\text{tr}[\rho^2]$. This has the following properties

- $1/d \leq \text{tr}[\rho^2] \leq 1$.
- $\text{tr}[\rho^2] = 1$ if and only if ρ is a pure state.
- $\text{tr}[\rho^2] = 1/d$ if and only if ρ is the maximally mixed state.

In general, any density operator can be written as a convex combination of pure states

$$\rho = \sum_i q_i |\Psi_i\rangle \langle \Psi_i|, \quad \sum_i q_i = 1, \quad q_i \geq 0 \forall i, \quad (2.3)$$

and hence can be physically realised by a source emitting a state from the ensemble $(q_i, |\Psi_i\rangle)$. The spectral decomposition, meaning the decomposition of ρ into its eigenbasis as detailed above, is a special case of such a decomposition where all the states in the ensemble are orthogonal. For a given state ρ , there may be multiple different ensembles that can be used to create it, with the spectral decomposition meaning at least one always exists. As the state captures all possible measurement outcomes, any two sources that lead to the same density operator will be physically indistinguishable.

Finally, we note that convex combinations of density operators are also density operators,

$$\rho = \sum_i q_i \rho_i \in \mathcal{D}(\mathcal{H}) \quad \text{if } \rho_i \in \mathcal{D}(\mathcal{H}) \forall i, \sum_i q_i = 1, q_i \geq 0 \forall i. \quad (2.4)$$

Physically, this means probabilistically selecting a source is also a valid quantum state.

States at a Glance

Definition 2.2.2 (Quantum states). *Quantum states are modelled by the set of density operators on a Hilbert space $\mathcal{H}$,*

$$\mathcal{D}(\mathcal{H}) := \{\rho : \rho \geq 0, \text{tr}[\rho] = 1\}. \quad (2.5)$$

These are the set of linear operators on $\mathcal{H}$ that are positive semi-definite and have unit trace.

2.3 Measurements

A measurement is a quantum operation that converts a quantum state into a classical probability distribution over measurement outcomes.

Observables are physically measurable quantities. In quantum mechanics, they are modelled by Hermitian operators on the Hilbert space. As stated above, a Hermitian operator $X \in \mathcal{L}(\mathcal{H})$ has all real eigenvalues and its eigenvectors can form a complete orthonormal basis. When measuring the observable X , the possible outcomes are the eigenvalues of X and the state after measurement will be supported within the eigenspace of X corresponding to the output eigenvalue. If the spectral decomposition of the observable is $X = \sum_i \lambda_i \Pi_i$, where Π_i is the projector onto the eigenspace of X with eigenvalue λ_i , then the probability of getting the outcome λ_i given the observable X is measured on $\rho \in \mathcal{D}(\mathcal{H})$ is

$$p(\lambda_i | X, \rho) = \text{tr}[\Pi_i \rho]. \quad (2.6)$$

The post measurement state given this outcome, ρ_i , is then

$$\rho_i = \frac{\Pi_i \rho \Pi_i}{p(\lambda_i | X, \rho)}. \quad (2.7)$$

It can be seen that if X is measured on ρ_i , then the outcome will be the same as before with certainty, $p(\lambda_j | X, \rho_i) = \delta_{ij}$ where δ_{ij} the the Kronecker delta function, as $\Pi_i \Pi_j = \delta_{ij}$. For this reason, the measurement of an observable is termed a *projective measurement*, as the state is projected into an eigenspace of the observable being measured.

However, not all measurements appear projective in reality. For example, if one considers measuring an observable on a state supported in a lower dimensional subspace, the measurement may not appear projective within that subspace. More generally, a measurement in quantum mechanics can be modelled by a set of m complex operators, $\mathbb{M} = \{M_i : M_i \in \mathcal{L}(\mathcal{H})\}_{i=0}^{m-1}$, such that

$$\sum_{i=0}^{m-1} M_i^\dagger M_i = \mathbb{I}. \quad (2.8)$$

Each operator in the set is associated to a different possible measurement outcome, with the probability

of getting the outcome i when performing the measurement $\mathbb{M}$ on the state $\rho \in \mathcal{D}(\mathcal{H})$ given by

$$p(i, | \mathbb{M}, \rho) = \text{tr}[M_i^\dagger M_i \rho]. \quad (2.9)$$

From this, it can be seen that Eq. (2.8) ensures that $\sum_i p_i = 1$. Given a measurement outcome i , the post measurement state, ρ_i , is then

$$\rho_i = \frac{M_i \rho M_i^\dagger}{p(i, | \mathbb{M}, \rho)}, \quad (2.10)$$

All generalised measurements can be physically realised by first applying a joint unitary on a state and an ancilla, followed by measuring an observable on the ancilla [12]. The measurement outcome is given by the output of the measurement on the ancilla, and the state is then left in the post measurement state. Colloquially, observables can be thought to model ideal closed system measurements, whilst generalised measurements model open system measurement.

A Positive Operator Valued Measure (POVM) is a generalised measurement where one does not care about the post measurement state, only the probability of getting the different outcomes. This might be the case because the post measurement states is unimportant for any future use, or, it might be because it physically does not exist, such as when a photon is measured. In this case, the photon is destroyed, leaving no photon for a post measurement state to exist in. In these cases, rather than considering the set of operators $\{M_x\}_{x=0}^{m-1}$, one can instead consider only the set $\{T_x = M_x^\dagger M_x\}_{x=0}^{m-1}$ which is enough to calculate the probabilities of the different possible measurement outcomes occurring. For each generalised measurement there is a unique POVM that corresponds to it. On the other hand, for each POVM there are infinitely many possible generalised measurements that could correspond to it [12].

Measurement at a Glance

Definition 2.3.1 (Observables). *An observable is modelled by a Hermitian operator. When measuring an observable $X \in \mathcal{L}(\mathcal{H})$, the possible outcomes are the eigenvalues of X and the state after measurement will be supported within the eigenspace of X corresponding to the output eigenvalue.*

Definition 2.3.2 (Generalised Measurements). *A generalised measurements is a set of m complex operators, $\{M_x : M_i \in \mathcal{L}(\mathcal{H})\}_{x=0}^{m-1}$, such that*

$$\sum_{x=0}^{m-1} M_x^\dagger M_x = \mathbb{I}. \quad (2.11)$$

Definition 2.3.3 (Positive Operator Value Measure (POVM)). *A POVM is a set of m operators, $\{T_x : T_x \in \mathcal{L}(\mathcal{H})\}_{x=0}^{m-1}$, such that*

$$\sum_{x=0}^{m-1} T_x = \mathbb{I}, \text{ and } T_x \geq 0 \forall x. \quad (2.12)$$

The first condition ensures that the sum of the probabilities of getting the different possible outcomes is one. The second condition ensures that all these probabilities are positive.

2.4 Dynamics

Closed systems — systems that do not interact with external systems — evolve via unitary dynamics:

$$\rho' = U\rho U^\dagger, \quad \rho, \rho' \in \mathcal{D}(\mathcal{H}), \quad (2.13)$$

where U is a unitary operator such that $UU^\dagger = U^\dagger U = \mathbb{I}$. As $V = U^\dagger$ is also a unitary, all closed system dynamics are reversible, $\rho = VU\rho U^\dagger V^\dagger$.

The evolution of open systems, that is systems that do interact with external systems, is described by a completely positive trace preserving (CPTP) linear maps from $\mathcal{D}(\mathcal{H}^d)$ to $\mathcal{D}(\mathcal{H}^{d'})$. Linearity of these means that if $\mathcal{E} \in \text{CPTP}$ then $\mathcal{E}(\rho + \sigma) = \mathcal{E}(\rho) + \mathcal{E}(\sigma)$; positivity means that positive operators are mapped to positive operators, if $\rho \geq 0$ then $\mathcal{E}(\rho) \geq 0$. Complete positivity then means that even if the maps acts only on a subsystem of a state, the state remains positive i.e., if $\rho \geq 0$ then $(\mathcal{I} \otimes \mathcal{E})(\rho) \geq 0$ where $\mathcal{I}(\cdot) = \mathbb{I}(\cdot)\mathbb{I}$ is the identity channel. Finally, trace preserving ensures that $\text{tr}[\mathcal{E}(\rho)] = \text{tr}[\rho]$. Taken together, these conditions ensure that CPTP maps, also known as quantum channels, take valid quantum states to valid quantum states. It can be seen that unitary channels, $U(\cdot)U^\dagger$, are a special case of CPTP maps.

There exists multiple different mathematical representations for a given quantum channel, defined as follows: a linear map $\mathcal{E} : \mathcal{H}^d \rightarrow \mathcal{H}^{d'}$ is a completely positive trace preserving map if and only if

- **Kraus Representation:** There exists a set of linear operators $\{E_i : \mathcal{H}^d \rightarrow \mathcal{H}^{d'}\}_i$, which for all $\rho \in \mathcal{D}(\mathcal{H})$ satisfies

$$\mathcal{E}(\rho) = \sum_i E_i \rho E_i^\dagger, \quad \text{and} \quad \sum_i E_i^\dagger E_i = \mathbb{I}. \quad (2.14)$$

The operators $\{E_i\}_i$ are called *Kraus Operators*.

- **Stinespring Representation:** There exists an isometry $V : \mathcal{H}^d \rightarrow \mathcal{H}^{d'} \otimes \mathcal{H}^{d_E}$ such that

$$\mathcal{E}(\rho) = \text{tr}_E[V\rho V^\dagger], \quad (2.15)$$

where V is an isometry if $V^\dagger V = \mathbb{I}$.

- **Choi-Jamiołkowski Representation:** There exists a matrix $\mathcal{J}_{AB}^\mathcal{E} \in \mathcal{D}(\mathcal{H}_A^d \otimes \mathcal{H}_B^{d'})$ defined as

$$\mathcal{J}_{AB}^\mathcal{E} = (\mathcal{I}_A \otimes \mathcal{E}_B)(|\Phi^+\rangle\langle\Phi^+|_{AB}), \quad (2.16)$$

where $|\Phi^+\rangle := d^{-1/2} \sum_i |ii\rangle$ is the maximally entangled state in $\mathcal{H}_A^d \otimes \mathcal{H}_B^d$, and the following two conditions hold

$$\mathcal{J}_{AB}^\mathcal{E} \geq 0 \quad \text{and} \quad d \text{tr}_B[\mathcal{J}_{AB}^\mathcal{E}] = \mathbb{I}/d. \quad (2.17)$$

The channel $\mathcal{N}$ can be recovered from $\mathcal{J}_{AB}^\mathcal{E}$ as

$$\mathcal{E}(\rho) = \text{tr}_B[(\rho_A^t \otimes \mathbb{I}_B)\mathcal{J}_{AB}^\mathcal{E}], \quad (2.18)$$

where $(\cdot)^t$ is the transpose in the basis used to define $|\Phi^+\rangle$.

Dynamics at a Glance

Definition 2.4.1 (Quantum Channels). *Dynamics in quantum information theory are modelled by quantum channels: completely positive, trace preserving, linear maps.*

2.5 Supermaps

All quantum states, measurements and channels can be generalised via the notion of *quantum operations*, which are linear, completely positive, trace non-increasing maps.

Definition 2.5.1 (Quantum Operations). A set of linear maps $\{E_i : \mathcal{H}_{\text{in}} \rightarrow \mathcal{H}_{\text{out}}\}_i$, is a quantum operation if

$$\mathcal{E}(\cdot) = \sum_i E_i(\cdot)E_i^\dagger, \text{ and } \sum_i E_i^\dagger E_i \leq \mathbb{I}, \quad (2.19)$$

such that $\mathcal{E} : \mathcal{H}_{\text{in}} \rightarrow \mathcal{H}_{\text{out}}$

The definition of quantum operations can be seen to be very similar to the Kraus decomposition of a quantum channel given in Eq. (2.14). And, indeed, if $\sum_i E_i^\dagger E_i = \mathbb{I}$, i.e., the bound is saturated, then the quantum operation $\mathcal{E}$ becomes trace-preserving and is therefore a quantum channel. Moreover, given a state $\rho = \sum_i p_i |\psi_i\rangle\langle\psi_i| \in \mathcal{D}(\mathcal{H}^d)$, this can be described by a quantum operation acting on a one-dimensional Hilbert space with operators $E_i : \mathcal{H}^1 \rightarrow \mathcal{H}^d = \sqrt{p_i}|\psi_i\rangle$. Lastly, a measurement can be seen to be a collection of quantum operations that sum to give a quantum channel, with the output states and probabilities of outcomes being calculated as detailed in Section 2.3.

Quantum supermaps are then linear, completely positive maps of quantum operations to quantum operations [40]. Linearity ensures that the maps are consistent with the probabilistic nature of quantum mechanics; complete positivity ensures that a supermap is still valid when only applied locally to part of a larger space. A supermap mapping a quantum channel to a quantum channel is specifically referred to as a *super-channel*.

The following result about the physical implementation of supermaps will be used throughout this thesis.

Theorem 2.5.1. Let $\mathcal{E} : \mathcal{H}_{\text{in}} \rightarrow \mathcal{H}_{\text{out}}$ and $\mathcal{T} : \mathcal{K}_{\text{in}} \rightarrow \mathcal{K}_{\text{out}}$ be quantum operations. There exists a supermap $\Pi : \mathfrak{D} \rightarrow \mathfrak{D}$, where $\mathfrak{D}$ is the set of all quantum operations, such that $\mathcal{T} = \Pi(\mathcal{E})$, if and only if there exists quantum channels $\mathcal{N} : \mathcal{K}_{\text{in}} \rightarrow \mathcal{H}_{\text{in}} \otimes \mathcal{H}_A$ and $\mathcal{M} : \mathcal{H}_{\text{out}} \otimes \mathcal{H}_A \rightarrow \mathcal{K}_{\text{out}}$ such that

$$\mathcal{T} = \Pi(\mathcal{E}) = \mathcal{M} \circ (\mathcal{E} \otimes \mathcal{I}) \circ \mathcal{N} \quad (2.20)$$

where $\mathcal{E} \otimes \mathcal{I} : \mathcal{H}_{\text{in}} \otimes \mathcal{H}_A \rightarrow \mathcal{H}_{\text{out}} \otimes \mathcal{H}_A$ with $\mathcal{H}_A$ an ancillary system, and $\mathcal{I}(\cdot) = \mathbb{I}(\cdot)\mathbb{I}$ is the identity channel. The channels $\mathcal{N}, \mathcal{M}$ will be referred to as pre-processing and post-processing channels respectively.

See [40] and the accompanying thesis for a proof.

Theorem 2.5.1 states that super maps can be implemented using an ancillary system, and a pre-processing and post-processing channel.

2.6 Norms

Norms are used ubiquitously throughout quantum information theory, and will be frequently used throughout this thesis. They are defined as follows, where $\mathbb{R}^+$ denotes the set of non-negative real numbers.

Definition 2.6.1 (Norm). Given a Hilbert space $\mathcal{H}$, a functional $\|\cdot\| : \mathcal{L}(\mathcal{H}_{\text{in}} \rightarrow \mathcal{H}_{\text{out}}) \rightarrow \mathbb{R}^+$ is a norm for all operators $A, B \in \mathcal{L}(\mathcal{H}_{\text{in}} \rightarrow \mathcal{H}_{\text{out}})$ if it satisfies the following conditions for $\lambda \in \mathbb{C}$:

1. $\|A\| \geq 0$ where $\|A\| = 0$ if and only if $A = 0$
 - Positivity
2. $\|\lambda A\| = |\lambda| \|A\|$
 - Scaling
3. $\|A + B\| \leq \|A\| + \|B\|$
 - Triangle Inequality

Norms can be thought of as functionals that measure the size of a linear operator.

2.6.1 Example Norms

Definition 2.6.2 (Schatten p -norm). *The Schatten p -norm of $A \in \mathcal{L}(\mathcal{H}_{\text{in}} \rightarrow \mathcal{H}_{\text{out}})$ is*

$$\|A\|_p := \left[\text{tr}[(A^\dagger A)^{\frac{p}{2}}] \right]^{\frac{1}{p}} \quad (2.21)$$

We will mainly use a few special cases of the Schatten p -norm, such as the $p = 1$ norm which is also referred to as the trace norm,

$$\|A\|_1 := \text{tr}[\sqrt{A^\dagger A}] = \sum_i \mu_i(A), \quad (2.22)$$

where $\{\mu_i(A)\}_i$ are the singular values of A such that $\|A\|_1$ is the sum over the singular values of A . Moreover, we will use the $p = \infty$ norm, also known as the infinity-norm or operator norm,

$$\|A\|_\infty := \lim_{p \rightarrow \infty} \|A\|_p = \max_i \{\mu_i(A)\}_i, \quad (2.23)$$

meaning that $\|A\|_\infty$ is the maximum singular value of A .

Another type of norm that we will make frequent use of are the Ky-Fan norms.

Definition 2.6.3 (Ky-Fan Norm). *The k th Ky-Fan norm of $A \in \mathcal{L}(\mathcal{H}_{\text{in}} \rightarrow \mathcal{H}_{\text{out}})$ is*

$$\|A\|_{k\text{-KY}} := \sum_{i=1}^k \mu_i^\downarrow(A), \quad (2.24)$$

where $\{\mu_i^\downarrow(A)\}_i$ are the singular values of A ordered in non-increasing order. The k th Ky-Fan norm is therefore the sum of the k largest singular values of A .

The singular values of an operator do not change under a pre-processing and post-processing unitary i.e., the singular values of A and UAV , where U and V are unitary operators, are the same. Therefore, norms that depend only on the singular values of an operator are unitarily invariant under all pre-processing and post-processing unitaries. For example,

$$\|A\|_1 = \|UAV\|_1 \quad \forall U, V \text{ such that } U \text{ and } V \text{ are unitary operators}, \quad (2.25)$$

as the $p = 1$ norm is just the sum of the singular values of A .

2.6.2 Dual Norms

Given a specific norm, the dual norm provides a method of creating another norm from it [41]. As the name suggests, dual norms act on the dual space of $\mathcal{H}$, that is, they act on the space of linear functionals on $\mathcal{H}$. Colloquially, the dual norm of a functional in $\mathcal{H}^*$ for the norm $\|\cdot\|$, is a measure of the maximum amount the functional can increase an element within the unit ball of $\|\cdot\|$. Whilst dual norms are norms on the dual space, through the Riesz Representation Theorem [42] all elements of the dual space $\mathcal{H}^*$ can be identified with elements of $\mathcal{H}$, allowing dual norms to be expressed only in terms of elements on $\mathcal{H}$.

they can be defined in terms of elements in or on $\mathcal{H}$ only, and without reference to elements within the dual space $\mathcal{H}^*$.

Definition 2.6.4 (Dual Norms). *Consider a vector space $\mathcal{H}$ with a norm $\|\cdot\|$. The dual norm of $\|\cdot\|$ for $Y \in \mathcal{L}(\mathcal{H}_{\text{in}} \rightarrow \mathcal{H}_{\text{out}})$ is*

$$\|Y\|^* := \sup_X \{ |\text{tr}[X^\dagger Y]| : \|X\| \leq 1, X \in \mathcal{L}(\mathcal{H}_{\text{out}} \rightarrow \mathcal{H}_{\text{in}}) \}. \quad (2.26)$$

Note, in general the dual norm is defined with respect to an arbitrary inner product; we have here used the Hilbert-Schmidt form of the inner product, which is the typically used inner product in quantum

information.

Finally, it is noted that taking the dual of a dual norm will return the original norm. This provides a method for stating an alternative but equivalent expression for a norm,

$$\|Y\| = \sup_X \{ |\text{tr}[X^\dagger Y]| : \|X\|^* \leq 1, X \in \mathcal{L}(\mathcal{H}_{\text{out}} \rightarrow \mathcal{H}_{\text{in}}) \}. \quad (2.27)$$

For example, as the dual of the one-norm is the infinity norm [41], an alternative but equivalent expression for the one-norm is

$$\|Y\|_1 = \sup_X \{ |\text{tr}[X^\dagger Y]| : \|X\|_\infty \leq 1, X \in \mathcal{L}(\mathcal{H}_{\text{out}} \rightarrow \mathcal{H}_{\text{in}}) \}. \quad (2.28)$$

2.7 Entropy Functions

Entropy functions are used throughout quantum information theory, often to bound an agent's ability to succeed in various operational tasks. Broadly, standard entropy functions describe the level of uncertainty in a probability distribution [8], whilst relative entropies determine the similarity between distributions [43]. All entropies are functionals from one or two elements in $\mathcal{H}$ to the real numbers, and an axiomatic approach to their development has lead to a plethora (infinite, really) number of different entropy functions [44, 45]. Although, it is when provided with an operational meaning that these functions tend to become relevant. We here detail a few entropy functions used in this thesis.

Definition 2.7.1 (von Neuman Entropy). *The von Neuman Entropy of a state $\rho \in \mathcal{D}(\mathcal{H})$ is*

$$S(\rho) := -\text{tr}[\rho \log(\rho)], \quad (2.29)$$

where logs will be considered to be base 2 unless specified otherwise. This is a generalisation of the Shannon entropy to quantum systems; if evaluated for a state in its eigenbasis, $\rho = \sum_i \lambda_i |\lambda_i\rangle\langle\lambda_i|$, then the von Neuman Entropy reduces to the Shannon entropy for the probability distribution over ρ 's eigenstates,

$$S(\rho) = -\sum_i \lambda_i \log(\lambda_i). \quad (2.30)$$

Definition 2.7.2 (Quantum Relative Entropy (Umegaki Entropy [46])). *The quantum relative entropy between the state $\rho \in \mathcal{D}(\mathcal{H})$ and $\sigma \geq 0$, where σ is a positive semidefinite operator such that $\text{supp}(\rho) \subseteq \text{supp}(\sigma)$, is*

$$D(\rho||\sigma) := \text{tr}[\rho \log(\rho) - \rho \log(\sigma)]. \quad (2.31)$$

If $\text{supp}(\rho) \not\subseteq \text{supp}(\sigma)$ then $D(\rho||\sigma) = \infty$ by definition.

The quantum relative entropy is seen as one of the most fundamental quantities in quantum information given it acts as a parent entropy, meaning other entropy functions can be recovered from it. For example,

$$-D(\rho||\mathbb{I}) = S(\rho) = -\text{tr}[\rho \log(\rho)], \quad (2.32)$$

meaning that the von Neuman entropy can be recovered from the relative entropy by inputting the identity in the second argument. Moreover, the quantum mutual information — a measure of the correlation between two systems A and B , can be found from the quantum relative entropy as

$$I(A|B) = D(\rho_{AB}||\rho_A \otimes \rho_B) = S(\rho_A) + S(\rho_B) - S(\rho_{AB}), \quad (2.33)$$

where $\rho_A = \text{tr}_B[\rho_{AB}]$ and $\rho_B = \text{tr}_A[\rho_{AB}]$.

Another relative entropy function we will make use of is the max-relative entropy, introduced in [47].

Definition 2.7.3 (Max Relative Entropy). *The max relative entropy between the state $\rho \in \mathcal{D}(\mathcal{H})$ and $\sigma \geq 0$, where σ a positive semidefinite operator such that $\text{supp}(\rho) \subseteq \text{supp}(\sigma)$, is*

$$D_{\max}(\rho||\sigma) := \log \min\{\lambda : \lambda \geq 1, \rho \leq \lambda\sigma\}. \quad (2.34)$$

If $\text{supp}(\rho) \not\subseteq \text{supp}(\sigma)$ then $D(\rho||\sigma) = \infty$ by definition.

The max relative entropy has been shown to have several operational interpretations [48–51], with another provided in this thesis (see Corollary 8.2.3). It can also be shown to be related to the quantum relative entropy as

$$D(\rho||\sigma) \leq D_{\max}(\rho||\sigma). \quad (2.35)$$

Both the quantum relative entropy and the max relative entropy have the following useful properties (note, we have stated the properties in terms of $D(\cdot||\cdot)$ only for clarity):

- $D(\rho||\sigma) \geq 0$, where $D(\rho||\sigma) = 0$ if and only if $\rho = \sigma$.
- $D(U\rho U^\dagger||U\sigma U^\dagger) = D(\rho||\sigma)$ where U is a unitary.
- $D(\rho||\sigma) \geq D(\mathcal{E}(\rho)||\mathcal{E}(\sigma))$ where $\mathcal{E} \in \text{CPTP}$,
 - This is known as the data processing inequality. Colloquially, it says that two states cannot become more distinguishable under deterministic linear dynamics.

The above properties also apply to other generalisations of relative entropies [45].

2.8 Stochastic Matrices and Majorisation

A stochastic matrix is a square matrix where all elements are non-negative real numbers ($\mathbb{R}^+$), and each row, column, or both rows and columns, are normalised. They act on vectors of probability distributions, mapping them to other probability distributions, and detail how the probabilities change under some stochastic process. Each element of the matrix represents the probability of transitioning from some initial state to a final state.

A stochastic matrix is said to be left stochastic if each row sums to one, right stochastic if each column sums to one, and doubly stochastic if each row and column sums to one. Doubly stochastic matrices play an important role in determining how mixed one probability distribution is as compared to another. Formally, consider two vectors of probability distributions:

$$\begin{aligned} \mathbf{p} &= (p_1, p_2, \dots, p_d) : p_i \in \mathbb{R}^+ \forall i \text{ and } \sum_i p_i = 1 \\ \mathbf{q} &= (q_1, q_2, \dots, q_d) : q_i \in \mathbb{R}^+ \forall i \text{ and } \sum_i q_i = 1. \end{aligned} \quad (2.36)$$

The probability distribution $\mathbf{q}$ is more mixed than the probability distribution $\mathbf{p}$ if there exists a doubly stochastic matrix M such that

$$\mathbf{q} = M\mathbf{p}, \text{ such that } \sum_i M_{ij} = 1, \sum_j M_{ij} = 1, M_{ij} \geq 0 \forall i, j, \quad (2.37)$$

where M_{ij} are the matrix elements of M . If there exists no such M between any two distributions, their ‘mixedness’ is considered to be incomparable. All doubly stochastic matrices have the maximally mixed probability vector as a fixed point, meaning that

$$\mathbf{I}M = \mathbf{I}, \quad M\mathbf{I}^t = \mathbf{I}^t, \quad (2.38)$$

where $\mathbf{I} := (1, 1, \dots, 1)$ is the all-one vector. Put alternatively, this means that doubly stochastic matrices act invariantly on the vector representing the maximally mixed distribution.

Moreover, the Birkhoff-von Neumann Theorem states that all doubly stochastic matrices can be written as a convex combination of permutations.

Theorem 2.8.1 (Birkhoff-von Neumann Theorem). *A real valued $n \times n$ square matrix M is a doubly stochastic matrix if and only if there exists a set of numbers $\{p_i\}_i$, where $p_i \geq 0 \forall i$ and $\sum_i p_i = 1$, such that*

$$M = \sum_i p_i P_i, \quad P_i \in \mathbb{P}_n \forall i \quad (2.39)$$

where $\mathbb{P}_n$ is the set of $n \times n$ permutation matrices.

See [52] for a proof of this theorem.

From Theorem 2.8.1 it can be understood why doubly stochastic matrices are considered to capture the notion of mixedness between probability vectors: the existence of a double stochastic matrix between two probability distributions means that the output distribution can be generated by probabilistic relabelling the input distribution. Put alternatively, the output distribution is equal to the input distribution up to some unknown relabelling of the possible outcomes. While each individual permutation is reversible, the overall transformation becomes irreversible when the applied permutation is unknown. Consequently, the output distribution can be interpreted as arising from a loss of information about which relabelling occurred. Equivalently, the existence of a doubly stochastic matrix relating two probability distributions implies that the output distribution can be generated by probabilistic relabelling of the input. As a result, doubly stochastic matrices model classical unbiased noisy channels: they preserve the maximally mixed distribution and therefore cannot introduce any preferred outcome, but act only to degrade the existing structure in the distribution.

2.8.0.1 Majorisation

Majorisation has found many uses throughout quantum information [53], quantum thermodynamics [17, 18] and beyond [54–56]. It is defined as follows.

Definition 2.8.1 (Majorisation). *Let $\mathbf{p}$ and $\mathbf{q}$ be d -dimensional vectors of probability distributions, as defined in Eq. (2.36). We say $\mathbf{p}$ majorises $\mathbf{q}$, denoted $\mathbf{p} \succ \mathbf{q}$, if*

$$\sum_{i=1}^k p_i^\downarrow \geq \sum_{i=1}^k q_i^\downarrow \quad \forall k \in [1, d], \quad (2.40)$$

where $p_i^\downarrow$ and $q_i^\downarrow$ are the components of $\mathbf{p}, \mathbf{q}$ respectively, ordered in non-increasing order.

Majorisation has been shown to be related to doubly stochastic matrices through the following theorem.

Theorem 2.8.2 (Hardy-Littlewood-Polya Theorem [54]). *Let $\mathbf{p}$ and $\mathbf{q}$ be d -dimensional vectors of probability distributions, as defined in Eq. (2.36). The following two conditions are equivalent*

$$\mathbf{p} \succ \mathbf{q} \iff \mathbf{q} = M\mathbf{p}, \quad M \in \text{Doubly stochastic}. \quad (2.41)$$

See [52], or the appendices of [19], for a proof of this theorem.

Using the Hardy-Littlewood-Polya Theorem it can be determined if one probability vector is more mixed than another by seeing if one majorises the other.

2.8.0.2 Unistochastic Matrices

Given some unitary matrix, a doubly stochastic matrix can always be found by taking the square magnitude of each matrix element. The set of doubly stochastic matrices that can be created in this way are called the unistochastic matrices, and they are a strict subset of the set of doubly stochastic matrices for matrices greater than 2×2 .

Definition 2.8.2 (Unistochastic Matrices). *A doubly stochastic matrix M is said to be unistochastic if there exists a unitary operator U such that $M_{ij} = |\langle i|U|j\rangle|^2$.*

Whilst the set of unistochastic matrices includes all permutations, it is not convex for matrices greater than 2×2 . Moreover, we note that given a unitary operator, the doubly stochastic matrix one gets from the unitary operator will depend on the basis in which it is decomposed.

2.9 The Discrete Weyl Operators

The discrete Weyl operators are a set of unitary operators that form a complete basis for the space of complex matrices. Within quantum information, they are used to define discrete Wigner functions [57], study non-stabilizerness [24], and define basis of bipartite maximally entangled states, among other things.

Definition 2.9.1 (The discrete Weyl operators). *The discrete Weyl operators are defined as*

$$W_{a,b} = X^a Z^b \in \mathcal{L}(\mathcal{H}^d), \quad a, b \in \{0, 1, \dots, d-1\}, \quad (2.42)$$

where

$$X = \sum_{j=0}^{d-1} |j+1 \bmod d\rangle \langle j|, \quad Z = \sum_{j=0}^{d-1} \Omega^j |j\rangle \langle j|, \quad \Omega = e^{\frac{2\pi i}{d}}, \quad (2.43)$$

such that the (a, b) th discrete Weyl operator is

$$W_{a,b} = \sum_c \Omega^{bc} |c+a \bmod d\rangle \langle c|. \quad (2.44)$$

The X and Z in the definition of the discrete Weyl operators can be thought of as generalised Pauli-operators in higher dimensional spaces. They are also referred to as shift and clock operators. We note that for qubit systems, the discrete Weyl operators are

$$W_{0,0} = \mathbb{I}, \quad W_{0,1} = \sigma_z, \quad W_{1,0} = \sigma_x, \quad W_{1,1} = -i\sigma_y, \quad (2.45)$$

where $\sigma_x, \sigma_y, \sigma_z$ are the qubit X, Y and Z Pauli-operators respectively.

The following identities for how the discrete Weyl operators are transformed via transposition, adjoint, and multiplication [10] are used in various places throughout this thesis:

$$\begin{aligned} (W_{a,b})^t &= \Omega^{-ab} W_{-a,b}, \\ (W_{a,b})^\dagger &= \Omega^{ab} W_{-a,-b}, \\ W_{a,b} W_{c,d} &= \Omega^{bc} W_{a+c, b+d} = \Omega^{bc-ad} W_{c,d} W_{a,b}. \end{aligned} \quad (2.46)$$

These trace and inner product identities are also used:

$$\text{tr}[W_{a,b}] = d \delta_{a,0} \delta_{b,0}, \quad \text{tr}[W_{a,b}^\dagger W_{c,d}] = d \delta_{a,c} \delta_{b,d}, \quad (2.47)$$

where $\delta_{a,b}$ is the Kronecker delta function.

We can now formalise the following two facts about the discrete Weyl operator mentioned in the introduction:

Theorem 2.9.1 (Complete Basis). *The set*

$$\left\{ \frac{1}{\sqrt{d}} W_{a,b} : a, b \in \{0, 1, \dots, d-1\} \right\}, \quad (2.48)$$

form a complete orthonormal basis for the space of $d \times d$ complex operators. Hence, if A is a $d \times d$ complex operator there always exists a set of complex coefficients $p_{a,b} \in \mathbb{C}$ such that

$$A = \frac{1}{\sqrt{d}} \sum_{a,b=0}^{d-1} p_{a,b} W_{a,b}. \quad (2.49)$$

See [10] for a proof of this theorem.

Theorem 2.9.2 (Generating Maximally Entangled Basis). *Let $|\Phi_{00}^+\rangle = d^{-1/2} \sum_{i=0}^{d-1} |ii\rangle \in \mathcal{H}^d$. The following set is an orthonormal basis for the space $\mathcal{H}^d \otimes \mathcal{H}^d$:*

$$\{|\Phi_{a,b}^+\rangle = (\mathbb{I} \otimes W_{a,b})|\Phi_{00}^+\rangle : a, b \in \{0, 1, \dots, d-1\}\} \quad (2.50)$$

See [58] for a proof.

We note that this holds for any basis of unitaries that are orthogonal under the Hilbert Schmit inner product.

3

Quantum Information Processing Tasks

In this chapter, a few fundamental quantum information processing tasks considered throughout this thesis are detailed. These are key primitives that are utilised in many places throughout the broad quantum information sciences. Namely, we consider quantum state discrimination, state exclusion and superdense coding. This is not an exhaustive list, and for more detail on some of these tasks and other vital quantum information processing primitives, the reader can consult [39].

3.1 Quantum State Discrimination

In classical physics, any two distinct states of a physical system are distinguishable — there always exists a measurement that can be made to tell the systems apart. In quantum mechanics, this is not generally the case, with the field of quantum state discrimination studying the fundamental limitations on our ability to distinguish quantum states [59]. As in many areas of information theory, it has proved fruitful to our understanding of distinguishability to phrase it in terms of a game.

Task 3.1.1 (Quantum State Discrimination). *Let $(p_x, \rho_x)_x$ be an ensemble, such that one gets the state ρ_x with probability p_x . Assume this ensemble is known to both a referee and a player. The referee then takes a state ρ_y from this ensemble with probability p_y , and gives it to the player. The player makes a measurement on the state and outputs an index g .*

If $g = y$ the player succeeds at the task, if $g \neq y$ the player fails at the task.

Quantum state discrimination therefore asks the player to correctly identify the state they have been given by the referee. To succeed in this task, one should typically consider performing a measurement with $N \geq |(p_x, \rho_x)_x|$ different possible outcomes, where $|(p_x, \rho_x)_x|$ is the number of different states in the ensemble, such that to each state a different measurement outcome can be assigned. As the post measurement state is not considered, this measurement can be modelled by a POVM (see Definition 2.3.3). To the state ρ_g from the ensemble a POVM element T_g is assigned, such that if the player gets the outcome associated to T_g they output the index g . The probability of outputting the index g given the player has been given the state ρ_g is then $\text{tr}[T_g \rho_g]$. This is therefore the probability that the player is successful in the task given the referee gave them ρ_g .

There exists multiple different ways in which one could claim to have performed optimally in the task of quantum state discrimination. If one wants to maximise the success probability, which is equivalent to minimising the average probability of error, the following optimisation problem can be solved,

$$P_{\text{suc}} = \max_{\{T_x\}_x} \sum_x p_x \text{tr}[T_x \rho_x] \quad (3.1)$$

$$\text{Subject to: } T_x \geq 0 \forall x, \sum_x T_x = \mathbb{I}, \quad (3.2)$$

such that all POVMs are optimised over [60]. This is a Semidefinite Program (SDP), meaning there exists efficient classical methods for finding the optimal solution [61]. In general there will not be a unique POVM that achieves this optimum [59], and the player can therefore choose the measurement strategy best suited to their experimental implementation.

If $P_{\text{suc}} = 1$ then the player is always able to identify which state they have been given by the referee with certainty. This is known to be possible if and only if the ensemble consists of mutually orthogonal states.

Theorem 3.1.2. $P_{\text{suc}} = 1$ if and only if $\text{tr}[\rho_y \rho_z] = \delta_{y,z} \forall \rho_y, \rho_z$ in the ensemble $(p_x, \rho_x)_x$.

See [39] for a proof.

An alternative strategy for claiming optimality is unambiguous state discrimination [59, 62, 63], where a measurement with $(N + 1)$ outcomes is used if $N = |(p_x, \rho_x)_x|$. The aim here is that the index g is output if and only if $g = x$, meaning the player outputs an index if and only if they are certain it relates to the state that they were given by the referee. This comes at the expense of the additional measurement outcome that gathers all the uncertain outcomes, with the player outputting that they are uncertain what state they have if they get this outcome, i.e., if the player gets the $(N + 1)$ th outcome they output ‘don’t know’ rather than an index.

It will not always be possible to find such a POVM, with characterisations for what sets of states this is possible for being detailed in [62, 63]. As with the maximum success probability, success in unambiguous state discrimination can be cast as an SDP [59]. Although, success now means minimising the probability of the player outputting ‘don’t know’.

For additional alternative strategies for state discrimination, see [59].

3.2 Quantum State Exclusion

The task of quantum state exclusion [64–66] arises from a natural relaxation of the task of quantum state discrimination — particularly in scenarios where deterministic quantum state discrimination is impossible. In this task, a weaker aim is considered, with the player instead being asked to determine states from the set they *have not* been given by the referee.

Task 3.2.1 (Quantum k -State Exclusion). Let $(p_x, \rho_x)_x$ be an ensemble, such that one gets the state ρ_x with probability p_x . Assume this ensemble is known to both a referee and a player. The referee then takes a state ρ_y from the ensemble with probability p_y and gives it to the player. The player makes a measurement on the state and outputs a set of indices $\mathbb{G}$ of length k , where $k \leq |(p_x, \rho_x)_x| - 1$.

If $y \notin \mathbb{G}$ the player succeeds at the task, if $y \in \mathbb{G}$ the player fails at the task.

Quantum state exclusion therefore asks the player to correctly identify k -states that they have *not* been given by the referee.

3.2.1 1-State Exclusion

If the player outputs a set containing only a single index, they are performing 1-state exclusion. If able to achieve this with certainty, we say they are able to perform *conclusive 1-state exclusion*. For this to be possible, there must exist a POVM $\{T_x\}_{x=1}^N$ such that

$$\text{tr}[T_x \rho_x] = 0 \quad \forall x \in \{1, \dots, N\}, \quad (3.3)$$

where $N = |(p_x, \rho_x)_x|$. Then, if the player gets the measurement outcome associated to a POVM element $T_g \in \{T_x\}_{x=1}^N$ they can output g knowing with certainty that the referee could not have given them ρ_g . In general, a POVM may not exist that can perform conclusive 1-state exclusion. Instead, state exclusion can be performed with some error, with this minimum error probability again being the solution to an SDP:

$$P_{\text{err}} = \min_{\{T_x\}_x} \sum_x \text{tr}[T_x \rho_x] \quad (3.4)$$

$$\text{Subject to: } T_x \geq 0 \quad \forall x, \quad \sum_x T_x = \mathbb{I}, \quad (3.5)$$

where P_{err} is the probability of making an error when performing exclusion, meaning the probability of outputting the index $g = x$. As above, the set of all POVMs is being optimised over. We note that if $P_{\text{err}} > 0$, then no POVM to perform conclusive 1-state exclusion exists.

3.2.2 k-State Exclusion

If the player instead outputs a set of $k > 1$ indices, $\mathbf{G} = \{g_i\}_{i=1}^k$, they are performing k -state exclusion. As above, if $g_i \neq x \forall g_i \in \mathbf{G}$ with certainty, we call this *conclusive k -state exclusion*. There are $\binom{N}{k}$ different sets of k labels the player could exclude, corresponding to all the different subsets of $\{1, \dots, N\}$ of length k . Performing k -state exclusion therefore means the player must aim to find a POVM with $\binom{N}{k}$ elements such that each measurement outcome allows the player to exclude a different subset of states from $(p_x, \rho_x)_x$ of length k (although, see Section 8.2 for a relaxation of this).

Interestingly, all k -state exclusion tasks can be recast as 1-state exclusion tasks by reformulating the set of states to be excluded [66]. Conceptually, this means all k -state exclusion tasks have a 1-state exclusion task that they are equivalent too, allowing all state exclusion tasks to be studied under the 1-state exclusion framework.

To see this, let $Y_{(N,k)}$ be the set of all subsets of the integers $\{1, 2, \dots, N\}$ of length k . The player aims to measure a POVM on a state ρ_y from the ensemble (p_x, ρ_x) , given to them by the referee, and output a set of labels $Y \in Y_{(N,k)}$ such that $y \notin Y$ i.e., the player identifies a set of k states they were not given. Such a measurement will be a POVM with $a = \binom{N}{k}$ terms as there are a subsets of $\{1, 2, \dots, N\}$ of length k . For each element of this POVM, $S = \{S_l\}_{l=1}^a$, the following equation must hold,

$$\text{tr}[S_Y \rho_y] = 0 \forall y \in Y, \quad (3.6)$$

for conclusive k -state exclusion to be possible. By defining

$$\rho_Y = \sum_{y \in Y} \tilde{\rho}_y, \quad \tilde{\rho}_y = p_y \rho_y, \quad (3.7)$$

the conclusive k -state exclusion task can be expressed as the following set of equations

$$\text{tr}[S_Y \rho_Y] = 0 \forall Y \in Y_{(N,k)}. \quad (3.8)$$

This is now the same form as the conditions for performing conclusive 1-state exclusion, as seen in Eq. (3.3). This is then the equivalent 1-state exclusion task to the k -state exclusion task.

As above, it is not always possible to perform conclusive k -state exclusion. Using the equivalent 1-state exclusion task, the minimum error probability of performing k -state exclusion can also be expressed as the solution to an SDP:

$$P_{\text{err}}^k = \min_{\{S_Y\}_Y} \sum_Y \text{tr}[S_Y \rho_Y] \quad (3.9)$$

$$\text{Subject to: } \sum_Y S_Y = \mathbb{I}, \quad S_Y \geq 0 \forall i, \quad (3.10)$$

where, as above, $P_{\text{err}}^k > 0$ if no POVM to perform conclusive k -state exclusion exists.

We note that if the player can perform k -state exclusion, they can also perform $0 \leq k' \leq k$ state exclusion by just output only a subset of $\mathbf{G}$. On the other hand, if the player cannot perform k -state exclusion, they also cannot perform $k' > k$ exclusion either.

3.2.3 Conclusive State Discrimination as $(N - 1)$ -State Exclusion

For an ensemble with N states, performing conclusive $(N - 1)$ -state exclusion is equivalent to performing deterministic quantum state discrimination: knowing for certain $N - 1$ states that you *do not* have is equivalent to knowing which state you *do* have. Put alternatively, $P_{\text{err}}^{N-1} = 0 \iff P_{\text{suc}} = 1$, from which the following modification of Theorem 3.1.2 can be made.

Theorem 3.2.2. $P_{\text{err}}^{N-1} = 0$ if and only if $\text{tr}[\rho_y \rho_z] = \delta_{y,z} \forall \rho_y, \rho_z$ in the ensemble $(p_x, \rho_x)_x$, where $N = |(p_x, \rho_x)_x|$.

Conclusive state discrimination can therefore be considered to be a special case of conclusive k -state exclusion. How the two tasks are related in the non-conclusive case, and for $k < N - 1$, remains an active area of research.

3.2.4 Sub-Channel Exclusion

A closely related task is sub-channel exclusion [4, 67]. Consider a collection of completely-positive trace-non-increasing linear maps, $\Xi = \{\zeta_x\}_{x=1}^N$, such that $\sum_{x=1}^N \zeta_x$ is a channel. Each map ζ_x is called a *sub-channel*, which is some quantum operation as defined in Definition 2.5.1. The collection of sub-channels Ξ is then called an *instrument*. In sub-channel exclusion, a player has a reference state ρ that they send to the referee. The referee then applies a sub-channel to ρ , where ζ_x is applied with some probability p_x . The output state is then returned to the player. The player measures a POVM on the state and outputs a label $g \in \{1, \dots, N\}$. They succeed if they output a label of a sub-channel that was not applied. As before, the player can output the label of a sub-channel not applied with certainty, they can output k labels, or they can output k labels with certainty. A special case of sub-channel exclusion concerns a collection of quantum channels that are applied probabilistically i.e., each sub-channel ζ_x is CPTP map that is applied to the reference state with probability p_x .

One can generalise the task of sub-channel exclusion further by letting the reference state now be some arbitrary bipartite state. The player then sends one of subsystems to the referee, who probabilistically applies a sub-channel, before returning it to the player. The player now makes a joint measurement on the whole system and then outputs some number of indices representing sub-channels they believe were not applied. Using this framework, it can be explored whether the existence of entanglement, or more generally correlations, between the subsystems can be used to enhance the players ability to perform sub-channel exclusion.

Finally, we note that sub-channel discrimination tasks [68] can be build in the same way.

3.3 Superdense Coding

Superdense coding is a quantum enhanced communication task [69]. The aim is for two spatially separated parties, Alice and Bob, to communicate one of a number of classical messages to each other by encoding the messages into a bipartite entangled quantum state. Before detailing the protocol, we consider the classical case — where we allow Alice and Bob to communicate using only states that are diagonal in some basis, and do not allow them to generate any coherence during the protocol.

3.3.1 Classical Communication

Assume that Alice would like to be able to communicate one of the d messages $\{a\}_{a=0}^{d-1}$ to Bob with certainty. To achieved this, they agree to assign each of the messages a physical state from the set $\{\rho_a\}_{a=0}^{d-1}$. If Alice want to send Bob the message $x \in \{a\}_{a=0}^{d-1}$, she then prepares the state ρ_x , and sends it to Bob. Assuming the channel from Alice to Bob is noiseless, this leaves Bob with the state ρ_x . To decode the message, Bob must then be able to tell which of the states from the set $\{\rho_a\}_{a=0}^{d-1}$ he has in his possession.

By phrasing communication in this way, it can be seen to be a state discrimination task. Namely, Alice — the referee — picks a state from the ensemble $(d^{-1}, \rho_a)_a$, where we assume that Alice will send each of the messages with equal probability. She then gives the state to Bob — the player. Bob correctly decodes the message, or succeeds in the state discrimination task, if he is able to correctly identify the state in his possession. As detailed in Theorem 3.1.2, Bob is only able to achieve this with certainty if all the states in which Alice's messages are encoded are orthogonal.

To send one of d messages with certainty, Alice must therefore send Bob a state supported within at least a d -dimensional Hilbert space, as this will ensure the existence of at least d -orthogonal states. Consider using the orthogonal set of states $\{|a\rangle\langle a| : |a\rangle\langle a| \in \mathcal{D}(\mathcal{H}^d)\}_{a=0}^{d-1}$, which is just some basis of $\mathcal{H}^d$.

Bob can then measure the non-degenerate observable

$$O_C = \sum_a \lambda_a |a\rangle\langle a|, \quad (3.11)$$

where if he gets the outcome λ_x he knows Alice encoded the message x with certainty (he knows he had the state $|x\rangle\langle x|$ with certainty). In total, Alice is able to deterministically send Bob one of d messages if using a classical system that has at least d perfectly distinguishable degrees of freedom.

3.3.2 Entanglement Enhanced Communication

We now assume that Alice would like to be able to communicate one of the d^2 messages $\{a\}_{a=0}^{d^2-1}$ to Bob with certainty, whilst still restricting herself to sending Bob a state supported within a d -dimensional Hilbert space. As before, Alice and Bob could agree to assign each of the messages a physical state from the set $\{\rho_a : \rho_a \in \mathcal{D}(\mathcal{H}^d)\}_{a=0}^{d^2-1}$ and perform a state discrimination task. However, there does not exist d^2 orthogonal states within this d -dimensional Hilbert. Hence, according to Theorem 3.1.2, they would not be able to communicate these d^2 messages with certainty.

Alice and Bob can therefore only achieve this task if they are given access to additional resources. Consider instead that they share a maximally entangled state of local dimension d , $|\Phi^+\rangle = d^{-1/2} \sum_i |ii\rangle$. Now, to each message $x \in \{a\}_{a=0}^{d^2-1}$, Alice and Bob can agree to assign a tuple (c, e) , where each tuple represents one of the d^2 discrete Weyl operators, $\{W_{c,e} : c, e \in \{0, 1, \dots, d-1\}\}$. If Alice would like to send the message x to Bob, which is assigned a tuple (a, b) , she then applies the unitary $W_{a,b}$ to her half of the maximally entangled state. Using Theorem 2.9.2, it can be seen that the state shared between Alice and Bob is now an element of the basis

$$\{|\Phi_{c,e}^+\rangle = (\mathbb{I} \otimes W_{c,e})|\Phi_{00}^+\rangle : c, e \in \{0, 1, \dots, d-1\}\}, \quad (3.12)$$

which therefore consists of d^2 orthogonal states. If Alice then sends her half of the maximally entangled state to Bob, such that Bob now has the whole state, due to Theorem 3.1.2 there will exist a measurement on the joint state that Bob can perform that will allow him to decode the message with certainty (allow him to determine the state he has in his possession with certainty). Specifically, Bob can measure the non-degenerate observable

$$O_{SD} = \sum_{c,e=0}^{d-1} \lambda_{c,e} |\Phi_{c,e}^+\rangle\langle\Phi_{c,e}^+|, \quad (3.13)$$

where if he gets the outcome $\lambda_{a,b}$ then he knows with certainty that Alice encoded the message x . Using this protocol, whilst Alice still sends Bob a state that has only d perfectly distinguishable degrees of freedom, she is able to send Bob one of d^2 messages. However, to achieve this, they needed to pre-share a maximally entangled state.

Task 3.3.1 (Superdense Coding). Assume Alice and Bob shared a bipartite maximally entangled state of local dimension d . Alice encodes one of d^2 messages into her half of the state by applying one of the d^2 discrete Weyl operators. She then send her half of the state to Bob. Bob then jointly measures the non-degenerate observable

$$O_{SD} = \sum_{c,e=0}^{d-1} \lambda_{c,e} |\Phi_{c,e}^+\rangle\langle\Phi_{c,e}^+|. \quad (3.14)$$

From the outcome of this measurement, Bob is able to decode which discrete Weyl operator Alice's performed, and hence decode her encoded message with certainty.

The task of superdense coding can therefore be considered to be an entanglement enhanced sub-channel discrimination tasks, where the sub-channels are the discrete Weyl operators applied with probability d^{-2} .

Quantum Resource Theories

“It is typically said that one man’s rubbish is another man’s treasure; in resource theoretic language, this reads that one agent’s free state is another agent’s resource” — Me

Quantum resource theories are used to systematically quantify certain properties of quantum objects [12, 13]. They achieve this via an operational approach, considering systems subject to some set of physical constraints, through which a rigorous mathematical framework for quantifying a given property can be developed. By tailoring the constraints, quantum resource theories can be applied to properties of both a pragmatic and fundamental nature, meaning they have broad applicability. Thus far they have proved a fruitful way to approach problems concerning a number of different quantum phenomena, with resource theories of entanglement [14], athermality [17, 19, 70], Bell-non locality [20], measurement [21], and non-Gaussianity [71], to name a few.

In this chapter, an overview of the resource theory structure and a discussion of their key features will be given. Following this, two examples of resource theories used throughout this thesis, namely the resource theories of informational non-equilibrium [18] and unspeakable coherence [16], will be detailed. Finally, resource preservability theories [26] will be discussed.

4.1 Overview

Value emerges from limitation. Consider an agent with access to an arbitrarily large amount of some object. To the agent, that object — or anything that it can be traded for — will have no value: they can use it, destroy it, or trade it, and always get more. If, however, the agent’s access to the object is limited, by either fundamental or practical physical constraints, then the object will have value, as it cannot be indefinitely replaced. What is considered valuable to an agent therefore depends on the physical constraints they are subject to. For example, consider that our agent is a typical human with a passion for scuba diving. When on land, they are able to freely consume oxygen; for all practical purposes, its supply is infinite, and it costs nothing to access. Oxygen therefore has no value to our agent when they are on land. Under the ocean, however, where access to oxygen is limited with dire consequences if it runs out, it becomes valuable. Two agents simultaneously subject to different physical constraints will therefore also have different ideas of what is valuable.

Using this notion, one can ascribe a given property value (with respect to some agent) by considering physical constraints that limit access to the property. Within these physically constrained environments, resource theories then aim to quantify the property by considering what can be done operationally with some object possessing it. On the other hand, one can consider what cannot be done without access to it. To see how this works, firstly consider there is a set of objects that our resource theory is defined over. This could be the set of quantum states, measurements or channels. On this set, one then considers a set of *allowed operations* that can be applied. These are the physical operations that can be applied to this set of objects given some imposed physical constraints. If the underlying objects of the resource theory are quantum states, these would be a subset of quantum channels, for example.

These allowed operations then split the set of all objects into two subsets: the set of *free* objects is a subset of the objects that is closed under the allowed operations, all remaining objects are said to be *resourceful*. Typically, the free objects can be arbitrarily generated by the agent within the physical constraints of the system (under allowed operations), justifying their description as free and ensuring they have no value to the agent. Any resourceful object will then have value as they are exactly the objects that cannot be arbitrarily generated within the physical constraints.

Given a property one wants to quantify, they can choose a set of allowed operations such that the set of free objects are those objects that contain no amount of that property. Hence, an object is resourceful only if it contains a non-zero amount of the property being considered. This property is then referred to as a *resource* within these physical constraints. The set of objects that contain the resource and those that do not have therefore been operationally separated via some physical operations, arising from a set of

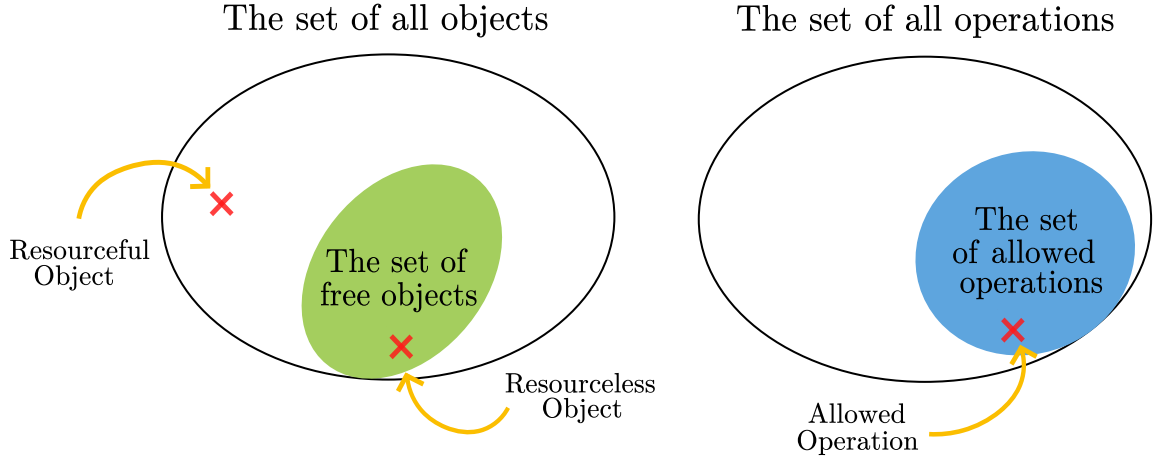


Figure 4.1: A schematic overview of a resource theory, showing the separation of objects into free and resourceful, and the separation of operations on those objects into allowed and not allowed.

physical constraints. One can then aim to quantify how much resource some object A has by comparing it to some object B within the free objects. Alternately, one can ask about the resource content of object A versus object B by considering what useful tasks can be achieved with A that cannot be achieved with B (within these physical constraints). See Fig. 4.1 for a schematic overview of resource theories.

We note that selecting allowed operations and free objects come hand in hand, and that either can be specified first. Moreover, as above, one can instead start at the property they want to quantify and then define either the allowed operations or free objects. Whether one starts at the property, allowed operations, or free objects will depend on the system they are modelling. However, the starting point of developing the theory does have the potential to alter the generated resource theory. For example, defining a set of allowed operations first determines a unique minimal set of free objects: the smallest non-trivial set that is closed under those operations. Although one may choose larger closed sets as the free set, the minimal closed set is uniquely fixed by the choice of allowed operations. Conversely, specifying the free set first determines a unique maximal set of allowed operations: the largest set of operations under which the free set is closed. One may choose smaller subsets of these operations as the allowed operations, provided they still satisfy the closure condition [13].

In this chapter, we detail some features about resource theories used throughout this thesis. We note that the features we discuss do not necessarily apply to all resource theories: part of the power of the resource theory framework is its flexibility. Here, we have focused on features applicable to resource theories used in this thesis. For a comprehensive overview of resource theories see [12].

Unless otherwise specified, and in the interest of clarity, in the remainder of this chapter we will discuss only static resource theories: resource theories defined over the set of quantum states. The allowed operations will then be some subset of the quantum channels.

4.2 Framework and Features

In this section, we will denote the set of free states of a resource theory as $\mathfrak{F} \subseteq \mathcal{D}(\mathcal{H})$ and the set of allowed operations as $\mathfrak{D} \subseteq \text{CPTP}$.

The following set of mathematical conditions are adhered to by most resource theories. To each condition given, a physical motivation is also provided.

Definition 4.2.1 (A Resource Theory). Let $\mathfrak{D} \subseteq \text{CPTP}$ be a set of quantum channels. Let $\mathfrak{F} \subseteq \mathcal{D}(\mathcal{H})$ be the set of quantum states $\mathfrak{D}$ maps to itself.

The tuple $\mathcal{R} = (\mathfrak{D}, \mathfrak{F})$ is a **quantum resource theory**, where $\mathfrak{D}$ is the set of allowed operations and

$\mathfrak{F}$ is the set of free states, if:

1. The identity channel, $\mathcal{I}(\cdot) = \mathbb{I}(\cdot)\mathbb{I}$, is in $\mathfrak{D}$, i.e. $\mathcal{I} \in \mathfrak{D}$.
 - **Physical interpretation:** If we do nothing we should not expect the state to become resourceful.
2. If there exists channels $\mathcal{E} \in \mathfrak{D}$ and $\mathcal{N} \in \mathfrak{D}$, then $\mathcal{E} \circ \mathcal{N} \in \mathfrak{D}$.
 - **Physical interpretation:** You cannot make something resourceful by applying successive allowed operations.
3. The set of allowed operations always includes the $\text{tr}(\cdot)$, i.e. $\text{tr}(\cdot) \in \mathfrak{D}$.
 - **Physical interpretation:** Disregarding a physical system can always be done for free.

Let $\mathfrak{D}_X$ be the allowed operations in some physical system X . The quantum resource theory $\mathcal{R}$ is said to admit a **tensor product structure** if:

4. For any physical systems A and B , and channel $\mathcal{E} \in \mathfrak{D}_A$, the channel $\mathcal{E}_A \otimes \mathcal{I}_B \in \mathfrak{D}_{AB}$, where $\mathcal{I}_B$ is the identity channel on system B .
 - **Physical interpretation:** Allowed operations are completely allowed. We cannot turn an allowed operation into something useful by just embedding the system in some higher dimensional space.
5. For any state $\sigma \in \mathfrak{F}$, the map $\mathcal{E}(\cdot) = (\cdot)_A \otimes \sigma_B \in \mathfrak{D}$.
 - **Physical interpretation:** Adding on something that is free is always an allowed operation. This means that by just adding something free we cannot make an object more resourceful.

It is often also assumed, or proved, that any free state of a resource theory can be generated from a trivial system under allowed operations [14, 18, 19, 24]. If so, for any given input state, any free state can be reached via some allowed operation: the current state can be traced out and the free state generated. Interestingly, this means that all free states can be reached from any other free state, implying that they can all be treated as equivalent. This idea is captured in the following additional condition on $\mathcal{R}$.

Definition 4.2.2. Consider a resource theory $\mathcal{R} = (\mathfrak{D}, \mathfrak{F})$. In addition to conditions 1 – 5, the following condition on $\mathcal{R}$ can also be assumed

6. $\tau \in \mathfrak{F} \iff \exists \Pi_\tau \in \mathfrak{D}$ where $\Pi_\tau(\rho) = \text{tr}[\rho]\tau \ \forall \ \rho \in \mathcal{D}(\mathcal{H})$.
 - **Physical interpretation:** For all free states there exists an allowed operation to generate it from any arbitrary input state. And, if a state can be reached via an allowed operation from any arbitrary input state, then it is a free state.

For a resource theory where both condition 2 and condition 6 hold, the so-called golden rule of resource theories can be shown to hold.

Theorem 4.2.1 (The Golden Rule). If $\tau \in \mathfrak{F}$ and $\mathcal{E} \in \mathfrak{D}$, then $\mathcal{E}(\tau) \in \mathfrak{F}$.

Proof. Condition 6 says that for all free states there exists an allowed operation that is a state preparation channel of that free state. Moreover, any state preparation channel in the set of allowed operations must prepare a free state. As a result of condition 6, any free state can now be defined via its respective state preparation channel, i.e., τ can be defined via $\Pi_\tau \in \mathfrak{D}$ where $\Pi_\tau(\cdot) = \text{tr}[\cdot]\tau$. Therefore, $\mathcal{E}(\tau)$ can now be seen to be the successive application of two allowed operations, namely $\mathcal{E} \circ \Pi_\tau$ on some arbitrary or trivial input state. Now, due to condition 2, this will also be an allowed operation as it is the concatenation of two allowed operations. Then, it can be seen that for all input states the channel $\mathcal{E} \circ \Pi_\tau$ will output the state $\mathcal{E}(\tau)$, meaning the allowed channel $\mathcal{E} \circ \Pi_\tau$ is a state preparation channel of $\mathcal{E}(\tau)$. Due to condition 6, all state preparation channels that are allowed operations must prepare free states. Hence, $\mathcal{E}(\tau) \in \mathfrak{F}$. $\square$

The golden rule says that given access to only free states and allowed operations, one cannot reach a state that is not within the free set. Hence, the golden rule ensures that value can be fairly attributed to

the resourceful states as they cannot be generated using only allowed operations and free states i.e., they cannot be generated within the physical constraints of the system.

As previously mentioned, not all resource theories fulfil these conditions. For example, in the state resource theory of non-locality there exists the property of super-activation: the tensor product of two free states can become resourceful [72]. Hence, condition 5 does not hold in this case. Whilst such resource theories are not considered in this thesis, this example is used to highlight the fact that the above conditions are not definitive axioms, but rather guidelines.

We now detail a few key features of resource theories and discuss their significance.

4.2.1 Allowed Operations

The largest set of CPTP operations that fulfil the criteria in Definition 4.2.1 are the resource non-generating operations.

Definition 4.2.3 (Resource Non-generating Operations). *Consider a set of free states $\mathfrak{F}$. An operation $\mathcal{E} \in \text{CPTP}$ is called resource non-generating if*

$$\mathcal{E}(\tau) \in \mathfrak{F} \quad \forall \tau \in \mathfrak{F}. \quad (4.1)$$

The set of all resource non-generating operations is then denoted by $\mathfrak{D}^{\max}$.

Any chosen set of allowed operations must not increase the resource content of free states, as per the criteria detailed in Definition 4.2.1. Hence, it must hold that $\mathfrak{D} \subseteq \mathfrak{D}^{\max}$ for any chosen set of allowed operations.

A physically well motivated set of allowed operations are those with a free dilation.

Definition 4.2.4 (Free Dilation). *A channel $\mathcal{E} \in \mathfrak{D}$ has a free dilation if there exists a state $\tau \in \mathfrak{F}$ and allowed unitary $V_{AE} \in \mathfrak{D}_{AE}$ such that*

$$\mathcal{E}(\cdot) = \text{tr}_E[V_{AE}((\cdot)_A \otimes \tau_E)V_{AE}^\dagger] \quad (4.2)$$

If a channel has a free dilation, it can be performed by appending a free state, applying a globally allowed unitary, and then performing the partial trace. Given it is easy to envision how such an operation could be performed within a lab, such operations are assumed to be physically well motivated.

Understanding the separation between a chosen physically motivated set of allowed operations and the set of resource non-generating operations has proven an interesting line of inquiry [12, 13, 73–75]. We note that in some cases, such as in some instances of the resource theory of athermality [17, 18] or the resource theory of non-stabilizers [24], the allowed operations are defined to be only those channels with a free dilation.

4.2.2 Convertibility Criteria

One of the central aims of all resource theories is to quantify how resourceful a given object is. However, within the framework we have thus far presented, only the binary question of whether $\rho \in \mathfrak{F}$ or $\rho \notin \mathfrak{F}$, i.e., is ρ free or resourceful, can be asked. A characterisation of when one state is more resourceful than another can therefore be defined operationally via the allowed operations of the resource theory.

Definition 4.2.5 (Resource Convertibility). *Consider a resource theory $R = (\mathfrak{D}, \mathfrak{F})$. A state ρ is not less resourceful than a state σ if there exists an allowed operation $\mathcal{E} \in \mathfrak{D}$ such that $\mathcal{E}(\rho) = \sigma$.*

The existence of an allowed operation between ρ and σ will be denoted $\rho \rightarrow \sigma$.

The logic for this definition is as follows: if there exists an allowed operation $\mathcal{E} \in \mathfrak{D}$ such that $\mathcal{E}(\rho) = \sigma$, then under the imposed physical restrictions there is nothing that can be achieved with σ that cannot be achieved with ρ . Namely, if one has ρ they can always convert it into σ before using it.

There will not always exist an allowed operation between two states. If so, the resource content of

the two objects is deemed incomparable. Moreover, if $\rho \rightarrow \sigma$ and $\sigma \rightarrow \rho$, then ρ and σ are said to have equal resource. In general, the convertibility question establishes a partial-order on the set of all states based on the existence of an allowed operation between them. This partial-order then allows the resource content of any two different states to be compared. Ideally, a simple characterisation of the existence of an allowed operation between two states would be found. In some important cases, such as for the manipulation of pure states under local operations and classical communication [53], energy-incoherent states under thermal operations [17], or for arbitrary states under noisy operations [18], it has. However, often it is too complicated to be simply solved.

4.2.3 Resource Measures

Resource measures offer an alternative method to the allowed operations for quantifying the resource within a state. They are function from the set of density operators to the non-negative real numbers that assign a numerical value to the resource content of a state.

Definition 4.2.6 (Resource Measures). *Consider a resource theory $\mathcal{R} = (\mathfrak{D}, \mathfrak{F})$. A function $M : \mathcal{D}(\mathcal{H}) \rightarrow \mathbb{R}^+$ is a resource measure of $\mathcal{R}$ if*

$$M(\rho) \geq M(\mathcal{E}(\rho)) \quad \forall \mathcal{E} \in \mathfrak{D}. \quad (4.3)$$

To be a resource measure, a function must therefore be monotonic under successive applications of operations from the set of allowed operations of the given resource theory. This is in keeping with the definition of one state being more resourceful than another given in Definition 4.2.5. Namely, if there exists an allowed operation between the states ρ and σ , then σ must not be more resourceful than ρ . For the function to be a good measure of the resource, we therefore want it to not be greater for ρ than σ i.e., we want the function to not increase as the resource decreases. Note, due to the requirement for monotonicity in a resource measure, they are often referred to as just *monotones* in the literature.

There exists a few additional desirable features of a resource measures, with faithfulness being used throughout this thesis.

Definition 4.2.7 (Faithfulness). *A resource measure $M : \mathcal{D}(\mathcal{H}) \rightarrow \mathbb{R}^+$ is faithful if*

$$M(\tau) = 0 \iff \tau \in \mathfrak{F}, \quad (4.4)$$

meaning the resource measure is zero if and only if the input state is a free state.

4.2.3.1 Complete Set of Monotones

A single resource measure does not in general capture the full partial-order applied to the set of all states by the allowed operations. Specifically, given a resource measure $M(\cdot)$ it could be the case that

$$M(\rho) \geq M(\sigma), \quad (4.5)$$

even if $\nexists \mathcal{E} \in \mathfrak{D}$ such that $\mathcal{E}(\rho) = \sigma$. To see why this occurs, one can colloquially consider a resource measure to be a projection of the partial order onto the real line. This inevitably means a lot of the structure of the partial order is lost. Despite this, there does exist sets of monotones that are able to fully capture the partial order. Such a set is called a *complete set of monotones*.

Definition 4.2.8 (Complete Set of Montones). *A set $\{M_i \mid M_i : \mathcal{D}(\mathcal{H}) \rightarrow \mathbb{R}^+\}_i$ is a complete set of monotones if*

$$M_i(\rho) \geq M_i(\sigma) \quad \forall i \iff \rho \rightarrow \sigma. \quad (4.6)$$

See Fig. 4.2 for a graphical depiction of a partial-order and a complete set of monotones. Note, it can be the case that a complete set of monotones contains an infinite number of functions, for example in [76].

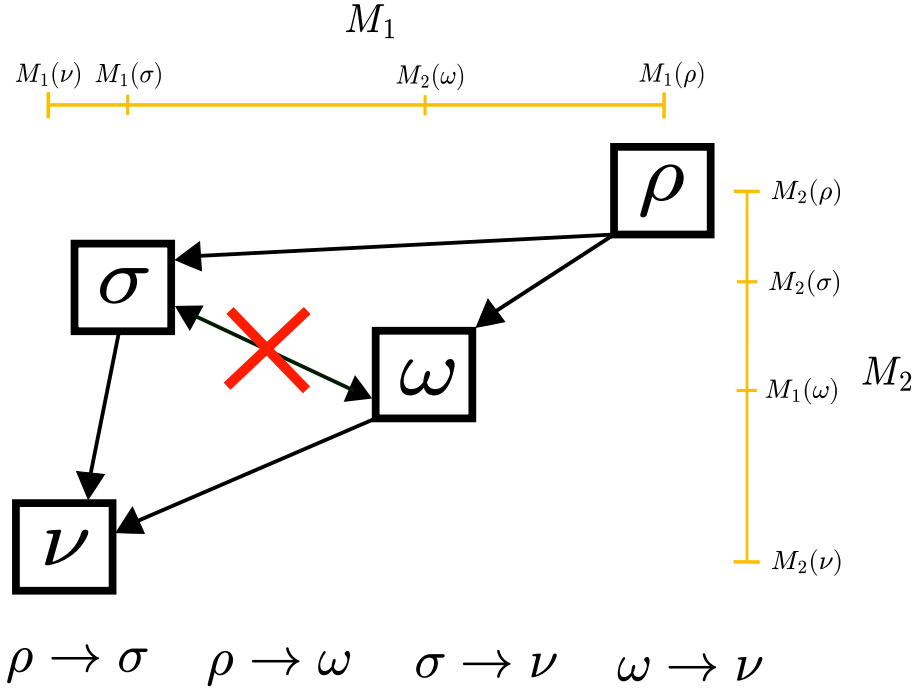


Figure 4.2: A graphical depiction of a complete set of monotones for a partial order. Each box represents a state, with an arrow between the states indicating the existence of an allowed operation between them. There then exists no allowed operation between σ and ω . The functions M_1 and M_2 can then be seen to be resource measures, with them both decreasing monotonically under the existence of an allowed operation. The M_1 resource measure can be seen to be a ‘projection’ of the partial order on the line of the X-axis, and M_2 the same for the Y-axis. Neither M_1 or M_2 capture the full partial order, with them swapping ω and σ . However, if both functions are evaluated and compared, then the partial order can be deduced from the two monotones.

4.2.3.2 Entropic Resource Measures

Entropic resource measures are a type of distance measure. In such measures, the aim is to quantify the resource content of some state by determining ‘how far’ it is from the set of free states. To do this, one firstly needs a notion of distance.

Definition 4.2.9 (Distance Function). *A function $d : \mathcal{D}(\mathcal{H}) \times \mathcal{D}(\mathcal{H}) \rightarrow \mathbb{R}^+$ is a distance measure if*

$$d(\rho, \sigma) \geq d(\mathcal{E}(\rho), \mathcal{E}(\sigma)) \geq 0 \quad \forall \rho, \sigma \in \mathcal{D}(\mathcal{H}), \mathcal{E} \in \text{CPTP}, \quad (4.7)$$

meaning it is contractive under quantum channels and non-negative.

We note that a distance measure can be more formally defined as a metric on a vector space, with a number of additional axioms to those given above needing to be satisfied. We omit those details here in the interest of clarity, but point the reader to the following references for more detail [41]

From this definition of a distance function, a natural resource quantifier can be defined as

$$M(\rho) = \inf_{\sigma \in \mathcal{F}} d(\rho, \sigma), \quad (4.8)$$

where the minimisation is over all free states. It can easily be seen from Definition 4.2.9 that $M(\cdot)$ will be monotonic under allowed operations and is therefore a valid resource measure, as per Definition 4.2.6.

In Section 2.7, it was stated that both the quantum relative entropy and the max-relative entropy are contractive under quantum channels, meaning they are both distance functions as defined in Definition 4.2.9. This leads to the following two entropic resource measures

$$M(\rho) = \inf_{\sigma \in \mathcal{F}} D(\rho || \sigma) \quad \& \quad N(\rho) = \inf_{\sigma \in \mathcal{F}} D_{\max}(\rho || \sigma). \quad (4.9)$$

Whilst not used in this thesis, such resource measures exists for all relative entropy functions.

4.2.4 Operational Interpretations

Operational interpretations link the abstract mathematics of resource theories to some physical realisable task. They may apply to the resource itself, determining broadly for what information processing tasks an advantage exists if one has access to the resource as compared to not having access. For example, under certain physical restrictions, having access to non-stabilizers allows non-classically-simulable quantum computation to be performed [24]. Moreover, having access to coherence allows quantum interference effects to be seen, whereas without it none can be seen [15, 16].

Alternatively, operational interpretations can apply to specific resource quantifiers. In this case, the operational interpretations are typically more targeted, putting a numerical value (the output of the quantifier) on ones ability to perform in some specific-task given access to a particular resourceful state. This has been done for resource quantifiers in general convex resource theories for the tasks of quantum state discrimination [48, 68] and quantum state exclusion [67]. It has also been done within particular resource theories, for tasks such as resource distillation [77, 78] and phase estimation [16]. This is by no means an exhaustive list, but gives an insight to range of tasks used to give resource quantifiers a grounding in practical reality.

4.3 Examples

4.3.1 The Resource theory of Informational Non-equilibrium

The informational contributions to thermodynamics can be isolated from the energetic ones — allowing them to be independently studied and quantified — by considering fully-degenerate Hamiltonians [18]. In the absence of energy differences, thermodynamic transformations must arise from information processing. In this regime, thermal equilibrium is described by the maximally mixed state, and all other states are considered to be in informational non-equilibrium. This notion coincides with purity when considering a fixed system size [18], allowing purity to also be studied within this framework. By understanding this special case of thermodynamics, insights can be gained into the general case, where both energy and information are considered.

The resource theory of informational non-equilibrium [18] is one method used to study this paradigm, where not being in the maximally mixed state is considered a resource. It is closely related to the resource theory of thermodynamics [19], with the two coinciding exactly when considering completely degenerate-Hamiltonians, $H = 0^*$.

Here, we will detail a few aspects of the theory used throughout this thesis. For a detailed overview see [18].

4.3.1.1 Free States and Allowed Operations

The free states in this paradigm — those with no resource — are states in informational equilibrium: that is, states in the maximally mixed state of some dimension.

Definition 4.3.1 (Free States). *The set of free states of the resource theory of informational non-equilibrium is*

$$\mathfrak{F} = \{\mathbb{I}/d\}, \quad (4.10)$$

where $d = \text{tr}[\mathbb{I}]$ is the size of the system.

It is assumed that one has access to a bath consisting of an infinite number of free states, meaning arbitrarily many can be accessed. Moreover, by tensoring maximally mixed states, or by taking the partial tracing over them, one can make a maximally mixed state of any finite dimension.

* It is often thought that they also coincide at $T = \infty$, however, the allowed operations in the resource theory of thermodynamics in this case still require energy conserving unitaries, rather than all unitaries as in the resource theory of informational non-equilibrium. So, whilst the free states align at $T = \infty$, the allowed operations do not.

Given some system S , it is then assumed that one can interact with a state from the bath via some global unitary, before some part of the joint system is discarded. Such operations are termed *noisy operations*.

Definition 4.3.2 (Allowed (Noisy) Operations). Let $\mathcal{E} : \mathcal{H}_S \rightarrow \mathcal{H}_{S'} \in \text{CPTP}$. The channel $\mathcal{E}$ is an allowed operation of the resource theory of informational non-equilibrium, $\mathcal{E} \in \mathfrak{D}$, if it can be decomposed as

$$\mathcal{E}(\rho_S) = \text{tr}_{B'} [U_{SB}(\rho_S \otimes \mathbb{I}_B/d)U_{SB}^\dagger], \quad (4.11)$$

where U_{SB} is any unitary on $\mathcal{H}_S \otimes \mathcal{H}_B$, and $\mathcal{H}_S \otimes \mathcal{H}_B = \mathcal{H}_{S'} \otimes \mathcal{H}_{B'}$ such that $d = \dim(\mathcal{H}_B)$.

It can easily be seen that inputting the free state $\mathbb{I}/d$ to any noisy operation will return a maximally mixed state, meaning noisy operations satisfy the golden rule of resource theories — allowed operations map free states to free states. In general, noisy operations form a strict subset of the set of unital channels, and, if the input and output dimension of the noisy operation is the same, they are a strict superset of the mixed unitary channels [18]. Both unital channels and mixed unitary channels are defined below.

Definition 4.3.3 (Unital Channel). Let $\mathcal{E} : \mathcal{H}_S \rightarrow \mathcal{H}_{S'} \in \text{CPTP}$, then $\mathcal{E}$ is a unital channel if

$$\mathcal{E}(\mathbb{I}_S/d_S) = \mathbb{I}_{S'}/d_{S'}, \quad (4.12)$$

where $d_S = \dim(\mathcal{H}_S)$ and $d_{S'} = \dim(\mathcal{H}_{S'})$.

Definition 4.3.4 (Mixed Unitary Channels). Let $\mathcal{E} : \mathcal{H}_S \rightarrow \mathcal{H}_S \in \text{CPTP}$, then $\mathcal{E}$ is a mixed unitary channel if there exists a probability vector $\mathbf{p}$ and set of unitaries $\{U_i : \mathcal{H}_S \rightarrow \mathcal{H}_S\}_i$ such that

$$\mathcal{E}(\cdot) = \sum_i p_i U_i(\cdot) U_i^\dagger, \quad (4.13)$$

where p_i are elements of $\mathbf{p}$, such that $p_i \geq 0$, $\sum_i p_i = 1$.

4.3.1.2 Convertibility Criteria

As detailed above, convertibility criteria within a resource theory asks: given two states ρ and σ , does there exist an allowed operation $\mathcal{E} \in \mathfrak{D}$ such that $\mathcal{E}(\rho) = \sigma$? i.e., does $\rho \rightarrow \sigma$? If so, it can be conclusively claimed that ρ is not less resourceful than σ .

To answer this question, the following fact is used:

Theorem 4.3.1. Let $\rho \in \mathcal{D}(\mathcal{H}_S)$ and $\sigma \in \mathcal{D}(\mathcal{H}_{S'})$. The follow propositions are equivalent

1. $\rho \rightarrow \sigma$ by a noisy operation,
2. $\rho \rightarrow \sigma$ by a unital channel,

and if $\dim(\mathcal{H}_S) = \dim(\mathcal{H}_{S'})$ then

3. $\rho \rightarrow \sigma$ by a mixed unitary channel.

See [18] for a proof.

Theorem 4.3.1 states that the well physically motivated noisy operations, according to Definition 4.2.4, have equal convertibility power to the mathematically simple unital channels or mixed unitary channels. Using this, if the input and output dimension of the noisy operation are the same, then the following convertibility criteria can be found.

Theorem 4.3.2. Let $\lambda(\omega)$ be a vector of eigenvalues of the state ω , and let $\rho, \sigma \in \mathcal{D}(\mathcal{H}_S)$. Then,

$$\rho \rightarrow \sigma \iff \lambda(\rho) \succ \lambda(\sigma) \quad (4.14)$$

meaning there exists a noisy operation between ρ and σ if and only if a vector of eigenvalues of ρ majorises a vector of eigenvalues of σ .

See [18] for a proof.

4.3.1.3 Resource Quantifiers

Distance-based resource quantifiers, such as entropic quantifiers, take a simple form within the resource theory of informational non-equilibrium given there is only a single free state (when the dimension is fixed). This means that no minimisation over a set of states is required, which is typically a complex calculation. The following entropic quantifier is used in this thesis.

Theorem 4.3.3. The function

$$\mathcal{P}(\rho) := D_{\max}(\rho || \mathbb{I}/d) \quad (4.15)$$

is a faithful resource quantifier of the resource theory of informational non-equilibrium.

Proof. This can be seen as, for every unital channel $\mathcal{E}$,

$$\mathcal{P}(\mathcal{E}(\rho)) = D_{\max}(\mathcal{E}(\rho) || \mathbb{I}/d) = D_{\max}(\mathcal{E}(\rho) || \mathcal{E}(\mathbb{I}/d)) \leq D_{\max}(\rho || \mathbb{I}/d) = \mathcal{P}(\rho), \quad (4.16)$$

where we have used the fact that $\mathcal{P}(\cdot)$ is a unital channel in conjunction with the data-processing inequality, detailed in Section 2.7. The faithfulness follows from the definition of the max-relative entropy given in Definition 2.7.3. $\square$

4.3.2 The Resource theory of Unspeakable Coherence

Coherence is a key quantum feature. Whilst it is necessary for many quantum-enhanced tasks — such as those in metrology [79], communication [80], computation [39], and more — it is highly non-trivial to generate, maintain, and manipulate. Consequently, as with many other quantum properties, it has proven fruitful to treat it as a resource [13, 15, 16].

Coherence comes in two forms: speakable and unspeakable [15, 16], with each necessitating a separate resource theory. In both cases, coherence is only meaningful when considered with respect to a known decomposition of the Hilbert space into subspaces, typically defined by the eigenspaces of some Hermitian operator. A state is said to be coherent with respect to this decomposition if it contains superpositions between vectors belonging to different subspaces. The distinction between speakable and unspeakable coherence then arises from either the physical restrictions imposed on the system (the allowed operations, see below), or the operational task being considered. A task involves unspeakable coherence if it depends on the eigenvalues of the operator defining the Hilbert-space decomposition — that is, when the eigenvalues are physically relevant such that the eigenspaces cannot be arbitrarily relabelled [16, 81]. If, on the other hand, the task depends only on the structure of the eigenspaces themselves and not the eigenvalues, meaning that the eigenvalues are labels that can be freely permuted, the coherence is speakable. For example, consider the following two states:

$$\frac{|0\rangle + |1\rangle}{\sqrt{2}} \quad \text{and} \quad \frac{|0\rangle + |2\rangle}{\sqrt{2}}. \quad (4.17)$$

If performing quantum computing with qutrits, both of these states are equally resourceful [15]. This is because one is free to relabel $1 \leftrightarrow 2$ without having any impact on computation. However, if performing phase estimation with these states, where $\{|l\rangle : l \in [0, 1, 2]\}$ are eigenstates of the number operator,

success in the task will depend on which of these states one has. Hence, in this second tasks, it matter not just that one has coherence, but between what physical states that coherence exists.

The notion of unspeakable coherence has attracted significant attention as it provides a natural framework for describing coherence manipulation under symmetry-constrained dynamics, such as those conserving energy or momentum [82–91]. As a result, it has been instrumental in building our understanding of the role of coherence in thermodynamics [82–84]. Moreover, unspeakable coherence has proven to be the relevant notion of coherence in the important tasks of quantum metrology [79], reference frame alignment [92], and work extraction [82], among others [93].

Unspeakable coherence is modelled as asymmetry with respect to a continuous transformation generated by the physically relevant observable decomposing the Hilbert space. For this reason, the resource theory of unspeakable coherence can be seen to be a special case of the resource theory of asymmetry [81, 94–99].

4.3.2.1 Free states and Allowed Operations

Firstly, consider a finite-dimensional system $\mathcal{H}$, and let $L \in \mathcal{L}(\mathcal{H})$ be a Hermitian operator, such that L describes a physical observable. The continuous transformation generated by L is then

$$U_{L,x} = e^{-ixL} : x \in \mathbb{R}, \quad (4.18)$$

which is a continuous family of unitary operators. A state $\rho \in \mathcal{D}(\mathcal{H})$ has no coherence with respect to the eigenspaces of L if and only if it is symmetric with respect to the continuous transformations generated by L ,

$$U_{L,x}\rho U_{L,x}^\dagger = \rho \quad \forall x \in \mathbb{R}. \quad (4.19)$$

The state ρ must be block-diagonal with respect to the eigenspaces of L for it to be symmetric under $U_{L,x}$, which is the definition of having no coherence with respect to the eigenspaces of L . Hence, the condition in Eq. (4.19) is equivalent to ρ commuting with L , i.e.,

$$U_{L,x}\rho U_{L,x}^\dagger = \rho \quad \forall x \in \mathbb{R} \iff [\rho, L] = 0. \quad (4.20)$$

A state $\rho \in \mathcal{D}(\mathcal{H})$ therefore has unspeakable coherence with respect to the eigenspaces of L , or simply is *coherent*, if and only if it is asymmetric with respect to this continuous transformations. The set of free states — those states which contain no resource — are states that contain no coherence between the eigenspaces of L . Hence, they are also termed the *incoherent states*.

Definition 4.3.5 (Free States). *The set of free states of the resource theory of unspeakable coherence — or the incoherent states — is*

$$\mathfrak{F} = \{\rho : U_{L,x}\rho U_{L,x}^\dagger = \rho \quad \forall x \in \mathbb{R}\}. \quad (4.21)$$

Or, equivalently,

$$\mathfrak{F} = \{\rho : [\rho, L] = 0\}. \quad (4.22)$$

The allowed operations of the resource theory are those CPTP maps that are translationally-covariant with respect to the continuous transformations generated by L .

Definition 4.3.6 (Allowed Operations). *Let $\mathcal{E} \in \mathcal{H}_S \rightarrow \mathcal{H}_S \in \text{CPTP}$. The channel $\mathcal{E}$ is an allowed operation of the resource theory of unspeakable coherence if*

$$\mathcal{U}_{L,x} \circ \mathcal{E} = \mathcal{E} \circ \mathcal{U}_{L,x} \quad \forall x \in \mathbb{R}, \quad (4.23)$$

where $\mathcal{U}_{L,x}(\cdot) = U_{L,x}(\cdot)U_{L,x}^\dagger$.

Physically, being translationally-covariant with respect to L means that one can apply the channel $\mathcal{E}$ before or after a transformation $\mathcal{U}_{L,x}(\cdot)$ and get the same outcome. We note that this definition could be extended to channels with different input and output spaces by altering the generator being considered

in each space [16], however, we do not consider such cases within this thesis.

The minimum requirement to be a set of allowed operations is that they must be resource non-generating. By letting $\rho \in \mathfrak{F}$, this can be seen to hold for the translationally-covariant channels defined in Definition 4.3.6 as

$$\mathcal{U}_{L,x} \circ \mathcal{E}(\rho) = \mathcal{E} \circ \mathcal{U}_{L,x}(\rho) = \mathcal{E}(\rho) \quad \forall x \in \mathbb{R}, \quad (4.24)$$

as $\mathcal{U}_{L,x}(\rho) = \rho \quad \forall x \in \mathbb{R}$ if $\rho \in \mathfrak{F}$. Therefore, $\mathcal{E}(\rho) \in \mathfrak{F}$ as it is symmetric under $\mathcal{U}_{L,x}(\cdot)$. Hence, translationally-covariant operations map free states to free states and are therefore resource non-generating.

If one considers only unitary dynamics $\mathcal{E}(\cdot) = V(\cdot)V^\dagger$, the condition for $\mathcal{E}$ to be an allowed operation simplifies to $[V, e^{-iLx}] = 0 \quad \forall x \in \mathbb{R}$, which holds if and only if $[U, L] = 0$. Hence, a unitary V is an *allowed unitary* if and only if it is block diagonal with respect to the eigenspaces of L , i.e., $V = \sum_k V_k \Pi_k$, where each V_k is an independent unitary and $\{\Pi_k\}_k$ are the projectors onto the individual eigenspaces of $L = \sum \lambda_k \Pi_k$, where $\lambda_k \neq \lambda_l$ if $k \neq l$, such that $V_k V_l^\dagger = \delta_{kl} \Pi_k$.

4.3.2.2 Bipartite Systems

This framework can be expanded to a bipartite system AB by considering $L_{AB} = L_A \otimes \mathbb{I}_B + \mathbb{I}_A \otimes L_B$. If $[\rho_{AB}, L_{AB}] = 0$, then the bipartite state ρ_{AB} is incoherent between the eigenspaces of L_{AB} ; also, a bipartite allowed unitary V_{AB} satisfies $[V_{AB}, L_{AB}] = 0$. If L_{AB} is non-degenerate, then all allowed unitaries would be diagonal in the joint eigenbasis of L_{AB} . Such unitaries cannot facilitate any interaction between the two subsystems. If L_{AB} does contain degeneracies, then the allowed unitaries are arbitrary unitaries within the degenerate subspaces of L_{AB} . Physically, this means that whatever property L_{AB} represents can be freely exchanged between A and B , as long as its total value is conserved. E.g., if L_{AB} is the total Hamiltonian, then $[U_{AB}, L_{AB}] = 0$ ensures the total energy is conserved under the dynamics U_{AB} [16], but if L_{AB} has degeneracies then energy can be moved between the A and B subsystems.

4.3.2.3 Modes of Coherence

In this section, we consider our physical observable to be the non-degenerate, truncated number operator $L = \sum_{n=0}^{d-1} n |n\rangle\langle n|$.

Under the allowed operations of the resource theory of unspeakable coherence, quantum states can be decomposed into a set of modes [98]. Each mode is a set of components of the state that transform independently due to the symmetry of the covariant dynamics.

Definition 4.3.7 (Modes of Coherence). *Let j be an integer. The j th mode of a d -dimensional state ρ , with respect to $L = \sum_{n=0}^{d-1} n |n\rangle\langle n|$, is given by*

$$\rho^{(j)} := \sum_{n=0}^{d-1} |n+j\rangle\langle n+j| \rho |n\rangle\langle n|, \quad (4.25)$$

where we take $|n+j\rangle = 0$ if $n+j \geq d$.

The j th mode $\rho^{(j)}$ is an operator supported in the j th mode subspace $\mathcal{B}^{(j)} := \text{span}\{|n+j\rangle\langle n+j|\}_{n=0}^{d-1}$. By definition, we have $-d+1 \leq j \leq d-1$ and

$$\rho = \sum_{j=-d+1}^{d-1} \rho^{(j)}. \quad (4.26)$$

As an example, let $L = \sum_{n=0}^2 n |n\rangle\langle n|$ be a Hermitian operator and $\rho \in \mathcal{D}(\mathcal{H}^3)$ a qutrit state. Below, all

the modes of ρ with respect to L are visualised when decomposing ρ into the eigenbasis of L :

$$\rho = \begin{bmatrix} p_{00} & p_{01} & p_{02} \\ p_{10} & p_{11} & p_{12} \\ p_{20} & p_{21} & p_{22} \end{bmatrix} \quad \text{where} \quad \begin{array}{c|c} \text{Mode} & \text{Colour} \\ \hline \rho^{(0)} & \text{red} \\ \rho^{(1)} & \text{orange} \\ \rho^{(2)} & \text{yellow} \\ \rho^{(-1)} & \text{blue} \\ \rho^{(-2)} & \text{green} \end{array} \quad (4.27)$$

such that, for example,

$$\rho^{(0)} = \begin{bmatrix} p_{00} & 0 & 0 \\ 0 & p_{11} & 0 \\ 0 & 0 & p_{22} \end{bmatrix} \quad \text{and} \quad \rho^{(2)} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ p_{20} & 0 & 0 \end{bmatrix}. \quad (4.28)$$

We note that since ρ is a state — meaning $\rho^\dagger = \rho$ — it holds that

$$[\rho^{(j)}]^\dagger = \rho^{(-j)} \quad \forall j. \quad (4.29)$$

One therefore only needs to consider the positive modes. In addition, it can be seen that $\rho = \rho^{(0)}$ if and only if ρ is incoherent i.e.,

$$\rho = \rho^{(0)} \iff [\rho, L] = 0. \quad (4.30)$$

A key and powerful fact about modes is that they transform independently under allowed operations.

Theorem 4.3.4 (Mode Independence). *If there exists an allowed operation $\mathcal{E}$ between the states $\rho \in \mathcal{D}(\mathcal{H})$ and $\sigma \in \mathcal{D}(\mathcal{H})$, such that $\sigma = \mathcal{E}(\rho)$, then*

$$\sigma^{(j)} = \mathcal{E}(\rho^{(j)}) \quad \forall j, \quad (4.31)$$

meaning that σ 's j th mode depends only on ρ 's j th mode.

See [98] for a proof.

The modes therefore transform independently under translationally-covariant (symmetry constrained) dynamics. Consequently, $\rho^{(j)} = 0$ implies $\sigma^{(j)} = 0$, which can be generalised to

$$\text{modes}(\sigma) \subseteq \text{modes}(\rho), \quad (4.32)$$

where $\text{modes}(\rho) := \{j \in \mathbb{Z} \mid \rho^{(j)} \neq 0\}$ is the set of all integers labelling modes for which ρ has a nonzero component, if there exists an allowed operation $\mathcal{E}$ such that $\sigma = \mathcal{E}(\rho)$.

In the bipartite setting, the global modes of a bipartite product state are given by a combination of local modes.

Definition 4.3.8 (Bipartite Modes of Coherence). *The j th mode of a bipartite state $\rho_A \otimes \rho_B$ of local dimensions d , with respect to $L_{AB} = L_A \otimes \mathbb{I}_B + \mathbb{I}_A \otimes L_B$ where $L_X = \sum_{n=0}^{d-1} |n\rangle\langle n|_X$ for $X \in \{A, B\}$, is given by*

$$(\rho_A \otimes \rho_B)^{(j)} = \sum_k \rho_A^{(k)} \otimes \rho_B^{(j-k)}. \quad (4.33)$$

It can therefore be seen that the j th mode subspace of the bipartite system reads

$$\mathcal{B}_{AB}^{(j)} := \text{span}\{|n+k, m+j-k\rangle\langle n, m|_{AB} \mid n, m, k \in \mathbb{N}\}. \quad (4.34)$$

4.3.3 Resource Measures

There exists resource measures for both the resource within the total state and the resource within a given mode. In this thesis, we consider only the resource within a given mode and hence here give just a quantifier for this. See [16] for a comprehensive overview of different resource quantifiers.

Theorem 4.3.5. *The function*

$$M^{(j)}(\rho) := \|\rho^{(j)}\|_1, \quad (4.35)$$

where $\rho^{(j)}$ is the j th mode of ρ , is a faithful resource quantifier of the resource in the j th mode.

Proof. The monotonicity condition can be seen from the fact that the one-norm is contractive for all operators under CPTP maps, such that

$$M^{(j)}(\mathcal{E}(\rho)) = \|\mathcal{E}(\rho^{(j)})\|_1 \leq \|\rho^{(j)}\|_1 = M^{(j)}(\rho). \quad (4.36)$$

To see that this measure is also faithful, it can be noted that there is zero resource in the j th mode if and only if $\rho^{(j)} = 0$. The one norm of this zero operator is then zero. $\square$

4.4 Resource Preservability Theories

Here, we discuss dynamical resource theories, where the underlying objects of the resource theory are quantum channels, rather than quantum states.

Given some static resource theory $\mathcal{R} = (\mathfrak{D}, \mathfrak{F})$, a resource preservability theory aims to answers the following question:

Given an allowed operation $\mathcal{E} \in \mathfrak{D}$, what is the ability of $\mathcal{E}$ to preserve the resource content of a system it acts upon?

Resource preservability theories, introduced in [26], look to answer this question by formulating a dynamical resource theory within the allowed operations of the static resource theory. Resource preservability theories are therefore induced by some static resource theory, and treat the ability of an allowed operation of the static resource theories to preserve the (static) resource as a dynamical resource.

As with any resource theory, resource preservability theories have a set of allowed operations — these will now be operations mapping quantum channels to quantum channels — and free objects, which will now be a set of quantum channels. Before we introduce these sets, will give the following definition of resource annihilating channels.

Definition 4.4.1 (Resource Annihilating Channels). *The set of resource annihilating channels within a given static resource theory $\mathcal{R} = (\mathfrak{D}, \mathfrak{F})$ is*

$$\mathfrak{a} := \{\mathcal{E} \in \mathfrak{D} : \mathcal{E}(\rho) \in \mathfrak{F} \forall \rho \in \mathcal{D}(\mathcal{H})\}. \quad (4.37)$$

By definition, for all possible input states into a resource annihilating channel the output state will be a free state. Every channel $\mathfrak{D} \setminus \mathfrak{a}$ is then said to be able to preserve the resource, as for all these channels there will exist at least one resourceful state that will retain some resource when output from the channel. Hence, the channel will have preserved some resource for at least that input state. Note, the term preserve can be justified as the channel is an allowed operation, meaning it cannot have generated any resource. Hence, all the resource within the output state must have also been in the input state.

4.4.0.1 Free States and Allowed Operations

The free objects — or free channels — of a resource preservability theory are those channels that cannot preserve any amount of resource i.e., the resource annihilating channels.

Definition 4.4.2 (Free Channels of Resource Preservability Theories). *The free channels of a resource preservability theory, induced by some static resource theory, are the resource annihilating channels of the given static resource theory.*

The allowed operations are built on the notion of super-channels [40], which are linear mappings of quantum channels to quantum channels, see Section 2.5 for more details. As with all allowed operations, they must be resource non-generating, meaning any channel output from an allowed operation of a resource preservability theory must not be able to preserve any more static-resource than the input channel. To achieve this, the pre-processing and post-processing channels used in the super-channel are allowed operations of the static resource theory. As these channels are static resource non-generating, they cannot enhance the resource preserving ability of the input channel. Moreover, to prevent the ancillary system in the definition of super-channels from being able to artificially provide preservability, a resource annihilating channel is included on this system.

Definition 4.4.3 (Allowed Operations of Resource Preservability Theories). *The allowed operations of a resource preservability theory, induced by a static resource theory $\mathcal{R} = (\mathfrak{D}, \mathfrak{F})$, are mappings $\Pi : \mathfrak{D} \rightarrow \mathfrak{D}$ of the form*

$$\Pi(\mathcal{E}) := \Lambda_b \circ (\mathcal{E} \otimes \Lambda_{\text{An}}) \circ \Lambda_a, \quad (4.38)$$

where $\Lambda_X \in \mathfrak{D} \forall X \in \{a, b\}$ and $\Lambda_{\text{An}} \in \mathfrak{a}$.

We note that Λ_a and Λ_b act on a larger system than Λ_{An} . Moreover, whilst we do not consider such resource theories in this thesis, we note that if the underlying static resource theory has super-activation, then Λ_{An} has a stricter requirement to be an absolutely resource annihilating channel [26]. See Fig. 4.3 for schematic overview of resource preservability theories, and see [26] for a complete overview of the details.

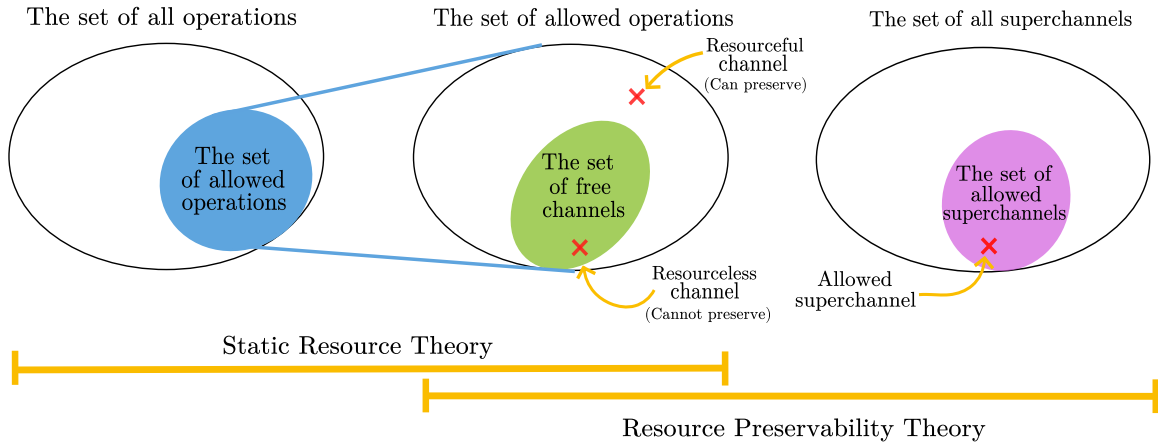


Figure 4.3: A schematic overview of a resource preservability theory, showing that the resource preservability theory is built on top of a static resource theory. The allowed objects can be seen to be quantum channels, whilst the allowed operations are now super-channels.

**RESULTS PART 1 — RESOURCE
PRESERVATION AND ENHANCEMENT**

5.1 Introduction

Resources are precious. Their value arises from their limitation, incentivising them to be efficiently utilised and *maintained*. Formally, an object is considered to be a resource if it can be used by some agent to overcome the physical constraints of a system. A resource therefore allows an agent to obtain an otherwise impossible advantage.

Within the quantum regime, the study of the limitations experienced by such an agent is the central aim of quantum resource theories [12, 13]. Initially, static resource theories (SRT) were the sole focus. In such SRTs, the primary objects in question are quantum states, and the resource is some property of these states. Resource theories have now been expanded beyond states to the dynamical regime [23, 26, 100–107], with the aim to assess and compare the ability of different quantum operations to be resourceful, therefore providing some advantage. Within these dynamical resource theories (DRT), the resourceful objects are quantum operations, with some property of the quantum operation being considered a resource.

One such important property is the ability of operations to preserve the static resource present in the state upon which they act. These *resource preservability theories*, introduced in [26] and Section 4.4, are built upon SRTs and apply structure to their set of allowed operations. In particular, not all allowed operations are equal. Some operations will completely preserve the resource, while others will completely destroy it. How well an allowed operation preserves the static-resource can itself be considered a type of dynamic-resource, and this is the focus of such DRTs.

Note that whilst intimately connected, the resource considered in the DRT and underlying SRT are fundamentally different and apply to different classes of objects — quantum channels and quantum states, respectively.

5.1.1 Motivation

The desire for resource preservation arises naturally in any resource theory — if a resource is a useful property, one wants to keep as much of it as possible. In the context of thermodynamics, the desire for preservability arises due to non-equilibrium states being considered a resource. For a given state, both its energy and the information an agent has about it will determine how resourceful it is [108]. The contribution arising from information can be studied in isolation by considering trivial Hamiltonians, where all energy levels are degenerate. In this regime, any thermodynamic transformation that occurs must arise solely from an information theoretic origin, given there can be no change in energy. The maximally mixed state is the thermal state in this picture, and all other states are considered to be in *informational non-equilibrium*. An understanding of this specific case can then be used to infer results about the general case, in which energetic considerations are necessary.

The study of informational non-equilibrium has been succinctly formalised through the resource theory of informational non-equilibrium [18, 109, 110]. Such SRTs often question the existence of transformations between states without any concern for the details of said transformations. This initially led to our understanding of thermodynamics having little to do with dynamics [19]; however, insight has since been gained into the dynamical aspects of thermodynamic resource theories. For example, the largest set of feasible dynamics has been investigated [75], and their consistency with the laws of thermodynamics [111] and practically relevant constraints assessed [112, 113]; their behaviour in the macroscopic limit has been considered [114], and their analogue in the continuous variable regime studied [115, 116]. Despite this, there has been no insight into how to describe the ability of a given dynamics to *preserve* informational non-equilibrium. Understanding what properties prevent thermalisation will not only enrich our foundational understanding of thermodynamics but also shed further light on the role of thermodynamics in information processing and transmission. In addition, most quantum dynamics used in practical applications, such as quantum computation [39] and quantum memories [117], benefit

from this lack of thermalisation, which is a prevalent type of decoherence. An improved understanding of these dynamics could therefore further our control over these systems.

In this chapter, we define and characterise the dynamical resource theory of informational non-equilibrium preservability to begin addressing this question. The informational non-equilibrium of a state will henceforth be referred to as the static-resource to distinguish it from the dynamic-resource.

In addition to improving our understanding of dynamics in thermodynamics, the resource theory of informational non-equilibrium preservability presented here represents the first attempt to build a resource preservability theory for a specific resource. Given that it is one of the simplest examples of a resource theory, the resource theory of informational non-equilibrium represents a good test-bed for new methods and ideas within resource theories in general, partly motivating its choice within this work. It is therefore hoped that this work will prove insightful for developing resource preservability theories of other resources, such as entanglement [14] or non-stabilizerness [24], in the future.

5.1.2 Framework

To make this chapter more self contained, we will firstly give a short reminder of the details of the resource theory of informational non-equilibrium, see [18] and Section 4.3.1 for more complete overviews.

5.1.2.1 The Resource Theory of informational non-equilibrium

The set of free states of a resource theory are those states that contain no resource. In the SRT of informational non-equilibrium, the only free state is the maximally mixed state, $\mathbb{I}/d_S$. Hence, the set of free states is $\{Y_S := \mathbb{I}_S/d_S\}$, where the subscript labels the subsystem and d is the system dimension.

All dynamics considered are quantum channels: completely positive trace-preserving linear maps. In a resource theory, dynamics allowed within the physical constraints are captured by the set of allowed operations, which are resource non-generating dynamics. Within the resource theory of informational non-equilibrium, these are taken to be the set of *noisy operations*, denoted here as $\mathfrak{D}^{\text{st}}$, given by

$$\mathcal{D}(\rho_S) = \text{tr}_{E'} \left[U_{SE} \left(\rho_S \otimes Y_E \right) U_{SE}^\dagger \right] \in \mathfrak{D}^{\text{st}}, \quad (5.1)$$

where U_{SE} is a unitary operator on a joint system SE^* . Physically, this means coupling the system S with a bath E in informational equilibrium, evolving the joint system under a global unitary, and then discarding some part of the joint system and environment, E' . The resource theory of informational non-equilibrium will be referred to as $\mathcal{R}^{\text{st}}$.

The existence of a noisy operation between two states ρ and σ is equivalent to the existence of a *unital* channel $\mathcal{N}$ such that $\sigma = \mathcal{N}(\rho)$, where a unital channel is a CPTP maps that takes the maximally mixed state on its input space to the maximally mixed state on its output space [18]. On the set of density operators, one can therefore say that noisy operations and unital channels have equal convertibility power. This is despite noisy operations being a strict subset of unital operations [18]. If ρ and σ are of equal dimension, this unital channel will be a *mixed unitary channel*.

5.1.2.2 The Dynamical Resource Theory of Informational Non-equilibrium Preservability

Any SRT induces a DRT of static resource preservability [26]. Here, we build the DRT of informational non-equilibrium upon $\mathcal{R}^{\text{st}}$. It aims to apply structure to the set of noisy operations (the allowed operations of $\mathcal{R}^{\text{st}}$) based on the channel's ability to preserve non-equilibrium. Formally, a noisy operation $\mathcal{N} \in \mathfrak{D}^{\text{st}}$ is considered resourceful in the DRT of informational non-equilibrium preservability if it can output some non-equilibrium state, $\mathcal{N}(\rho) \neq Y_S$ for some $\rho \in \mathcal{D}(\mathcal{H})$ i.e., there exists some state which, when input into the channel, does not output the free state. The free objects of this resource theory are therefore those channels that remove *all* of the resource from *any* state upon which they act. In this case, since the free set of the SRT is a single state, the only free channel in the DRT is the state preparation channel of Y_S .

* It will not be made explicit whether a state exists in the system or environment using a subscript S and E as it is mostly obvious from the context. Where it is not, the subscripts will be included.

Definition 5.1.1 (Free Channels). *The set of free channels in the resource theory of informational non-equilibrium preservability contains only the state preparation channel of the maximally mixed state*

$$\Lambda(\cdot) := \text{tr}[\cdot] Y_S, \quad (5.2)$$

as it is the channel that removes all resource of all possible input states.

The allowed operations of a dynamical resource theory are *super-channels* — linear mappings of quantum channels to quantum channels — that can only decrease the resource content of the channel [40]. Super-channels can be realised in the lab with a pre-processing and post-processing channel connected to a memory system [40]. Using this form and the structure of resource non-generating super-channels presented in [26] (see Section 4.4 also), the allowed super-channels in the dynamical resource theory of informational non-equilibrium preservability, denoted $\mathfrak{D}^{\text{dyn}}$, are defined as follows.

Definition 5.1.2 (Allowed Operations). *The allowed operations of the dynamical resource theory of informational non-equilibrium preservability, $\mathfrak{D}^{\text{dyn}}$, have the following form*

$$\Pi(\mathcal{N}) = \sum_{\kappa} p_{\kappa} \mathcal{E}_{\kappa} \circ \mathcal{N} \circ \mathcal{P}_{\kappa} \in \mathfrak{D}^{\text{dyn}}, \quad (5.3)$$

where $\sum_{\kappa} p_{\kappa} = 1$, $p_{\kappa} \geq 0 \forall \kappa$ and $\mathcal{E}_{\kappa}, \mathcal{P}_{\kappa} \in \mathfrak{D}^{\text{st}} \forall \kappa$.

It can be seen that the allowed super-channels of the dynamical preservability theory consists of a convex combination of a pre-processing and post-processing channels taken from the allowed operations of the underlying static resource theory, $\mathcal{R}^{\text{st}}$. It can be verified that these super-channels map noisy operations to noisy operations, with Λ only being mapped to itself. Further details on how the allowed super-channels of a resource preservability theory detailed in [26] are equivalent to those in Definition 5.1.2, see Section 5.8.1.

The physical reasoning behind this set of super-channels being allowed is that all parts come from the allowed set of $\mathcal{R}^{\text{st}}$. Given noisy operations only decrease informational non-equilibrium, the output states of $\Pi(\mathcal{N})$ cannot contain any more static-resource than the output states of $\mathcal{N}$. Super-channels in $\mathfrak{D}^{\text{dyn}}$ are therefore dynamic-resource non-increasing, as $\Pi(\mathcal{N})$ cannot preserve more informational non-equilibrium present in any state upon which it acts than $\mathcal{N}$. The dynamical resource theory of informational non-equilibrium preservability will be referred to as $\mathcal{R}^{\text{dyn}}$.

All resource theories aim to quantify how resourceful a given object is. Since allowed operations cannot generate any resource, we can understand this by studying which objects can be converted into each other. In $\mathcal{R}^{\text{st}}$, this characterisation takes the following form: given two quantum states ρ and σ , does there exist a noisy operation mapping $\rho \rightarrow \sigma$? Answering this establishes a pre-order on the set of all states based on the existence of a noisy operation between them. This pre-order then allows the state's resource content to be compared. For the SRT, where ρ and σ are of equal dimension, this convertibility question has been answered and is simple, being captured entirely by the mathematical notion of majorisation [18]. In what follows, we aim to answer this question for $\mathcal{R}^{\text{dyn}}$. This will establish a pre-order on the set of channels $\mathfrak{D}^{\text{st}}$ based on the existence of an allowed super-channel between them. Given these are resource non-generating, if there exists an allowed super-channel mapping $\mathcal{N} \rightarrow \mathcal{M}$, where $\mathcal{M}, \mathcal{N} \in \mathfrak{D}^{\text{st}}$, it will be known that $\mathcal{M}(\rho)$ will be 'closer' to Y_S than $\mathcal{N}(\rho)$ for all $\rho \in \mathcal{D}(\mathcal{H})$. In other words, $\mathcal{M}$ will not preserve informational non-equilibrium better than $\mathcal{N}$.

5.1.2.3 Characterising Allowed Super-channels by a Choi-state Representation

To answer the convertibility question, the Choi-Jamiołkowski isomorphism will be employed. This is a linear mapping between a quantum channel $\mathcal{N}$ and a bipartite quantum state $\mathcal{J}^{\mathcal{N}}$, given by

$$\mathcal{J}^{\mathcal{N}} = (\mathcal{I} \otimes \mathcal{N})(|\Phi^+\rangle\langle\Phi^+|), \quad (5.4)$$

where $|\Phi^+\rangle := \frac{1}{\sqrt{d}} \sum_i |ii\rangle$. If there exists an allowed operation $\Pi \in \mathfrak{D}^{\text{dyn}}$, so that $\mathcal{M} = \Pi(\mathcal{N})$, then the Choi-state of $\mathcal{M}$ is given by

$$\begin{aligned}
\mathcal{J}^{\mathcal{M}} &= (\mathcal{I} \otimes \sum_{\kappa} p_{\kappa} \mathcal{E}_{\kappa} \circ \mathcal{N} \circ \mathcal{P}_{\kappa}) |\Phi^+\rangle\langle\Phi^+| \\
&= \sum_{\kappa} p_{\kappa} (\mathcal{I} \otimes \mathcal{E}_{\kappa}) (\mathcal{I} \otimes \mathcal{N}) (\mathcal{I} \otimes \mathcal{P}_{\kappa}) |\Phi^+\rangle\langle\Phi^+| \\
&= \sum_{\kappa} p_{\kappa} (\mathcal{P}_{\kappa}^t \otimes \mathcal{I}) (\mathcal{I} \otimes \mathcal{E}_{\kappa}) (\mathcal{I} \otimes \mathcal{N}) |\Phi^+\rangle\langle\Phi^+| \\
&= \sum_{\kappa} p_{\kappa} (\tilde{\mathcal{P}}_{\kappa} \otimes \mathcal{E}_{\kappa}) (\mathcal{J}^{\mathcal{N}}),
\end{aligned} \tag{5.5}$$

where we define $\mathcal{P}^t := \sum_i K_i^t(\cdot)(K_i^t)^\dagger$, which is the transpose of a channel $\mathcal{P}$ with Kraus representation $\mathcal{P} = \sum_i K_i(\cdot)K_i^\dagger$, where $(\cdot)^t$ is the transpose operation. In the penultimate line we have relabelled $\mathcal{P}_{\kappa}^t \rightarrow \tilde{\mathcal{P}}_{\kappa}$ given the transpose of a unital channel is still a unital channel (see Section 8.6.3 for a proof). This gives the following Choi-state formulation of the allowed operations.

Definition 5.1.3 (Choi State Formalism of the Allowed Operations). *If there exists an allowed operation $\Pi \in \mathfrak{D}^{\text{dyn}}$ between $\mathcal{M}, \mathcal{N} \in \mathfrak{D}^{\text{dyn}}$, such that $\mathcal{M} = \Pi(\mathcal{N})$, then the corresponding Choi-states of the channels are related as*

$$\mathcal{J}^{\mathcal{M}} = \sum_{\kappa} p_{\kappa} (\mathcal{P}_{\kappa} \otimes \mathcal{E}_{\kappa}) (\mathcal{J}^{\mathcal{N}}), \tag{5.6}$$

where $\mathcal{E}_{\kappa}, \mathcal{P}_{\kappa} \in \mathfrak{D}^{\text{st}} \forall \kappa$.

In the remainder of this chapter, the dynamics of a quantum system of a constant finite dimension are considered. In this case, how far a state is from informational non-equilibrium is equivalent to asking how pure that state is [18]. As mentioned above, under this restriction one can consider $\mathfrak{D}^{\text{st}} = \mathfrak{U}_M$, where $\mathfrak{U}_M$ is the set of mixed unitary channels, as the convertibility power of the two sets of dynamics on the set of density operators is equal.

5.2 Structure

Firstly, we completely characterise the resource theory of informational non-equilibrium preservability for qubit channels, finding a necessary and sufficient condition for comparing the resourcefulness of two channels. Following this, we consider channels acting on arbitrary dimensional system in the restricted case of our underlying static resource theory containing only Weyl-covariant channels [10] as its allowed operations. We then find a complete set of monotones in both cases, before exploring some physically motivated monotones. An operational interpretation of the partial order set by the resource theory of informational non-equilibrium preservability is also found. The result presented here contribute to the manuscript [5].

5.3 Qubit Channels

Initially, qubit channels are focused on, such that $d_S = 2$ and $\mathfrak{D}^{\text{st}}$ are qubit mixed unitary channels. A characterisation of $\mathfrak{R}^{\text{dyn}}$ then looks to answer the following question:

Given two qubit channels $\mathcal{M}, \mathcal{N} \in \mathfrak{D}^{\text{st}}$, does there exist an allowed operation $\Pi \in \mathfrak{D}^{\text{dyn}}$ such that $\mathcal{M} = \Pi(\mathcal{N})$.

Using the form of the allowed operations given in Definition 5.1.3, the above question equivalently asks if there exists some convex combination of channels $\mathcal{P}_{\kappa}, \mathcal{E}_{\kappa} \in \mathfrak{D}^{\text{st}}$ that act on the subsystems of $\mathcal{J}^{\mathcal{N}}$ respectively to map $\mathcal{J}^{\mathcal{N}} \rightarrow \mathcal{J}^{\mathcal{M}}$. It is shown through the following result that this depends only upon the eigenvalues of the corresponding Choi-states. In what follows, λ and μ are vectors of eigenvalues of

the Choi states $\mathcal{J}^{\mathcal{N}}$ and $\mathcal{J}^{\mathcal{M}}$, respectively.

Result 5.3.1. *For every given $\mathcal{M}, \mathcal{N} \in \mathfrak{D}^{\text{st}}$, the following statements are equivalent:*

1. $\mathcal{M} = \Pi(\mathcal{N})$ for some allowed operation $\Pi \in \mathfrak{D}^{\text{dyn}}$.
2. $\boldsymbol{\mu} = \mathbb{D}\boldsymbol{\lambda}$, where $\mathbb{D} = \sum_{n,m=0}^1 p_{nm} \sigma_x^n \otimes \sigma_x^m$ with p_{nm} the elements of a probability vector and σ_x is the Pauli X .

Proof Sketch. Firstly, we prove that statement 1 implies statement 2. We simplify the form of the allowed operations through the following lemma:

Lemma 5.3.2. *The Choi-states of qubit unital channels can always be diagonalised in the Bell basis (maximally entangled basis) under local unitaries.*

The physical implication of this Lemma is that, given some channel $\mathcal{N} \in \mathfrak{D}^{\text{st}}$, there always exists an allowed operation that can map the given channel to another whose Choi-state is diagonal in the Bell basis. From this, it can be shown that the dynamic resourcefulness of all qubit unital channels can be captured through only the eigenvalues of their Choi-state. The Choi-state form of the allowed operations, detailed in Definition 5.1.3, can now be rephrased as $\boldsymbol{\mu} = \mathbb{B}\boldsymbol{\lambda}$, where $\mathbb{B}$ is in the convex hull of matrices with elements $B_{kl,nm} = |\langle \Phi_{kl} | U \otimes V | \Phi_{nm} \rangle|^2$ for some general unitaries U, V , where $|\Phi_{ij}\rangle = (\mathbb{I} \otimes W_{ij})|\Phi_{00}\rangle$ and $|\Phi_{00}\rangle := |\Phi\rangle$ are Bell basis states and W_{ij} are discrete Weyl operators (see Section 2.9 and [10]). Matrices of this type are a subset of the unistochastic matrices [118] — we coin this subset *the product-Bell unistochastic matrices*. The following lemma is then employed:

Lemma 5.3.3. *Product-Bell unistochastic matrices are a convex combination of local permutation matrices.*

The proof of this direction is then completed by noting that a convex combination of product-Bell unistochastic matrices remains a convex combination of local permutation matrices.

To show statement 1 implies statement 2, consider the channels $\tilde{\mathcal{N}}, \tilde{\mathcal{M}}$ whose Choi-states are diagonalised in the given Bell basis with the same eigenvalues as $\mathcal{J}^{\mathcal{N}}$ and $\mathcal{J}^{\mathcal{M}}$. This is guaranteed by Lemma 5.3.2. When statement 2 holds, one can write $\tilde{\mathcal{M}} = \alpha \tilde{\mathcal{N}} + (1 - \alpha) \tilde{\mathcal{N}}[\sigma_x(\cdot)\sigma_x]$ where $\alpha = \sum_{n=m} p_{nm}$, meaning there is an allowed operation converting $\mathcal{N}$ into $\mathcal{M}$, completing the proof. See Section 5.8.2 for a full derivation, including derivations of both these Lemmas. $\square$

Result 5.3.1 states that an allowed operation of $\mathcal{R}^{\text{dyn}}$ exists between $\mathcal{N}$ and $\mathcal{M}$ if and only if the vector of eigenvalues of the Choi-state of the channel $\mathcal{M}$ is in the convex hull of local permutations applied to the vector of eigenvalues of the Choi-state of the channel $\mathcal{N}$. This is reminiscent of the analogous result for $\mathcal{R}^{\text{st}}$ for channels of equal input and output dimension [18], which states that there exists an $\mathcal{N} \in \mathfrak{D}^{\text{st}}$ such that $\sigma = \mathcal{N}(\rho)$ if and only if $\lambda(\rho) \succ \lambda(\sigma)$, where $\lambda(\rho)$ ($\lambda(\sigma)$) is the spectrum of ρ (σ). Majorisation is then equivalent to the existence of a doubly stochastic matrix mapping the eigenvalues of ρ to σ [119]. Here, in contrast, we do not have all doubly stochastic matrices but only a subset, specifically those involving *local* permutations.

Mathematically, the question of the existence of an allowed operation between two channels can now be simplified to a convex optimisation problem that can be easily solved on conventional computing hardware. Physically, this allows a comparison of the ability of the allowed operations of $\mathcal{R}^{\text{st}}$ to preserve informational non-equilibrium/purity.

5.3.1 Example Channels

As an application of Result 5.3.1, we now consider the existence of an allowed operation of $\mathfrak{D}^{\text{dyn}}$ between the qubit depolarising and dephasing channels, defined as

$$\begin{aligned}\mathcal{D}_s^{\text{pol}}(\rho) &:= s\rho + (1-s)\text{tr}[\rho]\mathbb{I}/2, \\ \mathcal{D}_q^{\text{ph}}(\rho) &:= q\rho + (1-q)\sum_{n=0}^1 |n\rangle\langle n|\rho|n\rangle\langle n|,\end{aligned}\tag{5.7}$$

respectively. Both of these channels are unital i.e.,

$$\mathcal{D}_s^{\text{pol}}(\mathbb{I}) = \mathbb{I}, \mathcal{D}_q^{\text{ph}}(\mathbb{I}) = \mathbb{I} \quad \forall q, s \in [0, 1],\tag{5.8}$$

and therefore are within $\mathfrak{D}^{\text{st}}$. We note that if $s, q = 1$ then both channels are the identity channel; if $s, q = 0$ then the channels are the completely depolarising and completely dephasing channel respectively.

In what follows we will use the notation that

$$\mathcal{N} \rightarrow \mathcal{M} \text{ if } \exists \Pi \in \mathfrak{D}^{\text{dyn}} \text{ such that } \mathcal{M} = \Pi(\mathcal{N}),\tag{5.9}$$

and $\iff$ to mean if and only if. Now, as proved in Section 5.3.1, we have the following conditions for there to exist an allowed operation of $\mathfrak{D}^{\text{dyn}}$ mapping the depolarising channel to a depolarising channel, and the dephasing channel to a dephasing channel:

$$\begin{aligned}\mathcal{D}_s^{\text{pol}}(\cdot) \longrightarrow \mathcal{D}_{s'}^{\text{pol}}(\cdot) &\iff s' \leq s \\ \mathcal{D}_q^{\text{ph}}(\cdot) \longrightarrow \mathcal{D}_{q'}^{\text{ph}}(\cdot) &\iff q' \leq q.\end{aligned}\tag{5.10}$$

Physically, this means that if $s' \leq s$, then for all $\rho \in \mathcal{D}(\mathcal{H}^2)$ the static-resource of $\mathcal{D}_{s'}^{\text{pol}}(\rho)$ is less than or equal to the static-resource of $\mathcal{D}_s^{\text{pol}}(\rho)$, and likewise for $\mathcal{D}_q^{\text{ph}}(\rho)$. This is as expected, as increasing the probability of depolarising or dephasing decreases the amount of purity in the output state.

We then find the following requirement for the existence of an allowed operation mapping a depolarising channel to a dephasing channel:

$$\mathcal{D}_s^{\text{pol}}(\cdot) \rightarrow \mathcal{D}_q^{\text{ph}}(\cdot) \iff s = 1,\tag{5.11}$$

meaning there exists an allowed operation from $\mathfrak{D}^{\text{dyn}}$ mapping $\mathcal{D}_s^{\text{pol}}(\cdot)$ to $\mathcal{D}_q^{\text{ph}}(\cdot)$ only in the trivial case that $s = 1$, meaning that the depolarising channel is actually just the identity channel. Physically, this means that for all $s \neq 1$ and all q , one can always find at least one state $\rho \in \mathcal{D}(\mathcal{H}^2)$ such that $\mathcal{D}_q^{\text{ph}}(\rho)$ has more static resource than $\mathcal{D}_s^{\text{pol}}(\rho)$ (if dephasing a qubit in the computational basis one can just consider $\rho = |0\rangle\langle 0|$).

Finally, we find the following requirement for the existence of an allowed operation between the dephasing channel and depolarising channel:

$$\mathcal{D}_q^{\text{ph}}(\cdot) \rightarrow \mathcal{D}_s^{\text{pol}}(\cdot) \iff s \leq \frac{q}{4-q}.\tag{5.12}$$

meaning non-trivial allowed operations can exist between the dephasing and depolarising channels.

5.3.2 A Complete Set of Monotones

A monotone of a resource theory is a function $M : \mathcal{D}(\mathcal{H}) \rightarrow \mathbb{R}^+$ such that

$$M(\rho) \geq M(\mathcal{N}(\rho)) \quad \forall \mathcal{N} \in \mathfrak{D}^{\text{st}}, \rho \in \mathcal{D}(\mathcal{H}).\tag{5.13}$$

Whilst monotones can be considered to be resource measures, they do not capture the full partial order imposed by a resource theory. Hence, it may be the case that $M(\sigma) \leq M(\rho)$ even if no allowed operation exists between ρ and σ . A complete set of monotones $\{M_i\}$, on the other hand, does provide an alternative to the convertibility question by capturing the full partial order: an allowed operation exists such that $\rho \rightarrow \sigma$ if and only if $M_i(\sigma) \leq M_i(\rho) \forall i$. In this section, we use a geometrical interpretation of Result 5.3.1 to find a complete set of monotones for the resource theory of informational non-equilibrium preservability.

The set of local permutations acting on our vector of eigenvalues of the Choi-state of our input channel, λ , is $\mathcal{Q} = \{\mathbb{P}_i \lambda : \forall \mathbb{P}_i \in \mathfrak{P}\}$, where $\mathfrak{P}$ is the set of local qubit permutations. Result 5.3.1 can then be rephrased as stating that a matrix $\mathbb{D}$ exists, such that $\mu = \mathbb{D}\lambda$, if μ is in the convex hull of vectors in $\mathcal{Q}$.

Each element of $\mathcal{Q}$ lives in a three-dimensional subspace of $\mathbb{R}^4$ due to normalisation i.e., the trace of the Choi-state is one, so the sum of the components of each of the vectors in $\mathcal{Q}$ is one. The vectors $\mathcal{Q}$ within this lower dimensional subspace are given by $\tilde{\mathcal{Q}}$ — these are just the elements of $\mathcal{Q}$ with their final component discarded. Geometrically, the elements of $\tilde{\mathcal{Q}}$ form a simplex, as depicted in Fig. 5.1.

Each point in the simplex represents a qubit unital channel. The extreme points of the outer simplex are the identity channel and channels that are equivalent to the identity up to a Pauli operator. From these channels, it is possible to reach all other channels under allowed operations, as their convex hull is the whole space. The centre of the simplex, $\tilde{\Lambda}$, represents the free state preparation channel, Λ . This channel can be reached by any channel under allowed operations.

Using this geometrical representation, a complete set of monotones can readily be identified. In what follows, we assume that

$$\dim(\text{Span}\{\mathfrak{R}\}) = 3 (= d_S^2 - 1), \quad (5.14)$$

where

$$\mathfrak{R} = \{\tilde{\lambda}_i - \tilde{\lambda}_0 : \tilde{\lambda}_0 \in \tilde{\mathcal{Q}}, \forall \tilde{\lambda}_i \in \tilde{\mathcal{Q}}\}. \quad (5.15)$$

Note, in the first line we have included the general case to be considered below. This condition means that the set $\mathfrak{R}$ spans the whole space. Geometrically, this means that the shape of the convex hull of the points in $\tilde{\mathcal{Q}}$ is $d_S^2 - 1$ dimensional i.e., it does not live in a lower dimensional space. Colloquially, one can think of $\tilde{\mathcal{Q}}$ as plotting a set of points, and $\mathfrak{R}$ then capturing the directions between these points. If the directions between these points all exists in a lower dimensional subspace, then the shape created by joining up these points does too[†]. Under this assumption, the follow result can be proved.

Result 5.3.4. *There exists a set of $d_S^2 = 4$ linear inequalities that are a complete set of monotones of $\mathcal{R}^{\text{dyn}}$.*

Proof Sketch. With the vectors in $\tilde{\mathcal{Q}}$ satisfying Eq. (5.14), their convex hull forms a tetrahedron in $\mathbb{R}^3$. If the three-dimensional representation of the channel μ , given by $\tilde{\mu}$, lies inside this tetrahedron, it is then in the convex hull of the vectors in $\tilde{\mathcal{Q}}$. This is equivalent to μ being in the convex hull of the vectors in $\mathcal{Q}$ and hence an allowed operation between the channels represented by λ and μ existing. Each monotone then checks on what side of one of the four planes that encompass each of the four faces of the tetrahedron $\tilde{\mu}$ lies on. If for all four planes $\tilde{\mu}$ lies on the same side as $\tilde{\Lambda}$, it must be inside the tetrahedron — as $\tilde{\Lambda}$ always sits inside the tetrahedron — and an allowed operation will exist. See Section 5.8.5 for a complete proof, including a discussion of the case that Eq. (5.14) does not hold.

5.3.2.1 Qubit Example for the Complete Set of Monotones

For clarity, we here give an example of building this complete set of monotones for $\mathcal{R}^{\text{dyn}}$. The complete set of monotones will determine if there exists an allowed operations $\Pi \in \mathfrak{D}^{\text{dyn}}$, such that $\mathcal{M} = \Pi(\mathcal{N})$, where $\mathcal{M}, \mathcal{N} \in \mathfrak{D}^{\text{st}}$. We assume that all the vectors in $\tilde{\mathcal{Q}}$ satisfy Eq. (5.14).

Let the vector $\lambda \in \mathbb{R}^4$ contain the eigenvalues of the Choi-state $\mathcal{J}^{\lambda}$, $\lambda^t = [\lambda_{00}, \lambda_{01}, \lambda_{10}, \lambda_{11}]$, and let $\mu \in \mathbb{R}^4$ be the vector of eigenvalues of the Choi-state $\mathcal{J}^{\mu}$, $\mu^t = [\mu_{00}, \mu_{01}, \mu_{10}, \mu_{11}]$. The set $\tilde{\mathcal{Q}}$ is then

[†] When considering any ϵ level of noise in the description of a given channel, this condition is always expected to hold.

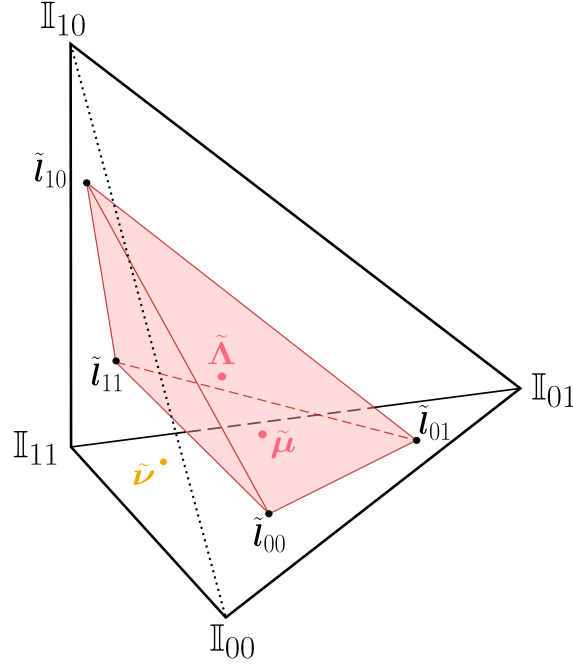


Figure 5.1: A simplex in $\mathbb{R}^3$ where each point represents the Choi-state of a qubit unital channel that is diagonal in the Bell basis. I_{ij} , the extreme points of the outer simplex, are the identity channel and channels that are equivalent to the identity channel up to a Pauli operator. The set of vectors $\tilde{\mathcal{V}}$ are plotted as points, $\tilde{I}_{ij}$, where each point represents a local permutation of λ . The convex hull of these vectors is shaded red with the free channel, represented by $\tilde{\Lambda}$, at the centre. $\tilde{\mu}$ sits within this convex hull, and hence, an allowed operation exists between the channel λ and μ . Then, as $\tilde{\nu}$ sits outside the convex hull, no allowed operation exists between the channel λ and ν .

given by

$$\tilde{\mathcal{V}} = \left\{ \begin{bmatrix} \lambda_{00} \\ \lambda_{01} \\ \lambda_{10} \end{bmatrix}, \begin{bmatrix} \lambda_{01} \\ \lambda_{00} \\ \lambda_{11} \end{bmatrix}, \begin{bmatrix} \lambda_{11} \\ \lambda_{10} \\ \lambda_{01} \end{bmatrix}, \begin{bmatrix} \lambda_{10} \\ \lambda_{11} \\ \lambda_{00} \end{bmatrix} \right\}, \quad (5.16)$$

with these vectors forming a tetrahedron when plotted in $\mathbb{R}^3$, as seen in Fig. 5.1, due to Eq. (5.14) being satisfied. To find each of the monotones, the planes encompassing each face of the tetrahedron must be found. In general, a plane is mathematically described by a vector perpendicular to all vectors within the plane. This can be found by taking the cross product of any two vectors within the plane.

The first face considered here is that in which the first three vectors of $\tilde{\mathcal{V}}$ terminate. By taking the difference of two vectors terminating in the plane, a vector within the plane can be found. Using this, the plane in which these three vectors terminate can be described by the vector

$$f_{00} = \left[\begin{bmatrix} \lambda_{00} \\ \lambda_{01} \\ \lambda_{10} \end{bmatrix} - \begin{bmatrix} \lambda_{01} \\ \lambda_{00} \\ \lambda_{11} \end{bmatrix} \right] \times \left[\begin{bmatrix} \lambda_{00} \\ \lambda_{01} \\ \lambda_{10} \end{bmatrix} - \begin{bmatrix} \lambda_{11} \\ \lambda_{10} \\ \lambda_{01} \end{bmatrix} \right], \quad (5.17)$$

which is graphically depicted in Fig. 5.2. Now, all vectors that terminate within the plane described by this vector have the same dot product with the vector (see Section 5.8.5 for more details on this). Labelling this dot product k , it can be seen that k can be found by taking the dot product of f_{00} with any of the three vectors,

$$f_{00} \cdot \begin{bmatrix} \lambda_{00} \\ \lambda_{01} \\ \lambda_{10} \end{bmatrix} = f_{00} \cdot \begin{bmatrix} \lambda_{01} \\ \lambda_{00} \\ \lambda_{11} \end{bmatrix} = f_{00} \cdot \begin{bmatrix} \lambda_{11} \\ \lambda_{10} \\ \lambda_{01} \end{bmatrix} = k, \quad (5.18)$$

By checking the dot product of f_{00} with $\tilde{\Lambda}$, the 'correct side' of the face for $\tilde{\mu}$ to be on can be found i.e., given $\tilde{\Lambda}$ is always inside the tetrahedron, the correct side of the plane to be on is the same side of the

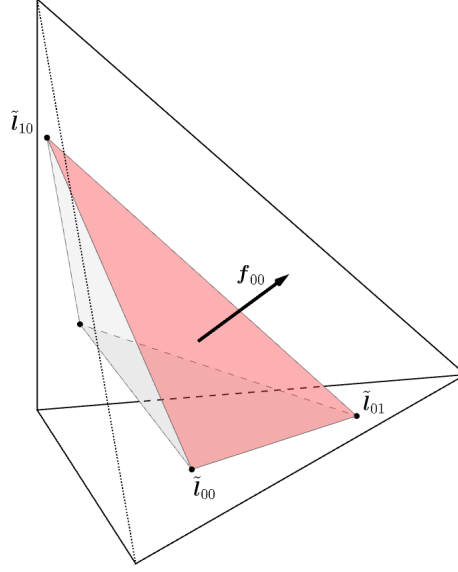


Figure 5.2: A single face of the simplex given by $\tilde{\mathcal{V}}$ is highlighted and the vector defining the plane encompassing that face, f_{00} , is shown. The plane contains three of the four vectors in $\tilde{\mathcal{V}}$ (Figure made with [120]).

plane that $\tilde{\Lambda}$ is on. Here, we assume without loss of generality that

$$f_{00} \cdot \tilde{\Lambda} = f_{00} \cdot \begin{bmatrix} 1/4 \\ 1/4 \\ 1/4 \end{bmatrix} \leq k, \quad (5.19)$$

as with the introduction of a minus sign where necessary one can ensure that the inequalities always point in this direction. Hence, any vector $a \in \mathbb{R}^3$ with $a \cdot f_{00} < l$ will be on the correct side of the plane, and hence the correct side of the face. As the three dimensional representation of the channel μ is given by $\tilde{\mu}^t = [\mu_{00}, \mu_{01}, \mu_{10}]$, one of the set of complete monotones is then

$$f_{00} \cdot \tilde{\mu} \leq k. \quad (5.20)$$

This assesses if $\tilde{\mu}$ is on the correct side of the hyperplane defined by f_{00} , such that it *could* be inside the tetrahedron.

The above procedure is then repeated for each combination of three vectors in $\tilde{\mathcal{V}}$ giving four inequalities characterised by the set of four vectors $\tilde{f} = \{f_{00}, f_{01}, f_{10}, f_{11}\}$. Using Table 5.1 for the values of the dot product, the complete set of monotones can then be summarised as

$$\{f_{00} \cdot \tilde{\mu} \leq k, f_{01} \cdot \tilde{\mu} \leq m, f_{10} \cdot \tilde{\mu} \leq o, f_{11} \cdot \tilde{\mu} \leq q\}. \quad (5.21)$$

These four inequalities are then a complete set of monotones, meaning that all four inequalities are satisfied if and only if there exists an allowed operation mapping $\mathcal{N} \rightarrow \mathcal{M}$.

Table 5.1: The values of the dot product of the elements of $\tilde{\mathfrak{V}}$, denoted by $\tilde{l}_{ij} \in \tilde{\mathfrak{V}}$, and the elements of $\mathfrak{f}$, denoted by $f_{ij} \in \mathfrak{f}$. The set of values $\{\alpha, \beta, \gamma, \delta\}$ are constants unrelated to each other.

	f_{00}	f_{01}	f_{10}	f_{11}
$\tilde{l}_{00}$	k	m	o	δ
$\tilde{l}_{01}$	k	m	γ	q
$\tilde{l}_{10}$	k	β	o	q
$\tilde{l}_{11}$	α	m	o	q

5.4 Arbitrary Finite Dimensional Systems

Here, the resource theory of informational non-equilibrium is extended to higher dimensional systems, such that $d_S = n$, in the important specialised case that $\mathfrak{S}^{\text{st}}$ is the set of *Weyl-covariant channels*, which we denote by $\mathfrak{U}_{\mathcal{W}}$. A channel $\mathcal{N}$ is Weyl-covariant if

$$\mathcal{N}(W_{ab}\rho W_{ab}^\dagger) = W_{ab}\mathcal{N}(\rho)W_{ab}^\dagger \quad \forall a, b \in [0, d_S - 1] \text{ \& } \rho \in \mathcal{D}(\mathcal{H}^n), \quad (5.22)$$

where W_{ab} are the discrete Weyl operators [10] (see Section 2.9 for more details). These are a physically relevant set of channels given that they contain the completely depolarising, completely dephasing and partial thermalisation channels. For all $\mathcal{N} \in \mathfrak{U}_{\mathcal{W}}$, there exists a probability vector p_{ab} such that

$$\mathcal{N}(\rho) = \sum_{ab} p_{ab} W_{ab} \rho W_{ab}^\dagger. \quad (5.23)$$

Given all discrete Weyl operators are unitary, it can easily be seen from here that Weyl-covariant channels are a subset of the mixed unitary channels, $\mathfrak{U}_{\mathcal{W}} \subseteq \mathfrak{U}_M$. Moreover, using the fact that the discrete Weyl operators can be used to generate a maximally entangled basis $|\Phi_{ab}\rangle = (\mathbb{I} \otimes W_{ab})|\Phi_{00}\rangle$ (See Theorem 2.9.2), it can be seen that the Choi-states of Weyl-covariant channel are diagonal in the Bell basis:

$$\begin{aligned} \mathcal{J}^{\mathcal{N}} &= (\mathcal{I} \otimes \mathcal{N})(|\Phi_{00}\rangle\langle\Phi_{00}|) \\ &= (\mathcal{I} \otimes \sum_{ab} p_{ab} W_{ab}(\cdot)W_{ab}^\dagger)(|\Phi_{00}\rangle\langle\Phi_{00}|) \\ &= \sum_{ab} p_{ab} (\mathbb{I} \otimes W_{ab})(|\Phi_{00}\rangle\langle\Phi_{00}|)(\mathbb{I} \otimes W_{ab}^\dagger) \\ &= \sum_{ab} p_{ab} |\Phi_{ab}\rangle\langle\Phi_{ab}|. \end{aligned} \quad (5.24)$$

Moreover, using the Choi-Jamiołkowski isomorphism, it can be seen that Choi-states that are diagonal in the Bell basis then correspond to Weyl-covariant channels. The physical meaning of the Choi-states of all qubit unital channels being diagonalisable under local unitaries, as detailed in Lemma 5.3.2, can now be seen: all qubit unital channels are equal to a Weyl-covariant channel up to a pre-processing and post-processing unitary (this is formalised in Lemma 5.8.1 in the proof of Result 5.3.1 in Section 5.8.2).

Using this, we can now characterise the dynamical resource theory in this special higher dimensional case. Here, the allowed operations of the resource theory of informational non-equilibrium preservability become a pre-processing and post-processing Weyl-covariant channel with shared randomness. We denote this by the set $\mathfrak{S}_{\mathcal{W}}^{\text{dyn}} \subseteq \mathfrak{S}^{\text{dyn}}$. The DRT considered for $d_S = n$ will be referred to as $\mathcal{R}_n^{\text{dyn}}$. In this picture, we get the following result that mirrors Result 5.3.1 for qubit channels.

Result 5.4.1. For every given $\mathcal{M}, \mathcal{N} \in \mathfrak{S}^{\text{st}} = \mathfrak{U}_{\mathcal{W}}$, the following statements are equivalent:

1. $\mathcal{M} = \Pi(\mathcal{N})$ for some allowed operation $\Pi \in \mathfrak{S}_{\mathcal{W}}^{\text{dyn}}$.
2. $\mu = \mathbb{D}'\lambda$, where $\mathbb{D}'$ is a convex combination of local *cyclic* permutation matrices.

Proof Sketch. From the fact that the Choi-states of Weyl-covariant channels are diagonal in the Bell basis, the proof of Result 5.4.1 is similar to that of Result 5.3.1. The difference arises from $\mathbb{B}$, the matrix relating the vector of eigenvalues of the Choi-state of the two channels, now being in the convex hull of matrices with elements $B_{kl,nm} = |\langle \Phi_{kl} | W_{ab} \otimes W_{cd} | \Phi_{nm} \rangle|^2$, where W_{ab} and W_{cd} are general discrete Weyl operators. This restricts the set of permutations to be only the set of local *cyclic* permutations. A complete derivation can be found in Section 5.8.4. $\square$

Following the same logic as in the qubit case, again assuming that Eq. (5.14) holds in the general case, a complete set of monotones can also be found in the higher dimensional case.

Result 5.4.2. There exists a set of $d_{\mathfrak{S}}^2$ inequalities that are a complete set of monotones of $\mathcal{R}_n^{\text{dyn}}$.

Proof Sketch. The logic is identical to Result 5.3.4, with a complete proof given in Section 5.8.5. $\square$

5.5 Operational Interpretation

In this section, we use the complete set of monotones developed for $\mathcal{R}^{\text{dyn}}$ and $\mathcal{R}_n^{\text{dyn}}$ to create an operational interpretation of the pre-order of the respective resource theories.

Firstly, we make a few observations that will be used in this section. Thus far throughout this chapter, we have been using Choi-states that are diagonal in a fixed basis, namely the Bell basis. As a result, we have been able to map the Choi-states to a vector of their eigenvalues,

$$\mathcal{J}^{\mathcal{N}} = (\mathcal{I} \otimes \mathcal{N})(|\Phi_{00}\rangle\langle\Phi_{00}|) \rightarrow \lambda \in \mathbb{R}^{d_{\mathfrak{S}}^2}. \quad (5.25)$$

Due to normalisation, the final dimension of the vector is redundant and has often been dropped without loss of information. The complete mapping consider has then been

$$\mathcal{J}^{\mathcal{N}} \rightarrow l_{00} \in \mathfrak{Q} \rightarrow \tilde{l}_{00} \in \tilde{\mathfrak{Q}}, \quad (5.26)$$

where $\mathfrak{Q} = \{\mathbb{P}_i \lambda : \forall \mathbb{P}_i \in \mathfrak{P}\}$, with $l_{00} = \lambda$, and $\tilde{\mathfrak{Q}}$ are the vectors in $\mathfrak{Q}$ with the final value in each vector discarded. We note that once $\mathcal{J}^{\mathcal{N}}$ has been found, then l_{00} can be found, from which the rest of $\mathfrak{Q}$ can be found by applying all of the possible local cyclic permutations to l_{00} . From this, all of $\tilde{\mathfrak{Q}}$ can then also be found.

Interestingly, it can be seen that there is an alternative way to generate the sets $\mathfrak{Q}$ and $\tilde{\mathfrak{Q}}$. Namely, they can be found by using different Bell states in the definition of the Choi-state, and following the same mapping given in Eq. (5.26). For example,

$$\begin{aligned} \mathcal{J}_{01}^{\mathcal{N}} &= (\mathcal{I} \otimes \mathcal{N})(|\Phi_{01}\rangle\langle\Phi_{01}|) \rightarrow l_{01} \in \mathfrak{Q} \rightarrow \tilde{l}_{01} \in \tilde{\mathfrak{Q}}, \\ \mathcal{J}_{10}^{\mathcal{N}} &= (\mathcal{I} \otimes \mathcal{N})(|\Phi_{10}\rangle\langle\Phi_{10}|) \rightarrow l_{10} \in \mathfrak{Q} \rightarrow \tilde{l}_{10} \in \tilde{\mathfrak{Q}}, \end{aligned} \quad (5.27)$$

and so on, where a subscript has been included on the Choi-state to indicate which Bell basis state was used in the Choi-state definition. Using a different Bell basis state leaves the eigenvalues invariant but permutes their position just as a local permutations does. Therefore, each element of $\tilde{\mathfrak{Q}}$ can be considered to be a mapping of the Choi-state of a channel $\mathcal{N} \in \mathfrak{S}^{\text{st}}$ to a vector in $\mathbb{R}^{d_{\mathfrak{S}}^2-1}$ where a different Bell state has been used in the Choi-state definition.

Now, using this notation, it can be seen that

$$\sum_{ij} \mathcal{J}_{ij}^{\mathcal{N}} = \mathbb{I}, \quad (5.28)$$

due to linearity, and the fact that $\sum_{ij} |\Phi_{ij}\rangle\langle\Phi_{ij}| = \mathbb{I}$ whilst $\mathcal{N}(\mathbb{I}) = \mathbb{I}$. Hence, the states $\mathcal{J}_{ij}^{\mathcal{N}}$ can be considered elements of a POVM, as they sum to the identity whilst $\mathcal{J}_{ij}^{\mathcal{N}} \geq 0 \forall i, j$ as $\mathcal{N} \in \text{CPTP}$.

We now present the operational interpretation of the partial order, again assuming that Eq. (5.14) holds.

Result 5.5.1. *Let $\mathcal{N}, \mathcal{M} \in \mathfrak{D}^{\text{st}}$ be given. Then there exists a set of d_S^2 many bipartite states $\{\sigma_{ij}\}_{ij}$ making the following two statements equivalent:*

1. $\mathcal{M} = \Pi(\mathcal{N})$ for some allowed operation $\Pi \in \mathfrak{D}^{\text{dyn}}$.
2. For every i, j it holds that

$$\langle\Phi_{ij}| (\mathcal{I} \otimes \mathcal{N})(\sigma_{ij}) |\Phi_{ij}\rangle \geq \langle\Phi_{ij}| (\mathcal{I} \otimes \mathcal{M})(\sigma_{ij}) |\Phi_{ij}\rangle. \quad (5.29)$$

Here, $|\Phi_{ij}\rangle$ is a state from the Bell basis, $|\Phi_{ab}\rangle = (\mathbb{I} \otimes W_{ab})|\Phi_{00}\rangle$.

Proof Sketch. Each of the vectors describing the hyper-planes used to build the complete set of monotones (in Result 5.3.4 and Result 5.4.2) can be used to generate the eigenvectors of a state that we assume to be diagonal in the Bell basis. Deciding on which side of the facets the vector representing the target channel sits can then be reformulated in terms of whether the probability of getting a particular outcome when measuring the POVM given in Eq. (5.28) is larger than some constant. This can then be rephrased in terms of condition (2) for each outcome. Therefore, each of the conditions in (2) check whether one of the monotones of the complete set of monotones is satisfied, meaning that an allowed operation exists if and only if all the conditions hold.

We note that the proof initially shows that if $\mathcal{M} = \Pi(\mathcal{N})$ for $\Pi \in \mathfrak{D}^{\text{dyn}}$ then condition (2) holds between $\mathcal{N}^\dagger$ and $\mathcal{M}^\dagger$, the adjoint of the channels $\mathcal{N}$ and $\mathcal{M}$ respectively. A separate lemma (Lemma 5.8.3) is then proved that states that if an allowed operation exists between two channels, then an allowed operation also exists between their adjoints, ensuring the equivalence of the two statements. See Section 5.8.6 for a full proof. $\square$

Result 5.5.1 can be interpreted as a type of state discrimination task: firstly, a referee chooses a state $\sigma_{ij} \in \{\sigma_{ij}\}_{ij}$, and distributes half of the bipartite state to Alice and half to Bob. Alice then sends her half to Bob via the channel $\mathcal{N}$ or $\mathcal{M}$, and Bob performs a Bell-state measurement i.e., he measures the non-degenerate observable $O = \sum_{ij} \lambda_{ij} |\Phi_{ij}\rangle\langle\Phi_{ij}|$. To each outcome of O , the referee and Bob will have assigned a state from $\{\sigma_{ij}\}_{ij}$, such that if he gets the outcome λ_{ij} Bob says that has the state σ_{ij} . Alice and Bob then succeed in the task if Bob gets the outcome λ_{ij} , when they are given the state σ_{ij} by the referee. If for all states in the set Bob's success probability of correctly identifying the state is at least as high when Alice applies $\mathcal{N}$ as when Alice applies $\mathcal{M}$, an allowed operation exists between $\mathcal{N}$ and $\mathcal{M}$. See Fig. 5.3 for a graphical depiction of the task. Geometrically, if for a given state σ_{ij} the success probability of Alice and Bob identifying the correct state is greater given Alice sends her half of the state to Bob via $\mathcal{N}$ rather than $\mathcal{M}$, then this is equivalent to checking that μ is on the correct side of one facet of the simplex. If the success probability is greater for all states, then this is equivalent to checking that μ is on the correct side of *all* facets of the simplex.

Here, the individual probabilities of successful discriminating a given state need to be considered, not just the average success probability as is typically considered in a state discrimination tasks (see [59] and Section 3.1). Whilst the average success probability of performing discrimination must be higher for $\mathcal{N}$ than $\mathcal{M}$, due to it being a weighted sum of the above conditions, this becomes only a necessary condition and does not capture the full partial order. Moreover, this state discrimination tasks differs in that it has a fixed measurement, rather than an optimisation over all possible measurements with the aim to maximise the probability of successful discrimination.

This result offers a route for the pre-order set by $\mathcal{R}^{\text{dyn}}$ and $\mathcal{R}_n^{\text{dyn}}$ to be experimentally investigated, meaning the ability of two channels to preserve informational non-equilibrium (or purity) for all possible states could be compared in the lab. To achieve this, one would need to play the state discrimination tasks for some number of rounds that depended upon the confidence with which they would like to assert that there exists an allowed operation between the channels. This would need to be done for both

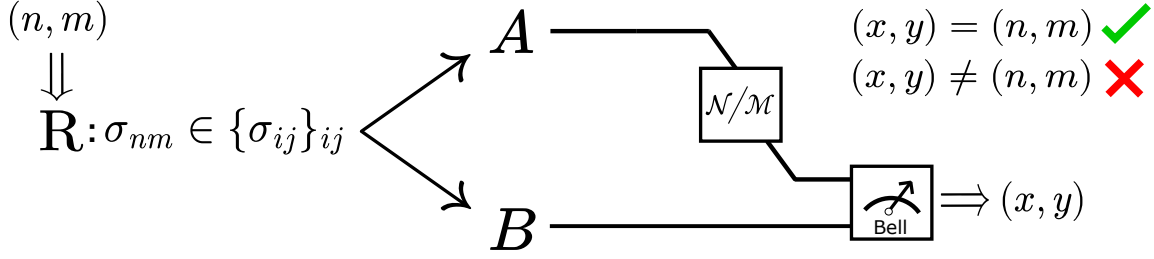


Figure 5.3: A graphical depiction of the operational interpretation of Result 5.5.1. The referee takes as an input a pair of indices (n, m) . They then prepare the bipartite state σ_{nm} , sending half to Alice and half to Bob. Alice then sends her half of the state σ_{nm} to Bob via the channel $\mathcal{N}$ or $\mathcal{M}$. Bob then makes a joint measurement in the Bell basis and outputs a pair of indices (x, y) . If $(x, y) = (n, m)$ then Alice and Bob succeed in the task, if $(x, y) \neq (n, m)$ then they fail. The probability that Alice and Bob succeed in the task cannot be worst when using $\mathcal{N}$ then $\mathcal{M}$ for every possible input state for there to exists an allowed operation between $\mathcal{N}$ and $\mathcal{M}$.

$\mathcal{N}$ and $\mathcal{M}$, and then a comparison of how successful Alice and Bob were at discriminating each of the individual states in the set would need to be made.

5.5.0.1 Qubit Example for the Operational Interpretation

In this section, an example of the formation of a set of states for the operational interpretation of the partial order for qubit channels in $\mathcal{R}^{\text{st}}$ is given.

As detailed in the proof of Result 5.5.1, the states used in the state discrimination tasks for the input channel $\mathcal{N}$ are constructed from the complete set of monotones for the channel $\mathcal{N}^+$. We then utilise the fact that an allowed operation existing between $\mathcal{N}^+$ and $\mathcal{M}^+$ implies the existence of an allowed operation between $\mathcal{N}$ and $\mathcal{M}$. Given this equivalence, we here construct the states from the complete set of monotones for the channel $\mathcal{N}$ so that we can utilise the example given in Section 5.3.2.1.

Recall from Section 5.3.2.1 that the vector $\lambda \in \mathbb{R}^4$ is a vector of eigenvalues of the Choi-state $\mathcal{J}^{\mathcal{N}}$, $\lambda^t = [\lambda_{00}, \lambda_{01}, \lambda_{10}, \lambda_{11}]$, and $\mathcal{Q}$ is the set of vectors that are the local permutations applied to λ , where we set $\lambda = I_{00} \in \mathcal{Q}$. The set $\tilde{\mathcal{Q}}$ is then this set of vectors with the final elements removed. Then, $\mu \in \mathbb{R}^4$ is a vector of eigenvalues of the Choi-state $\mathcal{J}^{\mathcal{M}}$, $\mu^t = [\mu_{00}, \mu_{01}, \mu_{10}, \mu_{11}]$. Moreover, detailed mathematically in Eq. (5.17) and graphical in Fig. 5.2, is the vector $f_{00} \in \mathbb{R}^3$ that characterises the plane encompasses one face of the tetrahedron. Here, we let

$$f_{00} = \begin{bmatrix} a \\ b \\ c \end{bmatrix} \in \mathfrak{f}, \quad (5.30)$$

where $\mathfrak{f}$ is as defined in Section 5.3.2.1. It was also seen in Eq. (5.18) that

$$f_{00} \cdot \tilde{l}_{00} = f_{00} \cdot \tilde{l}_{01} = f_{00} \cdot \tilde{l}_{10} = k, \quad (5.31)$$

where $\tilde{l}_{00}, \tilde{l}_{01}, \tilde{l}_{10} \in \tilde{\mathcal{Q}}$. Lastly, recall that if a channel represented by $\tilde{\mu}$ is on the same side of the plane f_{00} as $\tilde{\lambda}$ — such that it *could* be inside the simplex — then $f_{00} \cdot \tilde{\mu} \leq k$.

Now, the first of the set of states can be created by first embedding the vector f_{00} in a higher dimensional space,

$$F_{00} = \begin{bmatrix} a \\ b \\ c \\ 0 \end{bmatrix}. \quad (5.32)$$

The vector representing the state preparation channel of the maximally mixed state, scaled by some number r_{00} , is then added to F_{00} to generate a vector with all positive elements. Note, this introduces

significant freedom of choice. This vector is then normalised, giving

$$\mathbf{Q}_{00} = \frac{1}{w_{00}} \begin{bmatrix} a + r_{00}/4 \\ b + r_{00}/4 \\ c + r_{00}/4 \\ r_{00}/4 \end{bmatrix}, \quad (5.33)$$

where w_{00} is the normalisation.

The elements of $\mathbf{Q}_{00}$ are then taken to be the eigenvalues of a density operator, σ_{00} , that is diagonal in the Bell basis, such that

$$\sigma_{00} = \frac{1}{w_{00}} \begin{pmatrix} a + r_{00}/4 & 0 & 0 & 0 \\ 0 & b + r_{00}/4 & 0 & 0 \\ 0 & 0 & c + r_{00}/4 & 0 \\ 0 & 0 & 0 & r_{00}/4 \end{pmatrix}, \quad (5.34)$$

when σ_{00} is decomposed in the Bell basis. Given $\mathcal{J}_{00}^{\mathcal{N}}$ is also diagonal in the Bell basis, it holds that

$$\begin{aligned} \text{tr}[\mathcal{J}_{00}^{\mathcal{N}} \sigma_{00}] &= \mathbf{l}_{00} \cdot \mathbf{Q}_{00} \\ &= \frac{[a\lambda_{00} + b\lambda_{01} + c\lambda_{10}] + r_{00}/4 [\lambda_{00} + \lambda_{01} + \lambda_{10} + \lambda_{00}]}{w_{00}} \\ &= \frac{f_{00} \cdot \tilde{\mathbf{l}}_{00} + r_{00}/4}{w_{00}} \\ &= \frac{k + r_{00}/4}{w_{00}}, \end{aligned} \quad (5.35)$$

which can be interpreted as the probability of getting the outcome associated to the POVM element $\mathcal{J}_{00}^{\mathcal{N}}$ when measuring the POVM detailed in Eq. (5.28) on the state σ_{00} . This can be seen to be some constant that depends on the geometric interpretation i.e., it depends on k . Here, we let $\text{tr}[\mathcal{J}_{00}^{\mathcal{N}} \sigma_{00}] = p_k$, where p_k is a constant that depends on k .

We now consider the trace of the product of $\mathcal{J}_{00}^{\mathcal{M}}$ and σ_{00} :

$$\begin{aligned} \text{tr}[\mathcal{J}_{00}^{\mathcal{M}} \sigma_{00}] &= \boldsymbol{\mu} \cdot \mathbf{Q}_{00}, \\ &= \frac{[a\mu_{00} + b\mu_{01} + c\mu_{10}] + r_{00}/4 [\mu_{00} + \mu_{01} + \mu_{10} + \mu_{00}]}{w_{00}} \\ &= \frac{f_{00} \cdot \tilde{\boldsymbol{\mu}} + r_{00}/4}{w_{00}}. \end{aligned} \quad (5.36)$$

If $f_{00} \cdot \tilde{\boldsymbol{\mu}} \leq k$, then $\tilde{\boldsymbol{\mu}}$ is on the same side of the face as $\tilde{\mathbf{A}}$, and if $f_{00} \cdot \tilde{\boldsymbol{\mu}} > k$, then $\tilde{\boldsymbol{\mu}}$ is not on the same side of the face as $\tilde{\mathbf{A}}$. From here, it follows that if $\text{tr}[\mathcal{J}_{00}^{\mathcal{M}} \sigma_{00}] \leq p_k$ then $\tilde{\boldsymbol{\mu}}$ is on the same side of the face as $\tilde{\mathbf{A}}$, and if $\text{tr}[\mathcal{J}_{00}^{\mathcal{M}} \sigma_{00}] > p_k$ then $\tilde{\boldsymbol{\mu}}$ is not on the same side of the face as $\tilde{\mathbf{A}}$. Therefore, to be on the ‘correct side’ of the face it must hold that

$$\text{tr}[\mathcal{J}^{\mathcal{N}} \sigma_{00}] \geq \text{tr}[\mathcal{J}^{\mathcal{M}} \sigma_{00}]. \quad (5.37)$$

By then noting that $\text{tr}[\mathcal{J}^{\mathcal{N}} \sigma_{00}] = \langle \Phi_{00} | (\mathcal{I} \otimes \mathcal{N}^{\dagger})(\sigma_{00}) | \Phi_{00} \rangle$, a condition similar to that in Result 5.5.1 is found. Repeating this procedure, but starting with $\mathcal{N}^{\dagger}$, will give the correct condition.

This must now be repeated for the other hyper-planes characterised by the vectors in $\mathfrak{f}$, leading to the creation of a set of four states $\{\sigma_{ij}\}_{i,j=0}^1$. It can then be concluded that an allowed operation exists between $\mathcal{N}$ and $\mathcal{M}$ if and only if

$$\langle \Phi_{ij} | (\mathcal{I} \otimes \mathcal{N}(\sigma_{ij}) | \Phi_{ij} \rangle \geq \langle \Phi_{ij} | (\mathcal{I} \otimes \mathcal{M}(\sigma_{ij}) | \Phi_{ij} \rangle \quad \forall i, j \in \{0, 1\}. \quad (5.38)$$

Each condition therefore clearly evaluates one monotone of the complete set of monotones. Moreover, as detailed above, these conditions can be interpreted as a state discrimination task. Interestingly, it can be seen that only the diagonal elements of the set of states $\{\sigma_{ij}\}_{i,j=0}^1$ contribute and hence need to be fixed.

There is therefore freedom in the choice of all off diagonal elements.

Alternative Strategies

Whilst considering this set of conditions is favourable given it has an operational interpretation as a state discrimination task, they are not the only set of conditions of this form that can be used to verify the existence of an allowed operation. To see this, we note that from Eq. (5.31) and Eq. (5.27), it can be concluded that

$$\text{tr}[\mathcal{J}_{00}^{\mathcal{N}} \sigma_{00}] = \text{tr}[\mathcal{J}_{01}^{\mathcal{N}} \sigma_{00}] = \text{tr}[\mathcal{J}_{10}^{\mathcal{N}} \sigma_{00}] = p_k. \quad (5.39)$$

Then, given that $\{\mathcal{J}_{00}^{\mathcal{N}}, \mathcal{J}_{01}^{\mathcal{N}}, \mathcal{J}_{10}^{\mathcal{N}}, \mathcal{J}_{11}^{\mathcal{N}}\}$ is a POVM, as detailed in Eq. (5.28), it follows that

$$\begin{aligned} \text{Tr}[\mathcal{J}_{11}^{\mathcal{N}} \sigma_{00}] &= 1 - [\text{tr}[\mathcal{J}_{00}^{\mathcal{N}} \sigma_{00}] + \text{tr}[\mathcal{J}_{01}^{\mathcal{N}} \sigma_{00}] + \text{tr}[\mathcal{J}_{10}^{\mathcal{N}} \sigma_{00}]] \\ &= 1 - 3p_k. \end{aligned} \quad (5.40)$$

The same observations can be made for each of the states in the set $\{\sigma_{ij}\}_{ij=0}^1$. This means there are multiple different operational methods for evaluating each monotone of the complete set of monotones. Table 5.2 shows the complete range of possible combinations that can be used to assess each face of the simplex in this qubit case. Four values must be compared, namely $\{p_k, p_m, p_o, p_q\}$, one for each face of the simplex; each needs to be compared only once. Hence, another equally valid strategy of assessing the existence of an allowed operation is

$$\begin{aligned} p_k &= \langle \Phi_{00} | (\mathbb{I} \otimes \mathcal{N})(\sigma_{00}) | \Phi_{00} \rangle \geq \langle \Phi_{00} | (\mathbb{I} \otimes \mathcal{M})(\sigma_{00}) | \Phi_{00} \rangle, \\ p_m &= \langle \Phi_{00} | (\mathbb{I} \otimes \mathcal{N})(\sigma_{01}) | \Phi_{00} \rangle \geq \langle \Phi_{00} | (\mathbb{I} \otimes \mathcal{M})(\sigma_{01}) | \Phi_{00} \rangle, \\ p_o &= \langle \Phi_{00} | (\mathbb{I} \otimes \mathcal{N})(\sigma_{10}) | \Phi_{00} \rangle \geq \langle \Phi_{00} | (\mathbb{I} \otimes \mathcal{M})(\sigma_{10}) | \Phi_{00} \rangle, \\ 1 - 3p_q &= \langle \Phi_{00} | (\mathbb{I} \otimes \mathcal{N})(\sigma_{11}) | \Phi_{00} \rangle \leq \langle \Phi_{00} | (\mathbb{I} \otimes \mathcal{M})(\sigma_{11}) | \Phi_{00} \rangle. \end{aligned} \quad (5.41)$$

As long as each value is checked, the exact formation of the conditions used is irrelevant. A possible operational interpretation of these alternative conditions is left as an open question.

Table 5.2: The left most column, X , gives different elements of a POVM produced from the Choi-state of a channel $\mathcal{N}$ using different Bell basis states in the definition. The top most row, Y , gives the four different states generated from the vectors that are also used to generate the four monotones. The values in the table are $\text{Tr}(XY)$.

	σ_{00}	σ_{01}	σ_{10}	σ_{11}
$\mathcal{J}_{00}^{\mathcal{N}}$	p_k	p_m	p_o	$1 - 3p_q$
$\mathcal{J}_{01}^{\mathcal{N}}$	p_k	p_m	$1 - 3p_o$	p_q
$\mathcal{J}_{10}^{\mathcal{N}}$	p_k	$1 - 3p_m$	p_o	p_q
$\mathcal{J}_{11}^{\mathcal{N}}$	$1 - 3p_k$	p_m	p_o	p_q

5.6 Classical Capacity Quantifies Informational Non-Equilibrium Preservability

In addition to finding complete sets of monotones, it is important to find physically motivated monotones. Here, the *Holevo capacity* — a lower bound on the classical capacity of a quantum channel [10, 121–125] — is shown to be one such physically relevant monotone of both $\mathcal{R}^{\text{dyn}}$ and $\mathcal{R}_n^{\text{dyn}}$. Formally, the Holevo capacity is defined as follows.

Definition 5.6.1 (The Holevo capacity). *The Holevo capacity, a lower bound on the classical capacity of a quantum channel, is given by*

$$\chi(\mathcal{N}) = \sup_{\{p_i, \rho_i\}} \left\{ s\left(\sum_i p_i \mathcal{N}(\rho_i)\right) - \sum_i p_i s(\mathcal{N}(\rho_i)) \right\}, \quad (5.42)$$

where $s(\cdot)$ is the von-Neumann entropy, and the supremum is taken over the ensemble of quantum states $\{p_i, \rho_i\}$ [126–128].

The classical capacity provides one of a number of ways to quantify the ability of a quantum channel to transmit information. Informally, the classical capacity gives the number of classical bits that a single use of the channel can transmit between two parties. Whilst in some instances, the classical capacity is equal to the Holevo capacity [126, 127], in general, it is the asymptotic limit of the Holevo capacity for infinite uses of the channel [128]. Using the definition of the Holevo capacity, we get the following result.

Result 5.6.1. *The Holevo capacity is a faithful monotone of both $\mathcal{R}^{\text{dyn}}$ and $\mathcal{R}_n^{\text{dyn}}$, meaning that $\chi(\mathcal{N}) = 0$ if and only if $\mathcal{N} = \Lambda$, and*

$$\chi(\mathcal{N}) \geq \chi(\Pi(\mathcal{N})) \quad \forall \Pi \in \mathfrak{D}^{\text{dyn}}, \mathcal{N} \in \mathfrak{D}^{\text{st}}. \quad (5.43)$$

Proof Sketch. The unitary invariance of the Holevo capacity means that we can focus on only the Weyl-covariant channels that qubit unital channels are equivalent to up to a pre and post processing unitary channel; in the higher dimensional case we consider only Weyl-covariant channels anyway. We then use the fact that Weyl-covariant channels are a type of irreducibly covariant channels, for which the Holevo capacity can be rewritten in terms of the quantum relative entropy. Then, using the properties of the quantum relative entropy, such as convexity and data processing, the desired result can be achieved. $\square$

Result 5.6.1 means that $\mathcal{R}^{\text{dyn}}$ and $\mathcal{R}_n^{\text{dyn}}$ can be employed to compare the classical communication abilities of unital quantum channels in settings where non-equilibrium can only increase. One can compare $\mathcal{R}^{\text{dyn}}$ and $\mathcal{R}_n^{\text{dyn}}$ to the dynamical resource theory of classical communication [129], which has all state-preparation channels as the set of channels with no resource (the free set). This is due to these channels not being able to communicate any information. In the resource theories presented here, we have the additional thermodynamic constraint that the channels cannot drive the system out of equilibrium. This suggests that, conceptually, informational non-equilibrium preservability can be viewed as the ability to transmit classical information [129, 130] subject to additional thermodynamic constraints.

5.7 Conclusion

In this chapter, the dynamical resource theory of informational non-equilibrium preservability has been characterised for both qubit mixed unitary channels and Weyl-covariant channels. Through the presented framework, the ability of two allowed operations of $\mathcal{R}^{\text{st}}$ to preserve the informational non-equilibrium (purity) of input states can be compared for all states in the space of states. This characterisation acts as a proof of principle for resource preservability theories and suggests that others could be developed. The clear and immediate next step would be to extend these results to the resource theory of athermality. An understanding of the ability of *thermal operations* [17] to preserve non-equilibrium could further our knowledge of the dynamical aspects of quantum thermodynamics. Such understanding could also be utilised, for example, in the study of efficient thermal processes. Moreover, a characterisation of the DRT of information non-equilibrium for channels with differing input and output dimension would be another interesting extension of the above results. Additionally, resource preservability theories could also be built upon other successful SRTs of a more pragmatic nature. For instance, the set of local operations and classical communication could be studied and their ability to preserve entanglement quantified.

A link between the ability of a channel to preserve information non-equilibrium and classical capacity — a natural measure for classical communication — has been made. Further effort should be made to discover other physically relevant monotones of $\mathcal{R}^{\text{dyn}}$ in the hope of finding additional areas in which it is applicable.

5.8 Additional Material

5.8.1 Additional Details on the Allowed Operations of $\mathcal{R}^{\text{dyn}}$

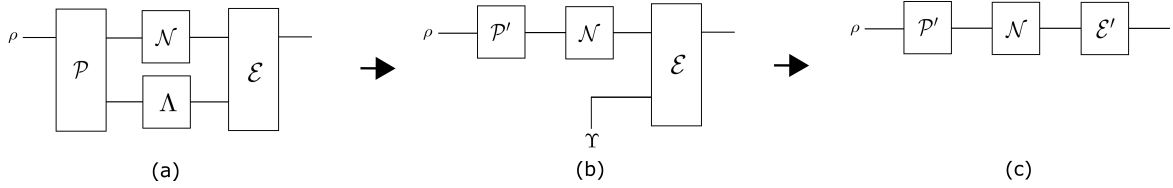


Figure 5.4: A quantum circuit diagram to depict how the allowed operations of resource preservability theories, defined in [26], become the allowed operations of the dynamical resource theory of informational non-equilibrium defined in the main text.

The allowed operations of resource preservability theories presented in [26], consist of a super-channel with a free channel in the memory system, see Fig. 5.4. a. When considering the underlying static resource theory to be the resource theory of informational non-equilibrium, the free super channels can be reformulated so that the memory system can be ignored.

This is a result of $\mathcal{R}^{\text{dyn}}$ having only one free channel (when considering a fixed dimension) — the state preparation channel of the maximally mixed state, $\Lambda(\rho) = \mathbb{I}/d_S \text{tr}[\rho]$. Therefore, an allowed operation $\Pi \in \mathfrak{D}^{\text{dyn}}$, such that $\mathcal{M} = \Pi(\mathcal{N})$, where $\mathcal{M} : \mathcal{K}_{in} \rightarrow \mathcal{K}_{out} \in \mathfrak{D}^{\text{st}}$, $\mathcal{N} : \mathcal{H}_{in} \rightarrow \mathcal{H}_{out} \in \mathfrak{D}^{\text{st}}$ can be written as

$$\mathcal{M} = \mathcal{E} \circ (\mathcal{N} \otimes \Lambda) \circ \mathcal{P}, \quad (5.44)$$

where $\mathcal{P} : \mathcal{K}_{in} \rightarrow \mathcal{H}_{in} \otimes \mathcal{H}_a$, $\mathcal{N} \otimes \Lambda : \mathcal{H}_{in} \otimes \mathcal{H}_a \rightarrow \mathcal{H}_{out} \otimes \mathcal{H}_a$ and $\mathcal{E} : \mathcal{H}_{out} \otimes \mathcal{H}_a \rightarrow \mathcal{K}_{out}$.

Consider a state $\rho \in \mathcal{K}_{in}$ input into the channel $\mathcal{M}$. The state after the application of $\mathcal{P}$ is $\sigma = \mathcal{P}(\rho) \in \mathcal{H}_{in} \otimes \mathcal{H}_a$. The memory system is then traced out by Λ , leaving $\sigma' = \text{Tr}_a(\sigma) \in \mathcal{H}_{in}$. Anything stored in the memory space is then lost. A new channel can be defined as $\mathcal{P}' = \text{tr}_a[\mathcal{P}] : \mathcal{K}_{in} \rightarrow \mathcal{H}_{in}$, such that $\sigma' = \mathcal{P}'(\rho)$. This channel can be seen to be a unital channel still and hence is still a noisy operation.

The state σ' is then input into the channel $\mathcal{N}$ giving $\omega = \mathcal{N}(\sigma')$. A maximally mixed state is then appended to ω and it is input into a channel $\mathcal{E}$, see Fig. 5.4.b. This appended maximally mixed state can be absorbed into the description of the channel $\mathcal{E}$ by considering it in the noisy operation formulation, see Definition 4.3.2. We can then define a new unital channel $\mathcal{E}' : \mathcal{H}_{out} \rightarrow \mathcal{K}_{out}$, see Fig. 5.4.c.

The allowed operation given in Eq. (5.44) can now be written as

$$\mathcal{M} = \mathcal{E}' \circ \mathcal{N} \circ \mathcal{P}', \quad (5.45)$$

where $\mathcal{P}' : \mathcal{K}_{in} \rightarrow \mathcal{H}_{in}$, $\mathcal{N} : \mathcal{H}_{in} \rightarrow \mathcal{H}_{out}$, $\mathcal{E}' : \mathcal{H}_{out} \rightarrow \mathcal{K}_{out}$ where $\mathcal{P}', \mathcal{N}, \mathcal{E}' \in \mathfrak{D}^{\text{st}}$. The channels $\mathcal{P}', \mathcal{E}'$ can then be relabelled to $\mathcal{P}, \mathcal{E}$ as they are still just some pre-processing and post-processing noisy operation.

When a random variable, κ , is shared between the pre-processing and post-processing channels with some probability p_κ , the channels applied depend on κ . The allowed operations become

$$\mathcal{M} = \sum_{\kappa} p_\kappa \mathcal{E}_\kappa \circ \mathcal{N} \circ \mathcal{P}_\kappa, \quad (5.46)$$

where $\mathcal{E}_\kappa, \mathcal{P}_\kappa \in \mathfrak{D}^{\text{st}}$. From this it can be seen that $\mathfrak{D}^{\text{dyn}}$ is a convex set.

5.8.2 Proof of Result 5.3.1

Result 5.3.1 For every given $\mathcal{M}, \mathcal{N} \in \mathfrak{D}^{\text{st}}$, the following statements are equivalent:

1. $\mathcal{M} = \Pi(\mathcal{N})$ for some allowed operation $\Pi \in \mathfrak{D}^{\text{dyn}}$.
2. $\mu = \mathbb{D}\lambda$, where $\mathbb{D} = \sum_{n,m=0}^1 p_{nm} \sigma_x^n \otimes \sigma_x^m$ with p_{nm} the elements of a probability vector and σ_x the Pauli X .

We will first prove that statement 1 implies statement 2. To do this, we will prove multiple Lemmas. We will then prove that statement 2 implies statement 1.

5.8.2.1 Proof that statement 1 implies statement 2

Proof of Lemma 5.3.2

Lemma 5.3.2 The Choi-states of qubit unital channels can always be digonalised in the Bell basis (maximally entangled basis) under local unitaries.

Proof. Given $\mathcal{J}^{\mathcal{N}}$ is a valid bipartite density operator it can always be written in Bloch form as

$$\mathcal{J}^{\mathcal{N}} = \frac{1}{4} (\mathbb{I} \otimes \mathbb{I} + (\beta \cdot \sigma) \otimes \mathbb{I} + \mathbb{I} \otimes (\gamma \cdot \sigma) + \sum_{ij} T_{ij} \sigma_i \otimes \sigma_j), \quad (5.47)$$

where $\beta, \gamma \in \mathbb{R}^3$, σ is a vector of the Pauli operators $\{\sigma_i\}_{i=1}^3$ defined by

$$\sigma_1 := |0\rangle\langle 1| + |1\rangle\langle 0| = X, \quad \sigma_2 := i|0\rangle\langle 1| - i|1\rangle\langle 0| = Y, \quad \sigma_3 := |0\rangle\langle 0| - |1\rangle\langle 1| = Z, \quad (5.48)$$

in the computation basis ($\{|0\rangle, |1\rangle\}$) and T_{ij} are the matrix elements of a real matrix $T \in \mathbb{R}^4 \otimes \mathbb{R}^4$.

From the definition of $\mathcal{J}^{\mathcal{N}}$ it is clear that the first marginal must be the maximally mixed state. Given $\mathcal{N}$ is a unital channel, the second marginal must also be the maximally mixed state. Hence, $\text{tr}_{\text{in}}[\mathcal{J}^{\mathcal{N}}] = \text{tr}_{\text{out}}[\mathcal{J}^{\mathcal{N}}] = \frac{1}{2}\mathbb{I}$, where $\{\text{in}, \text{out}\}$ denoted the bipartitions of the maximally entangled state in the definition of the Choi-state. Using these conditions it can be seen that $\beta \cdot \sigma = \gamma \cdot \sigma = 0$ as

$$\begin{aligned} \text{tr}_{\text{in}}[\mathcal{J}^{\mathcal{N}}] &= \frac{1}{4} (2\mathbb{I}_2 + 2\gamma \cdot \sigma) = \frac{1}{2}\mathbb{I}_2 \\ &\Rightarrow \gamma \cdot \sigma = 0. \\ \text{tr}_{\text{out}}[\mathcal{J}^{\mathcal{N}}] &= \frac{1}{4} (2\mathbb{I}_2 + 2\beta \cdot \sigma) = \frac{1}{2}\mathbb{I}_2 \\ &\Rightarrow \beta \cdot \sigma = 0. \end{aligned} \quad (5.49)$$

It was then shown in [131] that the matrix T , with elements T_{ij} , can always be digonalised under local unitary operations. We partly reproduce the proof here. Firstly, it can be shown that to every unitary transformation U there exists a uniquely associated rotation O such that

$$U\beta \cdot \sigma U^\dagger = (O\beta) \cdot \sigma. \quad (5.50)$$

For completeness, we prove this well known fact in Theorem 5.8.2 below. Using this, under local unitaries $\mathcal{J}^{\mathcal{N}}$ transforms as

$$\mathcal{J}^{\tilde{\mathcal{N}}} = (U_1 \otimes U_2) \mathcal{J}^{\mathcal{N}} (U_1 \otimes U_2)^\dagger, \quad (5.51)$$

such that the matrix T in the Pauli-decomposition of $\mathcal{J}^{\mathcal{N}}$ and the matrix T' in the Pauli-decomposition of $\mathcal{J}^{\tilde{\mathcal{N}}}$ are related as

$$T' = O_1 T O_2^t, \quad (5.52)$$

where O_1, O_2 are rotations associated to the unitaries U_1, U_2 respectively, and $(\cdot)^t$ is the transpose

operator. To see how this arises from Eq. (5.50), consider

$$\begin{aligned}
 (U_1 \otimes U_2) \left[\sum_{i,j=1}^3 T_{ij} \sigma_i \otimes \sigma_j \right] (U_1^\dagger \otimes U_2^\dagger) &= \sum_{i,j=1}^3 T_{ij} U_1 \sigma_i U_1^\dagger \otimes U_2 \sigma_j U_2^\dagger, \\
 &= \sum_{i,j=1}^3 T_{ij} \sum_{n=1}^3 \alpha_{ni} \sigma_n \otimes \sum_{m=1}^3 \gamma_{mj} \sigma_m \\
 &= \sum_{i,j,n,m=1}^3 \alpha_{ni} T_{ij} \gamma_{mj} (\sigma_n \otimes \sigma_m),
 \end{aligned} \tag{5.53}$$

where in the second line we have used the fact that the Pauli-operators and the identity form an operator basis for the two dimensional, complex square matrices, along with the fact that the Pauli-operators are traceless. Now, due to Eq. (5.50), the α 's and γ 's can be seen to be the components of a rotation matrix (See Theorem 5.8.2 for more details). Moreover, $\alpha_{in} T_{ij} \gamma_{jm}$ can be seen to be the components of a product of three matrices, namely,

$$O_1 = \sum_{n,l=1}^3 \alpha_{nl} |n\rangle \langle l|, \quad T = \sum_{i,j=1}^3 T_{ij} |i\rangle \langle j|, \quad O_2 = \sum_{m,k=1}^3 \gamma_{mk} |m\rangle \langle k|. \tag{5.54}$$

such that T' has the form of Eq. (5.52) i.e., T' is the conjugation of T by two rotation matrices.

From here, it can be concluded that there always exists two unitaries that induce rotations on T to ensure T' is diagonal, due to the singular valued decomposition. In summary, there always exists two local unitaries such that the Choi-state of a qubit unital channel under these unitaries becomes

$$\mathcal{J}^{\tilde{\mathcal{N}}} = \frac{1}{4} (\mathbb{I} \otimes \mathbb{I} + \sum_i T_{ii} \sigma_i \otimes \sigma_i). \tag{5.55}$$

In the computational basis $\mathcal{J}^{\tilde{\mathcal{N}}}$ is then

$$\mathcal{J}^{\tilde{\mathcal{N}}} = \frac{1}{4} \begin{pmatrix} 1+T_{33} & 0 & 0 & T_{11}-T_{22} \\ 0 & 1-T_{33} & T_{11}+T_{22} & 0 \\ 0 & T_{11}+T_{22} & 1-T_{33} & 0 \\ T_{11}-T_{22} & 0 & 0 & 1+T_{33} \end{pmatrix}. \tag{5.56}$$

This has the same form as a matrix that is diagonal in the Bell basis but decomposed in the computational basis, i.e., if

$$\rho = \frac{1}{2} \begin{pmatrix} p_{00} + p_{01} & 0 & 0 & p_{00} - p_{01} \\ 0 & p_{10} + p_{11} & p_{10} - p_{11} & 0 \\ 0 & p_{10} - p_{11} & p_{10} + p_{11} & 0 \\ p_{00} - p_{01} & 0 & 0 & p_{00} + p_{01} \end{pmatrix} \tag{5.57}$$

then ρ can equivalently be decomposed as

$$\rho = \sum_{ij} p_{ij} |\Phi_{ij}^+\rangle \langle \Phi_{ij}^+|, \tag{5.58}$$

where the states $|\Phi_{nm}^+\rangle$ are the Bell basis states, generated from $|\Phi_{00}^+\rangle = 1/\sqrt{d_S} \sum_i |ii\rangle$ and the discrete Weyl operators [10] as

$$|\Phi_{nm}^+\rangle = \mathbb{I} \otimes W_{nm} |\Phi_{00}^+\rangle, \tag{5.59}$$

due to Theorem. 2.9.2. Hence, $\mathcal{J}^{\tilde{\mathcal{N}}}$ can alternatively be written as

$$\mathcal{J}^{\tilde{\mathcal{N}}} = \sum_{nm} \lambda_{nm} |\Phi_{nm}^+\rangle \langle \Phi_{nm}^+|, \tag{5.60}$$

where λ_{ij} are the eigenvalues of $\mathcal{J}^{\tilde{\mathcal{N}}}$. These eigenvalues are real, non-negative and $\sum_{nm} \lambda_{nm} = 1$, given

that $\mathcal{J}^{\mathcal{N}}$ is a valid density operator. Moreover, there exists a constructive form for the relationships between these eigenvalues and the matrix elements $\{T_{ii}\}_1^3$. $\square$

Physical Implication of Lemma 5.3.2

We insert a quick aside from the proof to consider the physical implications of Lemma 5.3.2. A channel $\mathcal{M}$ is a Weyl-covariant channel if

$$\mathcal{M}(W_{ij}\rho W_{ij}^\dagger) = W_{ij}\mathcal{M}(\rho)W_{ij}^\dagger \quad \forall i, j \text{ \& } \rho \in \mathcal{D}(\mathcal{H}), \quad (5.61)$$

where W_{ij} are the discrete Weyl operators. Any Weyl-covariant channel can then be written as

$$\mathcal{M}(\rho) = \sum p_{ij} W_{ij} \rho W_{ij}^\dagger, \quad (5.62)$$

for some probability vector p_{ij} . On qubits, the discrete Weyl operators are $\{\mathbb{I}, \sigma_x, \sigma_z, -i\sigma_y\}$, where the Pauli operators are defined as in Eq. (5.48). Therefore, a qubit Weyl-covariant channel can always be written as

$$\mathcal{M}(\rho) = \sum_k p_k \sigma_k \rho \sigma_k, \quad (5.63)$$

for some probability vector p_k , which can also be refereed to as a *mixed Pauli-channel*. Using Lemma 5.3.2, we now prove the following Lemma.

Lemma 5.8.1. *Every qubit unital channel can be transformed into a Weyl covariant channel via a pre-processing and post-processing unitary channel.*

Proof. Consider a qubit unital channel $\mathcal{N}$ with Choi-state

$$\mathcal{J}^{\mathcal{N}} = (\mathcal{I} \otimes \mathcal{N})(|\Phi_{00}^+\rangle\langle\Phi_{00}^+|). \quad (5.64)$$

From Lemma 5.3.2, it is know that there always exists two unitaries U_1 and U_2 such that $\mathcal{J}^{\mathcal{N}}$ can be diagonalised in the Bell basis, i.e., there exists a U_1, U_2 such

$$\mathcal{J}^{\mathcal{N}} = (U_1 \otimes U_2) \mathcal{J}^{\tilde{\mathcal{N}}} (U_1 \otimes U_2)^\dagger, \quad (5.65)$$

where $\mathcal{J}^{\tilde{\mathcal{N}}}$ can be decomposed with respect to the Bell basis as

$$\mathcal{J}^{\tilde{\mathcal{N}}} = \sum_{ij} \lambda_{ij} |\Phi_{ij}^+\rangle\langle\Phi_{ij}^+|. \quad (5.66)$$

This is now a new Choi-state corresponding to the channel $\tilde{\mathcal{N}}$. Given the Choi-Jamiołkowski isomorphism, $\tilde{\mathcal{N}}$ will not be the same channel as $\mathcal{N}$. It can then be noted that, in all dimensions, Choi-states that are diagonal in the Bell basis correspond to Weyl-covariant channels. This can be seen from the fact that discrete Weyl operators can be used to generate Bell basis states, such that

$$\begin{aligned} \mathcal{J}^{\tilde{\mathcal{N}}} &= \sum_{ij} \lambda_{ij} |\Phi_{ij}^+\rangle\langle\Phi_{ij}^+| \\ &= \sum_{ij} \lambda_{ij} (\mathcal{I} \otimes W_{ij})(|\Phi_{00}^+\rangle\langle\Phi_{00}^+|)(\mathcal{I} \otimes W_{ij}^\dagger) \\ &= (\mathcal{I} \otimes \sum_{ij} \lambda_{ij} W_{ij}(\cdot)W_{ij}^\dagger)(|\Phi_{00}^+\rangle\langle\Phi_{00}^+|). \end{aligned} \quad (5.67)$$

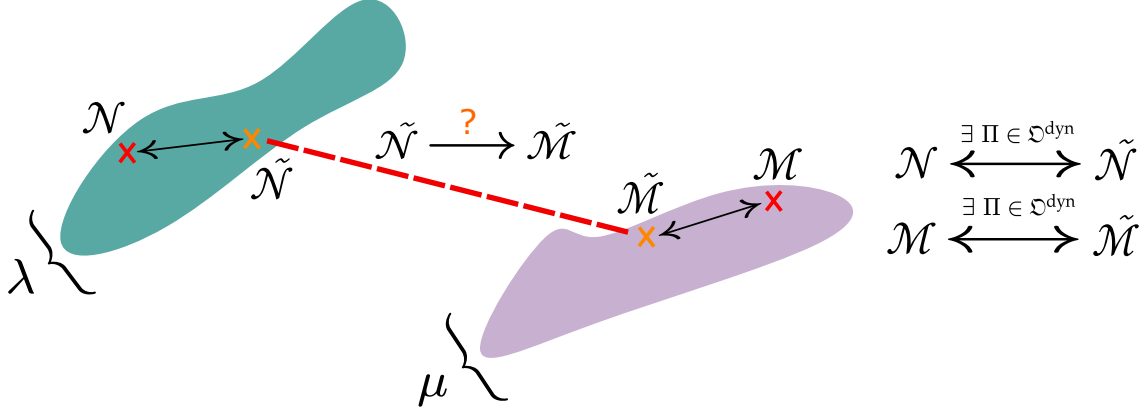


Figure 5.5: A graphical depiction of how Lemma 5.3.2 and Lemma 5.8.1 are used to prove Result 5.3.1. The cyan (purple) region represents the set of qubit unital channels whose Choi-states have eigenvalues λ (μ). All channels within a given region will have equal dynamical resource. Each region also has a Weyl-covariant channel, denoted on the diagram with an orange cross and a channel labelled with a tilde; the Choi-state of this Weyl-covariant channel is always diagonal in the Bell basis. Given a specific channel e.g., $\mathcal{N}$, in a region, there always exists an allowed operation that maps it to every other channel in the region, including the regions Weyl-covariant channel, $\tilde{\mathcal{N}}$. When asking if there exists an allowed operation in $\mathfrak{S}^{\text{dyn}}$ between two channels $\mathcal{N}$ and $\mathcal{M}$, given there always exists an allowed operation that maps the channels to the Weyl-covariant channels in their respective regions (and back), one can instead ask if there exists an allowed operation between the respective Weyl-covariant channels, $\tilde{\mathcal{N}}$ to $\tilde{\mathcal{M}}$.

Hence, $\tilde{\mathcal{N}}$ corresponds to a Weyl-covariant channel. Returning to Eq. (5.65), $\mathcal{J}^{\tilde{\mathcal{N}}}$ can be written as

$$\begin{aligned}\mathcal{J}^{\tilde{\mathcal{N}}} &= (U_1 \otimes U_2)(\mathcal{I} \otimes \mathcal{N})(|\Phi_{00}^+\rangle\langle\Phi_{00}^+|)(U_1 \otimes U_2)^\dagger \\ &= (\mathcal{I} \otimes U_2)(\mathcal{I} \otimes \mathcal{N})(\mathcal{I} \otimes U_1^\dagger)(|\Phi_{00}^+\rangle\langle\Phi_{00}^+|)(\mathcal{I} \otimes U_1^\dagger)^\dagger(\mathcal{I} \otimes U_2)^\dagger \\ &= (\mathcal{I} \otimes U_2 \circ \mathcal{N} \circ U_1^\dagger)(|\Phi_{00}^+\rangle\langle\Phi_{00}^+|)\end{aligned}\quad (5.68)$$

where $U_1^\dagger := U_1^\dagger(\cdot)(U_1^\dagger)^\dagger$ and $U_2 := U_2(\cdot)U_2^\dagger$ are unitary channels that apply the unitaries $U_1^\dagger$ and U_2 respectively. Again due to the Choi-Jamiołkowski isomorphism, it must therefore be the case that

$$\tilde{\mathcal{N}}(\rho) = U_2 \circ \mathcal{N} \circ U_1^\dagger(\rho). \quad (5.69)$$

Therefore, the diagonalisation of the Choi-states of qubit unital channels in the Bell basis is equivalent to fact that qubit unital channels are equal to a Weyl-covariant channel up to a pre-processing and post-processing unitary operation. Note, this result was discovered in [132] at the same time as ours. $\square$

Return to Proof that statement 1 implies statement 2

From Lemma 5.3.2, it can be seen that given an allowed operation $\mathcal{N} \in \mathfrak{S}^{\text{st}}$, there will always exists a pre-processing and post-processing unitary that will diagonalise its Choi-state. Equivalently, from Lemma 5.8.1, there will always exists a pre-processing and post-processing channel that will convert $\mathcal{N}$ into a Weyl covariant channel. As all unitaries are in $\mathfrak{S}^{\text{st}}$, and the allowed super channels of $\mathfrak{S}^{\text{dyn}}$ consist of a pre and post processing channel from $\mathfrak{S}^{\text{st}}$, for any $\mathcal{N} \in \mathfrak{S}^{\text{st}}$ there always exists a $\Pi \in \mathfrak{S}^{\text{dyn}}$ to covert it to a Weyl covariant channel. Moreover, there will also exists a $\Pi' \in \mathfrak{S}^{\text{dyn}}$ to covert it back. Hence, the resourcefulness of a channel $\mathcal{N} \in \mathfrak{S}^{\text{st}}$ and its corresponding Weyl-covariant channel $\tilde{\mathcal{N}} \in \mathfrak{S}^{\text{st}}$ are equivalent. This can be understood physically from the fact that unitaries do not change the static resource content on any state they act upon. In addition, it can be seen that any two channels $\mathcal{N}, \mathcal{Q} \in \mathfrak{S}^{\text{st}}$ that map to the same Weyl-covariant channel will also have the equivalent dynamical resource, as there will be a pre-processing and post-processing unitary channel that will map between them. This will occur if the eigenvalues of the Choi-states of $\mathcal{N}$ and $\mathcal{Q}$ are equivalent, as this defines the reachable Weyl-covariant channel.

This can be understood as separating $\mathfrak{S}^{\text{st}}$ into equivalence classes of channels with equal dynamical resource: if the eigenvalues of the Choi-states of two channels are equivalent, then the channels have equal dynamical resource and they belong to the same class. Within each equivalence class, there

will exist a Weyl-covariant channel which is always reachable via an allowed operation in $\mathfrak{D}^{\text{dyn}}$ from any other channel in the class. In fact, there exists an allowed operation between all channels within a given equivalence class. When determining if there exists an allowed operation of $\mathfrak{D}^{\text{dyn}}$ between two channels $\mathcal{N}, \mathcal{Q} \in \mathfrak{D}^{\text{st}}$, one can therefore instead just ask if there exists an allowed operation of $\mathfrak{D}^{\text{dyn}}$ between the dynamical-resource-equivalent Weyl-covariant channel of $\mathcal{N}, \mathcal{Q}$. See Fig. 5.5 for graphical depiction of these ideas.

Therefore, given we need only consider the existence of allowed operation of $\mathfrak{D}^{\text{dyn}}$ between Weyl-covariant channels, we can always assume that our Choi-states are diagonal in the Bell basis. We now need only consider the eigenvalues of the Choi-state, greatly reducing the number of parameters in consideration. For clarity, in the following proof we ignore the shared randomness. However, the same result is reached if it is included.

As a reminder, all qubit unital channels are mixed unitary channels. Consider now the general pre-processing and post-processing mixed unitary channels,

$$\mathcal{P}(\rho) = \sum_i p_i U_i \rho U_i^\dagger \in \mathfrak{D}^{\text{st}}, \quad \mathcal{E}(\rho) = \sum_j q_j V_j \rho V_j^\dagger \in \mathfrak{D}^{\text{st}}. \quad (5.70)$$

This gives rise to an allowed operation on $\mathcal{N} \in \mathfrak{D}^{\text{st}}$ of $\mathcal{E} \circ \mathcal{N} \circ \mathcal{P}$. The Choi-state formulation of this allowed operation, seen in Definition 5.1.3, is then

$$\begin{aligned} \mathcal{J}^{\mathcal{M}} &= (\mathcal{P}^t(\cdot) \otimes \mathcal{E}(\cdot)) \mathcal{J}^{\mathcal{N}} \\ &= \sum_{i,j} p_i q_j (U_i^t \otimes V_j) \mathcal{J}^{\mathcal{N}} (U_i^t \otimes V_j)^\dagger \\ &= \sum_{i,j} p_i q_j (U_i^t \otimes V_j) \left[\sum_{nm} \lambda_{nm} |\Phi_{nm}\rangle\langle\Phi_{nm}| \right] (U_i^t \otimes V_j)^\dagger \\ &= \sum_{i,j} \sum_{nm} p_i q_j \lambda_{nm} (U_i^t \otimes V_j) (|\Phi_{nm}\rangle\langle\Phi_{nm}|) (U_i^t \otimes V_j)^\dagger. \end{aligned} \quad (5.71)$$

Given operations in $\mathfrak{D}^{\text{dyn}}$ map channels in $\mathfrak{D}^{\text{st}}$ to channels in $\mathfrak{D}^{\text{st}}$, $\mathcal{J}^{\mathcal{M}}$ is also the Choi-state of a qubit unital channel and can therefore also be written diagonally in the Bell basis,

$$\mathcal{J}^{\mathcal{M}} = \sum_{kl} \mu_{kl} |\Phi_{kl}\rangle\langle\Phi_{kl}|, \quad (5.72)$$

such that

$$\mu_{kl} = \sum_{i,j} \sum_{nm} p_i q_j \lambda_{nm} |\langle\Phi_{kl}| U_i^t \otimes V_j |\Phi_{nm}\rangle|^2. \quad (5.73)$$

This can be interpreted as the product of a matrix and a vector, outputting another vector with components μ_{kl} . The existence of an allowed operation in $\mathfrak{D}^{\text{dyn}}$ can therefore be captured by the existence of a matrix $\mathbb{B}^{ij}$ such that

$$\boldsymbol{\mu} = \sum_{i,j} p_i q_j \mathbb{B}^{ij} \boldsymbol{\lambda}, \quad (5.74)$$

where $\boldsymbol{\lambda} \in \mathbb{R}^4$ is a vector of eigenvalues of $\mathcal{J}^{\mathcal{N}}$ and $\boldsymbol{\mu} \in \mathbb{R}^4$ is a vector of eigenvalues of $\mathcal{J}^{\mathcal{M}}$, i.e.,

$$\boldsymbol{\lambda} = \begin{bmatrix} \lambda_{00} \\ \lambda_{01} \\ \lambda_{10} \\ \lambda_{11} \end{bmatrix}, \quad \boldsymbol{\mu} = \begin{bmatrix} \mu_{00} \\ \mu_{01} \\ \mu_{10} \\ \mu_{11} \end{bmatrix}, \quad (5.75)$$

and $\mathbb{B}^{ij}$ is a matrix with elements $B_{nm,kl}^{ij} = |\langle\Phi_{kl}| U_i^t \otimes V_j |\Phi_{nm}\rangle|^2$. This means that the question of whether there exists an allowed operation in $\mathfrak{D}^{\text{dyn}}$ between two channels can now be succinctly asked in terms of the eigenvalues of the Choi-states of those channels. To complete the proof of this direction, Lemma 5.3.3 must be proved. This characterises the matrices of the form of $\mathbb{B}^{ij}$.

Proof of Lemma 5.3.3

Matrices of the form $\mathbb{B}_{ij}$ are a subset of the unistochastic matrices — the set of doubly stochastic matrices that can be produced by squaring the absolute values of the matrix elements of a unitary matrix. All unitary matrices give rise to a unistochastic matrix when this operation is performed. However, not all doubly stochastic matrices can be produced in this way i.e., the set of unistochastic matrices is a proper subset of the set of doubly stochastic matrices. Moreover, the doubly stochastic matrix one will get will depend on the basis in which the unitary matrix is decomposed when the square of the absolute value of the matrix elements is taken.

We coin the term product-Bell unistochastic matrix to describe $\mathbb{B}^{ij}$, as they are the subset of unistochastic matrices that can arise from squaring the absolute values of the matrix elements of a product unitary matrix decomposed in the Bell basis. Given $\mathbb{B}^{ij}$ is doubly stochastic, it can be written as a convex combination of permutation matrices in general [119]. Here, we have the additional structure of the unitary matrix being a tensor product of two unitaries and a fixed basis. We show in the following Lemma that this means that only the set of local permutations are required in the convex combination.

Lemma 5.3.3 *Product-Bell unistochastic matrices are a convex combination of local permutation matrices.*

Proof. Let $\mathbb{B}$ be a Product-Bell unistochastic matrix with matrix elements $B_{nm,kl} = |\langle \Phi_{kl} | U \otimes V | \Phi_{nm} \rangle|^2$. The unitaries U and V can be written in the Pauli operator basis as

$$U = \sum_{\alpha=0}^3 a_{\alpha} \sigma_{\alpha}, \quad V = \sum_{\beta=0}^3 b_{\beta} \sigma_{\beta}, \quad (5.76)$$

where $\sigma_0 := \mathbb{I}$, $\{\sigma_i\}_{i=1}^3$ are the Pauli operators as previously defined, and $a_{\alpha}, b_{\beta} \in \mathbb{C} \forall \alpha, \beta$. The matrix elements of $\mathbb{B}$ in the Bell basis are then

$$\begin{aligned} B_{kl,nm} &= |\langle \Phi_{kl} | U \otimes V | \Phi_{nm} \rangle|^2 \\ &= \left| \sum_{\alpha, \beta=0}^3 a_{\alpha} b_{\beta} \langle \Phi_{kl} | (\sigma_{\alpha} \otimes \sigma_{\beta}) | \Phi_{nm} \rangle \right|^2. \end{aligned} \quad (5.77)$$

Operators of the form $\mathbb{I} \otimes \sigma_i \forall i$ act on the Bell basis states to permute them to other Bell basis states multiplied by a phase term. Ignoring the phase term for now, the action of $\mathbb{I} \otimes \sigma_1$ and $\sigma_1 \otimes \mathbb{I}$ on the basis states is as follows,

$$\mathbb{I} \otimes \sigma_1 \text{ \& } \sigma_1 \otimes \mathbb{I} : |\Phi_{00}\rangle \rightarrow |\Phi_{10}\rangle, |\Phi_{01}\rangle \rightarrow |\Phi_{11}\rangle, |\Phi_{10}\rangle \rightarrow |\Phi_{01}\rangle, |\Phi_{11}\rangle \rightarrow |\Phi_{00}\rangle. \quad (5.78)$$

We will refer to this permutation of the Bell basis states as $\mathbb{P}_1$. The matrix of $\mathbb{P}_1$ decomposed in the Bell basis is then

$$\mathbb{P}_1 = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix} = \sigma_1 \otimes \sigma_0, \quad (5.79)$$

where ω_0 and ω_1 are the qubit permutations,

$$\sigma_0 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}. \quad (5.80)$$

Similarly we have

$$\begin{aligned} \mathbb{I} \otimes \sigma_2 \text{ \& } \sigma_2 \otimes \mathbb{I} : |\Phi_{00}\rangle &\rightarrow |\Phi_{11}\rangle, |\Phi_{01}\rangle \rightarrow |\Phi_{10}\rangle, |\Phi_{10}\rangle \rightarrow |\Phi_{01}\rangle, |\Phi_{11}\rangle \rightarrow |\Phi_{00}\rangle. \\ \mathbb{I} \otimes \sigma_3 \text{ \& } \sigma_3 \otimes \mathbb{I} : |\Phi_{00}\rangle &\rightarrow |\Phi_{01}\rangle, |\Phi_{01}\rangle \rightarrow |\Phi_{00}\rangle, |\Phi_{10}\rangle \rightarrow |\Phi_{11}\rangle, |\Phi_{11}\rangle \rightarrow |\Phi_{10}\rangle. \end{aligned} \quad (5.81)$$

which we call $\mathbb{P}_2$ and $\mathbb{P}_3$ respectively where, again with respect to the Bell basis, they have permutation

matrices

$$\mathbb{P}_2 = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix} = \sigma_1 \otimes \sigma_1, \quad (5.82)$$

and

$$\mathbb{P}_3 = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} = \sigma_0 \otimes \sigma_1. \quad (5.83)$$

Additionally, there is $\mathbb{I} \otimes \mathbb{I}$ that maps all basis states to themselves giving $\mathbb{P}_0 = \sigma_0 \otimes \sigma_0$. This set of permutations, $\mathfrak{P} = \{\mathbb{P}_0, \mathbb{P}_1, \mathbb{P}_2, \mathbb{P}_3\}$, are the local permutations given they can be broken down into a tensor product of qubit permutations.

It is then noted that combinations of permutations in $\mathfrak{P}$ are still in the set $\mathfrak{P}$ i.e., $\mathbb{P}_x \mathbb{P}_y \in \mathfrak{P}$ if $\mathbb{P}_x, \mathbb{P}_y \in \mathfrak{P}$. With this in mind, we then split up the Pauli operators in the description of $\mathbb{B}$ given in Eq. (5.77), showing that they can be written as combinations of permutation in $\mathfrak{P}$,

$$\begin{aligned} B_{kl,nm} &= \left| \sum_{\alpha,\beta=0}^3 a_\alpha b_\beta \langle \Phi_{kl} | (\sigma_\alpha \otimes \mathbb{I}) (\mathbb{I} \otimes \sigma_\beta) | \Phi_{nm} \rangle \right|^2, \\ &= \left| \sum_{\alpha,\beta=0}^3 a_\alpha b_\beta \theta_{\alpha,\beta} \langle \Phi_{kl} | \mathbb{P}_\alpha \circ \mathbb{P}_\beta | \Phi_{nm} \rangle \right|^2 \\ &= \left| \sum_{\gamma=0}^3 Q_\gamma \langle \Phi_{kl} | \mathbb{P}_\gamma | \Phi_{nm} \rangle \right|^2, \\ &= |\langle \Phi_{kl} | Q_0 \mathbb{P}_0 + Q_1 \mathbb{P}_1 + Q_2 \mathbb{P}_2 + Q_3 \mathbb{P}_3 | \Phi_{nm} \rangle|^2 \end{aligned} \quad (5.84)$$

where $\theta_{\alpha,\beta}$ is a phase factor arising from the permutations and in the third line a relabelling has taken place for clarity. The matrix $\mathbb{B}^{ij}$ decomposed in the Bell basis can now be seen to be a combination of the local permutation matrices,

$$\mathbb{B} = \begin{pmatrix} |Q_0|^2 & |Q_3|^2 & |Q_1|^2 & |Q_2|^2 \\ |Q_3|^2 & |Q_0|^2 & |Q_2|^2 & |Q_1|^2 \\ |Q_1|^2 & |Q_2|^2 & |Q_0|^2 & |Q_3|^2 \\ |Q_2|^2 & |Q_1|^2 & |Q_3|^2 & |Q_0|^2 \end{pmatrix} \quad (5.85)$$

which can be written as

$$\begin{aligned} \mathbb{B} &= |Q_0|^2 \mathbb{P}_0 + |Q_1|^2 \mathbb{P}_1 + |Q_2|^2 \mathbb{P}_2 + |Q_3|^2 \mathbb{P}_3, \\ &= |Q_0|^2 (\sigma_0 \otimes \sigma_0) + |Q_1|^2 (\sigma_1 \otimes \sigma_0) + |Q_2|^2 (\sigma_1 \otimes \sigma_1) + |Q_3|^2 (\sigma_0 \otimes \sigma_1). \end{aligned} \quad (5.86)$$

To be a convex combination it must be the case that

$$\sum_{\gamma=0}^3 |Q_\gamma|^2 = 1. \quad (5.87)$$

It can be seen from the matrix form of $\mathbb{B}^{ij}$ that a sum over all Q_γ is a sum over any row or column of the

matrix, such as

$$\begin{aligned}
\sum_{\gamma=0}^3 |Q_{\gamma}|^2 &= \sum_{n,m=0}^1 B_{00,nm} \\
&= \sum_{n,m=0}^1 |\langle \Phi_{00} | U_i \otimes V_j | \Phi_{nm} \rangle|^2 \\
&= \sum_{n,m=0}^1 \langle \Phi_{00} | (U_i \otimes V_j) (|\Phi_{nm}\rangle\langle\Phi_{nm}|) (U_i \otimes V_j)^{\dagger} | \Phi_{00} \rangle \\
&= \langle \Phi_{00} | (U_i \otimes V_j) \left[\sum_{n,m=0}^1 |\Phi_{nm}\rangle\langle\Phi_{nm}| \right] (U_i \otimes V_j)^{\dagger} | \Phi_{00} \rangle \\
&= 1.
\end{aligned} \tag{5.88}$$

as $\sum_{n,m=0}^1 |\Phi_{nm}\rangle\langle\Phi_{nm}| = \mathbb{I}$. Therefore, the matrix B is a convex combination of local permutations. The introduction of a dependence on the unitary in the mixed unitary channel, or shared randomness, does not effect this result as a convex combination of product-Bell unistochastic matrices remains a convex combination of local permutation matrices. $\square$

Inputting all this into the form of allowed operations found above leads to the conclusion that an allowed operation existing between $\mathcal{N} \rightarrow \mathcal{M}$ (statement 1) implies that there exists a matrix between λ and μ that is a convex combination of local permutations (statement 2). In the statement of Result 5.3.1, we then use that fact that $\sigma_0 = \sigma_1^0$, such that all permutations in $\mathfrak{P}$ can be generated from $\sigma_1^a \otimes \sigma_1^b$ for $a, b \in [0, 1]$. We then relabel $\sigma_1 \rightarrow \sigma_x$ to make it clear that σ_1 represents Pauli X.

To complete the proof of Result 1, the opposite direction must be shown.

5.8.2.2 Proof that statement 2 implies statement 1

Assuming statement 2 holds, we can write

$$\mu = \sum_{n,m=0}^1 p_{nm} \sigma_x^n \otimes \sigma_x^m \lambda, \tag{5.89}$$

where μ and λ are vectors of eigenvalues of the Weyl-covariant channels $\tilde{\mathcal{M}}, \tilde{\mathcal{N}} \in \mathfrak{D}^{\text{st}}$ respectively. Translating back to the Choi-states of the channels these vectors represent, we find

$$\mathcal{J}^{\tilde{\mathcal{M}}} = \sum_{n,m=0}^1 p_{nm} (\sigma_x^n \otimes \sigma_x^m) (\mathcal{J}^{\tilde{\mathcal{N}}}) (\sigma_x^n \otimes \sigma_x^m), \tag{5.90}$$

where $\mathcal{J}^{\tilde{\mathcal{M}}}$ and $\mathcal{J}^{\tilde{\mathcal{N}}}$ are Choi-states diagonal in the Bell basis, with eigenvalues given by the elements of μ and λ respectively. This can be re-written as

$$\begin{aligned}
\mathcal{J}^{\tilde{\mathcal{M}}} &= \sum_{n,m=0}^1 p_{nm} (\sigma_x^n \otimes \sigma_x^m) (\mathcal{I} \otimes \tilde{\mathcal{N}}) (|\Phi_{00}\rangle\langle\Phi_{00}|) (\sigma_x^n \otimes \sigma_x^m), \\
&= \sum_{n,m=0}^1 p_{nm} (\mathcal{I} \otimes \sigma_x^m) (\mathcal{I} \otimes \tilde{\mathcal{N}}) [(\mathcal{I} \otimes \sigma_x^n) (|\Phi_{00}\rangle\langle\Phi_{00}|) (\mathcal{I} \otimes \sigma_x^n)] (\mathcal{I} \otimes \sigma_x^m), \\
&= \sum_{n,m=0}^1 p_{nm} (\mathcal{I} \otimes \sigma_x^m [\cdot] \circ \tilde{\mathcal{N}} \circ \sigma_x^n [\cdot]) (|\Phi_{00}\rangle\langle\Phi_{00}|),
\end{aligned} \tag{5.91}$$

where $\sigma_x^m[\cdot]$ and $\sigma_x^n[\cdot]$ are channels applying σ_x^m and σ_x^n respectively, and we have used the fact that $\sigma_x^t = \sigma_x$. Hence, from the Choi-Jamiołkowski isomorphism, $\tilde{\mathcal{M}}$ is given by

$$\begin{aligned}\tilde{\mathcal{M}}(\rho) &= \sum_{nm} p_{nm} \sigma_x^m \circ \tilde{\mathcal{N}} \circ \sigma_x^n(\rho) \\ &= p_{00} \tilde{\mathcal{N}}(\rho) + p_{01} \sigma_x \tilde{\mathcal{N}}(\rho) \sigma_x + p_{10} \tilde{\mathcal{N}}(\sigma_x \rho \sigma_x) + p_{11} \sigma_x \tilde{\mathcal{N}}(\sigma_x \rho \sigma_x) \sigma_x.\end{aligned}\quad (5.92)$$

As $\tilde{\mathcal{N}}$ is a Weyl-covariant channel, and σ_x a discrete Weyl operator for qubits, the above decomposition of the channel can be written as

$$\begin{aligned}\tilde{\mathcal{M}}(\rho) &= (p_{00} + p_{11}) \tilde{\mathcal{N}}(\rho) + (p_{01} + p_{10}) \tilde{\mathcal{N}}(\sigma_x \rho \sigma_x) \\ &= \tilde{\mathcal{N}} \circ \mathcal{P},\end{aligned}\quad (5.93)$$

where we have used the fact that $\sigma_x \tilde{\mathcal{N}}(\rho) \sigma_x = \tilde{\mathcal{N}}(\sigma_x \rho \sigma_x)$ for a Weyl-covariant channel, and $\sigma_x^2 = \mathbb{I}$. We have then let $\mathcal{P}(\rho) = (p_{00} + p_{11}) \mathcal{I}(\rho) + (p_{01} + p_{10}) \sigma_x \rho \sigma_x$, where $\mathcal{I}(\cdot) = \mathbb{I}(\cdot) \mathbb{I}$. Therefore, if statement 2 holds then there always exists a mixed unitary pre-processing channel achieving the mapping $\mathcal{N} \rightarrow \mathcal{M}$. Hence, statement 2 implies statement 1. This completes the proof of Result 5.3.1.

Proof of Equation 5.50

Theorem 5.8.2. Let $\mathbf{n} \in \mathbb{R}^{3,+}$ be a Bloch vector of ρ . The Bloch vector of $\rho' = U\rho U^\dagger$ is then $O\mathbf{n}$, where O is a rotation matrix.

Proof. Let $\sigma = [\sigma_1, \sigma_2, \sigma_3]^t$ be a vector of Pauli operators. Then

$$\begin{aligned}U(\mathbf{n} \cdot \sigma)U^\dagger &= \sum_{j=1}^3 n_j U \sigma_j U^\dagger \\ &= \sum_{j=1}^3 n_j \sum_{i=1}^3 \alpha_{ij} \sigma_i \\ &= \sum_{i,j=1}^3 \alpha_{ij} n_j \sigma_i \\ &= (O\mathbf{n}) \cdot \sigma \quad \text{where } O = \sum_{i,j=1}^3 \alpha_{ij} |i\rangle \langle j|.\end{aligned}\quad (5.94)$$

In the above, n_j are the components of $\mathbf{n}$, and in the second line we have used the fact that the Pauli operators plus the identity form an operator basis. Note, as $\text{tr}[U \sigma_i U^\dagger] = 0 \forall i$ the coefficient for $\mathbb{I}$ in the decomposition of $U \sigma_i U^\dagger$ is zero.

To complete the proof we must now show that O is a rotation matrix. For this to hold it must be the case that $O^t O = O O^t = \mathbb{I}$, all entries must be real, and the determinant must be 1. We will prove each of these in turn. Firstly, consider

$$\begin{aligned}\text{tr}[(U \sigma_i U^\dagger)(U \sigma_j U^\dagger)] &= \text{tr}[\sigma_i \sigma_j] \\ &= \delta_{ij}.\end{aligned}\quad (5.95)$$

Alternatively,

$$\begin{aligned}
 \text{tr}[(U\sigma_i U^\dagger)(U\sigma_j U^\dagger)] &= \text{tr}\left[\sum_{k=1}^3 \alpha_{ki} \sigma_k \sum_{l=1}^3 \alpha_{lj} \sigma_l\right] \\
 &= \sum_{k,l=1}^3 \alpha_{ki} \alpha_{lj} \text{tr}[\sigma_k \sigma_l] \\
 &= \sum_{k=1}^3 \alpha_{ki} \alpha_{kj} \\
 &= \delta_{ij}
 \end{aligned} \tag{5.96}$$

where in the final line we have used Eq. (5.95). From this it can be seen that

$$\begin{aligned}
 O^t O &= \left(\sum_{i,j=1}^3 \alpha_{ji} |i\rangle \langle j|\right) \left(\sum_{k,l=1}^3 \alpha_{kl} |k\rangle \langle l|\right) \\
 &= \sum_{i,l=1}^3 \left(\sum_{k=1}^3 \alpha_{ki} \alpha_{kl}\right) |i\rangle \langle l| \\
 &= \sum_{i,l=1}^3 \delta_{il} |i\rangle \langle l| \\
 &= \mathbb{I}.
 \end{aligned} \tag{5.97}$$

Now, it can be seen that

$$\begin{aligned}
 \alpha_{ij}^* &= \text{tr}[\sigma_i U \sigma_j U^\dagger]^* \\
 &= \text{tr}[(\sigma_i U \sigma_j U^\dagger)^\dagger] \\
 &= \alpha_{ij},
 \end{aligned} \tag{5.98}$$

where we have used the cyclic properties of the trace. Hence, each component of the matrix is real. Lastly, to prove that O is a rotation matrix and not just an orthogonal matrix, it needs to be shown that the determinant is one. In general, it holds that the determinant of a three dimensional square matrix is given by

$$\text{Det}(O) \epsilon_{ijk} = \sum_{l,m,n=1}^3 \alpha_{li} \alpha_{mj} \alpha_{nk} \epsilon_{lmn}, \tag{5.99}$$

where

$$\epsilon_{ijk} = \begin{cases} 1 & \text{if } (i, j, k) \text{ is an even permutation of } (1, 2, 3) \\ -1 & \text{if } (i, j, k) \text{ is an odd permutation of } (1, 2, 3) \\ 0 & \text{if } i = j, \ j = k, \ k = i, \end{cases} \tag{5.100}$$

is the Levi-Civita symbol. By noting that

$$\sigma_i \sigma_j = \delta_{ij} \mathbb{I} + i \epsilon_{ijk} \sigma_k \tag{5.101}$$

it can be seen that

$$\begin{aligned}
 \text{tr}[U \sigma_i U^\dagger U \sigma_j U^\dagger U \sigma_k U^\dagger] &= \text{tr}[\sigma_i \sigma_j \sigma_k] \\
 &= 2i \epsilon_{ijk},
 \end{aligned} \tag{5.102}$$

where again the cyclic nature of the trace has been used and $\underline{i} = \sqrt{-1}$. Alternatively,

$$\begin{aligned} \text{tr}[U\sigma_i U^\dagger U\sigma_j U^\dagger U\sigma_k U^\dagger] &= \text{tr}\left[\left(\sum_{l=1}^3 \alpha_{li}\sigma_l\right)\left(\sum_{m=1}^3 \alpha_{mj}\sigma_m\right)\left(\sum_{n=1}^3 \alpha_{nk}\sigma_n\right)\right] \\ &= \sum_{l,m,n=1}^3 \alpha_{li}\alpha_{mj}\alpha_{nk}\text{tr}[\sigma_l\sigma_m\sigma_n] \\ &= 2\underline{i} \sum_{l,m,n=1}^3 \alpha_{li}\alpha_{mj}\alpha_{nk}\epsilon_{lmn}. \end{aligned} \quad (5.103)$$

Hence,

$$\epsilon_{ijk} = \sum_{l,m,n=1}^3 \alpha_{li}\alpha_{mj}\alpha_{nk}\epsilon_{lmn}, \quad (5.104)$$

which, from Eq. (5.99), means that $\text{Det}(O) = 1$, completing the proof. $\square$

5.8.3 Proofs of Example Channels

We here prove the claims made in Section 5.3.1 about the application of Result 5.3.1 to the depolarising and dephasing channels. We first define each channel, giving their Choi-state and a vector of the eigenvalues of their Choi-state.

Definition 5.8.1 (The qubit depolarising channel). *The qubit depolarising channel is defined as*

$$\mathcal{D}_s^{\text{pol}}(\rho) := s\rho + (1-s)\text{tr}[\rho]\mathbb{I}/2. \quad (5.105)$$

Its Choi-state is

$$\mathcal{J}_{AB}^{\mathcal{D}^{\text{pol}}} = s|\Phi^+\rangle\langle\Phi^+|_{AB} + (1-s)\mathbb{I}_{AB}/4. \quad (5.106)$$

A vector of eigenvalues of the Choi-state is then

$$\lambda^{\text{pol}} = \begin{bmatrix} s + (1-s)/4 \\ (1-s)/4 \\ (1-s)/4 \\ (1-s)/4 \end{bmatrix}. \quad (5.107)$$

Definition 5.8.2 (The qubit dephasing channel). *The qubit dephasing channel with respect to the basis $\{|0\rangle\langle 0|, |1\rangle\langle 1|\}$ is defined as*

$$\mathcal{D}_q^{\text{ph}}(\rho) := q\rho + (1-q) \sum_{n=0}^1 |n\rangle\langle n| \rho |n\rangle\langle n| \quad (5.108)$$

Its Choi-state is

$$\mathcal{J}_p^{\mathcal{D}^{\text{ph}}} = q|\Phi_{00}^+\rangle\langle\Phi_{00}^+| + (1-q)|\Phi_{01}^+\rangle\langle\Phi_{01}^+|. \quad (5.109)$$

See Section 8.6.2 for a full derivation of this Choi-state. A vector of eigenvalues of the Choi-state is then

$$\lambda^{\text{ph}} = \begin{bmatrix} \frac{1+q}{2} \\ \frac{1-q}{2} \\ 0 \\ 0 \end{bmatrix}. \quad (5.110)$$

We now consider the existence of an allowed operation $\Pi \in \mathfrak{D}^{\text{dyn}}$ between each type of channel in

turn.

5.8.3.1 Depolarising Channel to Depolarising Channel

Here, we consider both the input and the output channel to be a depolarising channel. Result 5.3.1 states that there exists an allowed operation $\mathcal{D}_s^{\text{pol}}(\cdot) \rightarrow \mathcal{D}_{s'}^{\text{pol}}(\cdot)$ if and only if there exists a matrix that is a convex combination of local permutations mapping λ^{pol} to μ^{pol} , where

$$\mu^{\text{pol}} = \begin{bmatrix} s' + (1 - s')/4 \\ (1 - s')/4 \\ (1 - s')/4 \\ (1 - s')/4 \end{bmatrix}, \quad (5.111)$$

such that μ^{pol} still represents a depolarising channel for some s' . Such an allowed operation exists if there exists a set of values $\{p_{00}, p_{01}, p_{10}, p_{11}\}$, where $\sum_{ij} p_{ij} = 1$ and p_{ij} is real and positive for all i, j such that

$$\begin{bmatrix} s' + (1 - s')/4 \\ (1 - s')/4 \\ (1 - s')/4 \\ (1 - s')/4 \end{bmatrix} = \begin{bmatrix} p_{00}s + (1 - s)/4 \\ p_{01}s + (1 - s)/4 \\ p_{10}s + (1 - s)/4 \\ p_{11}s + (1 - s)/4 \end{bmatrix}. \quad (5.112)$$

The left hand side of Eq. (5.112) has been generated by applying the set of local qubit permutations to λ^{pol} and then taking the convex combination of the resultant vectors. This gives the following four equations

$$\begin{aligned} p_{00} &= \frac{1}{4}(3\frac{s'}{s} + 1), \\ p_{01} &= p_{10} = p_{11} = \frac{1}{4}(1 - \frac{s'}{s}). \end{aligned} \quad (5.113)$$

By enforcing that $0 \leq p_{ij} \leq 1 \forall i, j$, it can be seen that there exists an allowed operation mapping $\mathcal{D}_s^{\text{pol}}(\cdot)$ to $\mathcal{D}_{s'}^{\text{pol}}(\cdot)$ if and only if $s' \leq s$.

5.8.3.2 Dephasing Channel to Dephasing Channel

Here, we consider both the input and the output channel to be a dephasing channel. Result 5.3.1 states that there exists an allowed operation $\mathcal{D}_q^{\text{ph}}(\cdot) \rightarrow \mathcal{D}_{q'}^{\text{ph}}(\cdot)$ if and only if there exists a matrix that is a convex combination of local permutations mapping λ^{ph} to μ^{ph} , where

$$\mu^{\text{ph}} = \frac{1}{2} \begin{bmatrix} 1 + q' \\ 1 - q' \\ 0 \\ 0 \end{bmatrix}. \quad (5.114)$$

such that μ^{ph} still represents a dephasing channel for some q' . As above, such an allowed operation exists if there exists a set of values $\{p_{00}, p_{01}, p_{10}, p_{11}\}$, where $\sum_{ij} p_{ij} = 1$ and p_{ij} is real and positive for all i, j , such that

$$\frac{1}{2} \begin{bmatrix} 1 + q' \\ 1 - q' \\ 0 \\ 0 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} p_{00}(1 + q) + p_{01}(1 - q) \\ p_{00}(1 - q) + p_{01}(1 + q) \\ p_{10}(1 - q) + p_{11}(1 + q) \\ p_{10}(1 + q) + p_{11}(1 - q) \end{bmatrix}. \quad (5.115)$$

This gives the following four equations

$$\begin{aligned} p_{00} &= \frac{1}{2} \left(1 + \frac{q'}{q}\right) \\ p_{01} &= \frac{1}{2} \left(1 - \frac{q'}{q}\right) \\ p_{10} &= p_{11} = 0. \end{aligned} \tag{5.116}$$

By enforcing that $0 \leq p_{ij} \leq 1 \forall i, j$, it can be seen that there exists an allowed operation mapping $\mathcal{D}_q^{\text{ph}}(\cdot)$ to $\mathcal{D}_{q'}^{\text{ph}}(\cdot)$ if and only if $q' \leq q$. To see this explicitly, set

$$\begin{aligned} 0 &\leq \frac{1}{2} \left(1 + \frac{q'}{q}\right) \leq 1 \\ -1 &\leq q'/q \leq 1, \end{aligned} \tag{5.117}$$

which becomes $0 \leq q' \leq q$ as $q, q' \geq 0$.

5.8.3.3 Depolarising Channel to Dephasing Channel

We now consider the case that the input channel is the depolarising channel whilst the output is the dephasing channel. An allowed operation $\mathcal{D}_s^{\text{pol}}(\cdot) \rightarrow \mathcal{D}_q^{\text{ph}}(\cdot)$ exists if and only if there exists a set of values $\{p_{00}, p_{01}, p_{10}, p_{11}\}$, where $\sum_{ij} p_{ij} = 1$ and p_{ij} is real and positive for all i, j , such that

$$\frac{1}{2} \begin{bmatrix} (1+q) \\ (1-q) \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} p_{00}s + (1-s)/4 \\ p_{01}s + (1-s)/4 \\ p_{10}s + (1-s)/4 \\ p_{11}s + (1-s)/4 \end{bmatrix}. \tag{5.118}$$

It can be seen immediately that the final two terms can be zero if and only if $s = 1$. Hence, there exists an allowed operation mapping $\mathcal{D}_s^{\text{pol}}(\cdot)$ to $\mathcal{D}_q^{\text{ph}}(\cdot)$ if and only if $s = 1$. This is the trivial case in which $\mathcal{D}_s^{\text{pol}}(\rho) = \mathcal{I}(\rho)$, meaning the channel performs no depolarising. We can therefore conclude that no allowed operation exists that can map any depolarising channel to any dephasing channel.

5.8.3.4 Dephasing Channel to Depolarising Channel

Lastly, we consider the case that the input channel is the dephasing channel whilst the output is the depolarising channel. An allowed operation $\mathcal{D}_q^{\text{ph}}(\cdot) \rightarrow \mathcal{D}_s^{\text{pol}}(\cdot)$ exists if and only if there exists a set of values $\{p_{00}, p_{01}, p_{10}, p_{11}\}$, where $\sum_{ij} p_{ij} = 1$ and p_{ij} is real and positive for all i, j , such that

$$\begin{bmatrix} s + (1-s)/4 \\ (1-s)/4 \\ (1-s)/4 \\ (1-s)/4 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} p_{00}(1+q) + p_{01}(1-q) \\ p_{00}(1-q) + p_{01}(1+q) \\ p_{10}(1-q) + p_{11}(1+q) \\ p_{10}(1+q) + p_{11}(1-q) \end{bmatrix}. \tag{5.119}$$

This gives the following four equations

$$\begin{aligned} p_{00} &= \frac{1}{4} \left(s + 1 + \frac{4s}{q}\right) \\ p_{01} &= \frac{1}{4} \left(s + 1 - \frac{4s}{q}\right) \\ p_{10} &= p_{11} = \frac{1}{4} (1-s). \end{aligned} \tag{5.120}$$

By enforcing $0 \leq p_{01} \leq 1$ in particular, it can be seen that there exists an allowed operation mapping $\mathcal{D}_q^{\text{ph}}(\cdot)$ to $\mathcal{D}_s^{\text{pol}}(\cdot)$ if and only if $s \leq q/(4-q)$. To see this,

$$\begin{aligned} 0 &\leq \frac{1}{4}\left(s+1-\frac{4s}{q}\right) \leq 1 \\ -1 &\leq s-\frac{4s}{q} \leq 3 \\ \frac{3}{1-\frac{4}{q}} &\leq s \leq -\frac{1}{1-\frac{4}{q}}, \end{aligned} \tag{5.121}$$

where the inequality has been flipped in the final line because $1 - \frac{4}{q} < 0$ for $q \in [0, 1]$. This then becomes $0 \leq s \leq q/(4-q)$ as $s \geq 0$. Whilst p_{00} also places restriction on the relationship between s and q , they are encompassed by the conditions imposed by p_{01} .

5.8.4 Proof of Result 5.4.1

Before we begin the proof, we provide a quick reminder of definitions of the discrete Weyl operators and Weyl-covariant channels. A channel is Weyl-covariant if

$$W_{ab}\mathcal{N}(\rho)W_{ab}^\dagger = \mathcal{N}(W_{ab}\rho W_{ab}^\dagger) \quad \forall a, b \text{ \& } \rho \in \mathcal{D}(\mathcal{H}), \tag{5.122}$$

where W_{ab} are the discrete Weyl operators,

$$W_{ab} = \sum_c \Omega^{bc} |c+a \bmod d_S\rangle \langle c|, \quad \Omega = e^{\frac{2\pi i}{d_S}}, \quad a, b \in \{0, 1, \dots, d_S-1\}. \tag{5.123}$$

It is noted that all the discrete Weyl operators are unitary. Moreover, they transform in the following ways by transposition, the adjoint and multiplication [10],

$$\begin{aligned} (W_{\alpha,\beta})^t &= \Omega^{-\alpha\beta} W_{-\alpha,\beta}, \\ (W_{\alpha,\beta})^\dagger &= \Omega^{\alpha\beta} W_{-\alpha,-\beta}, \\ W_{\alpha,\beta} W_{\gamma,\delta} &= \Omega^{\beta\gamma} W_{\alpha+\gamma,\beta+\delta} = \Omega^{\beta\gamma-\alpha\delta} W_{\gamma,\delta} W_{\alpha,\beta}. \end{aligned} \tag{5.124}$$

These identities are employed in the subsequent proof. See Section 2.9 for more details.

Result 5.4.1 For every given $\mathcal{M}, \mathcal{N} \in \mathfrak{S}^{\text{st}} = \mathfrak{U}_{\mathcal{W}}$, the following statements are equivalent:

1. $\mathcal{M} = \Pi(\mathcal{N})$ for some allowed operation $\Pi \in \mathfrak{S}_{\mathcal{W}}^{\text{dyn}}$.
2. $\mu = \mathbb{D}'\lambda$, where $\mathbb{D}'$ is a convex combination of local **cyclic** permutation matrices.

Proof. We will first prove that statement 1 implies statement 2. We will then prove that statement 2 implies statement 1.

5.8.4.1 Proof that Statement 1 implies statement 2

To begin, it is noted that Weyl-covariant channels are a subset of mixed unitary channels and can always be written in the following form

$$\mathcal{N}(\rho) = \sum_{ab} p_{ab} W_{ab} \rho W_{ab}^\dagger, \quad \mathcal{N} \in \mathfrak{U}_{\mathcal{W}}, \tag{5.125}$$

where p_{ab} are elements of a probability vector [10].

When a restriction is made such that the allowed operations of the underlying static resource theory of informational non-equilibrium are Weyl-covariant channels, $\mathfrak{S}^{\text{st}} = \mathfrak{U}_{\mathcal{W}}$, the allowed operations of the

resulting resource preservability theory, in their Choi-state formalism, are

$$\mathcal{J}^{\mathcal{M}} = (\mathcal{P} \otimes \mathcal{E})(\mathcal{J}^{\mathcal{N}}), \quad (5.126)$$

where $\mathcal{P}, \mathcal{E} \in \mathcal{U}_{\mathcal{W}}$ and $\mathcal{J}^{\mathcal{N}}, \mathcal{J}^{\mathcal{M}}$ are Choi-states of Weyl-covariant channels (note, we have ignored the shared randomness in this definition and for the remainder of the proof for clarity, but the same result is reached if it is included). Using the identities in Eq. 5.124 for the discrete Weyl operators and the fact that they map Bell basis states to Bell basis states, it can be seen that if $\mathcal{N}, \mathcal{P}, \mathcal{E} \in \mathcal{U}_{\mathcal{W}}$, then $\mathcal{M} \in \mathcal{U}_{\mathcal{W}}$ too.

Now, recall that the Choi-states of Weyl-covariant channel are diagonal in the Bell basis, such that the action of these channels can be captured through the eigenvalues of their Choi-state alone. By introduction two general Weyl-covariant channels,

$$\mathcal{P}(\rho) = \sum_{cd} p_{cd} W_{cd} \rho W_{cd}^{\dagger}, \quad \mathcal{E}(\rho) = \sum_{ef} q_{ef} W_{ef} \rho W_{ef}^{\dagger}, \quad (5.127)$$

the allowed operations can then be rewritten as

$$\mu = \sum_{cd} \sum_{ef} p_{cd} q_{ef} \mathbb{B}^{(cd,ef)} \lambda, \quad (5.128)$$

where $\lambda, \mu \in \mathbb{R}^{d_S^2}$ are vectors of eigenvalues of the Choi-states $\mathcal{J}^{\mathcal{N}}, \mathcal{J}^{\mathcal{M}}$ respectively and $\mathbb{B}^{(cd,ef)}$ is the matrix with elements $B_{kl,nm}^{cd,ef} = \left| \langle \Phi_{kl} | W_{cd} \otimes W_{ef} | \Phi_{nm} \rangle \right|^2$.

Mirroring the qubit case, to complete the proof we need to show that $\mathbb{B}^{(cd,ef)}$ is a permutation from the set of local cyclic permutations. We first note that $\mathbb{B}^{(cd,ef)}$ is a product-Bell unistochastic matrix as it is a unistochastic matrices that arises from squaring the absolute values of the matrix elements of a product unitary matrix decomposed in the Bell basis. Although, this is now a further subset where the unitaries are restricted to be discrete Weyl operators.

The matrix elements of $\mathbb{B}^{(cd,ef)}$ are given by

$$\begin{aligned} B_{kl,nm}^{cd,ef} &= \left| \langle \Phi_{kl} | W_{cd} \otimes W_{ef} | \Phi_{nm} \rangle \right|^2 \\ &= \left| \langle \Phi_{kl} | \mathbb{I} \otimes W_{ef} (W_{cd})^{\dagger} | \Phi_{nm} \rangle \right|^2 \\ &= \left| \langle \Phi_{kl} | \mathbb{I} \otimes \Omega^{-cd} W_{ef} W_{-c,d} | \Phi_{nm} \rangle \right|^2 \\ &= \left| \langle \Phi_{kl} | \mathbb{I} \otimes \Omega^{-cd-cf} W_{e-c,d+f} | \Phi_{nm} \rangle \right|^2 \\ &= \left| \langle \Phi_{kl} | \Phi_{n+e-c, m+d+f} \rangle \right|^2, \\ &= \delta_{k, n+e-c} \delta_{l, m+d+f}. \end{aligned} \quad (5.129)$$

Therefore, with respect to the computational basis, the matrix $\mathbb{B}^{(cd,ef)}$ can be decomposed as

$$\begin{aligned} \mathbb{B}^{(cd,ef)} &= \sum_{kl,nm} \delta_{k, n+e-c} \delta_{l, m+d+f} |k, l\rangle \langle n, m| \\ &= \sum_{kl,nm} \delta_{k, n+e-c} |k\rangle \langle n| \otimes \delta_{l, m+d+f} |l\rangle \langle m|, \\ &= \sum_{n,m} |n+e-c \bmod d_S\rangle \langle n| \otimes |m+d+f \bmod d_S\rangle \langle m| \\ &= \sum_n |n+e-c \bmod d_S\rangle \langle n| \otimes \sum_m |m+d+f \bmod d_S\rangle \langle m| \\ &= W_{e-c,0} \otimes W_{d+f,0}. \end{aligned} \quad (5.130)$$

We then note that $W_{a,0} = X_{d_S}^a$, where $X_{d_S}^a$ is a generalised Pauli X in d_S dimensions. Specifically,

$$X_{d_S}^a = \sum_n |n + a \bmod d_S\rangle \langle n|. \quad (5.131)$$

The set of cyclic (or even) permutations are then those permutations that rearrange elements in a single cycle. It can be seen that $X_{d_S}^a$ does exactly this, shifting each n to $n + a \bmod d_S$. Hence, $\mathbb{B}^{(cd,ef)}$ is the tensor product of two local cyclic permutations; an allowed operation therefore exists between $\mathcal{N} \rightarrow \mathcal{M}$ if there exists a matrix between λ and μ that is a convex combination of local cyclic permutations.

5.8.4.2 Proof that Statement 2 implies statement 1

Assuming statement 2 holds, we can write

$$\mu = \sum_{\alpha, \beta} p_{\alpha\beta} X_{d_S}^\alpha \otimes X_{d_S}^\beta \lambda, \quad (5.132)$$

where $\alpha, \beta \in [0, d_S - 1]$. Translating back to the Choi-states of the channels these vectors represent, we find

$$\mathcal{J}^\mathcal{M} = \sum_{\alpha, \beta} p_{\alpha\beta} (X_{d_S}^\alpha \otimes X_{d_S}^\beta) (\mathcal{J}^\mathcal{N}) (X_{d_S}^\alpha \otimes X_{d_S}^\beta), \quad (5.133)$$

where $\mathcal{J}^\mathcal{M}$ and $\mathcal{J}^\mathcal{N}$ are Choi-states of the Weyl covariant channels $\mathcal{M}$ and $\mathcal{N}$ — and are therefore diagonal in the Bell basis — with eigenvalues given by the elements of μ and λ respectively. The above can then be re-written as

$$\begin{aligned} \mathcal{J}^\mathcal{M} &= \sum_{\alpha\beta} p_{\alpha\beta} (X_{d_S}^\alpha \otimes X_{d_S}^\beta) (\mathcal{I} \otimes \mathcal{N}) (|\Phi_{00}\rangle\langle\Phi_{00}|) (X_{d_S}^\alpha \otimes X_{d_S}^\beta), \\ &= \sum_{\alpha\beta} p_{\alpha\beta} (\mathcal{I} \otimes X_{d_S}^\beta) (\mathcal{I} \otimes \mathcal{N}) [(\mathcal{I} \otimes (X_{d_S}^\alpha)^t) (|\Phi_{00}\rangle\langle\Phi_{00}|) (\mathcal{I} \otimes (X_{d_S}^\alpha)^t)] (\mathcal{I} \otimes X_{d_S}^\beta), \\ &= \sum_{nm} p_{nm} (\mathcal{I} \otimes X_{d_S}^\beta [\cdot] \circ \mathcal{N} \circ (X_{d_S}^\alpha)^t [\cdot]) (|\Phi_{00}\rangle\langle\Phi_{00}|), \end{aligned} \quad (5.134)$$

where $X_{d_S}^\beta [\cdot]$ and $(X_{d_S}^\alpha)^t [\cdot]$ are unitary channels that apply the unitaries $X_{d_S}^\beta$ and $(X_{d_S}^\alpha)^t$ respectively. Hence, from the Choi-Jamiołkowski isomorphism, we find that

$$\begin{aligned} \mathcal{M}(\cdot) &= \sum_{nm} p_{nm} X_{d_S}^\beta [\cdot] \circ \mathcal{N} \circ (X_{d_S}^\alpha)^t \cdot, \\ &= \sum_{nm} p_{nm} \mathcal{N} \circ X_{d_S}^\beta [\cdot] \circ (X_{d_S}^\alpha)^t \cdot, \\ &= \mathcal{N} \circ \mathcal{P}, \end{aligned} \quad (5.135)$$

where $\mathcal{P}(\cdot) = \sum_{nm} p_{nm} X_{d_S}^\beta [\cdot] \circ (X_{d_S}^\alpha)^t \cdot$ is a pre-processing Weyl covariant channel. In the penultimate line we have used the fact that $\mathcal{N}$ is a Weyl-covariant channel and that $X_{d_S}^\beta$ is a discrete Weyl operator for all β . Therefore, if statement 2 holds then there always exists a Weyl-covariant pre-processing channel that achieves that mapping $\mathcal{N} \rightarrow \mathcal{M}$. Hence, statement 2 implies statement 1, completing the proof. $\square$

5.8.5 Proof of Result 5.4.2

In this section, a detailed proof of Result 5.3.4 and Result 5.4.2 is given, including a discussion of when the shape of the convex hull of all vectors in $\tilde{\mathfrak{V}}$ exists in a lower dimensional space. We here prove Result 5.4.2, for which Result 5.3.4 is a special case.

Recall that $\mathfrak{V} = \{\mathbb{P}_i \lambda : \forall \mathbb{P}_i \in \mathfrak{P}\}$ where $\tilde{\mathfrak{V}}$ are the vectors in $\mathfrak{V}$ with the final value in each vector

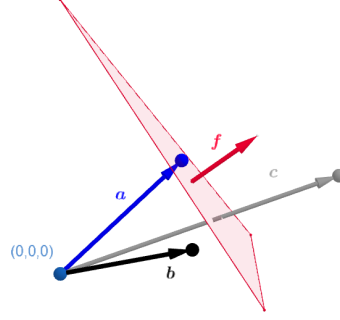


Figure 5.6: A section of a plane is shown in red. The plane is defined by the vector f that is perpendicular to all vector in the plane. The three vectors $\{a, b, c\}$ begin at the origin. The value of the dot product of these vectors with f is dependent on where these vectors determinate with respect to the plane described by f . (Figure made with [120]).

discarded due to normalisation. Moreover, recall that we assume that

$$\dim(\text{Span}\{\mathfrak{R}\}) = d_S^2 - 1, \quad (5.136)$$

where

$$\mathfrak{R} = \{\tilde{\lambda}_i - \tilde{\lambda}_0 : \tilde{\lambda}_0 \in \tilde{\mathfrak{L}}, \forall \tilde{\lambda}_i \in \tilde{\mathfrak{L}}\}. \quad (5.137)$$

Lemma 5.3.4 *There exists a set of d_S^2 linear inequalities that are a complete set of monotones of $\mathcal{R}_n^{\text{dyn}}$.*

Proof. If $\dim(\text{Span}\{\mathfrak{R}\}) = d_S^2 - 1$ the convex hull of vectors in $\tilde{\mathfrak{L}}$ form a simplex in $\mathbb{R}^{d_S^2-1}$; if $\tilde{\mu}$, the $\mathbb{R}^{d_S^2-1}$ representation of a channel $\mu \in \mathbb{R}^{d_S^2}$, is within the simplex, then it is within the convex hull of the vectors in $\tilde{\mathfrak{L}}$. This is equivalent to μ being in the convex hull of the vectors in $\mathfrak{L}$. If so, an allowed operation of $\mathfrak{D}^{\text{dyn}}$ exists between the channels represented by the vectors λ and μ . In summary, the convertibility question is reformulated as asking if a vector sits within the convex hull of permutations of another vector.

Each of the d_S^2 monotones is then associated to one facet of the simplex. Specifically, it assess if $\tilde{\mu}$ is on the side of the facet that places it inside the simplex, or the side of the facet that places it outside. To determine on what side of the facet μ sits, one needs to find the description of each of the hyper-planes that encompasses each of the facets. A hyper-plane is described by a vector $f \in \mathbb{R}^{d_S^2-1}$ that is perpendicular to all vectors in the hyper-plane.

Any vector then beginning at the origin and terminating on the hyper-plane has an equal value dot product with f i.e., $f \cdot a = k$ for some $a \in \mathbb{R}^{d_S^2-1}$ that terminates in the hyperplane. Any vector that terminates before the hyper-plane then has a dot product with f less than k , whilst any vector that terminates after the hyper-plane has a dot product with f greater than k . The value of the dot product of a vector with f therefore provides information about the geometry of the vector with respect to the plane f defines. See Fig. 5.6 for a graphical depiction in $\mathbb{R}^3$ of these vectors with respect to some plane.

Each hyper-plane encompassing a facet of the simplex has all but one of the vectors in $\tilde{\mathfrak{L}}$ terminate within it. A vector f can be found that is perpendicular to these $d_S^2 - 1$ vectors and l can then be calculated by taking the dot product of f with one of the vectors that terminate within it. The value of l can then be thought of as a distance like measure from the origin to one facet of the simplex.

To determine what side of the hyper-plane $\tilde{\mu}$ needs to be on such that it is inside the simplex, $\tilde{\Lambda} \cdot f$ can be checked. Given $\tilde{\Lambda}$ is always in the centre of the simplex, the side of the plane to be on is the side $\tilde{\Lambda}$ is on i.e., if $\tilde{\Lambda} \cdot f < k$ then one needs $\tilde{\mu} \cdot f < k$ for $\tilde{\mu}$ to be on the same side of the facet as $\tilde{\Lambda}$. Note, by playing with the signs of f and k , one can ensure that this is always checking an upper bound.

This must be repeated for all facets, meaning a set of vectors $\tilde{f}$ is found where each describes a different hyper-plane encompassing one facet of the simplex and is used to determine on which side of the facet $\tilde{\mu}$ sit. This means there are d_S^2 inequalities to check to determine. Once all have been checked, one knows for certain if $\tilde{\mu}$ sits inside the convex hull of $\tilde{\mathfrak{L}}$, and hence whether there exists an allowed

operation between the respective channels they represent. $\square$

We now consider the case that Eq. (5.136) does not hold, such that $\dim(\text{Span}\{\mathfrak{R}\}) < d_S^2 - 1$. Here, the number of monotones needed will vary with the geometry of the problem, with potentially both inequalities and equalities being needed. We demonstrate this through an example using qubit channels, $d_S = 2$, but leave a rigorous proof for future work.

If $\dim(\text{Span}\{\mathfrak{R}\}) = 2$, then the simplex formed by the vectors in $\tilde{\mathfrak{Q}}$ is a quadrilateral in $\mathbb{R}^3$. For an allowed operation to exist $\tilde{\mu}$ must sit in this quadrilateral. In this case one equality is needed to assess if $\tilde{\mu}$ is in the plane that contains this quadrilateral. Four inequalities are then needed to ensure $\tilde{\mu}$ sits on the correct side of the lines that make the boundaries of this quadrilateral. This leaves a total of four inequalities and one equality.

If the vectors in $\tilde{\mathfrak{Q}}$ form a line then it was found that two equalities and one inequality were needed. Hence, when $\dim(\text{Span}\{\mathfrak{R}\}) < d_S^2 - 1$ for general dimension channels the number of constraints needed will depend on the geometry created by $\tilde{\mathfrak{Q}}$.

5.8.6 Proof of Result 5.5.1

Before proving Result 5.5.1, we first prove the following Lemma that will be used in the proof. Recall that $\mathfrak{U}_M$ is the set of mixed unitary channels and $\mathfrak{U}_W$ the set of Weyl-covariant channels.

Lemma 5.8.3. *Let $\mathcal{M}, \mathcal{N} \in \mathfrak{D}^{\text{st}}$. Then the following two statements are equivalent for $\mathfrak{D}^{\text{dyn}}$:*

1. $\exists \Pi \in \mathfrak{D}^{\text{dyn}}$ such that $\mathcal{M} = \Pi(\mathcal{N})$.
2. $\exists \Theta \in \mathfrak{D}^{\text{dyn}}$ such that $\mathcal{M}^\dagger = \Theta(\mathcal{N}^\dagger)$.

Proof. Firstly, given the set of mixed unitary channels and the set of Weyl-covariant channels are subsets of the set of unital channels, this proof holds if $\mathfrak{D}^{\text{st}} = \mathfrak{U}_M$ or $\mathfrak{D}^{\text{st}} = \mathfrak{U}_W$.

Now, if $\mathcal{N}$ has Kraus decomposition $\mathcal{N}(\cdot) = \sum_i K_i(\cdot)K_i^\dagger$, such that $\sum_i K_i^\dagger K_i = \mathbb{I}$, then its adjoint $\mathcal{N}^\dagger$ is defined as $\mathcal{N}^\dagger(\cdot) = \sum_i K_i^\dagger(\cdot)K_i$. In general, if $\mathcal{N} \in \text{CPTP}$ then $\mathcal{N}^\dagger$ is a unital map, and vice versa. This can be seen as $\mathcal{N}^\dagger(\mathbb{I}) = \sum_i K_i^\dagger K_i = \mathbb{I}$. Using similar logic, it can also be seen that if $\mathcal{N}$ is a unital channel then $\mathcal{N}^\dagger$ is a unital channel too i.e., if $\mathcal{N} \in \mathfrak{D}^{\text{st}} \iff \mathcal{N}^\dagger \in \mathfrak{D}^{\text{st}}$.

Suppose that $\mathcal{M} = \Pi(\mathcal{N})$ for some $\Pi \in \mathfrak{D}^{\text{dyn}}$. This can be written as

$$\mathcal{M} = \sum_{\kappa} p_{\kappa} \mathcal{E}_{\kappa} \circ \mathcal{N} \circ \mathcal{P}_{\kappa}, \quad (5.138)$$

where $\mathcal{P}_{\kappa}, \mathcal{E}_{\kappa} \in \mathfrak{D}^{\text{st}}$. The adjoint of $\mathcal{M}$ is then

$$\mathcal{M}^\dagger = \sum_{\kappa} p_{\kappa} \mathcal{P}_{\kappa}^\dagger \circ \mathcal{N}^\dagger \circ \mathcal{E}_{\kappa}^\dagger, \quad (5.139)$$

where, using the above logic, $\mathcal{P}_{\kappa}^\dagger, \mathcal{E}_{\kappa}^\dagger \in \mathfrak{D}^{\text{st}}$. Hence, if an allowed operations exists between $\mathcal{N}$ and $\mathcal{M}$ then an allowed operations exists between $\mathcal{N}^\dagger$ and $\mathcal{M}^\dagger$. Likewise, if an allowed operations exists between $\mathcal{N}^\dagger$ and $\mathcal{M}^\dagger$, then the same proof shows that an allowed operations exists between $\mathcal{N}$ and $\mathcal{M}$. $\square$

We now return to the proof of Result 5.5.1. As in Result 5.3.4 and Result 5.4.2, the following assumption is also made here:

$$\dim(\text{Span}\{\mathfrak{R}\}) = d_S^2 - 1, \quad (5.140)$$

where

$$\mathfrak{R} = \{\tilde{\lambda}_i - \tilde{\lambda}_0 : \tilde{\lambda}_0 \in \tilde{\mathfrak{Q}}, \forall \tilde{\lambda}_i \in \tilde{\mathfrak{Q}}\}, \quad (5.141)$$

meaning the shape of the convex hull of the vectors in $\tilde{\mathfrak{Q}}$ does not exist in a lower dimensional space.

Result 5.5.1 Let $\mathcal{N} \in \mathfrak{D}^{\text{st}}$ be given. Then there exists a set of d_S^2 many bipartite states $\{\sigma_{ij}\}$ making the following two statements equivalent:

1. $\mathcal{M} = \Pi(\mathcal{N})$ for some $\Pi \in \mathfrak{D}^{\text{dyn}}$.
2. For every i, j we have $\langle \Phi_{ij} | (\mathcal{I} \otimes \mathcal{N})(\sigma_{ij}) | \Phi_{ij} \rangle \geq \langle \Phi_{ij} | (\mathcal{I} \otimes \mathcal{M})(\sigma_{ij}) | \Phi_{ij} \rangle$.

Here, $|\Phi_{ij}\rangle$ is a state from the Bell basis.

Proof. Firstly, we highlight that the set of d_S^2 many states $\{\sigma_{ij}\}$ arise from the geometrical interpretation of the allowed operations through vectors in $\mathbb{R}^{d_S^2-1}$. Specifically, each state is associated to one of the vectors characterising one of d_S^2 hyper-planes that encompass the d_S^2 facets of the simplex (as Eq. (5.140) is assumed to hold). Hence, each state is also associated to one of the d_S^2 monotones that form a complete set (see Result 5.3.4 and Result 5.4.2).

The hyper-planes that encompass each facet of the simplex are defined by a set of vectors $\mathfrak{f} = \{f_{ij} : f_{ij} \in \mathbb{R}^{d_S^2-1} \forall i, j\}$. From $\mathfrak{f}$, a set of vectors

$$\mathfrak{Q} = \{Q_{ij} : Q_{ij} \in \mathbb{R}^{d_S^2} \forall i, j\} \quad (5.142)$$

can be created, where

$$Q_{ij} = \frac{F_{ij} + r_{ij}\Lambda}{w_{ij}}, \quad (5.143)$$

such that $F_{ij} \in \mathbb{R}^{d_S^2}$ is a higher dimensional embedding of f_{ij} , where a zero has been appended to the vector in the added dimension i.e., $F_{ij}^t = [f_{ij}, 0]$; r_{ij} is a number that is sufficiently larger to ensure that all elements of Q_{ij} are positive; Λ is a vector of eigenvalues of the Choi-state of the state preparation channel of the maximally mixed state, and w_{ij} is a normalisation. As all elements are positive and they sum to one, each Q_{ij} can then be taken to be the eigenvalues of some state in the Bell basis, which we denoted σ_{ij} .

As the Choi-states of all channel we considered are diagonal in the Bell basis, and σ_{ij} is diagonal in the Bell basis too, it can be seen that

$$\begin{aligned} \text{Tr}[\mathcal{J}_{ij}^{\mathcal{N}} \sigma_{ij}] &= l_{ij} \cdot Q_{ij}, \\ &= \frac{l_{ij} \cdot F_{ij} + r_{ij} l_{ij} \cdot \Lambda}{w_{ij}}, \\ &= \frac{\tilde{l}_{ij} \cdot f_{ij} + r_{ij}/4}{w_{ij}}, \end{aligned} \quad (5.144)$$

where l_{ij} is a vector of eigenvalues of the Choi-state $\mathcal{J}_{ij}^{\mathcal{N}}$ and $\tilde{l}_{ij}$ is l_{ij} with the final element removed. The first term above simplifies due to the zero in the final dimension of F_{ij} and the second due to the elements of l_{ij} summing to one. Assuming that $\tilde{l}_{ij}$ lies within the hyper-planes characterised by f_{ij} , then $\tilde{l}_{ij} \cdot f_{ij} = k$ can be considered as a distance like measure from the origin to the facet of the simplex encompassed by the hyper-plane represented by f_{ij} , as consider in the proof of Result 5.3.4 and Result 5.4.2 in Section. 5.8.5.

Consider now

$$\text{Tr}[\mathcal{J}_{ij}^{\mathcal{M}} \sigma_{ij}] = \frac{\tilde{\mu}_{ij} \cdot f_{ij} + r_{ij}/4}{w_{ij}}, \quad (5.145)$$

where $\tilde{\mu}_{ij} \in \mathbb{R}^{d_S^2-1}$ is the vector representing the channel $\mathcal{M}$. If $\tilde{\mu}_{ij}$ falls on the same side of the hyper-plane represented by f_{ij} as the vector representing the free channel, Λ , then $\tilde{\mu}_{ij} \cdot f_{ij} < k$. Hence,

$$\begin{aligned} \text{Tr}(\mathcal{J}_{ij}^{\mathcal{M}} \sigma_{ij}) &< \frac{k + r_{ij}/4}{w_{ij}} \\ &= \text{Tr}(\mathcal{J}_{ij}^{\mathcal{N}} \sigma_{ij}) \end{aligned} \quad (5.146)$$

In order to evaluate if μ is on the correct side of each face of the simplex, the above must be true for all i, j .

We note that the indices in the above equations do not necessarily need to be same to reach the above result. Each facet of the simplex made from taking the convex hull of vectors in $\tilde{\mathcal{V}}$ has all but one of the vectors in $\tilde{\mathcal{V}}$ terminate within it. Therefore, for a given f_{ij} , for all but one vector $\tilde{l}_{nm} \in \tilde{\mathcal{V}}$ it holds that $\tilde{l}_{nm} \cdot f_{ij} = k$. This results in there being multiple different strategies for achieving this operational interpretation, which was mentioned for the qubit case in Section 5.5.0.1, but will not be explored in the general case.

To complete the proof, we now make a distinction between the $d_S = 2$ (qubit unital channels) and $d_S = n$ (Weyl-covariant channel) case.

$d_S = 2$ — qubit unital channels

Previously, it was shown that the Choi-state of all qubit unital channels can be diagonalised by product unitaries. Equivalently, all qubit unital channels can be converted into a Weyl-covariant channel by a pre-processing and post-processing unitary channel. When determining the states used in this operational interpretation, this diagonalisation needs to be ‘undone’. Explicitly, if

$$\mathcal{J}_{ij}^{\tilde{\mathcal{N}}} = (U_1 \otimes U_2) \mathcal{J}_{ij}^{\mathcal{N}} (U_1 \otimes U_2)^\dagger, \quad (5.147)$$

where $\mathcal{J}_{ij}^{\tilde{\mathcal{N}}}$ is diagonal in the Bell basis such that it is the Choi-state of a Weyl-covariant channel $\tilde{\mathcal{N}}$, and U_1/U_2 are unitaries, then the left hand side of Eq. (5.144) becomes

$$\begin{aligned} \text{Tr}[\mathcal{J}_{ij}^{\tilde{\mathcal{N}}} \sigma_{ij}] &= \text{Tr}((U_1 \otimes U_2) \mathcal{J}_{ij}^{\mathcal{N}} (U_1 \otimes U_2)^\dagger \sigma_{ij}) \\ &= \text{Tr}((U_1 \otimes U_2)^\dagger \sigma_{ij} (U_1 \otimes U_2) \mathcal{J}_{ij}^{\mathcal{N}}) \\ &= \text{Tr}(\tau_{ij} \mathcal{J}_{ij}^{\mathcal{N}}) \\ &= \text{Tr}((\tau_{ij})(\mathcal{I} \otimes \mathcal{N}) |\Phi_{ij}\rangle\langle\Phi_{ij}|) \\ &= \langle\Phi_{ij}| (\mathcal{I} \otimes \mathcal{N}^\dagger)(\tau_{ij}) |\Phi_{ij}\rangle, \end{aligned} \quad (5.148)$$

where $\tau_{ij} = (U_1 \otimes U_2)^\dagger \sigma_{ij} (U_1 \otimes U_2)$ and we have used the definition of the adjoint in the Hilbert-Schmitt inner product. Applying this same logic to $\mathcal{M}$ and $\tilde{\mathcal{M}}$, one can conclude that there exists a set of $d_S^2 = 4$ states $\{\tau_{ij}\}_{i,j=0}^1$ such that there exists an allowed operation $\Pi \in \mathfrak{S}^{\text{dyn}}$ where $\mathcal{M} = \Pi(\mathcal{N})$ if and only if

$$\langle\Phi_{ij}| (\mathcal{I} \otimes \mathcal{N}^\dagger)(\tau_{ij}) |\Phi_{ij}\rangle \geq \langle\Phi_{ij}| (\mathcal{I} \otimes \mathcal{M}^\dagger)(\tau_{ij}) |\Phi_{ij}\rangle \quad \forall i, j. \quad (5.149)$$

It is then known that an allowed operation exists between $\mathcal{N}^\dagger$ and $\mathcal{M}^\dagger$ only if there also exists an allowed operation between $\mathcal{N}$ and $\mathcal{M}$. Hence, by repeating the above process but starting with the channels $\mathcal{N}^\dagger$ and $\mathcal{M}^\dagger$, concludes the proof for qubit channels.

$d_S > 2$ — Weyl-covariant channel

The Choi-states of Weyl-covariant channels are already diagonal in the Bell basis and hence no diagonalisation needs to be ‘undone’, as in the qubit case. Hence, one does not need to alter the states $\{\sigma_{ij}\}_{ij}$, meaning this set of states can directly be used to verify the existence of an allowed operation between two channels. This completes the proof. $\square$

5.8.7 Proof of Result 5.6.1

We first prove the following Lemma about Weyl-covariant channels.

Lemma 5.8.4. *If $\mathcal{E}, \mathcal{P} \in \mathfrak{U}_{\mathcal{W}}$, then $\mathcal{E} \circ \mathcal{P}(\rho) = \mathcal{P} \circ \mathcal{E}(\rho)$.*

Proof. As $\mathcal{E}, \mathcal{P} \in \mathfrak{U}_{\mathcal{W}}$, they can be written as

$$\mathcal{E}(\rho) = \sum_{ab} p_{ab} W_{ab} \rho W_{ab}^\dagger, \quad \mathcal{P}(\rho) = \sum_{cd} q_{cd} W_{cd} \rho W_{cd}^\dagger. \quad (5.150)$$

From here it can be seen that

$$\mathcal{E} \circ \mathcal{P}(\rho) = \sum_{ab} \sum_{cd} p_{ab} q_{cd} W_{ab} W_{cd}(\rho) W_{cd}^\dagger W_{ab}^\dagger. \quad (5.151)$$

Using the identities detailed in Section 2.9 it can be seen that

$$W_{ab} W_{cd} = \Omega^{bc-ad} W_{cd} W_{ab}, \quad \Omega = e^{\frac{2\pi i}{n}}, \quad (5.152)$$

and

$$\begin{aligned} W_{cd}^\dagger W_{ab}^\dagger &= \Omega^{cd} W_{-c,-d} \Omega^{ab} W_{-a,-b} \\ &= \Omega^{cd} \Omega^{ab} \Omega^{ab-bc} W_{-a,-b} W_{-c,-d} \\ &= \Omega^{ab-bc} W_{a,b}^\dagger W_{c,d}^\dagger. \end{aligned} \quad (5.153)$$

Hence,

$$\begin{aligned} \mathcal{E} \circ \mathcal{P}(\rho) &= \sum_{ab} \sum_{cd} p_{ab} q_{cd} \left[\Omega^{bc-ad} W_{cd} W_{ab} \right] (\rho) \left[\Omega^{ab-bc} W_{a,b}^\dagger W_{c,d}^\dagger \right] \\ &= \mathcal{P} \circ \mathcal{E}(\rho). \end{aligned} \quad (5.154)$$

□

We now prove the main result, making use of this Lemma.

Result 5.3.1 *The Holevo capacity is a faithful monotone of both $\mathcal{R}_n^{\text{dyn}}$ and $\mathcal{R}_n^{\text{dyn}}$, meaning that $\chi(\mathcal{N}) = 0$ if and only if $\mathcal{N} = \Lambda$, and*

$$\chi(\mathcal{N}) \geq \chi(\Pi(\mathcal{N})) \quad \forall \Pi \in \mathfrak{S}^{\text{dyn}}, \quad \mathcal{N} \in \mathfrak{S}^{\text{st}}. \quad (5.155)$$

Proof. Firstly, it is noted that the Holevo capacity is invariant under pre-processing and post-processing unitary channels. To see this, consider the channel $\tilde{\mathcal{N}}(\rho) = V\mathcal{N}(U\rho U^\dagger)V^\dagger$, where U, V are unitary operators. The Holevo capacity of $\tilde{\mathcal{N}}$ is then

$$\chi(\tilde{\mathcal{N}}) = \sup_{\{p_i, \rho_i\}} \left\{ s\left(\sum_i p_i V\mathcal{N}(U\rho_i U^\dagger)V^\dagger\right) - \sum_i p_i s(V\mathcal{N}(U\rho_i U^\dagger)V^\dagger) \right\}. \quad (5.156)$$

By defining $\eta_i = U\rho_i U^\dagger$ and $\rho' = \sum_i p_i \eta_i$ the above becomes

$$\begin{aligned} \chi(\tilde{\mathcal{N}}) &= \sup_{\{p_i, \eta_i\}} \left\{ s(V\mathcal{N}(\rho')V^\dagger) - \sum_i p_i s(V\mathcal{N}(\eta_i)V^\dagger) \right\}, \\ &= \sup_{\{p_i, \eta_i\}} \left\{ s\left(\sum_i p_i \mathcal{N}(\eta_i)\right) - \sum_i p_i s(\mathcal{N}(\eta_i)) \right\}, \\ &= \chi(\mathcal{N}), \end{aligned} \quad (5.157)$$

where we have used the fact that the von Neumann entropy is unitarily invariant. We then recall that the ability to diagonalise the Choi-states of qubit unital channels physically means that they are equivalent

to a Weyl-covariant channel up to some pre-processing and post-processing unitary channels, as seen in Lemma 5.8.1. Therefore, the above results shows that the Holevo capacity of a qubit unital channel can equally be assessed through the Weyl covariant channel that it is equivalent to up to some pre-processing and post-processing unitary channels. Hence, only the Holevo capacity of Weyl-covariant channels needs to be considered. Within the remainder of the proof all channels are considered to be Weyl-covariant. This ensures that this proof applies to both $\mathcal{R}^{\text{dyn}}$ and $\mathcal{R}_n^{\text{dyn}}$.

Irreducibly covariant channels, of which Weyl-covariant channels are a subset, are channels that are covariant with respect to the irreducible unitary representation of some group. For such channels, the Holevo capacity is simplified to be linearly proportional to the minimum output entropy [125–127]. Specifically, it becomes

$$\chi(\mathcal{N}) = \ln(d_S) - \min_{\rho} s(\mathcal{N}(\rho)), \quad (5.158)$$

where the minimisation is over all states ρ , and d_S is the output dimension of the channel. This can then be rewritten as

$$\chi(\mathcal{N}) = \max_{\rho} D(\mathcal{N}(\rho) || Y), \quad (5.159)$$

where Y is the maximally mixed state, $Y = \mathbb{I}/d$, and $S(\rho || \sigma) := \text{tr}[\rho(\log_2 \rho - \log_2 \sigma)]$ is the quantum relative entropy.

Consider now a channel $\mathcal{M}$, where there exists an allowed operations $\Pi \in \mathfrak{D}^{\text{dyn}}$, such that $\mathcal{M} = \Pi(\mathcal{N})$, where $\mathcal{M}, \mathcal{N} \in \mathfrak{D}^{\text{st}}$. The Holevo capacity of $\mathcal{M}$ is given by

$$\begin{aligned} \chi(\mathcal{M}) &= \max_{\rho} D(\mathcal{M}(\rho) || Y), \\ &= \max_{\rho} D\left(\sum_{\kappa} p_{\kappa} \mathcal{E}_{\kappa} \circ \mathcal{N} \circ \mathcal{P}_{\kappa}(\rho) || Y\right), \\ &= \max_{\rho} D\left(\sum_{\kappa} p_{\kappa} \mathcal{E}_{\kappa} \circ \mathcal{N} \circ \mathcal{P}_{\kappa}(\rho) || \sum_{\kappa} p_{\kappa} \mathcal{E}_{\kappa} \circ \mathcal{N} \circ \mathcal{P}_{\kappa}(Y)\right), \\ &\leq \sum_{\kappa} p_{\kappa} \max_{\rho} D(\mathcal{E}_{\kappa} \circ \mathcal{N} \circ \mathcal{P}_{\kappa}(\rho) || Y). \end{aligned} \quad (5.160)$$

In the third line we have used the fact that $\mathcal{N}$ and $\mathcal{E}_{\kappa}, \mathcal{P}_{\kappa} \forall \kappa$ are unital, and in the final line we have used the fact that the quantum relative entropy is jointly convex in its arguments. By then using Lemma 5.8.4, which states that Weyl-covariant channels commute with Weyl-covariant channels, a channel $\mathcal{Q}_{\kappa} = \mathcal{E}_{\kappa} \circ \mathcal{P}_{\kappa}$ can be defined such that

$$\begin{aligned} \chi(\mathcal{M}) &\leq \sum_{\kappa} p_{\kappa} \max_{\rho} D(\mathcal{Q}_{\kappa} \circ \mathcal{N}(\rho) || Y), \\ &= \sum_{\kappa} p_{\kappa} \max_{\rho} D(\mathcal{Q}_{\kappa} \circ \mathcal{N}(\rho) || \mathcal{Q}_{\kappa}(Y)), \\ &\leq \sum_{\kappa} p_{\kappa} \max_{\rho} D(\mathcal{N}(\rho) || Y) \\ &= \max_{\rho} D(\mathcal{N}(\rho) || Y), \end{aligned} \quad (5.161)$$

where we have used the fact that $\mathcal{Q}_{\kappa}(\cdot)$ is a unital channel, and in the penultimate line the data processing inequality has been used, i.e.,

$$D(\mathcal{Q}_{\kappa} \circ \mathcal{N}(\rho) || \mathcal{Q}_{\kappa}(Y)) \leq D(\mathcal{N}(\rho) || Y) \quad \forall \mathcal{Q}_{\kappa} \in \text{CPTP}. \quad (5.162)$$

Putting this together, one gets that

$$\chi(\mathcal{M}) \leq \chi(\mathcal{N}) \quad \text{if } \mathcal{M} = \Pi(\mathcal{N}) \quad (5.163)$$

Hence, the Holevo capacity is a monotone of $\mathcal{R}^{\text{dyn}}$ and $\mathcal{R}_n^{\text{dyn}}$. If $\mathcal{N} = \Lambda$, then

$$\begin{aligned}\chi(\Lambda) &= \max_{\rho} D(\Lambda(\rho)||Y), \\ &= D(Y||Y) \quad \forall \rho \in \mathcal{D}(\mathcal{H}), \\ &= 0.\end{aligned}\tag{5.164}$$

If $\mathcal{N} \neq \Lambda$, then there exists at least one state $\rho \in \mathcal{D}(\mathcal{H})$ such that $\mathcal{N}(\rho) \neq Y$. Given $D(\rho||\sigma) = 0$ if and only if $\rho = \sigma$, it can be concluded that $\chi(\mathcal{N}) = 0$ if and only if $\mathcal{N} = \Lambda$, such that the Holevo capacity is a faithful monotone. This completes the proof. $\square$

6.1 Introduction

Given the defined usefulness of resources, it is natural to want to increase the amount one has. Such questions have often been considered via *resource distillation* [13], where several copies of a less resourceful state are converted into fewer copies of a more resourceful state, with the help of an arbitrary (possibly infinite) supply of free states. Additionally, these protocols may also require global operations that act simultaneously on many copies of the system and free states simultaneously. Practically implementing resource distillation can, therefore, be highly expensive in labs, and likely well beyond the ability of contemporary quantum technological platforms (e.g., Refs. [133–135]). As being able to concentrate multiple noisy objects into a single, more resourceful object is crucial for the successful operation of quantum technologies, we here focus on the *simplest*—and hence *most practically feasible*—setting for which resource enhancement can be studied.

We coin the term *resource concentration* to describe this paradigm: given two copies of a state, $\rho_A \otimes \rho_B$, can we enhance the resource in system A via a global resource non-increasing unitary? Colloquially, resource concentration problems ask whether an agent, given access to two uncorrelated copies of a state, can transfer resources from one system to the other under allowed operations. More formally, they can be viewed as a minimalist distillation protocol. Understanding this limiting case sets a lower bound on what resource enhancement is possible, complementing the upper bounds set by considering asymptotic distillation protocols [85]. Moreover, it allows a bottom-up understanding of what is necessary for performing resource enhancement.

Note, the task of resource concentration has been defined without access to any free states and with the ability to only perform unitary dynamics i.e., it is closed system dynamics. In addition to being more practically relevant, this allows us to do complete accounting of all information changes. This differs from resource distillation protocols that only concern the input and output states, ignoring any “junk” produced in the process. In the concentration scenario, we can analyse this junk, and see what it is possible to learn from it.

In this chapter, we first present a general framework for the task of resource concentration, before applying it within the resource theory of informational non-equilibrium and the resource theory of unspeakable coherence. In both contexts, we present results on the maximum amount of resource that can be concentrated, discuss the notion of bound-resource—states for which no resource can be concentrated—and investigate the power of initial correlations. We then aim to apply what we have learned from the concentration scenario to gain an understanding about more general resource enhancement scenarios.

6.1.1 Motivation

Quantum information processing protocols are usually designed assuming access to a specific resourceful state: for example, quantum teleportation [136] and super-dense coding [69] consider access to maximally entangled bipartite states, while quantum computation assumes access to pure states and, depending on the architecture, to magic states [25] too. These states are all pure states in their respective resource theories, namely the resource theories of bipartite entanglement [14], informational non-equilibrium (purity) [18], and non-stabilizerness [24] respectively. However, in reality, noise in the system — whether inherent to the hardware or as a result of our limited technological ability — will prevent perfect access to these states. If one performs their desired protocol using some noisy variant of the resource state, rather than the ideal resource state, the noise in the resource state will be transferred to noise in the output of the protocol.

Given a threshold level of noise that can be tolerated, one can calculate the resourceful state σ necessary to at least achieve it. If then limited to a source only able to produce a less resourceful state ρ , to successfully meet the threshold one must produce σ from the source of ρ 's. This must be done within

the physical restrictions of the system i.e., under allowed operations. Such problems have been studied in the context of resource distillation [13], where the aim is to convert some number of copies of a less resourceful state into fewer copies of a more resourceful state: $\rho^{\otimes n} \rightarrow \sigma^{\otimes m}$ where $m \leq n$. The ability to achieve this is captured by the *distillation rate*, which quantifies the number of target states σ that can be obtained per copy of the initial state ρ . In the asymptotic limit, the distillation rate is given by

$$r(\rho \rightarrow \sigma) = \sup \left\{ r : \lim_{n \rightarrow \infty} \inf_{\Lambda \in \mathcal{F}} D(\Lambda(\rho^{\otimes n}), \sigma^{\otimes \lfloor rn \rfloor}) = 0 \right\}, \quad (6.1)$$

where $D(\cdot, \cdot)$ is some distance measure * (see Definition 4.2.9). Here, one assumes access to an infinite number of initial states, and then assumes the ability to perform an allowed operation on all of the input states simultaneously. If considering allowed operations with a free dilation, then access to an infinite number of free states is therefore also assumed. Since one is maximising all these available freedoms i.e., the number of copies of ρ , the order of interaction and the number of free states, the asymptotic distillation rate is the optimal rate that can ever be achieved to (deterministically) convert copies of ρ into σ under allowed operations. The asymptotic rate therefore considers resource enhancement of ρ in the extreme [†]

This raises the question of what it is possible to achieve in the opposite limit: what amount of resource enhancement is achievable when all available freedoms are minimised? In this limit, the smallest number of copies that can be feasibly considered is two. This is in contrast to resource distillation, where one often considers access to an infinite number of initial states. The minimum number of free states that can be considered when performing resource enhancement is zero; in resource distillation access to an infinite number is considered. In this minimal limit, one is therefore left with just two copies of ρ as an input — fixing the order of interaction at two. On these two copies, we then consider applying an allowed operation to increase the resource content in one of the two subsystem. We can simplify the resource enhancement further by considering applying only unitaries from the set of allowed operations (note, if considering a resource theory with a free dilation, the restriction to no free states would then restrict the set of allowed operations to be only unitaries anyway).

By considering this minimal limiting case — which can be considered to be a minimal resource distillation scenario — we can characterise the worst case resource enhancement scenario: what is still possible to achieve despite being given access to as little freedoms as possible. Once the problem of resource enhancement has been stripped back and understood, one can start to build it back up, by incrementally reintroducing the available freedoms e.g., allowing access to three states but still restricting the order of interaction to two; allowing initial correlations or entanglement between the states; or allowing access to one free state. Then, by consider these various combinations, one can aim to isolate the necessary ingredients for performing resource enhancement to some desired level.

In addition, considering the minimal scenario brings the investigation of resource enhancement firmly within the capabilities of contemporary quantum hardware. For example, the memory of current devices is very limited, meaning they lack the ability to simultaneously store a large number of copies of ρ . By restricting to only two copies, we minimise this issue [‡]. Moreover, our ability to physically implement a given allowed operations on real hardware is probably very limited, as this would likely require appending a free state to the input system before applying a joint operation on the system and free state. Not only does this further increase the necessary memory requirements, but the generation of free states is likely to be difficult in reality. Removing the ability to append free states, and restricting to only allowed unitaries, prevents these considerations.

6.1.2 Framework

In addition to considering this minimal case, we relax the problem of resource enhancement via the following observation: in certain cases, the performance in some information processing protocol that

* The trace distance is usually used, but one could also consider other measures. For example, the distance to σ within each subsystem could be used, therefore assessing how locally close the output state is to σ in each subsystem rather than how close the global state is to multiple copies of σ

[†] We note that one could also add the further freedom of having access to some catalytic state.

[‡] Of course, if ρ is high dimensional i.e., it consists of many qubits, lack of memory remains a problem

depends on a resource can be characterised by a monotone $R(\cdot)$ of that resource. For example, in teleportation, the fidelity of teleportation — a measure of the noise in the teleportation channel — is captured entirely by the entanglement fraction of the initial shared state [137]. Therefore, if aiming to improve their quality of teleportation, given a source producing ρ one can aim to apply an allowed operation to some number of copies of ρ to optimally increase the entanglement fraction within some subspace. The state in this subspace can then be used in the protocol, and will provide an improvement over ρ if the entanglement fraction has been increased. This notion is in contrast to resource distillation, where one considers a specific target output state to be reached.

For instance, given $(n + 1)$ -copies of ρ as an input, the Hilbert space can be bi-partitioned into $\mathcal{H}_A \otimes \mathcal{H}_B$ such that $\rho_A \otimes \rho_B^{\otimes n}$. The aim is then to optimally increase the resource content in system A under allowed operations, which can be captured by the following figure of merit

$$\Delta R_n := \max_{\Lambda \in \mathfrak{D}} R(\text{tr}_B [\Lambda(\rho_A \otimes \rho_B^n)]) - R(\rho_A), \quad (6.2)$$

where $\mathfrak{D}$ is the set of allowed operations of the resource theory being considered[§]. In such problems, one can set the number of copies available and restrict the allowed operations — for example, to only unitaries — depending on the context.

To apply the given information processing protocol, however, the specific state output within the subspace will need to be known. Hence, one can first calculate ΔR_n , therefore determining by how much the protocol can be improved given access to n copies of ρ and $\mathfrak{D}$. From here, the necessary dynamics in $\mathfrak{D}$ to achieve this increase must be found, from which the output state in the subspace can also be found. Considering resource enhancement in this way prevents the need to consider an initial target state. One can instead search for the set of reachable states given the tools available i.e., number of states and allowed operations, that will achieve the optimal performance in the operational task of interest.

We can now introduce the framework of resource concentration, which models the smallest scenarios in which resource enhancement can be credibly discussed.

Definition 6.1.1 (Resource Concentration). *Let $\rho \in \mathcal{D}(\mathcal{H}^d)$. A resource concentration problem asks whether there exists a bipartite resource non-increasing unitary, $V_{ab} \in \mathfrak{D}$, such that*

$$\Delta R(\rho) := \max_{V_{AB}} R(\sigma_A) - R(\rho_A) > 0, \quad (6.3)$$

where $R(\cdot)$ is some resource monotone, the maximisation is taken over all bipartite resource non-increasing unitaries V_{AB} , and

$$\sigma_A = \text{tr}_B [V_{AB}(\rho_A \otimes \rho_B)V_{AB}^\dagger]. \quad (6.4)$$

If $\Delta R(\rho) > 0$, we say that the resource can be concentrated.

In the remainder of this chapter we will apply the framework of resource concentration to the resource theories of informational non-equilibrium and unspeakable coherence. In each case, we then begin to build on the concentration scenario by allowing initial correlations and entanglement, or by increasing the number of input states. From this, we can start to isolate the contributes of these features to the task of resource enhancement.

6.1.3 Structure

Firstly, resource concentration will be considered in the resource theory of informational non-equilibrium. We will consider a resource quantifier for which equalities on our ability to concentrate informational non-equilibrium can be found. The notion of bound resources is then introduced. These ideas will then be solidified through a qutrit example. Finally, we will push beyond the concentration scenario, considering the ability of both initial correlations and multiple initial copies to aid in local resource enhancement.

In the second half, resource concentration will be considered in the resource theory of informational non-equilibrium. The problem is fully solved for qubits, which is then used to develop a multi-qubit

[§] The max is considered over sup to ensure the allowed operation is achievable.

coherence amplification protocol. We then place two incomparable and fundamental bounds on our ability to concentrate coherence in the general case — that is, for states of arbitrary dimension — before again solidifying the concepts through a qutrit example. Finally, we again push beyond the concentration scenario, and consider the ability of initial correlations to aid in the local enhancement of unspeakable coherence.

The first half of this Chapter is based on work published in [3]. Result 6.2.1 and Corollary 6.2.2 were proved by Chung-Yun Hsieh, but are included here for necessity and completeness. Also, the contrapositive of Result 6.2.5 was pointed out to me by Ludovico Lami at the 2025 Quantum Resource Theories Workshop. The remainder of the included results are proved by me.

The second half of this chapter is based on work in the manuscript [1]. This work was led by me, but Result 6.3.5 was proved by Chung-Yun Hsieh.

6.2 Informational Non-equilibrium Concentration

In this section we consider resource concentration within the resource theory of information non-equilibrium [18]. See Section 4.3.1 for an overview of the resource theory of information non-equilibrium.

6.2.1 Quantifier

To propose a resource concentration problem, one must first pick an resource quantifier. For informational non-equilibrium and a state $\rho \in \mathcal{D}(\mathcal{H}^d)$, we consider the max-relative entropy [47] between ρ and the the maximally mixed state

$$\mathcal{P}(\rho) := D_{\max}(\rho \parallel \mathbb{I}^{(d)} / d), \quad (6.5)$$

where $D_{\max}(\rho \parallel \sigma) := \log_2 \min\{\lambda \geq 0 \mid \rho \leq \lambda \sigma\}$ is the *max-relative entropy* [47], widely used for its numerical feasibility and operational relevance [48–51]. See Definition 2.7.3. Explicitly, this can be seen to be

$$\mathcal{P}(\rho) = \log_2 (d \|\rho\|_{\infty}), \quad (6.6)$$

where $\|\rho\|_{\infty}$ is the infinity norm, which is the maximum eigenvalue of ρ . The aim is then to investigate

$$\Delta\mathcal{P}(\rho) := \max_{V_{AB}} \mathcal{P}(\sigma_A) - \mathcal{P}(\rho), \quad (6.7)$$

where $\sigma_A = \text{tr}_B[V_{AB}(\rho_A \otimes \rho_B)V_{AB}^\dagger]$, and the maximisation is taken over all bipartite unitaries, as all unitaries are allowed operations in the resource theory of informational non-equilibrium. Within this section, whenever we discuss concentration it will be with respect to $\Delta\mathcal{P}(\rho)$, unless specified otherwise.

6.2.2 Limits on Concentration

In the section, we consider the limitations on the amount of informational non-equilibrium that it is possible to concentrate.

Firstly, if $\rho = \mathbb{I}/d$ — which is the only free state in the resource theory of informational non-equilibrium when considering a fixed dimension — no resource can be concentrated, $\Delta\mathcal{P}(\mathbb{I}/d) = 0$. This is a result of all allowed operations of the resource theory of informational non-equilibrium being unital. Physically, $\mathbb{I}/d$ has no resource, so there is no resource to move from the A system into the B system.

In general, given an arbitrary input state, the amount of informational non-equilibrium that can be concentrated is captured by the following result.

Result 6.2.1. For $\rho \in \mathcal{D}(\mathcal{H}^d)$,

$$\Delta\mathcal{P}(\rho) = \log_2 \left(\frac{\|\rho \otimes \rho\|_{d\text{-KY}}}{\|\rho\|_\infty} \right), \quad (6.8)$$

where $\|\cdot\|_{d\text{-KY}}$ is the d th Ky-Fan norm, which is the sum of the d largest singular values.

Proof Sketch. Using Eq. (6.6), it can be seen that to find $\Delta\mathcal{P}(\rho)$ one must first find

$$\|\sigma_A\|_\infty = \max_{V_{AB}} \left\| \text{tr}_B [V_{AB}(\rho_A \otimes \rho_B)V_{AB}^\dagger] \right\|_\infty. \quad (6.9)$$

This can be shown to be equal to the maximisation over the trace of rank- d projectors acting on $\rho_A \otimes \rho_B$, which in turn can be shown to be the d th Ky-Fan norm. Inputting this into Eq. (6.6) and taking away $\mathcal{P}(\rho)$ completes the proof. See Section 6.5.1 for the complete proof. $\square$

From Result 6.2.1, it can be seen that $\Delta\mathcal{P}(\rho) > 0$ if and only if $\|\rho \otimes \rho\|_{d\text{-KY}} > \|\rho\|_\infty$. From here, the observation that this can never hold for qubits leads to the following corollary.

Corollary 6.2.2. If $\rho \in \mathcal{D}(\mathcal{H}^2)$, then $\Delta\mathcal{P}(\rho) = 0$, meaning no concentration is possible in the simplest case of two qubits.

Proof. Firstly, note that the singular values of a Hermitian matrix are the absolute value of the eigenvalues. Moreover, all density operators have real and positive eigenvalues, hence, their singular values are equal to their eigenvalues.

Now, the spectrum of all qubit states, ρ , can be written as $\text{spec}(\rho) = \{\lambda, 1 - \lambda\}$, where without loss of generality $1/2 \leq \lambda \leq 1$, such that the spectrum is ordered in non-increasing order. Therefore, $\|\rho\|_\infty = \lambda$. The spectrum of $\rho \otimes \rho$ in non-increasing order is then

$$\text{spec}^\downarrow(\rho \otimes \rho) = \{\lambda^2, \lambda(1 - \lambda), \lambda(1 - \lambda), (1 - \lambda)^2\}. \quad (6.10)$$

Hence, the 2th Ky-Fan norm of $\rho \otimes \rho$, which is the sum of the 2 largest singular values of $\rho \otimes \rho$, can be read off as

$$\begin{aligned} \|\rho \otimes \rho\|_{d\text{-KY}} &= \lambda^2 + \lambda(1 - \lambda) \\ &= \lambda. \end{aligned} \quad (6.11)$$

Therefore, $\|\rho \otimes \rho\|_{d\text{-KY}} = \|\rho\|_\infty \forall \rho \in \mathcal{D}(\mathcal{H}^2)$, meaning $\Delta\mathcal{P}(\rho) = 0$ and no informational non-equilibrium can be concentrated, completing the proof. $\square$

Corollary 6.2.2 states that the structure of quantum theory forbids the concentration of informational non-equilibrium with respect to $\mathcal{P}(\cdot)$ in two qubits. Interestingly, Corollary 6.2.2 can be extended beyond just this resource quantifier. To see this, note that the informational non-equilibrium (purity) of a qubit can be characterised by its largest eigenvalue, i.e. if $\rho \in \mathcal{D}(\mathcal{H}^2)$ then up to local unitaries

$$\rho = \|\rho\|_\infty |0\rangle\langle 0| + (1 - \|\rho\|_\infty) |1\rangle\langle 1|. \quad (6.12)$$

Any function of $\|\rho\|_\infty$ that is non-increasing under allowed operations of the resource theory of informational non-equilibrium will therefore fully characterise the resource content of ρ . Re-examining the proof of Corollary 6.2.2, it says that it is not possible to increase the maximum eigenvalue of a qubit in the concentration scenario. As a result, any resource monotone that is a function of the maximum eigenvalue of a qubit cannot be increased under concentration. This is detailed in Proposition 6.2.3.

Proposition 6.2.3. If $\rho \in \mathcal{D}(\mathcal{H}^2)$, let $R(\cdot)$ be a function of $\|\rho\|_\infty$ that is monotonic under the allowed operations of the resource theory of informational non-equilibrium, then

$$\Delta R(\rho) := \max_{V_{AB}} R(\sigma_A) - R(\rho_A) = 0 \quad \forall \rho \in \mathcal{D}(\mathcal{H}^2), \quad (6.13)$$

where $\sigma_A = \text{tr}_B[V_{AB}(\rho_A \otimes \rho_B)V_{AB}^\dagger]$ and the maximisation is taken over all bipartite unitaries V_{AB} .

Beyond qubits, Result 6.2.1 can be used to shown that there exists states in all dimensions for which informational non-equilibrium can be concentrated. This can easily be seen by considering an effective qubit in a higher dimensional system,

$$\rho = p |0\rangle\langle 0| + (1 - p) |1\rangle\langle 1| \in \mathcal{D}(\mathcal{H}^d), \quad (6.14)$$

where $d > 2$ and $1/2 \leq p < 1$, such that $\|\rho\|_\infty = p$. It can then be seen that

$$\|\rho \otimes \rho\|_{d\text{-KY}} \geq p^2 + 2p(1 - p) > \|\rho\|_\infty, \quad (6.15)$$

meaning there always exists some state such that $\Delta\mathcal{P}(\rho) > 0$.

Lastly, we consider when it is possible to create a maximally resourceful state via concentration. Within the resource theory of informational non-equilibrium, a state is maximally resourceful if it is a pure state. Therefore, to perform *perfect resource concentration* the output state in the A system, σ_A , must be a pure state, $\text{tr}[\sigma_A^2] = 1$. The following result captures when it is possible to achieve this, and is independent of the specific resource quantifier being used.

Result 6.2.4. If $\rho \in \mathcal{D}(\mathcal{H}^d)$ and $\text{rank}(\rho) = r$, then there exists a unitary to perform perfect resource concentration, such that $\text{tr}[\sigma_A^2] = 1$, if and only if $r^2 \leq d$.

Proof. In the proof of Result 6.2.1, it was shown that

$$\|\sigma_A\|_\infty = \max_{V_{AB}} \left\| \text{tr}_B[V_{AB}(\rho_A \otimes \rho_B)V_{AB}^\dagger] \right\|_\infty = \|\rho \otimes \rho\|_{d\text{-KY}}. \quad (6.16)$$

For σ_A to be pure it must be the case that $\|\sigma_A\|_\infty = 1$ and hence that $\|\rho \otimes \rho\|_{d\text{-KY}} = 1$. If $\text{rank}(\rho) = r$, then ρ has r eigenvalues and therefore $\rho \otimes \rho$ has r^2 . If $r^2 \leq d$, then the d th Ky-Fan norm of $\rho \otimes \rho$ is the sum over all its eigenvalues and — due to normalisation — is 1. Hence, $\|\sigma_A\|_\infty = 1$ and σ_A is a pure state. If $r^2 > d$, then the sum of the d -largest eigenvalues of $\rho_A \otimes \rho_B$ will be less than one, as the total sum of the eigenvalues is one, all the eigenvalues are positive, and only a subset is now here being summed over. Hence, under the optimal unitary $\|\sigma_A\|_\infty < 1$, and thus σ_A will not be a pure state. $\square$

To consider the specific unitaries able to achieve perfect concentration, let $\rho = \sum_{i=0}^{r-1} \lambda_i |i\rangle\langle i|$, such that $\rho_A \otimes \rho_B = \sum_{i,j=0}^{r-1} \lambda_i \lambda_j |ij\rangle\langle ij|$. For each of the r^2 eigenstates of $\rho_A \otimes \rho_B$ with non-zero eigenvalue, one needs to apply a unitary that performs the mapping $|ij\rangle \rightarrow |0n\rangle$ for each i, j and some $n \in [0, d-1]$ (where the choice of 0 in the A system is arbitrary). When the B system is then traced out, this will leave the A system in the pure state $|0\rangle$. From this construction, one can see the reason for this upper bound on the rank of ρ limiting the ability to perform perfect concentration — there are only d eigenstates of $\rho_A \otimes \rho_B$ of the form $|0n\rangle$. Hence, if the state has a larger support, the eigenvalues cannot all be unitarily assigned to a basis state with the same label in the A system. Lastly, in the concentration problem with respect to $\mathcal{P}(\cdot)$, we note that if $r^2 \leq d$ such that perfect resource concentration can be performed, then

$$\begin{aligned} \Delta\mathcal{P}(\rho) &= \log_2(d) - \log_2(d\|\rho\|_\infty) \\ &= -\log_2(\|\rho\|_\infty). \end{aligned} \quad (6.17)$$

6.2.3 Bound Resources

When considering resource enhancement via distillation, the idea of bound states has frequently emerged [138, 139]. These are objects of the resource theory that contain non-zero resource, and non-maximal resource, but from which a more resourceful state cannot be created via allowed operations. Here, we investigate the notion of bound states in the resource concentration scenario. Intuitively, we expect to find the same — or more — bound resources in the concentration scenario than in distillation, as we are using a more restricted set of allowed operations. The definition of bound states in the concentration framework is given in Definition 6.2.1.

Definition 6.2.1 (Bound Resources). *Given a resource concentration problem, a state $\rho \in \mathcal{D}(\mathcal{H}^d)$ is said to have bound resource if it is neither maximally resourceful, nor a free state, but no resource can be concentrated.*

We note that, given a resource concentration problem is defined with respect to a specific monotone, it could be possible for a state to have bound resource for one concentration problem and not for another, even if the underlying resource is the same. Whilst an interesting direction, we do not explore this idea in the work presented here.

Returning to our current resource concentration problem, it can be seen that all qubit states that are not pure or maximally mixed fit the definition of having bound resource given in Definition 6.2.1. It can also be seen that this bound resource in qubits is very robust, as no small changes in the state can take you from a state for which no informational non-equilibrium could be concentrated to one that it can. In general, the following results characterise all the states that have bound resource within the resource concentration problem with respect to $\mathcal{P}(\rho)$.

Result 6.2.5. *A state has bound resource within the resource concentration problem with respect to $\mathcal{P}(\rho)$, i.e., $\Delta\mathcal{P}(\rho) = 0$, if and only if it is of the form*

$$\rho(p, |\psi\rangle) = p |\psi\rangle\langle\psi| + \frac{1-p}{d} \mathbb{I} \in \mathcal{D}(\mathcal{H}^d), \quad (6.18)$$

for some $0 < p < 1$ and $|\psi\rangle \in \mathcal{H}^d$. Such states are exactly those that are $(d-1)$ -fold degeneracy in their second largest eigenvalue, where the second largest eigenvalue is non-zero.

Proof Sketch. The forward direction of this proof can be achieved by direct calculation. The contrapositive is proved by first lower bounding the d th Ky-Fan norm of $\rho \otimes \rho$ by the maximum eigenvalue of ρ , $\|\rho \otimes \rho\|_{d\text{-KY}} \geq \|\rho\|_\infty$, and then showing that only if the spectrum of ρ is $(d-1)$ -fold degenerate in its second largest eigenvalue does the equality hold. See Section 6.5.2 for a full proof, including a graphical depiction of the contrapositive part Fig. 6.8. $\square$

It can be seen from Result 6.2.5 that whenever a dephasing channel, $\mathcal{D}_q^{\text{ph}}(\cdot) := q\mathcal{I}(\cdot) + (1-q)\text{tr}[\cdot]\mathbb{I}/d$ where $\mathcal{I}$ is the identity channel, is applied to a pure state it will output a state with bound resources if $0 < p < 1$. Moreover, if a dephasing channel is applied to a state that already has bound resource, it will output another state with bound resource, as $\mathcal{D}_q^{\text{ph}}(\rho(p, |\psi\rangle)) = \rho(qp, |\psi\rangle)$. This implies that any dephasing noise in a system cannot be combated via resource concentration: a system where the informational non-equilibrium has been lowered via a dephasing channel must have its resource content increased via some more expensive resource enhancement procedure than concentration. Although, it can be seen that bound resources in states with dimension greater than two are not robust. Given a state $\rho(p, |\psi\rangle)$, small perturbations in the state would likely lead to one with unbound resource.

6.2.4 Qutrit Example

In this section we go through an example of performing concentration of informational non-equilibrium with two qutrits. We explore both the ‘junk’ in the B system and the necessary creation of correlations between the A and B systems when performing the optimal concentration unitary. We see that even in this smallest non-trivial case of resource concentration, there is a rich structure to the resource

enhancement.

Firstly, consider a state $\rho \in \mathcal{D}(\mathcal{H}^3)$ decomposed with respect to its eigenbasis,

$$\rho = \lambda_0 |0\rangle\langle 0| + \lambda_1 |1\rangle\langle 1| + \lambda_2 |2\rangle\langle 2|, \quad (6.19)$$

such that the eigenvalues are ordered in decreasing order: $\lambda_0 \geq \lambda_1 \geq \lambda_2$. We then note that qutrits, along with qubits, are the only states for which the d th Ky-Fan norm can be specified given only the spectrum of ρ in non-increasing order, denoted $\text{spec}^\downarrow(\rho) = \{\lambda_0, \lambda_1, \lambda_2\}$. To see this, for qutrits the largest three eigenvalues of $\rho \otimes \rho$ will always be $\lambda_0\lambda_0, \lambda_0\lambda_1, \lambda_1\lambda_0$. If one is instead considering $\omega \in \mathcal{D}(\mathcal{H}^4)$, with $\text{spec}^\downarrow(\omega) = \{\mu_0, \mu_1, \mu_2, \mu_3\}$, then the three largest eigenvalues of $\omega \otimes \omega$ will take the same form as for a qutrit. However, without additional information it is not known if the 4th largest eigenvalue is $\mu_0\mu_2$ or $\mu_1\mu_2$. This is true for all $d > 3$, and means the d th Ky-Fan norm cannot be specified. Therefore, qutrits are the only example of a non-trivial concentration problem that can be solved analytically in full generality.

Returning to qutrits, Result 6.2.1 implies that

$$\Delta\mathcal{P}(\rho) = \log_2(\lambda_0 + 2\lambda_1). \quad (6.20)$$

Interestingly, it can be seen that the unitary U_{AB}^{opt} that performs the mapping $|10\rangle \leftrightarrow |02\rangle$, and leaves all other basis states untouched, is able to achieve this optimal concentration for all input states. Hence, for qutrits, U_{AB}^{opt} is a universal optimal concentrating unitary.

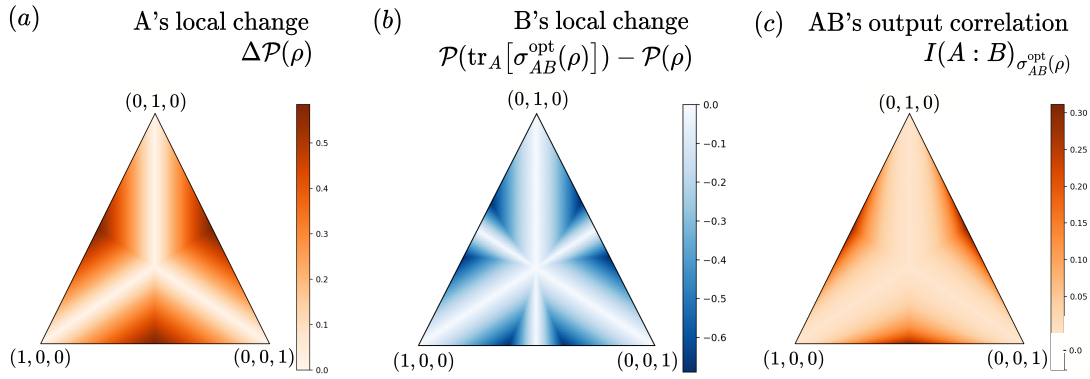


Figure 6.1: A plot of Eq. (6.20) for each qutrit state in state space. Each point in the triangle represents a $(\lambda_0, \lambda_1, \lambda_2)$ — the eigenvalues of a qutrit state — with the colour giving the change of informational non-equilibrium as measured by $\mathcal{P}(\cdot)$ under U_{AB}^{opt} . (a) Optimal increase in A according to Eq. (6.20). States with bound resources are those on the white lines running from the corners to the centre of the triangle. (b) Change in B when A achieves the optimal change $\Delta\mathcal{P}(\rho)$. (c) The mutual information between A and B after the optimal concentration has been performed. States for which no correlations are created have been explicitly highlighted in white, and can be seen to coincide with the states possessing bound resource.

In Fig. 6.1.a, Eq. (6.20) for all qutrit states in the state space is plotted on a 2-simplex. It can be seen that states that are a convex combination of the pure eigenstates (the corners of the simplex) and the maximally mixed state (the middle of the simplex) have $\Delta\mathcal{P}(\rho) = 0$. These are exactly the states with bound resource, i.e., those given in Result 6.2.5. It can also be seen that the maximum change in local resource occurs when the initial state is a maximally mixed effect qubit i.e, $\text{rank}(\rho) = 2$ with both eigenvalues equal to 2^{-1} .

Moreover, letting $\sigma_{AB}^{\text{opt}} = U_{AB}^{\text{opt}}(\rho_A \otimes \rho_B)U_{AB}^{\text{opt},\dagger}$, we can investigate the change in resource in the B system via $\mathcal{P}(\text{tr}_A[\sigma_{AB}^{\text{opt}}]) - \mathcal{P}(\rho)$. In Fig. 6.1.b, this is plotted for all qutrit states. Surprisingly, it can be seen that for states with the largest change in resource in the A system — maximally mixed effect qubits — there is zero change in resource of the B system. This is despite the optimal concentrating dynamics being resource non-increasing. This apparent paradox is easily resolved by considering the full spectrum of $\text{tr}_A[\sigma_{AB}^{\text{opt}}]$. From here, it can be seen that whilst the states maximum eigenvalue stays the same, and hence $\Delta\mathcal{P}(\rho)$ remains unchanged, the rest of the support becomes more mixed. See Fig. 6.2 for

a graphical depiction. Therefore, this can be seen to be just a feature of the resource quantifier being used, arising from $\mathcal{P}(\rho)$ dependence on only the maximum eigenvalue ρ , which does not fully characterise the resource content of ρ .

Lastly, we look to quantify the build up of correlations between the A and B system under U_{AB}^{opt} , the optimal concentrating unitary. To do this, we use the quantum mutual information, a widely used correlation measure. For a bipartite state $\eta_{AB} \in \mathcal{D}(\mathcal{H}_A \otimes \mathcal{H}_B)$, the quantum mutual information is defined as

$$I(A : B)_{\eta_{AB}} := S(\eta_A) + S(\eta_B) - S(\eta_{AB}), \quad (6.21)$$

where $S(\eta) = -\text{tr}[\eta \log_2(\eta)]$ and η_A, η_B are the marginals of η_{AB} in the A and B system respectively. We find the following relationship between optimal concentration and the mutual information.

Result 6.2.6. *Two qutrit optimal concentration always generates correlations between the A and B systems: $I(A : B)_{\eta_{AB}} > 0$ if $\Delta\mathcal{P}(\rho) > 0$.*

Proof Sketch. This can be proved by direction calculation. See Fig. 6.1.c for a graphical proof — a plot of $I(A : B)_{\eta_{AB}}$ for all qutrit states in the state space which can be seen to be non-zero for all but the bound states. See Section 6.5.3 for the analytical details behind Fig. 6.1.c. $\square$

From Result 6.2.6, counter-intuitively, one must increase global correlation and local purity simultaneously. Since a pure state cannot be correlated with any other system, this result means that it is impossible to map a non-pure qutrit state to a pure state in the resource concentration scenario with two qutrits. This is confirmed by Result 6.2.4, as $r^2 \not\leq d^2$ for qutrits, where r is the rank of the state, other than in the trivial case that $r = 1$ (such that the qutrit is already a pure state). From here, it can be seen that Result 6.2.4 therefore implies that the necessary creation of correlations under optimal concentration for all states will only hold for qutrits: beyond qutrits, there always exists non-trivial states that can be perfectly concentrated, meaning the output state is in a pure state and hence cannot be correlated with another system.

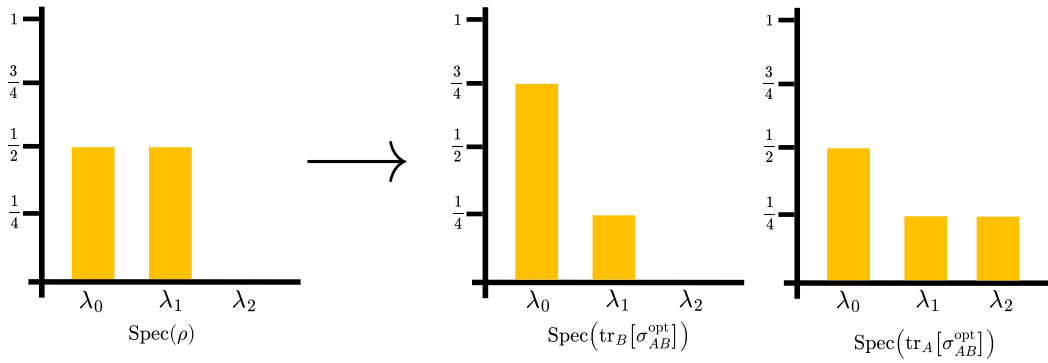


Figure 6.2: A graphical deception of how the maximum change in resource in the A system coincides with zero change in the resource in the B system, despite the dynamics being resource non-increasing. The spectrums of the input and output states in each of the subsystem is plotted, and it can be seen that the maximum eigenvalue in the input system and output B -system are the same, meaning $\mathcal{P}(\cdot)$ remains unchanged, whilst the maximum eigenvalue in the output A system has increased. However, the rest of the support in the B system has become more mixed.

6.2.5 Initial Correlations and Entanglement

Here, we expand the resource concentration scenario to consider states that are initially correlated or entangled, aiming to understand the role these play on our ability to increase the local resource content. We still consider only bipartite systems and resource non-increasing bipartite unitaries. Hence, despite not considering inputs of the form $\rho_A \otimes \rho_B$, in this section we still refer to the task as resource concentration.

It can easily be seen that initial correlations can activate concentration, even if the amount of initial

correlations is vanishingly small. Consider an initial isotropic state,

$$p |\Phi^+\rangle\langle\Phi^+|_{AB} + (1-p)\mathbb{I}_{AB}/d^2 \in \mathcal{D}(\mathcal{H}_A^d \otimes \mathcal{H}_B^d), \quad (6.22)$$

where $|\Phi^+\rangle = (1/\sqrt{d}) \sum_{i=0}^{d-1} |ii\rangle$ is the bipartite maximally entangled state and $0 < p \leq 1$. Both of the local subsystems are maximally mixed, which is a state for which no informational non-equilibrium can be concentrated, $\Delta\mathcal{P}(\mathbb{I}/d) = 0$. However, if the unitary that performs the mapping $|\Phi^+\rangle \leftrightarrow |00\rangle_{AB}$ is applied, then the output state is no longer maximally mixed in both its subsystems. Hence, some informational non-equilibrium has been concentrated. Physically, the global correlations in the isotropic state have been converted into local resource. This is possible for all $p \neq 0$, meaning even the weakest initial correlations can be used to activate concentration.

This idea can be extended to arbitrary states, $\rho = \sum_i \lambda_i |i\rangle\langle i|$, using the purification of ρ . This is the pure state $|\rho\rangle_{AB} = \sum_i \sqrt{\lambda_i} |ii\rangle$ that has ρ as both its marginals. In parallel to the isotropic state above, one can then consider the state

$$p |\rho\rangle\langle\rho|_{AB} + (1-p)\rho_A \otimes \rho_B \in \mathcal{D}(\mathcal{H}_A^d \otimes \mathcal{H}_B^d), \quad (6.23)$$

where $0 \leq p \leq 1$. Applying the unitary that performs the mapping $|\rho\rangle_{AB} \leftrightarrow |00\rangle_{AB}$ will therefore change the local resource content, even if ρ is a state with bound resource. Hence, global correlation can activate bound resources. As a result, in both qubit and higher dimensional systems, the notion of bound resource is not robust to even vanishingly small global correlations.

In a similar direction, we now aim to quantitatively investigate to what extent initial entanglement can enhance the ability to perform concentration. To do this, we again consider the purification of ρ as, firstly, it is guaranteed to be entangled if ρ is not a pure state[¶]. Moreover, since both marginals of $|\rho\rangle_{AB}$ are ρ , comparing our ability to concentrate with $|\rho\rangle_{AB}$, as oppose to $\rho_A \otimes \rho_B$, removes any variation arising from the local state. This allows us to isolate any additional ability to concentrate coming from the entanglement. To quantify this entanglement advantage, we first consider the following figure of merit,

$$\Delta\mathcal{P}_{\text{Ent}}(\rho) := \max_{V_{AB}} \mathcal{P}(\text{tr}_B[V_{AB} |\rho\rangle\langle\rho|_{AB} V_{AB}^\dagger]) - \mathcal{P}(\rho_A), \quad (6.24)$$

where the maximisation is over all bipartite unitaries V_{AB} . This is a measure of the amount of information non-equilibrium that can be concentrated given the entangled state $|\rho\rangle_{AB}$. As there always exists a unitary V_{AB} that performs the mapping $|\rho\rangle_{AB} \leftrightarrow |00\rangle_{AB}$, then

$$\max_{V_{AB}} \left\| \text{tr}_B[V_{AB} |\rho\rangle\langle\rho|_{AB} (V_{AB}^\dagger)] \right\|_\infty = 1, \quad (6.25)$$

meaning a locally pure state can always be output. Hence, $\Delta\mathcal{P}_{\text{Ent}}(\rho) = -\log_2(\|\rho\|_\infty) \forall \rho \in \mathcal{D}(\mathcal{H})$. We then consider

$$\begin{aligned} \Delta\mathcal{P}_{\text{Ent}}(\rho) - \Delta\mathcal{P}(\rho) &= -\log_2(\|\rho\|_\infty) - \log_2(\|\rho \otimes \rho\|_{d-KY} / \|\rho\|_\infty) \\ &= -\log_2(\|\rho \otimes \rho\|_{d-KY}), \end{aligned} \quad (6.26)$$

which is the difference between the optimal amount of informational non-equilibrium that can be concentrated using $|\rho\rangle_{AB}$ and $\rho_A \otimes \rho_B$. We therefore purpose this as a quantitative measure of the entanglement advantage in concentration, as summarised in Proposition 6.2.7.

Proposition 6.2.7. *Given a state $\rho \in \mathcal{D}(\mathcal{H}^d)$, a measure of the entanglement advantage when performing concentration is*

$$\mathbb{E}(\rho) := -\frac{1}{d} \log_2(\|\rho \otimes \rho\|_{d-KY}). \quad (6.27)$$

This measure has the following properties

[¶] If the partial trace over a pure state is pure, the global state is a product state over the partition traced over.

- $0 \leq \mathbb{E}(\rho) \leq 1$.
- If $\text{rank}(\rho) = r$, then $\mathbb{E}(\rho) = 0$ if and only if $r^2 \leq d$.
- $\mathbb{E}(\rho) = 1$ if and only if $\rho = \mathbb{I}/d$.

We will quickly review the presented properties of this measure. If $\text{rank}(\rho) = r$ and $r^2 \leq d$, then one can perform perfect concentration with ρ , as seen in Result 6.2.4. This means it is always possible to create a locally pure state. Hence, there is no entanglement advantage for such states, as the creation of a local pure state is what is always achievable with $|\rho\rangle_{AB}$. The largest possible advantage comes when $\rho = \mathbb{I}/d$, as given two initially uncorrelated copies of this state, no concentration is possible.

As a draw back of this measure, we note that for two different mixed states the entanglement in their purification can be different (as measured by the Schmidt number, for example). Hence, the entanglement available for conversion into local informational non-equilibrium is not necessarily equal for different input states. Hence, its unclear if one can fairly compare $\mathbb{E}(\cdot)$ for different input states. Although, a full analysis of this measure was beyond the scope of this work, and it has been left for the future.

6.2.6 Concentration with n -initial States

Here, we again expand on the resource concentration scenario, now considering an agent's ability to optimally increase the local resource content — as measured by $\mathcal{P}(\cdot)$ — when given access to an initial state $\rho_A \otimes \rho_B^{\otimes n}$. Hence, there is one copy of ρ in system A and n copies of ρ in system B . We also assume the ability to perform interactions on all $n + 1$ copies simultaneously. We aim to understand how having multiple copies of ρ , and being able to perform a global unitary on all copies, affects our ability to locally increase the resource content in some subsystem.

Firstly, we define

$$\Delta\mathcal{P}^n(\rho) := \max_{V_{AB}} \mathcal{P}(\sigma_A^n) - \mathcal{P}(\rho), \quad \sigma_A^n = \text{tr}_B[V_{AB}(\rho_A \otimes \rho_B^{\otimes n})V_{AB}^\dagger], \quad (6.28)$$

Such that $\Delta\mathcal{P}^n(\rho)$ quantifies the maximum change in informational non-equilibrium that can be achieved given access to $n + 1$ copies of the state $\rho \in \mathcal{D}(\mathcal{H}^d)$. We note that in this notion we have $\Delta\mathcal{P}^1(\rho) := \Delta\mathcal{P}(\rho)$, where $\Delta\mathcal{P}(\rho)$ is as defined in Eq. (6.7). The following corollary determines the value of $\Delta\mathcal{P}^n(\rho)$ for all n .

Corollary 6.2.8. For $\rho \in \mathcal{D}(\mathcal{H}^d)$,

$$\Delta\mathcal{P}^n(\rho) = \log_2 \left(\frac{\|\rho \otimes \rho^{\otimes n}\|_{d^n - \text{KY}}}{\|\rho\|_\infty} \right), \quad (6.29)$$

where $\|\cdot\|_{d^n - \text{KY}}$ is the d^n th Ky-Fan norm, which is the sum of the d^n largest singular values.

Proof. As before, to find $\mathcal{P}(\sigma_A^n)$ one must find the maximum eigenvalue of σ_A^n :

$$\|\sigma_A^n\|_\infty = \max_{V_{AB}} \left\| \text{tr}_B[V_{AB}(\rho_A \otimes \rho_B^{\otimes n})V_{AB}^\dagger] \right\|_\infty. \quad (6.30)$$

At the end of the proof of Result 6.2.1, detailed in Section 6.5.1, it was shown that

$$\max_{V_{AB}} \left\| \text{tr}_B[V_{AB}(\rho_A \otimes \omega_B)V_{AB}^\dagger] \right\|_\infty = \|\rho \otimes \omega\|_{d_B - \text{KY}}, \quad (6.31)$$

where $d_B = \dim(\omega)$. Applying this to the above we see that

$$\|\sigma_A^n\|_\infty = \|\rho \otimes \rho^{\otimes n}\|_{d^n - \text{KY}}, \quad (6.32)$$

where $d^n = \dim(\rho^{\otimes n})$. Substituting σ^n into $\mathcal{P}^n(\cdot)$ and subtracting $\mathcal{P}(\rho)$ completes the proof. $\square$

From Corollary 6.2.8, it can be seen that there is an advantage to using $n + 1$ copies of a state ρ , over

using two copies, when performing local resource enhancement, if and only if

$$\Delta\mathcal{P}^n(\rho) - \Delta\mathcal{P}(\rho) > 0 \iff \|\rho \otimes \rho^{\otimes n}\|_{d^n - KY} > \|\rho \otimes \rho\|_{d - KY}. \quad (6.33)$$

It is noted that if there is an advantage in the $n = 2$ case, then there will be an advantage for all $n > 2$ over the concentration scenario, as one could always at least apply the unitary that gives an advantage in the $n = 2$ case (utilising only two of the n copies). With this in mind, we first consider qubits, $\rho = \lambda |0\rangle\langle 0| + (1 - \lambda) |1\rangle\langle 1|$ with $0 \leq \lambda \leq 1$, when $n = 2$. It can be seen that

$$\|\rho \otimes \rho^{\otimes 2}\|_{4 - KY} = \lambda^2(3 - 2\lambda). \quad (6.34)$$

Then, as $\|\rho \otimes \rho\|_{d - KY} = \|\rho\|_\infty = \lambda$ for qubits, it is the case that

$$\Delta\mathcal{P}^2(\rho) - \Delta\mathcal{P}(\rho) > 0 \iff \lambda^2(3 - 2\lambda) > \lambda. \quad (6.35)$$

To see that this holds, the left hand side of Eq. (6.35) can be factorised, from which it can be verify that $\lambda^2(3 - 2\lambda) > \lambda$ for all $2^{-1} < \lambda < 1$. Hence, $\Delta\mathcal{P}^2(\rho) - \Delta\mathcal{P}(\rho) > 0$ for all qubits that are not maximally mixed ($\lambda = 2^{-1}$) or pure ($\lambda = 1$). Therefore, using the above logic, $\Delta\mathcal{P}^n(\rho) - \Delta\mathcal{P}(\rho) > 0$ for all qubits that are not maximally mixed or pure, meaning there is always an advantage to having $n \geq 2$ copies of ρ .

6.2.6.1 Bound Resources with $(n + 1)$ -initial States

In this section, we will consider an extension of the notion of bound informational non-equilibrium to the $(n + 1)$ -copy case.

A state was defined to have bound informational non-equilibrium in Definition 6.2.1 if ρ was not maximally resourceful, nor minimally resourceful, but $\Delta\mathcal{P}(\rho) = 0$. This can firstly be extended to the $(n + 1)$ -copy case by asking if there exists states for which $\Delta\mathcal{P}^n(\rho) = 0$. This would mean that despite having access to $n + 1$ copies of ρ , no resource can be locally concentrated. To do this, we need only consider states that are bound in the concentration scenario, as if there were states that only became bound as the number of copies increased one could just perform an operation on a fewer number of copies instead.

In addition, the notion of bound informational non-equilibrium can also be expanded upon by considering when having an additional copy of the state does not facilitate additional local resource enhancement, i.e, when does

$$\Delta\mathcal{P}^{n+1}(\rho) = \Delta\mathcal{P}^n(\rho). \quad (6.36)$$

This would mean that the ability to increase the local resource content is equal when given access to $(n + 1)$ or $(n + 2)$ copies of ρ . It can be seen from Corollary 6.2.8 that this occurs if and only if

$$\|\rho \otimes \rho^{\otimes n+1}\|_{d^{n+1} - KY} = \|\rho \otimes \rho^{\otimes n}\|_{d^n - KY}. \quad (6.37)$$

To explore both of these scenarios, we will make use of the graphical aid in Fig. 6.3. This will be used to gain an intuition for the employed proof strategies of the following results. In Fig. 6.3, a grid is presented, where each cell of the grid represents an eigenvalue of $\rho \otimes \rho \otimes \rho$, where $\rho \in \mathcal{D}(\mathcal{H}^3)$. This is an extension of the grid notion considered in Fig. 6.8 in the proof of Result 6.2.5.

On the y -axis of the grid, the spectrum of ρ is presented in non-increasing order, and on the x -axis of the grid, the spectrum of $\rho \otimes \rho$ is presented in non-increasing order, where

$$\begin{aligned} \text{spec}^\downarrow(\rho) &= \{\lambda_i(\rho) : \lambda_i(\rho) \geq \lambda_j(\rho) \forall i \leq j\}_{i=0}^{d-1} \\ \text{spec}^\downarrow(\rho \otimes \rho^{\otimes k}) &= \{\lambda_i(\rho \otimes \rho^{\otimes k}) : \lambda_i(\rho \otimes \rho^{\otimes k}) \geq \lambda_j(\rho \otimes \rho^{\otimes k}) \forall i < j\}_{i=0}^{d^{k+1}-1}. \end{aligned} \quad (6.38)$$

such that $\lambda_j(\sigma)$ denotes the $(j + 1)$ th largest eigenvalues of σ .

The d th Ky-fan norm of $\rho \otimes \rho$ in the qutrit example presented in Fig. 6.8 is given by

$$\|\rho \otimes \rho\|_{3\text{-KY}} = \lambda_0(\rho \otimes \rho) + \lambda_1(\rho \otimes \rho) + \lambda_2(\rho \otimes \rho). \quad (6.39)$$

A lower bound on the 9-Ky-Fan norm of $\rho \otimes \rho \otimes \rho$ can be found by taking the sum of any $d^2 = 9$ cells in the grid. It can be seen that by summing over the $d^2 = 9$ purple-highlighted cells in Fig. 6.3 gives the following lower-bound on the 9-Ky-Fan norm:

$$\|\rho \otimes \rho \otimes \rho\|_{9\text{-KY}} \geq \|\rho \otimes \rho\|_{3\text{-KY}}, \quad (6.40)$$

where we have used the fact that $\sum_i \lambda_i = 1$ due to normalisation. In the general d -dimensional case, this would then read

$$\|\rho \otimes \rho \otimes \rho\|_{d^2\text{-KY}} \geq \|\rho \otimes \rho\|_{d\text{-KY}}. \quad (6.41)$$

Finally, it can be noted that the sum over the cells in the dashed-cyan rectangle in Fig. 6.3 is equal to $\|\rho\|_\infty = \lambda_0$, again due to normalisation. As $\|\rho \otimes \rho\|_{d\text{-KY}} \geq \|\rho\|_\infty \forall d$, it can be concluded that in general

$$\|\rho \otimes \rho \otimes \rho\|_{d^2\text{-KY}} \geq \|\rho \otimes \rho\|_{d\text{-KY}} \geq \|\rho\|_\infty. \quad (6.42)$$

Via this method, this can be generalised further to

$$\|\rho \otimes \rho^{\otimes k}\|_{d^k\text{-KY}} \geq \|\rho \otimes \rho^{\otimes j}\|_{d^j\text{-KY}} \quad \forall k \geq j. \quad (6.43)$$

Using these tools, we now prove the following result about bound-informational non-equilibrium in the n -copy case.

Result 6.2.9. *Let $\rho \in \mathcal{D}(\mathcal{H}^d)$ be a state that is not maximally resourceful nor minimally resourceful. Then $\Delta\mathcal{P}^n(\rho) > 0$.*

Proof Sketch. For $\Delta\mathcal{P}^n(\rho) > 0$ to hold, it must be the case that $\|\rho \otimes \rho^{\otimes n}\|_{d^{n+1}\text{-KY}} > \|\rho\|_\infty$. As argued above, it is known that $\|\rho \otimes \rho^{\otimes n}\|_{d^{n+1}\text{-KY}} \geq \|\rho\|_\infty \forall n$. To prove the result, we first prove that this is a strict equality in the example presented in Fig. 6.8, before considering the general case. Here, it can be seen that this bound is an equality if the cells in the cyan rectangle are the 9-largest cells in the grid. To show that this is never true, and it is always a strict inequality, one needs to show that there exists a cell outside this cyan rectangle that is always larger than a cell within it. This can be done by comparing the smallest cell within it (blue-highlighted) to the largest outside it (orange-highlighted), depicted in inequality (A) below the grid. This idea is then extended beyond the example and shown to hold for all n and arbitrary dimension states. See Section 6.5.4 for a full proof. $\square$

Result 6.2.9 shows that there are no longer any bound resources when $n \geq 2$ i.e., it is always possible to increase the informational non-equilibrium locally via global unitary dynamics if the state is not maximally mixed nor is it a pure state. This shows a significant advantage for the $(n+1)$ -copy case over the concentration scenario, where there exists states with bound informational non-equilibrium in every dimension.

We now look to begin characterising when an additional copy of ρ does not facilitate additional local resource enhancement, meaning that the amount of local resource that can be increased is equal given $(n+1)$ -copies of the state, or $(n+2)$ -copies. We first present a necessary and sufficient condition on the spectrum of $\rho \otimes \rho^{\otimes n}$ and ρ for this to occur, before using it to fully characterise for what states $\Delta\mathcal{P}^2(\rho) = \Delta\mathcal{P}(\rho)$, meaning there is no advantage to having 3 copies of ρ , as opposed to 2 copies.

Lemma 6.2.10. Let $\rho \in \mathcal{D}(\mathcal{H}^d)$. A necessary and sufficient condition for

$$\|\rho \otimes \rho^{\otimes(n+1)}\|_{d^{n+1}-KY} = \|\rho \otimes \rho^{\otimes n}\|_{d^n-KY} \quad (6.44)$$

is

$$\frac{\lambda_{d^n}(\rho \otimes \rho^{\otimes n})}{\lambda_{d^n-1}(\rho \otimes \rho^{\otimes n})} \leq \frac{\lambda_d(\rho)}{\lambda_0(\rho)}, \quad (6.45)$$

meaning that the ratio of the $(d^n + 1)$ th and (d^n) th largest eigenvalues of $\rho \otimes \rho^{\otimes n}$ must be less than the ratio of the d th and largest eigenvalues of ρ .

Proof Sketch. We again consider the example presented in Fig. 6.8, before extending to the general case. Above, it is argued that $\|\rho \otimes \rho \otimes \rho\|_{d^2-KY} \geq \|\rho \otimes \rho\|_{d-KY}$. In Fig. 6.8, this can be seen by picking the sum over the purple highlighted cells to be a lower bound on $\|\rho \otimes \rho \otimes \rho\|_{d^2-KY}$. To characterise when this is a strict equality, one then needs to find when there exists no cell outside the set of purple highlighted cells that is larger than a cell within it. This can be done by comparing the smallest cell within the set (green highlighted) to the largest outside it (red highlighted), depicted in inequality (B) below the grid. If the green highlighted cell is larger than or equal to the red highlighted cell then $\|\rho \otimes \rho \otimes \rho\|_{9-KY} = \|\rho \otimes \rho\|_{3-KY}$. If the green highlighted cell is less than the red highlighted cell then $\|\rho \otimes \rho \otimes \rho\|_{9-KY} > \|\rho \otimes \rho\|_{3-KY}$. This idea is then extended beyond the example and shown to hold for general n and arbitrary dimension states. See Section 6.5.5 for a full proof. $\square$

We note that as $\|\rho \otimes \rho^{\otimes(n+1)}\|_{d^{n+1}-KY} \geq \|\rho \otimes \rho^{\otimes n}\|_{d^n-KY}$, if the condition given in Eq. (6.109) in Lemma 6.2.10 does not hold, then $\|\rho \otimes \rho^{\otimes(n+1)}\|_{d^{n+1}-KY} > \|\rho \otimes \rho^{\otimes n}\|_{d^n-KY}$, with the inequality becoming strict.

Here, using Lemma 6.2.10, we characterise an example of when having $(n + 2)$ copies of ρ offers no advantage over having $(n + 1)$ copies.

Result 6.2.11. Let $\rho \in \mathcal{D}(\mathcal{H}^d)$ be a state that is not maximally resourceful nor minimally resourceful. Then $\Delta\mathcal{P}^2(\rho) = \Delta\mathcal{P}(\rho)$ if and only if ρ is $\sqrt{d}$ -fold degenerate in its largest eigenvalues, and $(d - \sqrt{d})$ -fold degenerate in its second largest eigenvalue. If the state is not of this form, then $\Delta\mathcal{P}^2(\rho) > \Delta\mathcal{P}(\rho)$.

Proof Sketch. Firstly, it can be seen from Corollary 6.2.8 that $\Delta\mathcal{P}^2(\rho) = \Delta\mathcal{P}(\rho)$ if and only if $\|\rho \otimes \rho^{\otimes 2}\|_{d^2-KY} = \|\rho \otimes \rho\|_{d-KY}$. One can then employ Lemma 6.2.10 to shown that if the d th largest eigenvalue of $\rho \otimes \rho$ is strictly less than the square of the largest eigenvalue of ρ , then $\|\rho \otimes \rho^{\otimes 2}\|_{d^2-KY} > \|\rho \otimes \rho\|_{d-KY}$, meaning that $\Delta\mathcal{P}^2(\rho) > \Delta\mathcal{P}(\rho)$. One therefore only just needs to consider the case that the d th largest eigenvalue of $\rho \otimes \rho$ is equal to the largest eigenvalue of ρ squared, as it cannot be less due to the ordering of the eigenvalues in non-decreasing order, and will in general be larger. By considering what the spectrum of ρ must be for this to occur, and then applying Lemma 6.2.10 to all possible variations, it can be seen that it must be the case that ρ is $\sqrt{d}$ -fold degenerate in its largest eigenvalues, and $(d - \sqrt{d})$ -fold degenerate in its second largest eigenvalue. Note, if $\sqrt{d}$ is not an integer, then no state exists in that dimension such that $\Delta\mathcal{P}^2(\rho) = \Delta\mathcal{P}(\rho)$. This is in contrast to bound states, where there existed bound states in all dimensions. See Section 6.5.6 for a full proof. $\square$

Result 6.2.11 shows that for $n = 2$ it is not always advantageous — in terms of the ability to increase the local resource content — to have more copies of the state: it is not possible to do worse, but having an additional copy does not allow more resource to be locally enhanced. We conjecture that a similar phenomenon should exist for larger n if $\sqrt[n]{n}$ is an integer, such that the initial state ρ can be $\sqrt[n]{n}$ -fold degenerate in its largest eigenvalue. However, we leave a full exploration of the potential n -copy advantage in the concentration problem to future work. Although, we note that this is a curious outcome. As, given the necessary fine-tuning of the spectrum, as an agent gets more copies of ρ they will in general be able to increase the amount of informational non-equilibrium they can locally concentrate. Then, at some $(n + 1)$, given another copy the agent will reach a plateau, unable to concentrate more than they

did with n copies. But, given $(n + 2)$ copies, they will again be able to increase the local informational non-equilibrium content further.

6.3 Unspeakable Coherence Concentration

In this section we consider resource concentration within the resource theory of unspeakable coherence [16]. A brief overview of the resource theory of unspeakable is first given, as it is necessary to understand our first result in this section. See Section. 4.3.2 or [16] for a more complete overview.

6.3.1 Framework

Let $L = \sum_k a_k \Pi_k$ be a Hermitian linear operator acting on a finite dimensional Hilbert space, where Π_k is the projector onto the eigenspace of L with eigenvalue a_k . The free states of the resource theory of unspeakable coherence, with respect to the decomposition of the Hilbert space defined by L , are the states that are symmetric with respect to the continuous translation generated by L , meaning $\rho \in \mathcal{D}(\mathcal{H})$ is free if

$$U_{L,x} \rho U_{L,x}^\dagger = \rho \quad \forall x \in \mathbb{R}, \quad (6.46)$$

where $U_{L,x} = e^{-ixL}$. It can be seen that this is equivalent to $[\rho, L] = 0$. The allowed operations are the CPTP maps that are translationally covariant with respect to $U_{L,x}$,

$$\forall x \in \mathbb{R} : \mathcal{E} \circ U_{L,x} = U_{L,x} \circ \mathcal{E}, \quad (6.47)$$

where $\mathcal{E} \in \text{CPTP}$ and $U_{L,x}(\cdot) = U_{L,x}(\cdot)U_{L,x}^\dagger$. If one considers only unitary dynamics $\mathcal{E} = V(\cdot)V^\dagger$, Eq. (6.47) simplifies to $\forall x \in \mathbb{R} : [V, U_x] = 0$, which holds if and only if $[V, L] = 0$. Hence, a unitary V is an allowed operation if and only if it is block diagonal with respect to the eigenspaces of L , such that $V = \sum_k V_k \Pi_k$, where each V_k is an independent unitary, $V_k V_k^\dagger = \mathbb{I} \quad \forall k$ and $V_k V_l^\dagger = \delta_{k,l} \Pi_k$. The following statements are therefore all equivalent within this context: a state ρ has coherence between the eigenspaces of L ; it is asymmetric with respect to the transformation $U_x(\cdot)$; it is resourceful.

6.3.1.1 Bipartite Systems

As detailed in Section 4.3.2, this framework can be expanded to a bipartite space over AB by considering the following generator of translation on the bipartite system

$$L_{AB} = L_A \otimes \mathbb{I}_B + \mathbb{I}_A \otimes L_B. \quad (6.48)$$

Hence, if $[\rho_{AB}, L_{AB}] = 0$ then ρ_{AB} has no coherence between the eigenspaces of L_{AB} . A unitary V_{AB} on the bipartite space is then resource non-increasing, meaning it does not generate coherence between the eigenspaces of L_{AB} , if and only if $[V_{AB}, L_{AB}] = 0$. See Section 6.6.1 for more details on the differing global and local symmetries. If L_{AB} contains no degeneracies, then all resource non-increasing unitaries on the bipartite space are diagonal in the joint eigenbasis of L_{AB} . Such unitaries do not facilitate any interaction between the two subsystems, meaning no coherence concentration is possible. If L_{AB} does contain degeneracies, then the resource non-generating unitaries are then those that apply arbitrary unitaries within the degenerate subspaces of L_{AB} .

In the remainder of this chapter, we consider our local generator of translation to be the non-degenerate, truncated number operator $L = \sum_{n=0}^{d-1} n |n\rangle\langle n|$. If the local generator were degenerate, coherence could be freely generated within the degenerate eigenspaces, which could obscure our understanding of whether coherence has locally increased due to it moving from the B system into the A system. Considering this local generator also ensures that L_{AB} contains degeneracies such that it is possible to perform coherence concentration.

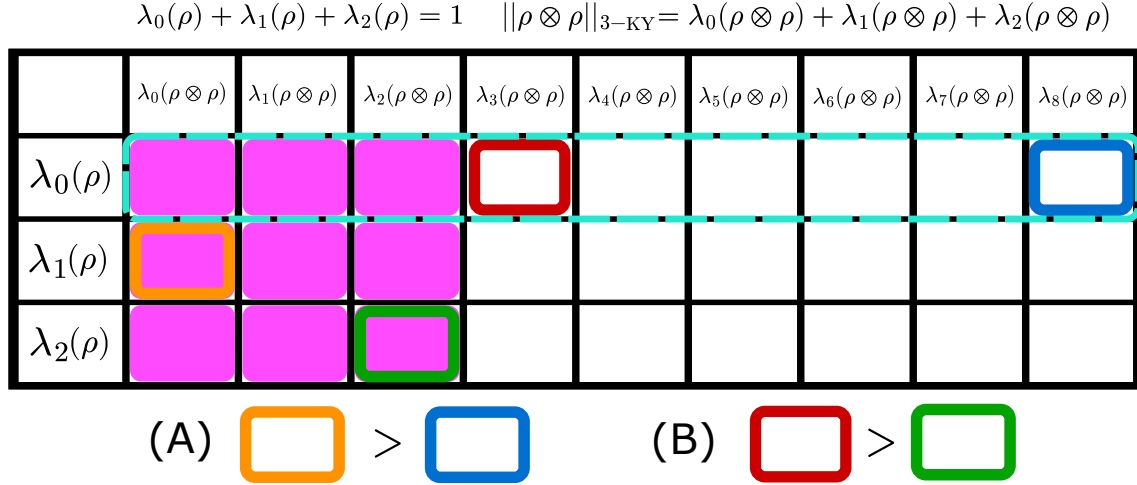


Figure 6.3: A graphical depiction of the eigenvalues of $\rho \otimes \rho \otimes \rho$, showing how lower bounds on the d^2 Ky-fan norm can be found, and highlighting the comparison made to perform the proofs in Lemma 6.2.10 and Result 6.2.9.

6.3.1.2 Modes

When considering symmetric dynamics, quantum states can be decomposed into a set of modes [98]. Each mode is a set of components of the state that transform independently due to the symmetry of the dynamics; if there exists an allowed operation $\mathcal{E}$ such that $\sigma = \mathcal{E}(\rho)$, then $\sigma^{(j)} = \mathcal{E}(\rho^{(j)})$, where $\rho^{(j)}$ ($\sigma^{(j)}$) is the j th mode of ρ (σ). Hence, under symmetric dynamics, the j th mode of the output state depends only on the j th mode of the input state. From this, it follows that if $\rho^{(j)} = 0$ then $\sigma^{(j)} = 0$, which can be generalised to $\text{modes}(\sigma) \subseteq \text{modes}(\rho)$, where $\text{modes}(\rho) := \{j \in \mathbb{Z} \mid \rho^{(j)} \neq 0\}$ is the set of all integers labelling modes for which ρ has a nonzero component. We note that since ρ is a state — meaning $\rho^\dagger = \rho$ — it holds that

$$[\rho^{(j)}]^\dagger = \rho^{(-j)} \quad \forall j. \quad (6.49)$$

One therefore only needs to consider the positive modes.

With respect to our chosen L , the j th mode of ρ is given by

$$\rho^{(j)} := \sum_{n=0}^{d-1} |n+j\rangle\langle n+j| \rho |n\rangle\langle n|, \quad (6.50)$$

which is an operator supported in the j th mode subspace $\mathcal{B}^j = \text{Span}\{|n+j\rangle\langle n| : \forall n\}$. By definition, we have $\rho = \sum_{j=-d+1}^{d-1} \rho^{(j)}$. It can be seen that $\rho = \rho^{(0)} \Leftrightarrow [\rho, L] = 0$, meaning ρ has no coherence with respect to the eigenspaces of L if and only if it is supported only in the 0th mode subspace.

The notion of modes can also be expanded to bipartite systems; the global modes of a bipartite system are given by the following combination of local modes [98]:

$$(\rho_A \otimes \rho_B)^{(j)} = \sum_k \rho_A^{(k)} \otimes \rho_B^{(j-k)}. \quad (6.51)$$

From this definition, it can be seen that, in this case, the j th mode subspace of the bipartite Hilbert space is given by

$$\mathcal{B}_{AB}^{(j)} = \text{Span}\{|n+k, m+j-k\rangle\langle n, m| : \forall n, m, k\}. \quad (6.52)$$

Using this, we now present the following Lemma.

Lemma 6.3.1. Let $\sigma_{AB} \in \mathcal{D}(\mathcal{H}_A \otimes \mathcal{H}_B)$ and $\sigma_A = \text{tr}_B[\sigma_{AB}]$, then

$$\sigma_A^{(j)} = \text{tr}_B[\sigma_{AB}^{(j)}], \quad (6.53)$$

meaning the j th local mode depends only on the j th global mode.

Proof. The j th mode of the bipartite state σ_{AB} is given by

$$\sigma_{AB}^{(j)} = \sum_k \sum_{n,m} q_{n+k, m+j-k, n, m} |n+k, m+j-k\rangle \langle n, m|, \quad (6.54)$$

which is some operator supported in $\mathcal{B}_{AB}^{(j)}$. Taking the partial trace over the B system of this operator gives

$$\text{tr}_B[\sigma_{AB}^{(j)}] = \sum_n \tilde{q}_{n+j, n} |n+j\rangle \langle n| \in \mathcal{B}_A^{(j)}, \quad (6.55)$$

where $\tilde{q}_{n+j, n} = \sum_m q_{n+k, m+j-k, n, m}$ and $\mathcal{B}_A^{(j)}$ is the j th mode subspace of the A system. From the definition of $\tilde{q}_{n+j, n}$, it can be seen that the matrix elements associated to each basis state are those one would expect from taking the partial trace of the whole system, $\tilde{q}_{n+j, n} = \langle n+j| \text{tr}_B[\sigma_{AB}] |n\rangle$. As the operator exists only in $\mathcal{B}_A^{(j)}$ and has the correct matrix elements, it is $\sigma_A^{(j)}$. $\square$

Given the dimension of the bipartite space is d^2 , as compared to the local dimension of d , there are more available modes in the bipartite space than there are locally. Tracing over a global mode with $j > (d-1)$ therefore equals zero, with them making no contribution to the local modes in either subsystem.

6.3.2 Quantifier

We are now in a position to introduce our resource quantifier such that the concentration problem can be defined. Here, we choose to consider concentrating coherence onto a particular mode of the local state; for a state $\rho \in \mathcal{D}(\mathcal{H})$, the amount of coherence in the j th mode can be quantified by

$$M^{(j)}(\rho) := \|\rho^{(j)}\|_1 = \sum_n |p_{n+j, n}|, \quad (6.56)$$

where $\rho^{(j)}$ is j th mode of ρ , and the equality holds only due to our L being non-degenerate. For all $j \neq 0$, this family of measures are monotonic under allowed operations, and faithful i.e., $M^{(j)}(\rho^{(j)}) \geq M^{(j)}(\sigma^{(j)}) \forall j \neq 0$ if $\exists \mathcal{E} \in \mathfrak{D}$ such that $\mathcal{E}(\rho) = \sigma$, and $M^{(j)}(\rho) = 0 \forall j \neq 0 \iff [\rho, L] = 0$. It is also faithful for a specific mode: $M^{(j)}(\rho) = 0 \iff \rho^{(j)} = 0$. See Section 4.3.2 for a proof of this.

We consider this *mode concentration* to demonstrate that an agents capacity to locally increase unspeakable concentration depends not only on the existence of coherence, but also on how it is distributed across the modes. If considering the j th mode, the aim is therefore to investigate the following figure of merit:

$$\Delta M_A^{(j)} := \max_{V_{AB}} M^{(j)}(\sigma_A) - M^{(j)}(\rho_A), \quad (6.57)$$

where $\sigma_A = \text{tr}_B[V_{AB}(\rho_A \otimes \rho_B)V_{AB}^\dagger]$, with the maximisation over all resource non-increasing unitaries i.e., $V_{AB} : [V_{AB}, L_{AB}] = 0$.

6.3.3 Limits on Concentration: Qubits

We first focus on performing coherence concentration using two qubits, where we can set the local generator to be $L = |1\rangle\langle 1|$. Each qubit has only two modes, with support on the 1st mode, $|1\rangle\langle 0|$, indicating the presence of coherence. This 1st mode then only has a single component—the single off-diagonal element of ρ . This is the sole parameter of a qubit state that indicates coherence with respect to the eigenspaces of L . Hence, any coherence measure on a qubit will depend only on this parameter. Therefore, whilst we use $\Delta M^{(j)}$, these results can be applied to *any* qubit coherence measure.

Any qubit state ρ can be completely characterised by $p_{00} := \langle 0|\rho|0\rangle$ and $p_{01} := \langle 0|\rho|1\rangle$, such that initially $M^{(1)}(\rho_A) = |p_{01}|$. From here, we can explicitly write down the analytical form for the optimal coherence concentration possible.

Result 6.3.2. For a state $\rho \in \mathcal{D}(\mathcal{H}^2)$, we have

$$\Delta M_A^{(1)} = |p_{01}| \left(\sqrt{1 + (2p_{00} - 1)^2} - 1 \right), \quad (6.58)$$

which is achieved by applying a unitary rotation of angle

$$\cos^{-1} \left(\frac{1}{\sqrt{1 + (2p_{00} - 1)^2}} \right) \quad (6.59)$$

in the $\{|01\rangle, |10\rangle\}$ degenerate subspace of L_{AB} .

Proof Sketch. In the two-qubit case, an allowed unitary applies an arbitrary qubit unitary in the $\{|01\rangle, |10\rangle\}$ degenerate subspace of L_{AB} . This unitary can be decomposed into two Z-rotations (with respect to $|01\rangle$ and $|10\rangle$), a Y-rotation, and some global phase. The Z-rotations and global phase can be shown to only decrease local coherence, leaving only the Y-rotation able to increase local coherence. One can apply this rotation to two copies of ρ , and then use a geometric argument to find the optimal rotation angle. See Section 6.6.1.1 for a complete proof. $\square$

Result 6.3.2 implies that if ρ has no initial coherence, i.e. $p_{01} = 0$, then no coherence can be concentrated, as expected. Interestingly, it can also be seen that if $p_{00} = 1/2$ (i.e., the diagonal of ρ is maximally mixed in the eigenbasis of L), then no coherence can be concentrated. Put alternatively, if ρ sits on the plane through the equator of the Bloch sphere defined by L , then no coherence can be concentrated. Hence, qubit states having $p_{00} = 1/2$ and $p_{01} > 0$ carry *bound coherence*—they do not have concentratable coherence, despite having non-zero and non-maximal initial coherence [3]. This is another example of a bound resource, as was introduced in Definition 6.2.1 and shown to exist for informational non-equilibrium in Section 6.2.3 and Section 6.2.6.1.

6.3.4 Qubit Concatenation

Result 6.3.2 fully solves the coherence concentration problem for two qubits. We now assume access to many copies of some qubit, but still aim to maximally increase the coherence of single qubit. Whilst this puts us outside of the resource concentration framework [3], we utilise Result 6.3.2 to develop a multi-qubit resource amplification protocol. Specifically, we consider applying sequential optimal qubit concentration unitaries, feeding the output of two optimal concentration processes into the input of the next (see Fig. 6.6 for an example of the protocol).

We equate such a concatenation protocol to a greedy optimisation algorithm: the locally optimal choice is always performed, potentially at the expense of reaching the global optimum. Performing a global operation on all available copies of ρ simultaneously will be at least as powerful, and likely more efficient in terms of the local resource increase per number of copies of ρ used. However, the approach presented here offers a fully constructive method for generating a multi-copy coherence amplification process, with all performed unitaries being effective qubit unitaries in the $\{|01\rangle, |10\rangle\}$ degenerate subspace of L_{AB} .

Assuming the input qubits are characterised by p_{00}^0 and p_{01}^0 , after applying m steps of this concatenation protocol, the output qubit state of the $(m + 1)$ th step is characterised by the following result.

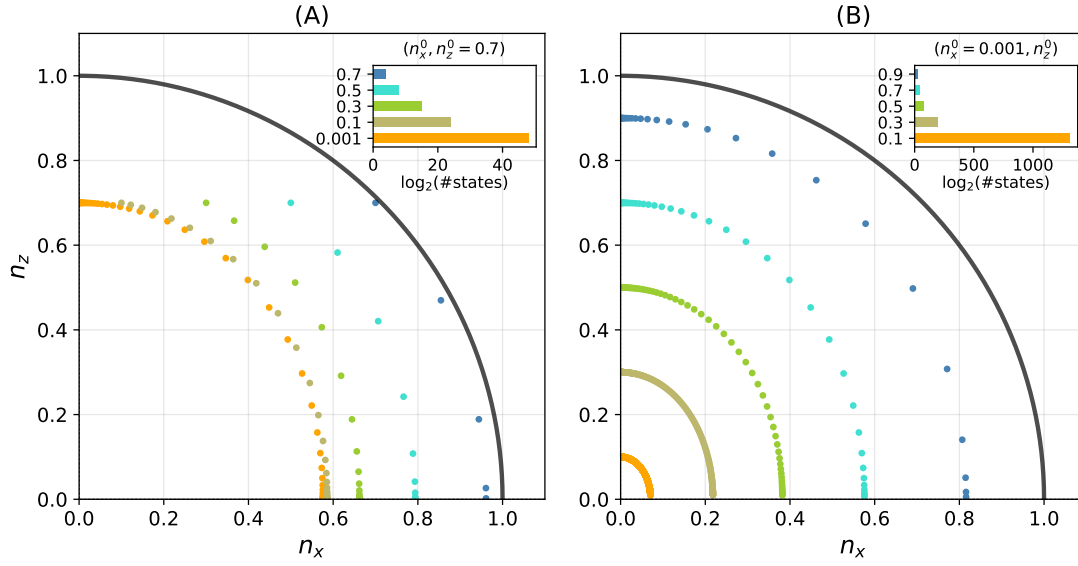


Figure 6.4: Qubit concatenation paths through the Bloch sphere. (A) Paths for $n_z^0 = 0.7$ and various n_x^0 . (B) Paths for $n_x^0 = 0.001$ and various n_z^0 . Inserts show the $\log_2(\# \text{ input states})$ —the number of steps—needed to converge to within 0.001 of $n_z = 0$, which dictates the maximum reachable coherence.

Result 6.3.3. *The output qubit state of the $(m + 1)$ th step of the qubit concatenation protocol will be*

$$\begin{aligned} p_{00}^{m+1} &= p_{00}^m - \frac{2(2p_{00}^m - 1)|p_{01}^m|^2}{1 + (2p_{00}^m - 1)^2}; \\ p_{01}^{m+1} &= |p_{01}^m| \sqrt{1 + (2p_{00}^m - 1)^2}, \end{aligned} \quad (6.60)$$

Proof Sketch. This is proved via direction calculation: applying the optimal concentrating unitary detailed in Result 6.3.2 to two copies of some arbitrary state allows functions detailing the output states two characterising parameters (p_{00} and p_{01}) to be found. These function take as input the characterising parameters of the input state. If one then considers performing concentration with two copies of this output state, they would get the same functions. However, they will now take as input the characterising parameters of the previous output state. One can then consider repeating this process arbitrarily many times, with the previous output characterising parameters always being used as the next input characterising parameters. This is the definition of a recurrence relation, leading to the above equations. See Section 6.6.2 for a complete proof. $\square$

Note, after the m th step, 2^m copies of ρ will have been consumed. Hence, each additional layer leads to an exponential increase in the number of initial states need.

The equations in Result 6.3.3 are coupled, non-linear recurrence relations. As a first step, we find the fixed points of these recurrence relations: values that remain unchanged when the recurrence is applied (i.e., $p_{00}^{m+1} = p_{00}^m$ and $p_{01}^{m+1} = p_{01}^m$). Physically, they represent qubit states for which no coherence can be concentrated, with $p_{00}^m = 1/2$ (part of the condition for bound coherence seen above) and $p_{01}^m = 0$ (no initial coherence) both found to be fixed points.

By setting $p_{00} = (1 + n_z)/2$ and $p_{01} = (n_x + in_y)/2$, where $\mathbf{n} = (n_x, n_y, n_z)$ is the Bloch vector, we plot the output states from each step of the concatenation on the Bloch sphere in Fig. 6.4. This is done for multiple different input states. As Z-rotations are locally allowed, we can set $n_y = 0$ without loss of generality, leaving $M^{(1)}(\rho) = |n_x|$ to completely characterise the coherence of ρ . Since the z-axis and x-axis are fixed points of the recurrence relations, as $p_{00} = 1/2 \rightarrow n_z = 0$ and $p_{01} = 0 \rightarrow n_x = 0$, we also only need to consider one quarter of the $n_y = 0$ plane of the Bloch sphere, as the same behaviour will be replicated in the other quarters.

It can be seen in Fig. 6.4 that even when given access to infinity many copies of ρ , the concatenation

protocol still cannot reach a maximally coherent state ($n_x = 1$). In the following result, the maximum increase in coherence that can be achieved as $m \rightarrow \infty$ is shown to depend on the purity of input state ρ .

Result 6.3.4. For an initial qubit state ρ , let $\sigma_{(m)}$ be the qubit state output from the m th step of the concatenation protocol. Then $M^{(1)}(\sigma_{(m)}) \leq \sqrt{2\text{tr}(\rho^2) - 1}$

Proof Sketch. It can be shown that under the optimal concentrating unitary, the purity of the state in the A subsystem is reduced. Hence, under each step of the concatenation protocol, the purity is reduced. Relating the Bloch vector to purity via its 2-norm and using the formula of $M^{(1)}(\rho)$ completes the proof. See Section 6.6.3 for a full proof. $\square$

In Fig. 6.5, we plot the vector field of the recurrence relations in Result 6.3.3. It can be noted from Fig. 6.5 that the change in n_z and n_x in a single step is highly dependent on the input state, with an n_x or n_z close to the fixed points leading to smaller changes. Interestingly, the physical reasoning behind why differs in different locations: when close to the n_z axis, such that n_x is close to zero, the small step size results from both input states into the concentration having low coherence, and hence there not being much coherence to relocate from system B into system A . When close to the n_x axis, however, the step sizes remain small, despite both input states having a (potentially) large amount of coherence. The small step size here instead arises from the fact that increasing coherence when the Bloch vector points predominantly along the x -axis is equivalent to increasing the purity of the state. Since it is impossible to increase the local purity given two qubits and unitary operations, see [3] or Corollary 6.2.2, each step size is the small change in coherence that can be managed whilst not increasing the purity.

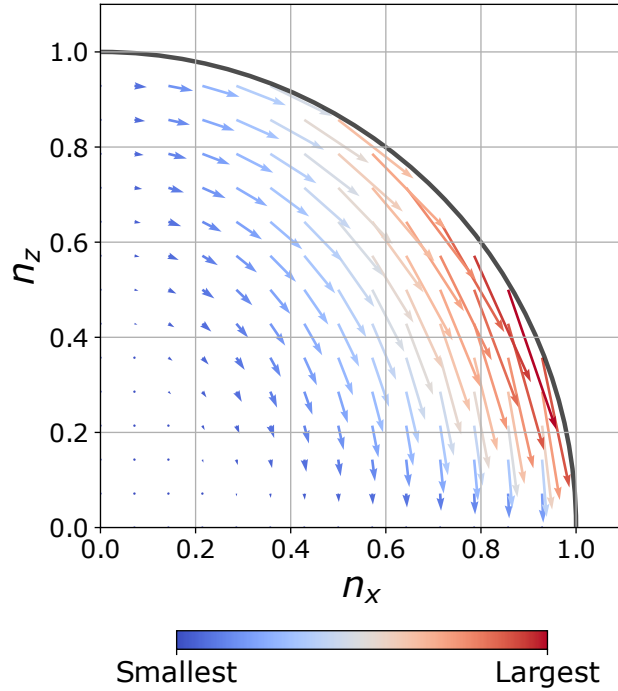


Figure 6.5: Vector field for the recurrence relations in Result 6.3.3. The length of the vector and its colour detail how much a given point will change under the recurrence relation. For example, a short blue arrow means a small change will occur, a long red arrow means a large change will occur.

The above analysis reveals an interesting feature of coherence amplification. From Eq. (6.60), and exemplified in Fig. 6.4 (B), even for an initial qubit state with an *arbitrarily small* amount of coherence, coherence can still be amplified via the concatenation protocol. Unlike in the asymptotic and catalytic case [85, 90], this is not arbitrary amplification, as the achievable coherence is bounded by Result 6.3.4. However, there does exist states for which the *ratio* of the final to initial coherence is *unbounded*.

Result 6.3.5. For any number of states $n = 2^N$ where $N \in \mathbb{N}$ and $\epsilon > 0$, there exists an initial state ρ (dependent on n, ϵ) such that

$$M^{(1)}(\rho) < \frac{1}{n} \ \& \ \frac{M^{(1)}(\sigma_{(m)})}{M^{(1)}(\rho)} > 2^{-\epsilon} \sqrt{n} \ \forall \ m \geq \log_2(n). \quad (6.61)$$

Proof Sketch. This can be proved by assuming an appropriate initial state, and then using the recurrence relations from Result 6.60 to calculate the output state from m layers of qubit concatenation. By finding an appropriate lower bound on the characterising parameters of the state after m layers, the result can be proved. See Section 6.6.4 for a full proof. $\square$

Therefore, if n is very large, then one has a large number of initial states with low initial coherence. Result 6.3.5 then states that $\log_2(n)$ number of steps of the concatenation protocol is sufficient to amplify the ratio of the coherence $n^{3/2}$ times. As n can be arbitrarily large, the ratio of initial to final coherence can be unbounded when given access to arbitrarily many copies of the initial state. This finding is complementary to those reported in Ref. [85, 90], where arbitrary amplifications of unspeakable coherence with unbounded rates are demonstrated in the asymptotic and catalytic transformation. Here, we show that, rather surprisingly, in a very limited setting where only single-qubit states and effective qubit unitaries are used, an unbounded input-output ratio can already be found.

6.3.4.1 Qubit Concatenation Example

In this section, we give an example of a two-layer qubit concatenation protocol given in Fig. 6.6 to cement the ideas introduced above. It can be seen that the protocol takes four copies of the state ρ as input. A

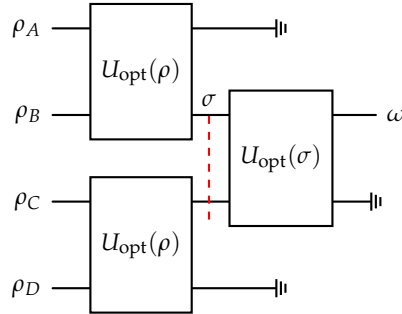


Figure 6.6: A quantum circuit diagram of the qubit concatenation protocol for four input states.

two qubit unitary, denoted by $U_{\text{opt}}(\rho)$, is applied to the first and second copy, in spaces A and B , and the third and fourth copies in spaces C and D . Here, $U_{\text{opt}}(\rho)$ is the optimal concentrating unitary for ρ ; i.e., if $p_{00}^0 = \langle 0|\rho|0\rangle$ and $p_{01}^0 = \langle 0|\rho|1\rangle$, then

$$U_{\text{opt}}(\rho) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos \theta_{\text{opt}} & -\sin \theta_{\text{opt}} & 0 \\ 0 & \sin \theta_{\text{opt}} & \cos \theta_{\text{opt}} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad (6.62)$$

where $\theta_{\text{opt}} = \cos^{-1} \left(1 / \sqrt{1 + (2p_{00}^{(0)} - 1)^2} \right)$. Let $\sigma_B = \text{tr}_A [U_{\text{opt}}(\rho)(\rho_A \otimes \rho_B)U_{\text{opt}}(\rho)^\dagger]$ be the state output into space B from the 1st concentration step, which is also the state output into space C . Let $p_{00}^1 = \langle 0|\sigma|0\rangle$ and $p_{01}^1 = \langle 0|\sigma|1\rangle$, which can be explicitly calculated via the recurrence relations in Result 6.3.3. From here, the optimal unitary for performing coherence concentration given two copies of σ , denoted by $U_{\text{opt}}(\sigma)$, can be found and is of the same form as above. Applying $U_{\text{opt}}(\sigma)$ to the states in wires B and C will output an even more coherent state $\omega_B = \text{tr}_A [U_{\text{opt}}(\sigma)(\sigma_B \otimes \sigma_C)U_{\text{opt}}(\sigma)^\dagger]$. One can then repeat this process for an arbitrary number of layers.

6.3.5 Limits on Concentration: Beyond Qubits

We now investigate the general concentration case, considering states with arbitrary finite local dimension.

Firstly, it was clear in the qubit case that if ρ is a free state, meaning it has support only on the 0th mode, then no coherence concentration is possible — as expected. It is here argued that this also holds in the d -dimensional case, as it is expected too. If $\rho = \rho^{(0)}$, then $\rho_{AB} := \rho_A \otimes \rho_B = \rho_{AB}^{(0)}$. Hence, $\sigma_{AB} = \mathcal{E}(\rho_{AB}) = \sigma_{AB}^{(0)}$ for all allowed operations $\mathcal{E} \in \mathfrak{D}$. Then, using Lemma 6.3.1, it can be seen that $\sigma_A := \text{tr}_B[\sigma_{AB}] = \sigma_A^{(0)}$. Therefore, there exists no allowed operation, not just no allowed unitary, such that σ_A has coherence (support on a non-zero mode) if ρ is a free state.

Now, in general, when aiming to concentrate coherence on the j th local mode, given that modes transform independently under allowed operations, in conjunction with Lemma 6.3.1, one only needs to consider the j th global mode. However, not all components of the global mode will contribute to the local mode. Specifically, one can separate $\mathcal{B}_{AB}^{(j)}$ — the j th mode subspace — into two subspaces,

$$\begin{aligned}\mathcal{V}_{\text{in}}^{(j)} &:= \text{span}\{|n+j, m\rangle\langle n, m|\}_{n,m} \\ \mathcal{V}_{\text{out}}^{(j)} &:= \text{span}\{|n+k, m+j-k\rangle\langle n, m|\}_{n,m,k \neq j},\end{aligned}\tag{6.63}$$

where $\mathcal{B}_{AB}^{(j)} = \mathcal{V}_{\text{in}}^{(j)} \cup \mathcal{V}_{\text{out}}^{(j)}$ and $\mathcal{V}_{\text{in}}^{(j)} \cap \mathcal{V}_{\text{out}}^{(j)} = \emptyset$, such that the basis operators in $\mathcal{V}_{\text{in}}^{(j)}$ contribute to the local mode, whereas those in $\mathcal{V}_{\text{out}}^{(j)}$ do not. The aim of mode concentration can therefore be loosely summarised as maximising the coefficients of the basis operators in $\mathcal{V}_{\text{in}}^{(j)}$ under allowed unitaries. It is noted that the projector onto $\mathcal{V}_{\text{in}}^{(j)}$ reads $\Pi_A^{(j)} \otimes \Pi_B^{(0)}$, where $\Pi_X^{(k)}$ is the projector onto the j th local mode ($X = A, B$). Using this, we find the following upper bound on the ability to concentrate coherence in the j th local mode.

Result 6.3.6. Let $d^{(j)} := \dim(\mathcal{V}_{\text{in}}^{(j)})$. For any state ρ , we have

$$\Delta M_A^{(j)} \leq \|(\rho_A \otimes \rho_B)^{(j)}\|_{d^{(j)}-\text{KF}} - \|\rho^{(j)}\|_1.\tag{6.64}$$

Proof Sketch. The figure of merit $M_A^{(j)}(\sigma_A)$ can be seen to be upper bounded by the sum of the absolute values of the matrix elements of $\sigma_{AB}^{(j)}$ associated to the basis operators in $\mathcal{V}_{\text{in}}^{(j)}$. All these matrix elements sit along a diagonal of length $\tilde{d}$. It can separately be shown that for any operator, the sum of the absolute values of the matrix elements along any diagonal of length k is upper-bounded by the k th Ky-Fan norm of the operator. Using the unitary invariance of the Ky-Fan norm and taking away $M^{(j)}(\rho_A)$ completes the proof. See Section 6.6.5 for a full proof. $\square$

Here, $\|P\|_{\alpha\text{-KF}}$ is the α -Ky-Fan norm [140], which is the sum of the α largest singular values of the operator P . Hence, as for informational non-equilibrium concentration [3], the Ky-Fan norm emerges as a tool for quantifying how much of the global resource content of a state can be relocated into a subsystem (See Result 6.2.1).

We now present an alternative, and more technical, bound on coherence concentration. We achieve this by noting that a given global (bipartite) mode can be separated further into independently transforming components under allowed unitaries. Within the j th mode, we introduce the $(\lambda_c + j, \lambda_c)$ left-right-degenerate (LR-D) operator of the mode as $\Pi_{c+j}(\rho \otimes \rho)^{(j)}\Pi_c$, where Π_c is the projection onto the eigenspace of L_{AB} with eigenvalue λ_c (i.e., $L_{AB} = \sum_c \lambda_c \Pi_c$ with spectrum $\{\lambda_c\}_c$). Under any bipartite allowed unitary V_{AB} , the $(\lambda_c + j, \lambda_c)$ LR-D operator of the j th mode then evolves independently as $V_{c+j}(\Pi_{c+j}(\rho \otimes \rho)^{(j)}\Pi_c)V_c$, where V_c is the unitary that acts within the c th degenerate subspace of L_{AB} . One can then perform a similar analysis to Result 6.3.6 and define the $(\lambda_c + j, \lambda_c)$ th LR-D subspace, $\mathcal{B}_{AB}^{(j,c)}$, and

$$\begin{aligned}\mathcal{V}_{\text{in}}^{(j,c)} &:= \text{span}\{|n+j, m\rangle\langle n, m|\}_{n+m=c} \\ \mathcal{V}_{\text{out}}^{(j,c)} &:= \text{span}\{|n+k, m+j-k\rangle\langle n, m|\}_{n+m=c, k \neq j}\end{aligned}\tag{6.65}$$

with $\mathcal{B}_{AB}^{(j,c)} = \mathcal{V}_{\text{in}}^{(j,c)} \cup \mathcal{V}_{\text{out}}^{(j,c)}$, $\mathcal{V}_{\text{in}}^{(j,c)} \cap \mathcal{V}_{\text{out}}^{(j,c)} = \emptyset$ and $\mathcal{B}_{AB}^{(j,c)} \subset \mathcal{B}_{AB}^{(j)}$. The aim of mode concentration is again to maximise the coefficients associated with the basis operators in $\mathcal{V}_{\text{in}}^{(j,c)}$. Applying this approach leads to the following upper bound:

Result 6.3.7. Let $d^{(j,c)} := \dim(\mathcal{V}_{\text{in}}^{(j,c)})$. For any state ρ , we have

$$\Delta M_A^{(j)} \leq \sum_{c=0}^{2d-2-j} \|\Pi_{c+j}(\rho \otimes \rho)^{(j)} \Pi_c\|_{d^{(j,c)}-\text{KF}} - \|\rho^{(j)}\|_1. \quad (6.66)$$

Proof Sketch. The proof technique is the same as for Result 6.3.6, but now applied individually to each LR-D subspace. See Supplementary Material 6.6.6 for proof. $\square$

Interestingly, it can be shown that the bounds from Results 6.3.6 and 6.3.7 are incomparable — each provides a tighter upper bound in a variety of cases considered. For $d = 3$ and $d = 4$, the bounds are calculated for a sample of random states varying in rank (see Fig. 6.7). It is firstly noted that for $d = 3$, both bounds are equal for all tested pure (i.e., rank-one) states. For states of higher ranks in $d = 3$, and states of all rank in $d = 4$, we see that there is not a clear comparison of the two upper bounds. This can be confirmed in the insert, which shows the number of sampled states for which one bound is greater than the other. In general, it can be seen that Result 6.3.7 performs better for $d = 3$, and Result 6.3.6 performs better for $d = 4$. We conjecture that these bounds will not coincide for all d , and that in general one should just take whichever of the two bounds is smallest to be an upper bound on the concentrateable coherence.

6.3.5.1 Qutrit Example

Here we will run through an example of mode concentration using two qutrits. Graphical depictions will be used where appropriate to make our explanations as clear as possible.

Let $\rho \in \mathcal{D}(\mathcal{H}^3)$ be an arbitrary qutrit, and let

$$L = \sum_{n=0}^2 n |n\rangle\langle n| = 0 |0\rangle\langle 0| + 1 |1\rangle\langle 1| + 2 |2\rangle\langle 2|, \quad (6.67)$$

be the Hermitian operator defining the eigenspaces between which we aim to increase coherence. With respect to the eigenbasis of L , ρ can be decomposed as

$$\rho = \begin{bmatrix} p_{00} & p_{01} & p_{02} \\ p_{10} & p_{11} & p_{12} \\ p_{20} & p_{21} & p_{22} \end{bmatrix}. \quad (6.68)$$

Our aim will be to concentrate coherence onto the 2nd local mode. To do so, one need only consider applying an allowed unitary to the 2nd global mode and then performing the partial trace. The 2nd global mode is given by

$$(\rho_A \otimes \rho_B)^{(2)} = \rho_A^{(0)} \otimes \rho_B^{(2)} + \rho_A^{(1)} \otimes \rho_B^{(1)} + \rho_A^{(2)} \otimes \rho_B^{(0)}, \quad (6.69)$$

which, with respect to the eigenbasis of $L_{AB} = L_A \otimes \mathbb{I}_B + \mathbb{I} \otimes L_B$, is given by

$$(\rho_A \otimes \rho_B)^{(2)} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ p_{00}p_{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ p_{10}p_{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & p_{10}p_{21} & 0 & p_{11}p_{20} & 0 & 0 & 0 & 0 & 0 \\ p_{20}p_{00} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & p_{20}p_{11} & 0 & p_{21}p_{10} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & p_{20}p_{22} & 0 & p_{21}p_{21} & 0 & p_{22}p_{20} & 0 & 0 \end{bmatrix} \quad (6.70)$$

Space	Colour
$\mathcal{V}_{\text{in}}^{(2)}$	●
$\mathcal{V}_{\text{out}}^{(2)}$	●

where we have highlighted the coefficients associated with basis operators in $\mathcal{V}_{\text{in}}^{(2)}$, (pink dots) i.e., the basis operators whose coefficients contribute to the local mode. In this case, the 2nd local mode has a single component, given by the sum of the coefficients associated with the basis operators in $\mathcal{V}_{\text{in}}^{(2)}$. Hence, $M^{(2)}(\sigma_A)$ is the absolute value of the sum of these components. Optimally concentrating coherence on the 2nd mode therefore means maximising the absolute value of the sum of the coefficients associated with the basis operators in $\mathcal{V}_{\text{in}}^{(2)}$. An upper-bound on $M^{(2)}(\sigma_A)$ is then given by the sum of the maximised absolute values of each coefficient, due to the triangle inequality.

Example for Result 6.3.6

We now detail the logical steps that led to Result 6.3.6 in the main text. Firstly, we relax the problem by assuming that any pre-processing and post-processing unitary can be applied to $(\rho_A \otimes \rho_B)^{(2)}$, not just an allowed unitary. From here, Lemma 6.6.2 can be employed to show that the sum over the absolute value of the matrix coefficients that give the 2nd local mode is upper-bounded by the 3rd Ky-Fan norm of this operator. This is because we are summing over a diagonal of length $\dim(\mathcal{V}_{\text{in}}^{(2)}) = 3$. As $M^{(2)}(\sigma_A)$ is upper-bounded by the sum of the absolute values of these matrix coefficients, which is further upper-bounded by the 3rd Ky-Fan norm of this operator, one can therefore conclude that $M^{(2)}(\sigma_A) \leq \|(\rho_A \otimes \rho_B)^{(2)}\|_{3\text{-KF}}$.

Qutrit Example for Result 6.3.7

We now consider in detail splitting $(\rho_A \otimes \rho_B)^{(2)}$ up into the individual LR-D subspaces that evolve independently under allowed unitaries, and hence go through the logical steps that led to Result 6.3.7. Firstly, any allowed unitary on the bipartite system can be decomposed as $V_{AB} = \sum_c V_c \Pi_c$, where each V_k is an independent unitary acting within a degenerate subspace of

$$\begin{aligned} L_{AB} &= L_A \otimes \mathbb{I}_B + \mathbb{I}_A \otimes L_B \\ &= \sum_c \lambda_c \Pi_c, \end{aligned} \quad (6.71)$$

where $\{\lambda_c\}_c$ is the spectrum of L_{AB} . Any allowed unitary can therefore be decomposed with respect to

the eigenbasis of L_{AB} as

$$V_{AB} = \begin{bmatrix} \text{yellow} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \text{magenta} & 0 & \text{magenta} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \text{orange} & 0 & \text{orange} & 0 & \text{orange} & 0 & 0 \\ 0 & \text{magenta} & 0 & \text{magenta} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \text{orange} & 0 & \text{orange} & 0 & \text{orange} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \text{cyan} & 0 & \text{cyan} & 0 \\ 0 & 0 & \text{orange} & 0 & \text{orange} & 0 & \text{orange} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \text{cyan} & 0 & \text{cyan} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \text{purple} \end{bmatrix} \quad \begin{array}{c|c} \text{Unitary} & \text{Colour} \\ \hline V_0 & \text{yellow} \\ V_1 & \text{magenta} \\ V_2 & \text{orange} \\ V_3 & \text{cyan} \\ V_4 & \text{purple} \end{array} \quad (6.72)$$

where each matrix element associated with a given independent unitary is represented by a different colour. This unitary is then applied to $(\rho_A \otimes \rho_B)^{(2)}$. We now represent $(\rho_A \otimes \rho_B)^{(2)}$ while highlighting all the different LR-D subspaces within the mode,

$$(\rho_A \otimes \rho_B)^{(2)} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ p_{00}p_{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ p_{10}p_{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & p_{10}p_{21} & 0 & p_{11}p_{20} & 0 & 0 & 0 & 0 & 0 \\ p_{20}p_{00} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & p_{20}p_{11} & 0 & p_{21}p_{10} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & p_{20}p_{22} & 0 & p_{21}p_{21} & 0 & p_{22}p_{20} & 0 & 0 \end{bmatrix} \quad \begin{array}{c|c} \text{LR-D Space} & \text{Colour} \\ \hline (2,0) & \text{purple} \\ (3,1) & \text{cyan} \\ (4,2) & \text{orange} \end{array} \quad (6.73)$$

where the $(c+j, c)$ LR-D subspace of the j th mode is defined as $\Pi_{c+j}(\rho_A \otimes \rho_B)^{(j)}\Pi_c$. The benefit of this formulation is that each LR-D subspace evolves independently and unitarily when a global allowed unitary is applied to the mode. More specifically, each of the LR-D subspaces evolve as

$$V_2\Pi_2(\rho_A \otimes \rho_B)^{(2)}\Pi_0V_0 = \begin{bmatrix} \text{orange} & \text{orange} & \text{orange} \\ \text{orange} & \text{orange} & \text{orange} \\ \text{orange} & \text{orange} & \text{orange} \end{bmatrix} \begin{bmatrix} p_{00}p_{20} \\ p_{10}p_{10} \\ p_{20}p_{00} \end{bmatrix} \begin{bmatrix} \text{yellow} \end{bmatrix} \quad (6.74)$$

$$V_3\Pi_2(\rho_A \otimes \rho_B)^{(2)}\Pi_0V_1 = \begin{bmatrix} \text{cyan} & \text{cyan} \\ \text{cyan} & \text{cyan} \end{bmatrix} \begin{bmatrix} p_{10}p_{21} & p_{11}p_{20} \\ p_{20}p_{11} & p_{21}p_{10} \end{bmatrix} \begin{bmatrix} \text{magenta} & \text{magenta} \\ \text{magenta} & \text{magenta} \end{bmatrix} \quad (6.75)$$

$$V_4\Pi_2(\rho_A \otimes \rho_B)^{(2)}\Pi_0V_2 = \begin{bmatrix} \text{purple} \end{bmatrix} \begin{bmatrix} p_{20}p_{22}, & p_{21}p_{21}, & p_{22}p_{20} \end{bmatrix} \begin{bmatrix} \text{orange} & \text{orange} & \text{orange} \\ \text{orange} & \text{orange} & \text{orange} \\ \text{orange} & \text{orange} & \text{orange} \end{bmatrix} \quad (6.76)$$

In the above, we have highlighted what coefficients within each LR-D subspace belong to basis states in $\mathcal{V}_{\text{in}}^{(2)}$, again using a pink dot. The aim when performing mode concentration is still to maximise the absolute value of the sum of the coefficients associated with these basis operators. What this framework provides is a finer level of detail over how these coefficients can evolve.

To find the bound given in Result 6.3.7, one applies the same methodology as for Result 6.3.6, but this time to each LR-D subspace individually. It can be seen, as above, that an upper bound on $M^{(2)}(\sigma_A)$ is given by the sum of the absolute values of the matrix coefficients associated with basis operators in $\mathcal{V}_{\text{in}}^{(2)}$. Using Lemma 6.6.2, this can be upper-bounded by sum over the k th Ky-Fan norm of all the $(c+j, c)$ -LR-D operators, where k is the number of basis states in $\mathcal{V}_{\text{in}}^{(2)}$ upon which a given $(c+j, c)$ -LR-D operator has support.

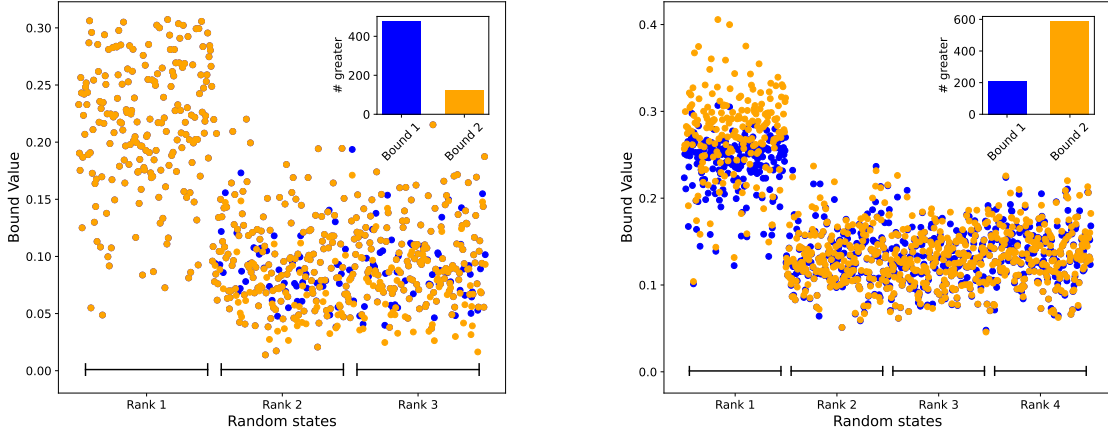


Figure 6.7: The value of Result 6.3.6 (bound 1) and Result 6.3.7 (bound 2) plotted for a sample of states in $d = 3$ (Left) and $d = 4$ (Right) for all possible ranks. The insert shows the number of sampled states for which a given bound is greater than the other.

6.3.6 Initial Correlations

In this final section, we expand the concentration problem beyond two initially uncorrelated copies, and ask when initial correlations can aid in local coherence enhancement, and when they cannot. In AB with equal local dimension d , consider a state $\rho_{AB} = \rho_{AB}^{(0)} + \sum_{i \geq d} \rho_{AB}^{(i)}$ such that there exists at least one $k \geq d$ where $\rho_{AB}^{(k)} \neq 0$ and $\rho_{AB}^{(k)} = 0 \forall k \in [1, d-1]$. We now show that such a mode distribution implies correlations exist between A and B .

Lemma 6.3.8. *Let $\rho_{AB} \in \mathcal{D}(\mathcal{H}_A \otimes \mathcal{H}_B)$ be a bipartite state in AB with equal local dimension $d < \infty$. Then ρ_{AB} is not a product state, and hence has correlation, if*

$$\text{modes}(\rho_{AB}) \neq \{0\} \text{ but } [1, d-1] \cap \text{modes}(\rho_{AB}) = \emptyset. \quad (6.77)$$

Namely, ρ_{AB} satisfies

$$\rho_{AB} = \rho_{AB}^{(0)} + \sum_{i \geq d} \rho_{AB}^{(i)} \quad \& \quad \sum_{i \geq d} \rho_{AB}^{(i)} \neq 0. \quad (6.78)$$

Proof. Suppose the opposite, that $\rho_{AB} = \rho_A \otimes \rho_B$ and

$$\rho_A \otimes \rho_B = \rho_{AB}^{(0)} + \sum_{i \geq d} \rho_{AB}^{(i)} \quad \& \quad \sum_{i \geq d} \rho_{AB}^{(i)} \neq 0. \quad (6.79)$$

Due to Lemma 6.3.1, it can then be concluded that $\rho_A = \text{tr}_B(\rho_{AB}^{(0)}) = \rho_A^{(0)}$ and $\rho_B = \text{tr}_A(\rho_{AB}^{(0)}) = \rho_B^{(0)}$, meaning that both ρ_A and ρ_B would need to be states supported only on the 0th mode. However, using

the fact that the bipartite modes of a tensor product of states are given by [98]:

$$(\rho_A \otimes \rho_B)^{(j)} = \sum_k \rho_A^{(k)} \otimes \rho_B^{(j-k)} \quad \forall j, \quad (6.80)$$

it can be concluded that $\rho_{AB} = \rho_A^{(0)} \otimes \rho_B^{(0)} = \rho_{AB}^{(0)}$, meaning that $\sum_{i \geq d} \rho_{AB}^{(i)} = 0$, in contradiction with Eq. (6.79). $\square$

Now, applying any allowed operation to such a state will map it to an output state supported on the same modes. Then, using Lemma 6.3.1, it can be concluded that both local states are supported only on the 0th mode (as the partial trace over modes with $k \geq d$ equals zero, making no contribution to the reduced states). Hence, both local states are always symmetric, regardless of the applied allowed operation. Therefore, despite the global state having coherence between the eigenspaces of L_{AB} , there exists no allowed operation—not just no allowed unitary—that can increase the local coherence. Furthermore, this argument is *independent* of the choice of coherence measure. We thus summarise this as the following no-go result.

Result 6.3.9. *If $[1, d-1] \cap \text{modes}(\rho_{AB}) = \emptyset$, then there exists no allowed operation to concentrate coherence with respect to any coherence measure.*

As an explicit example, consider the two-qubit isotropic state,

$$\rho_{AB}^{\text{Iso}} = p |\Phi^+\rangle\langle\Phi^+|_{AB} + (1-p)\mathbb{I}/2, \quad (6.81)$$

where $0 \leq p \leq 1$ and $|\Phi^+\rangle_{AB} = (|00\rangle + |11\rangle)_{AB}/\sqrt{2}$. As this state has support only on the 0th and 2nd bipartite modes, no coherence can be locally concentrated for all $p \geq 0$, despite there existing correlation between A and B for all $p > 0$. This also means global entanglement — even maximal entanglement — may not always be exchangeable for local coherence, despite the ability to perform global allowed operations.

6.4 Conclusions

6.4.1 Informational Non-equilibrium

By applying the framework of resource concentration to the resource theory of informational non-equilibrium, we have proved the feasibility of the resource concentration framework, demonstrating that there is plenty of interesting physics to be explored even within this minimal case of resource enhancement. For example, thermodynamically, solving the problem of informational non-equilibrium concentration is calculating the highest locally extractable work given access to $\rho \otimes \rho$ and joint unitary operations. By performing optimal unitaries, a local system is maximally driven from equilibrium, hence becoming maximally thermodynamically resourceful, and one can then extract work from it via, e.g., Szilard engine [141]. An obvious and important follow-up to this work is to apply the resource concentration framework to the more general thermodynamic setting with non-degenerate Hamiltonians, leading to the concentration of athermality [17, 19].

6.4.2 Unspeakable Coherence

A complete characterisation of unspeakable coherence concentration was found in the qubit case, and upper bounds were given beyond qubits. From this, we detailed a completely constructive, multi-qubit coherence amplification protocol that uses only effective qubit unitaries. It was shown to be able to unboundedly amplify the ratio of the input-output coherence for certain initial states—a finding that is complementary to the recently reported unbounded coherence amplification in the asymptotic and catalytic regime [85]. It would be instructive to quantitatively understand how far from the global optimal this protocol is, as it would inform whether the increased practicality of the concatenation protocol is worth the reduction in coherence concentrated. In addition, it would be interesting to introduce finitely

many incoherent states into the concentration protocol, and assess whether more coherence enhancement can then be achieved given access to these incoherent states.

Finally, a family of states with bound-coherence was shown to exist, i.e., states that are not maximally coherent, nor incoherent, but for which no allowed unitary exists to perform coherence concentration. We leave it open to determine if such states exist in higher dimensions.

6.4.3 General

Through this work, the viability of the resource concentration framework has been demonstrated. For both resource theories considered, it was shown that this minimal possible case of resource enhancement could be well understood, and in some cases fully solved. Exact dynamics for achieving these optimal increases in local resource content were also found, giving an understanding of how these concentration protocols could be physically implement in the lab.

Moreover, in both examples, an understanding of the resource concentration scenario was shown to provide a useful bottom up approach for understanding more general resource enhancement scenarios. For example, for both informational non-equilibrium and unspeakable coherence, the local resource enhancing power of initial correlations and entanglement were investigated using knowledge gained from the concentration scenario. Moreover, for informational non-equilibrium, an understanding of the concentration scenario was used to consider the case of resource enhancement given $n + 1$ initial copies of the state and the ability to perform global operations on all copies simultaneously. For unspeakable coherence, it was used to develop a fully constructive coherence amplification protocol, where it was assumed that one had access to an infinite number of states but were only able to act on two copies simultaneously. These initial examples demonstrate the power of understanding the minimal case for developing an understanding of the general case. It would now be interesting to generalise further by introducing a finite number of free states, to see whether this improves one's ability to enhance the local resource content. If so, access to free states could be considered a resource for the task of local resource enhancement.

6.5 Additional Material - Informational Non-equilibrium Concentration

6.5.1 Proof of Result 6.2.1

Result 6.2.1 For $\rho \in \mathcal{D}(\mathcal{H}^d)$,

$$\Delta\mathcal{P} = \log_2 \left(\frac{\|\rho \otimes \rho\|_{d-\text{KY}}}{\|\rho\|_\infty} \right), \quad (6.82)$$

where $\|\cdot\|_{d-\text{KY}}$ is the d th Ky-Fan norm, which is the sum of the d largest singular values.

Proof. Using Eq. (6.6), it can be seen that to find $\Delta\mathcal{P}(\rho)$ one must find

$$\begin{aligned} \|\sigma_A\|_\infty &= \max_{V_{AB}} \left\| \text{tr}_B [V_{AB}(\rho_A \otimes \rho_B) V_{AB}^\dagger] \right\|_\infty \\ &= \max_{V_{AB}, |\phi\rangle_A} \text{tr} \left[|\phi\rangle\langle\phi|_A (\text{tr}_B [V_{AB}(\rho_A \otimes \rho_B) V_{AB}^\dagger]) \right] \\ &= \max_{V_{AB}, |\phi\rangle_A} \text{tr} [V_{AB}^\dagger (|\phi\rangle\langle\phi|_A \otimes \mathbb{I}_B) V_{AB} (\rho_A \otimes \rho_B)] \\ &\leq \max_{\Pi_{AB}^d} \text{tr} [\Pi_{AB}^d (\rho_A \otimes \rho_B)] \end{aligned} \quad (6.83)$$

where in the final line we have used the fact that $V_{AB}^\dagger (|\phi\rangle\langle\phi|_A \otimes \mathbb{I}_B) V_{AB}$ is a rank- d projector, resulting in the final inequality.

A rank- d projector is a linear operator mapping the total Hilbert space onto some d -dimensional subspace (which also squares to itself). One can find a set of d -orthonormal basis kets, $\{|\kappa_n\rangle\}_{n=1}^d$, that Span this subspace, such that

$$\Pi_{AB}^d = \sum_{n=1}^d |\kappa_n\rangle\langle\kappa_n|_{AB}. \quad (6.84)$$

One can then note that there exists a unitary, $\tilde{V}_{AB}^\dagger$, that performs the mapping

$$\tilde{V}_{AB}^\dagger |0\rangle_A \otimes |n\rangle = |\kappa_n\rangle, \quad (6.85)$$

and leaves all other basis states unchanged. Therefore,

$$\tilde{V}_{AB}^\dagger (|0\rangle\langle 0|_A \otimes \mathbb{I}) \tilde{V}_{AB} = \Pi_{AB}^d, \quad (6.86)$$

and the inequality in the final line of Eq. (6.83) becomes an equality, as there always exists a unitary to create the maximum projection. Therefore, we are left to find

$$\|\sigma_A\|_\infty = \max_{\Pi_{AB}^d} \text{tr} [\Pi_{AB}^d (\rho_A \otimes \rho_B)]. \quad (6.87)$$

Now, let $\rho = \sum_{i=0}^{d-1} \lambda_i^\downarrow |i\rangle\langle i|$, where $\lambda_i^\downarrow$ are the eigenvalues of ρ ordered in non-increasing order. The tensor product of two copies of ρ is then

$$\rho_A \otimes \rho_B = \sum_{i,j=0}^{d-1} \lambda_i^\downarrow \lambda_j^\downarrow |ij\rangle\langle ij|, \quad (6.88)$$

such that

$$\|\sigma_A\|_\infty = \max_{\Pi_{AB}^d} \sum_{i,j=0}^{d-1} \lambda_i^\downarrow \lambda_j^\downarrow \langle ij | \Pi_{AB}^d | ij \rangle. \quad (6.89)$$

This is maximised when Π_{AB}^d is a projection onto the d -dimensional subspace spanned by the d -

eigenvectors of $\rho_A \otimes \rho_B$ with the d -largest associated eigenvalues. To see this, note that $0 \leq \langle ij | \Pi_{AB}^d | ij \rangle \leq 1$ and $\sum_{ij} \langle ij | \Pi_{AB}^d | ij \rangle = d$. To maximise the summation, one wants $\langle ij | \Pi_{AB}^d | ij \rangle = 1$ for the d largest eigenvalues, such that $\langle ij | \Pi_{AB}^d | ij \rangle = 0$ for all others.

The sum over the d -largest eigenvalues (singular-values) is then exactly the definition of d th Ky-Fan norm, such that

$$\|\sigma_A\|_\infty = \|\rho \otimes \rho\|_{d-KY}. \quad (6.90)$$

Hence,

$$\begin{aligned} \Delta \mathcal{P} &= \log_2 (d \|\rho \otimes \rho\|_{d-KY}) - \log_2 (d \|\rho\|_\infty) \\ &= \log_2 \left(\frac{\|\rho \otimes \rho\|_{d-KY}}{\|\rho\|_\infty} \right), \end{aligned} \quad (6.91)$$

completing the proof. $\square$

We note that nothing in the proof technique of the above result required that $\rho_A = \rho_B$, nor that the state in the B system even be the same dimension as the state in the A system. One can therefore be more general here, and state that

$$\begin{aligned} \|\sigma_A\|_\infty &= \max_{V_{AB}} \left\| \text{tr}_B [V_{AB} (\rho_A \otimes \omega_B) V_{AB}^\dagger] \right\|_\infty \\ &= \|\rho \otimes \omega\|_{d_B-KY}, \end{aligned} \quad (6.92)$$

where $d_B = \dim(\omega)$.

6.5.2 Proof of Result 6.2.5

Result 6.2.5 *A state has bound resource if and only if it is of the form*

$$\rho(p, |\psi\rangle) = p |\psi\rangle\langle\psi| + \frac{1-p}{d} \mathbb{I} \in \mathcal{D}(\mathcal{H}^d), \quad (6.93)$$

for some $0 < p < 1$ and $|\psi\rangle \in \mathcal{H}^d$. Such states are exactly those that are $(d-1)$ -fold degeneracy in their second largest eigenvalue, where the second largest eigenvalue is non-zero.

Proof. A state ρ has bound resource if and only if $\|\rho \otimes \rho\|_{d-KY} = \|\rho\|_\infty$. The forward direction of the result can simply be shown via direct calculation. The maximum eigenvalue of $\rho(p, |\psi\rangle)$ is

$$\|\rho(p, |\psi\rangle)\|_\infty = p + \frac{1-p}{d}. \quad (6.94)$$

The d th Ky-Fan norm of $\rho \otimes \rho$ can then be seen to be

$$\begin{aligned} \|\rho(p, |\psi\rangle) \otimes \rho(p, |\psi\rangle)\|_{d-KY} &= \left(p + \frac{1-p}{d} \right)^2 + (d-1) \frac{1-p}{d} \left(p + \frac{1-p}{d} \right) \\ &= \left(p + \frac{1-p}{d} \right) \left(p + \frac{1-p}{d} + (d-1) \frac{1-p}{d} \right) \\ &= \|\rho(p, |\psi\rangle)\|_\infty. \end{aligned} \quad (6.95)$$

To see the contrapositive, let $\rho \in \mathcal{D}(\mathcal{H}^d)$ now be some generic state where $\text{spec}^\downarrow(\rho) = \{\lambda_i : \lambda_i \geq \lambda_{i+1} \forall i\}_{i=0}^{d-1}$ is the spectrum of ρ ordered in non-increasing order, such that $\|\rho\|_\infty = \lambda_0$. The eigenvalues of $\rho \otimes \rho$ are then products of the eigenvalues of ρ , such that $\text{spec}(\rho \otimes \rho) = \{\lambda_i \lambda_j\}_{i,j=0}^{d^2-1}$.

The d th Ky-Fan norm of $\rho \otimes \rho$ is the sum over the d largest elements of $\text{spec}(\rho \otimes \rho)$. It can therefore be lower bounded by summing over any d elements of $\text{spec}(\rho \otimes \rho)$. Choosing the d elements that contain

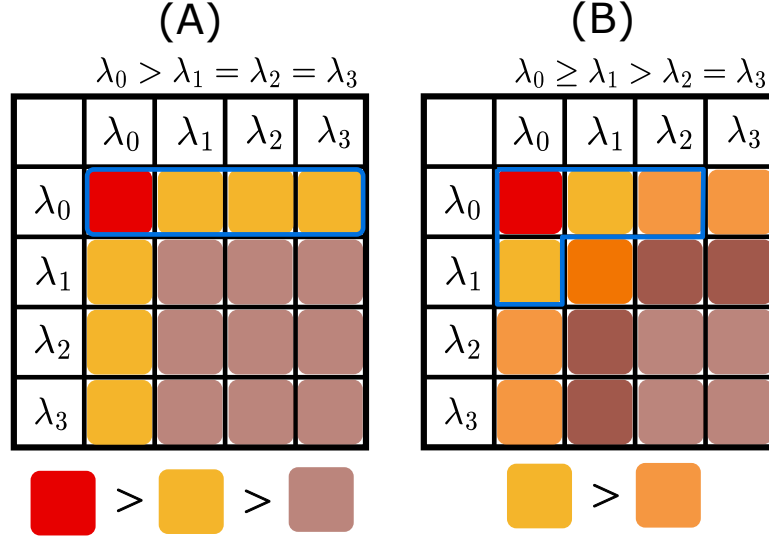


Figure 6.8: A graphical depiction of the proof of Result 6.2.5 for a state $\rho \in \mathcal{D}(\mathcal{H}^4)$ with $\text{spec}^\downarrow(\rho) = \{\lambda_i\}_{i=0}^3$, where $\lambda_i \geq \lambda_j$ if $i < j$ such that $\|\rho\|_\infty = \lambda_0$. Each cell of the grid represents an eigenvalue of $\rho \otimes \rho$, such that a cell holds the product of two of the eigenvalues of ρ . The lighter the colour of the cell, the greater the value it holds. Taking the sum of any four cells in the grid will give a lower bound on $\|\rho \otimes \rho\|_{d-\text{KY}}$, as the d th Ky-Fan norm of $\rho \otimes \rho$ is the sum over the states largest d eigenvalues. **In (A)**, the joint eigenvalues of a state that is $(d-1)$ -degenerate in its second largest eigenvalue is depicted. It can be seen that by taking the sum of the cells inside the blue rectangle that $\|\rho \otimes \rho\|_{d-\text{KY}} \geq \lambda_0$, where we have used the fact that $\sum_i \lambda_i = 1$ due to normalisation. Then, it can be noted that the cells in the blue rectangle are the four largest cells, due to the relationships between the joint eigenvalues depicted below the grid. Hence, this lower bound is tight, such that $\|\rho \otimes \rho\|_{d-\text{KY}} = \lambda_0$. (Note, one could have also take the sum over the first column). From this, it can be concluded that states that are $(d-1)$ -degenerate in their second largest eigenvalue have bound informational non-equilibrium. **In (B)**, a state that breaks this degeneracy, with $\lambda_0 \geq \lambda_1 > \lambda_2 = \lambda_3$, is depicted. The sum over the first row of cells (or the first column) stills gives the lower-bound $\|\rho \otimes \rho\|_{d-\text{KY}} \geq \lambda_0$, as in case (A). However, due to the relationships between the joint eigenvalues, again depicted below the grid, the broken degeneracy now means that $\lambda_1 \lambda_0 > \lambda_0 \lambda_3$. Hence, a better lower-bound on $\|\rho \otimes \rho\|_{d-\text{KY}}$ can be found by summing over the cells in the new blue rectangle. Due to $\lambda_1 \lambda_0 > \lambda_0 \lambda_3$, this sum is strictly larger than λ_0 (the sum of the first row), and hence gives the bound $\|\rho \otimes \rho\|_{d-\text{KY}} > \lambda_0$. Therefore, a state that is not $(d-1)$ -degenerate in its smallest eigenvalue will not have bound informational non-equilibrium.

λ_0 then gives

$$\begin{aligned}
 \|\rho \otimes \rho\|_{d-\text{KY}} &\geq \lambda_0^2 + \lambda_0 \lambda_1 + \lambda_0 \lambda_2 + \dots + \lambda_0 \lambda_{d-1} \\
 &= \lambda_0 \left(\sum_{i=0}^{d-1} \lambda_i \right) \\
 &= \|\rho\|_\infty,
 \end{aligned} \tag{6.96}$$

where in the third line we have used the fact that $\text{tr}[\rho] = 1$ such that $\sum_{i=0}^{d-1} \lambda_i = 1$.

If ρ is of the form $\rho(p, |\psi\rangle)$, then its spectrum is such that $\lambda_0 > \lambda_1 = \lambda_2 = \dots = \lambda_{d-1}$, as it is $(d-1)$ -fold degenerate in its second largest eigenvalue. Such states therefore saturate the above bound. If one now assumes the spectrum of ρ is such that $\lambda_0 > \lambda_1 > \lambda_2 = \lambda_3 = \dots = \lambda_{d-1}$, breaking the $(d-1)$ -fold degeneracy, then it can be seen that $\lambda_1 \lambda_0 > \lambda_0 \lambda_{d-1}$. The lower bound on the d th Ky-Fan norm can then be updated to

$$\begin{aligned}
 \|\rho \otimes \rho\|_{d-\text{KY}} &\geq \lambda_0^2 + \lambda_0 \lambda_1 + \lambda_0 \lambda_2 + \dots + \lambda_1 \lambda_0 \\
 &> \lambda_0^2 + \lambda_0 \lambda_1 + \lambda_0 \lambda_2 + \dots + \lambda_0 \lambda_{d-1} \\
 &= \|\rho\|_\infty,
 \end{aligned} \tag{6.97}$$

meaning if the $(d-1)$ -fold degeneracy is broken, then $\|\rho \otimes \rho\|_{d-\text{KY}} > \|\rho\|_\infty$ and informational non-equilibrium can always be concentrated. See Fig. 6.8 for graphical depiction of this proof. $\square$

6.5.3 Proof of Result 6.2.6

Result 6.2.6 Two qutrit optimal concentration always generates correlations between the A and B systems: $I(A : B)_{\eta_{AB}} > 0$ if $\Delta\mathcal{P}(\rho) > 0$.

Proof. First, we have

$$\begin{aligned}\sigma_A^{\text{opt}}(\rho) &:= \text{tr}_B [\sigma_{AB}^{\text{opt}}(\rho)] \\ &= \lambda_0 (\lambda_0 + 2\lambda_1) |0\rangle\langle 0| + [\lambda_1\lambda_1 + (1 - \lambda_2)\lambda_2] |1\rangle\langle 1| + \lambda_2 |2\rangle\langle 2|;\end{aligned}\quad (6.98)$$

and

$$\begin{aligned}\sigma_B^{\text{opt}}(\rho) &:= \text{tr}_A [\sigma_{AB}^{\text{opt}}(\rho)] \\ &= \lambda_0 (\lambda_0 + 2\lambda_2) |0\rangle\langle 0| + \lambda_1 |1\rangle\langle 1| + [\lambda_2\lambda_2 + (1 - \lambda_1)\lambda_1] |2\rangle\langle 2|.\end{aligned}\quad (6.99)$$

Also, since $S[\sigma_{AB}^{\text{opt}}(\rho)] = S(\rho \otimes \rho) = 2S(\rho)$, we have

$$I(A : B)_{\sigma_{AB}^{\text{opt}}(\rho)} = S[\sigma_A^{\text{opt}}(\rho)] + S[\sigma_B^{\text{opt}}(\rho)] - 2S(\rho), \quad (6.100)$$

which is strictly positive if $\Delta\mathcal{P}(\rho) = \log_2(\lambda_0 + 2\lambda_1) > 0$, as shown in Fig. 6.1.c $\square$

6.5.4 Proof of Result 6.2.9

Result 6.2.9 Let $\rho \in \mathcal{D}(\mathcal{H}^d)$ be a state that is not maximally resourceful nor minimally resourceful. Then $\Delta\mathcal{P}^n(\rho) > 0$.

Proof. We will first prove the result for the example in Fig. 6.8 to gain some intuition, and then extend it to the general case.

A lower bound on the 9-Ky-Fan norm of $\rho \otimes \rho \otimes \rho$, where $\rho \in \mathcal{D}(\mathcal{H}^3)$, can be found by summing over any $d^2 = 9$ cells in the grid. By summing over the $d^2 = 9$ cells in the cyan-rectangle, one can concluded that

$$\|\rho \otimes \rho \otimes \rho\|_{9\text{-KY}} \geq \|\rho\|_{\infty}. \quad (6.101)$$

For this to become an equality, the cells inside the cyan-rectangle must be equal to the $d^2 = 9$ largest cells in the grid. If, instead, there exists a cell outside of the cyan-rectangle that is larger than one within it, then this becomes a strict inequality. To see that this is always true, consider comparing the blue-highlighted cell against the orange-highlighted cell, as depicted in inequality (A) below the grid. In Fig. 6.8, this reads

$$\lambda_1(\rho)\lambda_0(\rho \otimes \rho) > \lambda_0(\rho)\lambda_8(\rho \otimes \rho), \quad (6.102)$$

where $\lambda_0(\rho \otimes \rho)$ is the largest eigenvalue of $\rho \otimes \rho$ and $\lambda_8(\rho \otimes \rho)$ is the smallest. In the general n -copy and d -dimensional case, a comparison of the equivalent grid cells would give

$$\lambda_1(\rho)\lambda_0(\rho \otimes \rho^{\otimes n}) > \lambda_0(\rho)\lambda_{[d^{n+1}-1]}(\rho \otimes \rho^{\otimes n}), \quad (6.103)$$

where here, $\lambda_0(\rho \otimes \rho^{\otimes n})$ is the largest eigenvalue of $\rho \otimes \rho^{\otimes n}$ and $\lambda_{[d^{n+1}-1]}(\rho \otimes \rho^{\otimes n})$ is the smallest. Now, the maximum and minimum eigenvalue of $\rho \otimes \rho^n$ are always

$$\begin{aligned}\lambda_0(\rho \otimes \rho^n) &= \lambda_0(\rho)^{n+1} \\ \lambda_{[d^{n+1}-1]}(\rho \otimes \rho^n) &= \lambda_{d-1}(\rho)^{n+1},\end{aligned}\quad (6.104)$$

where $\lambda_0(\rho)$ and $\lambda_{d-1}(\rho)$ are the largest and smallest eigenvalues of ρ respectively. Inputting this into Eq. (6.103) gives

$$\lambda_1(\rho)\lambda_0(\rho)^{n+1} > \lambda_0(\rho)\lambda_{d-1}(\rho)^{n+1}, \quad (6.105)$$

which becomes

$$\lambda_1(\rho)\lambda_0(\rho)^n > \lambda_d(\rho)^{n+1}. \quad (6.106)$$

If $\lambda_0(\rho) > \lambda_d(\rho)$ and $\lambda_1(\rho) \neq 0$, this always holds. Hence, any state that is not maximally mixed, such that $\lambda_0(\rho) = \lambda_d(\rho)$, or pure, such that $\lambda_1(\rho) \neq 0$, must therefore have

$$\|\rho \otimes \rho^{\otimes n}\|_{d^{n+1}-KY} > \|\rho\|_{\infty}. \quad (6.107)$$

This means that $\Delta\mathcal{P}^n(\rho) > 0$ if ρ is not maximally mixed or pure, completing the proof. $\square$

6.5.5 Proof of Lemma 6.2.10

Lemma 6.2.10 *Let $\rho \in \mathcal{D}(\mathcal{H}^d)$. A necessary and sufficient condition for*

$$\|\rho \otimes \rho^{\otimes(n+1)}\|_{d^{n+1}-KY} = \|\rho \otimes \rho^{\otimes n}\|_{d^n-KY} \quad (6.108)$$

is

$$\frac{\lambda_{d^n}(\rho \otimes \rho^{\otimes n})}{\lambda_{d^n-1}(\rho \otimes \rho^{\otimes n})} \leq \frac{\lambda_d(\rho)}{\lambda_0(\rho)}, \quad (6.109)$$

meaning that the ratio of the $(d^n + 1)$ th and (d^n) th largest eigenvalues of $\rho \otimes \rho^{\otimes n}$ must be less than the ratio of the d th and largest eigenvalues of ρ .

Proof. As a reminder, we define

$$\begin{aligned} \text{spec}^\downarrow(\rho) &= \{\lambda_i(\rho) : \lambda_i(\rho) \geq \lambda_j(\rho) \forall i \leq j\}_{i=0}^d \\ \text{spec}^\downarrow(\rho \otimes \rho^{\otimes k}) &= \{\lambda_i(\rho \otimes \rho^{\otimes k}) : \lambda_i(\rho \otimes \rho^{\otimes k}) \geq \lambda_j(\rho \otimes \rho^{\otimes k}) \forall i < j\}_{i=0}^{d^{k+1}} \end{aligned} \quad (6.110)$$

such that $\lambda_i(\rho)$ is $(i + 1)$ th largest eigenvalue of ρ and $\lambda_i(\rho \otimes \rho^{\otimes k})$ is the $(i + 1)$ th largest eigenvalue of $\rho \otimes \rho^{\otimes k}$.

We will first prove the result for the example in Fig. 6.8 to gain some intuition, and then extend it to the general case.

A lower bound on the 9-Ky-Fan norm of $\rho \otimes \rho \otimes \rho$, where $\rho \in \mathcal{D}(\mathcal{H}^3)$, can be found by summing over any $d^2 = 9$ cells in the grid. By summing over the purple highlighted cells in the grid, one gets the lower bound

$$\|\rho \otimes \rho \otimes \rho\|_{9-KY} \geq \|\rho \otimes \rho\|_{3-KY}. \quad (6.111)$$

For this to become an equality, one must show that there exists no non-purple cell in the grid that is greater than one of the purple cells. As, if so, the sum over the purple cells is equal to the sum over the largest $d^2 = 9$ cells, which is exactly the 9-Ky-Fan norm. Using the ordering of the eigenvalues in non-increasing order, one can compare the largest cell that is not purple (red highlighted) to the smallest cell that is purple (green highlighted), which gives

$$\lambda_0(\rho)\lambda_3(\rho \otimes \rho) \leq \lambda_2(\rho)\lambda_2(\rho \otimes \rho), \quad (6.112)$$

where $\lambda_0(\rho)$ and $\lambda_2(\rho)$ are the largest and smallest eigenvalues of ρ respectively, and $\lambda_2(\rho \otimes \rho)$ and $\lambda_3(\rho \otimes \rho)$ are the $d = 3$ th and $(d + 1) = 4$ th largest eigenvalues of $\rho \otimes \rho$. If this condition holds, one includes the green highlighted cell in the sum over the $d^2 = 9$ -largest cells, rather than the red highlighted cell. As all purple-highlighted cells are greater than the red-highlighted cell, and all the non-purple cells are less than the red-highlighted cell — due to the ordering of the eigenvalues — the sum of the purple highlighted cells is the sum over the largest $d^2 = 9$ -cells. Hence, this is a sufficient condition for Eq. (6.42) to become an equality. The condition can also be seen to be necessary, as if it does not hold, one should always include the red-highlighted over the green-highlighted cell, which mean that Eq. (6.42) will become a strict inequality.

By imagining the equivalent grid seen in Fig. 6.3 for the n copy case, where the y-axis remains the same, but the x-axis now contains the d^{n+1} eigenvalues of $\rho \otimes \rho^n$, the result can be extended to the general n -copy case and d -dimensional case. In this case, Eq. (6.112) becomes

$$\lambda_0(\rho)\lambda_{d^n}(\rho \otimes \rho) \leq \lambda_d(\rho)\lambda_{d^n-1}(\rho \otimes \rho), \quad (6.113)$$

where $\lambda_0(\rho)$ and $\lambda_d(\rho)$ are the largest and smallest eigenvalues of ρ respectively, and $\lambda_{d^n-1}(\rho \otimes \rho)$ and $\lambda_{d^n}(\rho \otimes \rho)$ are the d th and $(d+1)$ th largest eigenvalues of $\rho \otimes \rho$ respectively. This completes the proof. $\square$

6.5.6 Proof of Result 6.2.11

Result 6.2.11 Let $\rho \in \mathcal{D}(\mathcal{H}^d)$ be a state that is not maximally resourceful nor minimally resourceful. Then $\Delta\mathcal{P}^2(\rho) = \Delta\mathcal{P}(\rho)$ if and only if ρ is $\sqrt{d}$ -fold degenerate in its largest eigenvalues, and $(d - \sqrt{d})$ -fold degenerate in its second largest eigenvalue. If the state is not of this form, then $\Delta\mathcal{P}^2(\rho) > \Delta\mathcal{P}(\rho)$.

Proof. It can be seen that $\Delta\mathcal{P}^2(\rho) = \Delta\mathcal{P}(\rho)$ if and only if $\|\rho \otimes \rho^{\otimes 2}\|_{d^2-KY} = \|\rho \otimes \rho\|_{d-KY}$. Lemma 6.2.10 states that this equality holds in this case if and only if

$$\frac{\lambda_d(\rho \otimes \rho)}{\lambda_{d-1}(\rho \otimes \rho)} \leq \frac{\lambda_d(\rho)}{\lambda_0(\rho)}. \quad (6.114)$$

Where $\lambda_0(\rho)$ and $\lambda_d(\rho)$ are the largest and smallest eigenvalues of ρ respectively. From here, it can be seen that the largest eigenvalue of $\rho \otimes \rho$, $\lambda_0(\rho \otimes \rho)$, is therefore $\lambda_0(\rho)^2$. As we assume a non-increasing order on the eigenvalues, it can be concluded that

$$\lambda_0(\rho)^2 \geq \lambda_{d-1}(\rho \otimes \rho), \quad (6.115)$$

meaning the d th largest eigenvalue of $\rho \otimes \rho$ is always less than or equal to $\lambda_0(\rho)^2$, the largest eigenvalue of $\rho \otimes \rho$. We then also note that

$$\lambda_d(\rho \otimes \rho) \geq \lambda_0(\rho)\lambda_d(\rho), \quad (6.116)$$

which can be concluded via a counting argument.

We now assume that $\lambda_0(\rho)^2 > \lambda_{d-1}(\rho \otimes \rho)$. Inputting this assumption, and Eq. (6.116) into the left-hand side of Eq. (6.114) gives

$$\frac{\lambda_d(\rho \otimes \rho)}{\lambda_{d-1}(\rho \otimes \rho)} > \frac{\lambda_d(\rho \otimes \rho)}{\lambda_0(\rho)^2} \geq \frac{\lambda_0(\rho)\lambda_d(\rho)}{\lambda_0(\rho)^2} = \frac{\lambda_d(\rho)}{\lambda_0(\rho)}, \quad (6.117)$$

meaning Eq. (6.114) is never satisfied. Therefore, if $\lambda_0(\rho)^2 > \lambda_{d-1}(\rho \otimes \rho)$, then $\|\rho \otimes \rho^{\otimes 2}\|_{d^2-KY} > \|\rho \otimes \rho\|_{d-KY}$, meaning that $\Delta\mathcal{P}^2(\rho) > \Delta\mathcal{P}(\rho)$.

We must now deal only with the cases where $\lambda_0(\rho)^2 = \lambda_{d-1}(\rho \otimes \rho)$, as $\lambda_{d-1}(\rho \otimes \rho)$ cannot be lower than $\lambda_0(\rho)^2$ due to the ordering of the eigenvalues in non-decreasing order. To do this, we consider a state ρ that is n -fold degenerate in its largest eigenvalue, and $(d - n)$ -fold degenerate in its second largest eigenvalue. We write the spectrum of this state compactly as

$$\text{Spec}(\rho) = n[\lambda_0(\rho)] + (d - n)[\lambda_1(\rho)] : \lambda_0(\rho) > \lambda_1(\rho), \quad (6.118)$$

meaning the spectrum contains n eigenvalues $\lambda_0(\rho)$ and $(d - n)$ eigenvalues $\lambda_1(\rho)$. Using this notation, the spectrum of $\rho \otimes \rho$ is then

$$\text{Spec}(\rho \otimes \rho) = n^2[\lambda_0(\rho)] + 2n(d - n)[\lambda_0(\rho)\lambda_1(\rho)] + (d - n)^2[\lambda_1^2(\rho)]. \quad (6.119)$$

For such a state with only two distinct eigenvalues, Eq. (6.114) becomes

$$\frac{\lambda_d(\rho \otimes \rho)}{\lambda_{d-1}(\rho \otimes \rho)} \leq \frac{\lambda_1(\rho)}{\lambda_0(\rho)}. \quad (6.120)$$

We will now consider an exhaustive set of scenarios for different values of n , breaking the degeneracy of the second largest eigenvalue where necessary.

$n^2 = d$ and full degeneracy on remaining spectrum: If $n^2 = d$, then from Eq. (6.119) it can be seen that $\lambda_{d-1}(\rho \otimes \rho) = \lambda_0(\rho)^2$ and $\lambda_d(\rho \otimes \rho) = \lambda_0(\rho)\lambda_1(\rho)$. Therefore,

$$\frac{\lambda_d(\rho \otimes \rho)}{\lambda_{d-1}(\rho \otimes \rho)} = \frac{\lambda_1(\rho)}{\lambda_0(\rho)}. \quad (6.121)$$

Hence, if ρ is $\sqrt{d}$ -degenerate in its largest eigenvalue and $(d - \sqrt{d})$ -degenerate on its remaining spectrum, then $\Delta\mathcal{P}^2(\rho) = \Delta\mathcal{P}(\rho)$ as Eq. (6.114) holds.

$n^2 = d$ and not fully degeneracy on remaining spectrum: Consider now a slightly perturbed spectrum of ρ :

$$\text{Spec}(\rho) = n[\lambda_0(\rho)] + k[\lambda_1(\rho)] + (d - n - k)[\lambda_2(\rho)], \quad \lambda_0(\rho) > \lambda_1(\rho) > \lambda_2(\rho). \quad (6.122)$$

In this case, Eq. (6.114) becomes

$$\frac{\lambda_d(\rho \otimes \rho)}{\lambda_{d-1}(\rho \otimes \rho)} \leq \frac{\lambda_2(\rho)}{\lambda_0(\rho)}. \quad (6.123)$$

As $n^2 = d$ still, then one has $\lambda_{d-1}(\rho \otimes \rho) = \lambda_0(\rho)^2$ and $\lambda_d(\rho \otimes \rho) = \lambda_0(\rho)\lambda_1(\rho)$. Therefore,

$$\frac{\lambda_d(\rho \otimes \rho)}{\lambda_{d-1}(\rho \otimes \rho)} = \frac{\lambda_1(\rho)}{\lambda_0(\rho)} > \frac{\lambda_2(\rho)}{\lambda_0(\rho)}, \quad (6.124)$$

as $\lambda_1(\rho) > \lambda_2(\rho)$. Therefore, if ρ is $\sqrt{d}$ -degenerate in its largest eigenvalue, but the remaining spectrum is not fully degenerate, then $\Delta\mathcal{P}^2(\rho) > \Delta\mathcal{P}(\rho)$.

$n^2 > d$ and arbitrary remaining spectrum: In this case, Eq. (6.114) is as originally stated. As $n^2 > d$, then one has $\lambda_{d-1}(\rho \otimes \rho) = \lambda_0(\rho)^2$ and $\lambda_d(\rho \otimes \rho) = \lambda_0(\rho)^2$. Therefore,

$$\frac{\lambda_d(\rho \otimes \rho)}{\lambda_{d-1}(\rho \otimes \rho)} = 1. \quad (6.125)$$

The only state ρ that has the ratio of its largest and smallest eigenvalues greater than or equal to one is $\rho = \mathbb{I}/d$ i.e., if $\lambda_0(\rho) > \lambda_d(\rho)$ then Eq. (6.114) does not hold, meaning $\Delta\mathcal{P}^2(\rho) > \Delta\mathcal{P}(\rho)$ also in this case.

$n^2 < d$ and arbitrary remaining spectrum: In this case, $\lambda_0(\rho)^2 > \lambda_{d-1}(\rho \otimes \rho)$, which is proved above.

This captures all cases for which $\|\rho \otimes \rho^2\|_{d^2-KY} = \|\rho \otimes \rho\|_{d-KY}$, meaning $\Delta\mathcal{P}^2(\rho) = \Delta\mathcal{P}(\rho)$ if and only if ρ is $\sqrt{d}$ -fold degenerate in its largest eigenvalues, and $d - \sqrt{d}$ -fold degenerate in its second largest eigenvalue, and not maximally mixed or pure. This completes the proof. $\square$

6.6 Additional Material - Unspeakable Coherence Concentration

6.6.1 Relationship between Local and Global Symmetry

A few observations can be made about the relationship between the local and global (bipartite) symmetry and hence the local and global coherence. Firstly, we prove the following Lemma.

Lemma 6.6.1. Consider a finite-dimensional bipartite system AB . Let ρ_{AB} be a bipartite state in AB and L_A and L_B be Hermitian operators acting on systems A and B , respectively. Then

$$[\rho_{AB}, L_{AB}] = 0 \implies [\rho_A, L_A] = 0, \quad (6.126)$$

where $\rho_A = \text{tr}_B[\rho_{AB}]$ and $L_{AB} = L_A \otimes \mathbb{I}_B + \mathbb{I}_A \otimes L_B$, where L_A, L_B are non-degenerate, truncated number operators.

Proof. As L_X is a Hermitian (linear) operator ($X = A, B$), its eigenstates form an orthonormal basis, such that L_X can be written as $L_X = \sum_n \lambda_n^{(X)} |n\rangle\langle n|_X$, where $nm = \delta_{nm} \forall n, m$. The bipartite operator L_{AB} can then be written in the joint eigenbasis as

$$L_{AB} = \sum_{i,j} \left(\lambda_n^{(A)} + \lambda_m^{(B)} \right) |n\rangle\langle n|_A \otimes |m\rangle\langle m|_B. \quad (6.127)$$

As $[\rho_{AB}, L_{AB}] = 0$, ρ_{AB} is block diagonal with respect to the eigenspaces of L_{AB} , namely,

$$\rho_{AB} = \sum_{n,m,s,t} p_{nmst} |n\rangle\langle m|_A \otimes |s\rangle\langle t|_B, \text{ where } p_{nmst} > 0 \text{ if and only if } \lambda_n^{(A)} + \lambda_s^{(B)} = \lambda_m^{(A)} + \lambda_t^{(B)}, \quad (6.128)$$

and $p_{nmst} = \langle ns | \rho_{AB} | mt \rangle$. The reduced state in system A is then

$$\rho_A = \text{tr}_B(\rho_{AB}) = \sum_{n,m,t} p_{nmst} |n\rangle\langle m|_A. \quad (6.129)$$

From here, it can be seen that $p_{nmst} > 0$ if and only if $\lambda_n^{(A)} = \lambda_m^{(A)}$. In particular, this already implies that ρ_A is block diagonal with respect to the eigenbasis of L_A , meaning that $[\rho_A, L_A] = 0$ and completing the proof. $\square$

Lemma 6.6.1 means that global symmetry implies local symmetry. However, the reverse is not true: the presence of local symmetry does not imply global symmetry. To see this, consider $\rho_{AB} = |\Phi^+\rangle\langle\Phi^+|_{AB}$ with $|\Phi^+\rangle_{AB} = (1/\sqrt{2})(|00\rangle + |11\rangle)_{AB}$, and $L_X = \lambda_0 |0\rangle\langle 0|_X + \lambda_1 |1\rangle\langle 1|_X$ for $\lambda_0 \neq \lambda_1$ ($X = A, B$). Locally both systems of ρ_{AB} are maximally mixed and hence symmetric, while the global state is not symmetric with respect to L_{AB} .

In the specific case that ρ_{AB} is a product state, $\rho_{AB} = \rho_A \otimes \rho_B$, then local symmetry in both subsystem ($[\rho_A, L_A] = 0$ and $[\rho_B, L_B] = 0$) does imply global symmetry ($[\rho_A \otimes \rho_B, L_{AB}] = 0$). This can then be extended to hold for any convex mixture of product states that are locally symmetric.

Using these ideas, we now argue that one cannot exploit the differing global and local symmetries in the concentration scenario to increase the local coherence unless there is some initial local coherence present. If ρ_{AB} is a product state and both local systems contain no coherence, then ρ_{AB} is incoherent. Since V_{AB} cannot increase coherence, $V_{AB}(\rho_A \otimes \rho_B)V_{AB}^\dagger$ must also be incoherent. Hence, according to Lemma 6.6.1, the output local systems will be incoherent, too.

6.6.1.1 Proof of Result 6.3.2

We first state Result 6.3.2 here for clarity.

Result 6.3.2 For a state $\rho \in \mathcal{D}(\mathcal{H}^2)$, we have

$$\Delta M_A^{(1)} = |p_{01}| \left(\sqrt{1 + (2p_{00} - 1)^2} - 1 \right), \quad (6.130)$$

which is achieved by applying a unitary rotation of angle

$$\cos^{-1} \left(\frac{1}{\sqrt{1 + (2p_{00} - 1)^2}} \right) \quad (6.131)$$

in the $\{|01\rangle, |10\rangle\}$ degenerate subspace of L_{AB} .

Proof. Firstly, we note that an arbitrary single-qubit state ρ can be written as

$$\rho = \begin{bmatrix} p_{00} & p_{01} \\ p_{01}^* & 1 - p_{00} \end{bmatrix}, \quad (6.132)$$

where the decomposition is with respect to the eigenbasis of L , and $p_{00} = \langle 0 | \rho | 0 \rangle$, $p_{01} = \langle 0 | \rho | 1 \rangle$. To two copies of ρ , we apply an allowed unitary V_{AB} on it (i.e., $[V_{AB}, L_{AB}] = 0$). The most general form of such a unitary is

$$V_{AB} = \begin{bmatrix} e^{i\omega_0} & 0 & 0 & 0 \\ 0 & U_{00} & U_{01} & 0 \\ 0 & U_{10} & U_{11} & 0 \\ 0 & 0 & 0 & e^{i\omega_1} \end{bmatrix}, \quad (6.133)$$

where $0 \leq \omega_i \leq 2\pi$ for $i \in \{0, 1\}$ and

$$\begin{bmatrix} U_{00} & U_{01} \\ U_{10} & U_{11} \end{bmatrix} = e^{-i\alpha} \begin{bmatrix} e^{-i\phi_1} & 0 \\ 0 & e^{i\phi_1} \end{bmatrix} \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} e^{-i\phi_2} & 0 \\ 0 & e^{i\phi_2} \end{bmatrix} \quad (6.134)$$

is a general two-qubit unitary in the $\{|01\rangle, |10\rangle\}$ degenerate subspace, which has been decomposed into a combination of two rotations about the Z-axis (defined by the states $\{|01\rangle, |10\rangle\}$), a rotation about the Y-axis, and a phase factor, α . Hence, the two-qubit unitary V_{AB} can be written as a product of unitaries,

$$V_{AB} = S(\omega_0)T(\omega_1)R^z(2\phi_1)R^y(2\theta)R^z(2\phi_2) \quad (6.135)$$

where

$$S(\omega_0) = \begin{bmatrix} e^{i\omega_0} & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad T(\omega_1) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & e^{i\omega_1} \end{bmatrix}, \quad R^z(2\phi) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & e^{-i\phi} & 0 & 0 \\ 0 & 0 & e^{i\phi} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad (6.136)$$

$$R^y(2\theta) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta & 0 \\ 0 & \sin \theta & \cos \theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix},$$

with all decompositions with respect to the eigenbasis of L_{AB} .

To achieve coherence concentration, we need to evaluate the output state $\sigma_A = \text{tr}_B[V_{AB}(\rho_A \otimes \rho_B)V_{AB}^\dagger]$, which is characterised by $q_{00} := \langle 0 | \sigma_A | 0 \rangle$ and $q_{01} := \langle 0 | \sigma_A | 1 \rangle$. To this end, we now sequentially apply each unitary in Eq. (6.135) to $\rho_A \otimes \rho_B$, assess each's ability to increase local coherence. Firstly, if applying $S(\omega_0)$,

$$|q_{01}| = |p_{01}(1 - p_{00} + e^{i\omega_0}p_{00})| \leq |p_{01}|(|1 - p_{00}| + |e^{i\omega_0}p_{00}|) = |p_{01}|, \quad (6.137)$$

where the equality occurs when $e^{i\omega_0} = 1$. Hence, under unitaries of this form, we have $\Delta M^{(j)} \leq 0 \forall \omega_0$, with equality if $\omega_0 = 2\pi n$ for integers n . The same can be shown for $T(\omega_1)$, meaning that when looking to concentrate coherence, it is always sufficient to set $\omega_0 = 0 = \omega_1$ and hence apply the identity in the $|00\rangle$ and $|11\rangle$ subspaces.

Secondly, when applying $R^z(2\phi)$ to $\rho_A \otimes \rho_B$ and taking the partial trace, it can be seen that $|q_{01}| = |p_{01}| \forall \phi$, meaning that $\Delta M^{(j)} = 0$ for all unitaries of this form. Hence, when looking to concentrate coherence, it is sufficient to always set $\phi = 0$.

It is noted that $[S(\omega_0), R^z(2\phi_1)R^y(2\theta)R^z(2\phi_1)] = 0$ and $[T(\omega_1), R^z(2\phi_1)R^y(2\theta)R^z(2\phi_1)] = 0$, meaning that changing the order in which these unitaries are applied cannot help one concentrate coherence. One therefore only has to consider the local coherence increasing power of $R^y(2\theta)$.

Applying $R^y(2\theta)$ to $\rho_A \otimes \rho_B$ and tracing out the B system then gives

$$q_{01} = p_{01} [\cos \theta + (2p_{00} - 1) \sin \theta]. \quad (6.138)$$

From here, the maximum $M^{(j)}(\sigma_A)$ that can be achieved under allowed unitaries is given by

$$\max_{V_{AB}} M^{(j)}(\sigma_A) = \max_{\theta} |p_{01} [\cos \theta + \sin \theta (2p_{00} - 1)]| = |p_{01}| \max_u |u \cdot v|, \quad (6.139)$$

where

$$u = \begin{bmatrix} \cos \theta \\ \sin \theta \end{bmatrix}, \quad v = \begin{bmatrix} 1 \\ 2p_{00} - 1 \end{bmatrix}. \quad (6.140)$$

The dot product will be maximised when u points in the same direction as v . Hence, the maximum is achieved by

$$u_{\text{opt}} = \frac{1}{\sqrt{1 + (2p_{00} - 1)^2}} \begin{bmatrix} 1 \\ 2p_{00} - 1 \end{bmatrix}, \quad (6.141)$$

from which the optimal angle of rotation to increase the local coherence can be found:

$$\theta_{\text{opt}} := \cos^{-1} \left(\frac{1}{\sqrt{1 + (2p_{00} - 1)^2}} \right). \quad (6.142)$$

Finally, using u_{opt} , the maximum $\Delta M_A^{(j)}$ can be found to be

$$\Delta M_A^{(j)} := \max_{V_{AB}} M^{(j)}(\sigma_A) - M^{(j)}(\rho_A) = |p_{01}| \left(\sqrt{1 + (2p_{00} - 1)^2} \right) - |p_{01}|, \quad (6.143)$$

completing the proof. $\square$

6.6.2 Proof of Result 6.3.3

Here, we will show how the coupled non-linear recurrence relationships given in Result 6.3.3 are found.

Result 6.3.3 *The output qubit state of the $(m + 1)$ th step of the qubit concatenation protocol will be*

$$\begin{aligned} p_{00}^{m+1} &= p_{00}^m - \frac{2(2p_{00}^m - 1)|p_{01}^m|^2}{1 + (2p_{00}^m - 1)^2}; \\ p_{01}^{m+1} &= |p_{01}^m| \sqrt{1 + (2p_{00}^m - 1)^2}, \end{aligned} \quad (6.144)$$

Proof. Assuming one initially has two copies of the state ρ :

$$\begin{aligned}\rho_A \otimes \rho_B &= \begin{bmatrix} p_{00} & p_{01} \\ p_{01}^* & (1-p_{00}) \end{bmatrix} \otimes \begin{bmatrix} p_{00} & p_{01} \\ p_{01}^* & (1-p_{00}) \end{bmatrix} \\ &= \begin{bmatrix} p_{00}^2 & p_{00}p_{01} & p_{01}p_{00} & p_{01}^2 \\ p_{00}p_{01}^* & p_{00}(1-p_{00}) & |p_{01}|^2 & p_{01}(1-p_{00}) \\ p_{01}^*p_{00} & |p_{01}|^2 & (1-p_{00})p_{00} & (1-p_{00})p_{01} \\ (p_{01}^*)^2 & p_{01}^*(1-p_{00}) & (1-p_{00})p_{01}^* & (1-p_{00})^2 \end{bmatrix}. \end{aligned} \quad (6.145)$$

To this bipartite state, the following optimal concentrating unitary, found in Result 6.3.2, is applied,

$$U_{\text{opt}}(\rho) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos \theta_{\text{opt}} & -\sin \theta_{\text{opt}} & 0 \\ 0 & \sin \theta_{\text{opt}} & \cos \theta_{\text{opt}} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad (6.146)$$

where we have

$$\cos \theta_{\text{opt}} = \frac{1}{\sqrt{1 + (2p_{00} - 1)^2}}, \quad \sin \theta_{\text{opt}} = \frac{2p_{00} - 1}{\sqrt{1 + (2p_{00} - 1)^2}}, \quad (6.147)$$

which are dependent on ρ . The B system is then traced over, giving an output state

$$\sigma_A = \text{tr}_B [U_{\text{opt}}(\rho)(\rho_A \otimes \rho_B)U_{\text{opt}}(\rho)^\dagger] =: \begin{bmatrix} q_{00} & q_{01} \\ q_{01}^* & (1-q_{00}) \end{bmatrix}. \quad (6.148)$$

By explicitly performing this calculation, it can be seen that

$$\begin{aligned}q_{00} &= p_{00}^2 + p_{00}(1-p_{00})\cos^2 \theta_{\text{opt}} - |p_{01}|^2 \sin \theta_{\text{opt}} \cos \theta_{\text{opt}} - |p_{01}|^2 \sin \theta_{\text{opt}} \cos \theta_{\text{opt}} \\ &= p_{00} - 2|p_{01}|^2 \sin \theta_{\text{opt}} \cos \theta_{\text{opt}}, \end{aligned} \quad (6.149)$$

whilst

$$\begin{aligned}q_{01} &= p_{00}p_{01} \sin \theta_{\text{opt}} + p_{01}p_{00} \cos \theta_{\text{opt}} + p_{01}(1-p_{00}) \cos \theta_{\text{opt}} - (1-p_{00})p_{01} \sin \theta_{\text{opt}} \\ &= p_{01} \left[\cos \theta_{\text{opt}} + \sin \theta_{\text{opt}}(2p_{00} - 1) \right]. \end{aligned} \quad (6.150)$$

Inputting $\cos \theta_{\text{opt}}$ and $\sin \theta_{\text{opt}}$ into these equation then gives

$$\begin{aligned}q_{00} &= p_{00} - \frac{2(2p_{00} - 1)|p_{01}|^2}{1 + (2p_{00} - 1)^2}; \\ q_{01} &= p_{01} \sqrt{1 + (2p_{00} - 1)^2}. \end{aligned} \quad (6.151)$$

The output state σ , as characterised by q_{00} and q_{01} , therefore depends only on p_{00} and p_{01} from ρ . We then note that local Z -rotations are always allowed, such that we can set $p_{01} \rightarrow |p_{01}|$ without loss of generality. One can now consider taking two copies of σ and repeating the same process. This would output the state

$$\omega_A = \text{tr}_B [U_{\text{opt}}(\sigma)(\sigma_A \otimes \sigma_B)U_{\text{opt}}(\sigma)^\dagger] =: \begin{bmatrix} l_{00} & l_{01} \\ l_{01}^* & (1-l_{00}) \end{bmatrix}, \quad (6.152)$$

where, using the same calculation as above, it can be seen that

$$\begin{aligned}l_{00} &= q_{00} - \frac{2(2q_{00} - 1)|q_{01}|^2}{1 + (2q_{00} - 1)^2}; \\ l_{01} &= |q_{01}| \sqrt{1 + (2q_{00} - 1)^2}. \end{aligned} \quad (6.153)$$

Again, the output state ω , as characterised by l_{00} and l_{01} , can be seen to depend on q_{00} and q_{01} from σ . This process can then be repeated an arbitrary finite number of times, where each time the parameters characterising the output state can be formulated in terms of the parameters characterising the input state. This, therefore, takes the form of recurrence relations, which here are coupled and non-linear functions where the next value of the function depends on some combination of the previous value. We note that here the outputs only depend on the previous values, and not any further back in the sequence. By labelling the input state from the m th layer as p_{00}^m and p_{01}^m , the outputs from the $m + 1$ layer can therefore be given by

$$\begin{aligned} p_{00}^{m+1} &= p_{00}^m - \frac{2(2p_{00}^m - 1)|p_{01}^m|^2}{1 + (2p_{00}^m - 1)^2}; \\ p_{01}^{m+1} &= |p_{01}^m| \sqrt{1 + (2p_{00}^m - 1)^2}, \end{aligned} \quad (6.154)$$

which completes the proof. $\square$

6.6.3 Proof of Result 6.3.4

This section contains a proof of Result 6.3.4, which places an upper bound on the achievable final coherence in the qubit concatenation protocol by the purity of the input state.

Result 6.3.4 *Given an initial qubit state ρ , let $\sigma_{(m)}$ be the qubit state output after the m th step of the concatenation protocol. Then*

$$M^{(1)}(\sigma_m) \leq \sqrt{2\text{tr}(\rho^2) - 1} \quad (6.155)$$

Proof. Firstly, if ρ is already incoherent, we must have $M^{(1)}(\sigma_{(m)}) = 0 \forall m$ and the bound holds immediately. Hence, it suffices to focus on ρ with some initial coherence.

Now, let $\mathbf{n}^0 = (n_x^0, n_y^0, n_z^0) \in \mathbb{R}^3$ be ρ 's Bloch vector, which has 2-norm $\|\mathbf{n}^0\|_2 = \sqrt{(n_x^0)^2 + (n_y^0)^2 + (n_z^0)^2}$. Let $p_{00}^0 := \langle 0 | \rho | 0 \rangle = (1 + n_z^0)/2$ and $p_{01}^0 := \langle 0 | \rho | 1 \rangle = (n_x^0 + in_y^0)/2$. As stated in the main text, since Z-rotations are all locally allowed, we can set $n_y = 0$ without loss of generality. Also, as we assume ρ is coherent, we must have $n_x^0 \neq 0$. This means ρ 's purity, $\text{tr}(\rho^2)$, can be expressed as

$$\text{tr}[\rho^2] = (p_{00}^0)^2 + (1 - p_{00}^0)^2 + 2|p_{01}^0|^2 = \frac{1 + \|\mathbf{n}^0\|_2^2}{2}. \quad (6.156)$$

Furthermore, the recurrence relations, given in Result 6.3.3, can be rewritten in terms of the Bloch vector as

$$\begin{aligned} n_z^{m+1} &= n_z^m - \frac{n_z^m (n_x^m)^2}{1 + (n_z^m)^2}; \\ |n_x^{m+1}| &= |n_x^m| \sqrt{1 + (n_z^m)^2}, \end{aligned} \quad (6.157)$$

where n_z^m and n_x^m are the Z and X components of $\sigma_{(m)}$'s Bloch vector, denoted by $\mathbf{n}^m$. We note that $n_y^m = 0 \forall m$ such that $M^{(1)}(\sigma_{(m)}) = |n_x^m|$. Now, a direct calculation shows that

$$\|\mathbf{n}^{m+1}\|_2^2 = \|\mathbf{n}^m\|_2^2 + \frac{(n_z^m n_x^m)^2}{1 + (n_z^m)^2} \left[(n_z^m)^2 + \frac{(n_x^m)^2}{1 + (n_z^m)^2} - 1 \right]. \quad (6.158)$$

It is then noted that

$$(n_z^m)^2 + \frac{(n_x^m)^2}{1 + (n_z^m)^2} = \frac{(n_z^m)^2 + (n_x^m)^2 + (n_z^m)^4}{1 + (n_z^m)^2} \leq \frac{1 + (n_z^m)^4}{1 + (n_z^m)^2} = 1 - (n_z^m)^2 \leq 1 \quad \forall m, \quad (6.159)$$

as $(n_z^m)^2 + (n_x^m)^2 \leq 1$ due to normalisation and $1 + (n_z^m)^4 = [1 + (n_z^m)^2][1 - (n_z^m)^2]$. As

$$(n_z^m n_x^m)^2 / [1 + (n_z^m)^2] \geq 0 \forall m, \quad (6.160)$$

Eq. (6.158), Eq. (6.159) and Eq. (6.160) jointly imply that

$$\|\mathbf{n}^{m+1}\|_2 \leq \|\mathbf{n}^m\|_2 \quad \forall m. \quad (6.161)$$

Using Eq. (6.156), the definition of the 2-norm, and recalling that the coherence of $\sigma_{(m)}$ reads $M^{(1)}(\sigma_{(m)}) = |n_x^m|$, we obtain

$$M^{(1)}(\sigma_{(m)}) = |n_x^m| \leq \|\mathbf{n}^m\|_2 \leq \|\mathbf{n}^0\|_2 = \sqrt{2\text{tr}(\rho^2) - 1}, \quad (6.162)$$

which is the desired result. $\square$

As a remark, we have the following observation:

If the initial qubit state ρ is coherent, then, no matter how small the initial coherence is, we must have $n_z^\infty = 0$.

Namely, a vanishing amount of coherence possessed by ρ suffices to asymptotically erase the Z component of the Bloch vector.

Proof. To see this, first, from the above proof, we observe that $n_z^{m+1} \leq n_z^m$ and $|n_x^{m+1}| \geq |n_x^m| \forall m \in \mathbb{N}$, meaning that $\{|n_x^m|\}_m$ is monotonic non-decreasing and $\{n_z^m\}_m$ is monotonic non-increasing when $m \rightarrow \infty$. Hence, by the monotone convergence theorem (see, e.g., Ref. [142]), we must have

$$\lim_{m \rightarrow \infty} n_z^m = \inf_m n_z^m =: n_z^\infty \quad \& \quad \lim_{m \rightarrow \infty} |n_x^m| = \sup_m |n_x^m| =: n_x^\infty. \quad (6.163)$$

Now we argue that $n_z^\infty = 0$. To see this, we apply $m \rightarrow \infty$ on both sides of $|n_x^{m+1}| = |n_x^m| \sqrt{1 + (n_z^m)^2}$ [i.e., the second line in Eq. (6.157)], obtaining $(n_x^\infty)^2 = (n_x^\infty)^2 [1 + (n_z^\infty)^2]$ (which is true because both limits $\lim_{m \rightarrow \infty} n_z^m$ and $\lim_{m \rightarrow \infty} |n_x^m|$ exist). This then implies that $|n_x^\infty| n_z^\infty = 0$. Hence, we must have either $n_z^\infty = 0$ or $|n_x^\infty| = 0$. However, if $n_x^\infty = 0$, then, since $\{|n_x^m|\}_m$ is monotonic non-decreasing, we must have $n_x^0 = 0$, meaning that the initial state ρ is already incoherent, in contradiction to our assumption. Consequently, we must have $n_z^\infty = 0$. $\square$

6.6.4 Proof of Result 6.3.5

Here a proof of Result 6.3.5 is given.

Result 6.3.5 *For any number of states $n = 2^N$ where $N \in \mathbb{N}$ and $\epsilon > 0$, there exists an initial state ρ (dependent on n, ϵ) such that*

$$M^{(1)}(\rho) < \frac{1}{n} \quad \& \quad \frac{M^{(1)}(\sigma_{(m)})}{M^{(1)}(\rho)} > 2^{-\epsilon} \sqrt{n} \quad \forall m \geq \log_2(n). \quad (6.164)$$

Proof. For a given $N \in \mathbb{N}$, we firstly choose an initial qubit state ρ with Bloch vector $(n_x^0, n_y^0 = 0, n_z^0)$ satisfying

$$0 < |n_x^0|^2 \ll \frac{1}{2^{2N}} \times \frac{1}{2^N} \times \min\{1 - |n_z^0|^2, |n_z^0|^2\}, \quad (6.165)$$

where the notation “ $x \ll y$ ” means that $x/y \sim O(1)$ is significantly smaller than 1. As a result, this state has a vanishing amount of coherence, as

$$M^{(1)}(\rho) = |n_x^0| \ll \frac{1}{2^N}. \quad (6.166)$$

For convenience, let us define, for $m \in \mathbb{N} \cup \{0\}$,

$$A_m := |n_x^m|^2 \quad \& \quad B_m := |n_z^m|^2. \quad (6.167)$$

Then Eq. (6.165) becomes

$$0 < A_0 \ll \frac{1}{2^{2N}} \times \frac{1}{2N} \times \min\{1 - B_0, B_0\}. \quad (6.168)$$

Also, using Eq. (6.157), we have that, for every $m \in \mathbb{N} \cup \{0\}$,

$$\begin{aligned} B_{m+1} &= B_m \left(1 - \frac{A_m}{1 + B_m}\right)^2 = B_0 \prod_{k=0}^m \left(1 - \frac{A_k}{1 + B_k}\right)^2; \\ A_{m+1} &= A_m (1 + B_m) = A_0 \prod_{k=0}^m (1 + B_k). \end{aligned} \quad (6.169)$$

Using the fact that $0 \leq B_k \leq 1 \forall k$, we can combine Eqs. (6.168) and (6.169) to obtain

$$A_m \leq A_0 \prod_{k=0}^m (2) = 2^m A_0 \leq 2^N A_0 \ll \frac{1}{2^N} \times \frac{1}{2N} \times \min\{1 - B_0, B_0\} \quad \forall m \leq N. \quad (6.170)$$

Where in the first inequality we have set $B_k = 1$, and in the second we have used the fact that $m \leq N$. Finally, we have subbed in Eq. (6.168).

Furthermore, using the fact that $A_0 \leq A_k \leq 1$ and $B_0 \geq B_k \geq 0 \forall k$, Eq. (6.169) also implies that, for every $m \leq N$ (note, the justification details why we get to the next line),

$$\begin{aligned} B_0 &\geq B_{m+1} && \text{as Eq. (6.169)} \\ &= B_0 \prod_{k=0}^m \left(1 - \frac{A_k}{1 + B_k}\right)^2 && \text{as } B_k \geq 0 \forall k \\ &\geq B_0 \prod_{k=0}^m (1 - A_k)^2 && \text{as } A_m \geq A_{m-1} \\ &\geq B_0 \prod_{k=0}^m (1 - A_m)^2 = B_0 (1 - A_m)^{2m} && \text{as } m \leq N \\ &\geq B_0 (1 - A_m)^{2N} && \text{as } (1 + x)^n = \sum_{k=0}^n \binom{n}{k} x^k \text{ ignoring negative on } A_m \\ &\geq B_0 \left(1 - \sum_{k=1}^{2N} (2N A_m)^k\right) && \text{as sum of a geometric series} \\ &= B_0 \left(1 - 2N A_m \times \frac{1 - (2N A_m)^{2N}}{1 - 2N A_m}\right) && \text{as } (2N A_m)^{2N} \geq 0 \\ &\geq B_0 \left(1 - \frac{2N A_m}{1 - 2N A_m}\right) && \text{as } 0 \leq 2N A_m \ll 2^{-N} < 1/2 \text{ due to Eq. (6.170)} \\ &\geq B_0 - 4N A_m, \end{aligned} \quad (6.171)$$

giving in total $B_{m+1} \geq B_0 - 4N A_m$. This means that, as long as $m \leq N$, we have $B_{m+1} \approx B_0$, due to Eq. (6.170).

Now, by setting $m = N - 1$, Eqs. (6.169) and (6.171) jointly suggest that (below, we define $A_{-1} := 0$,

so that $B_0 \geq B_0 - 4NA_{-1}$

$$\begin{aligned}
\log_2 \frac{A_N}{A_0} &= \sum_{k=0}^{N-1} \log_2(1 + B_k) && \text{as } B_{k+1} \geq B_0 - 4NA_k \ \forall k \leq N \text{ from Eq. (6.171)} \\
&\geq \sum_{k=0}^{N-1} \log_2(1 + B_0 - 4NA_{k-1}) && \text{as } A_k \leq A_N \ \forall k \leq N \ \& \ A_{-1} := 0 \\
&\geq \sum_{k=0}^{N-1} \log_2(1 + B_0 - 4NA_N) && (6.172) \\
&= N \log_2(1 + B_0 - 4NA_N) \\
&= N \log_2[2 - (1 - B_0 + 4NA_N)] && \text{as } 4NA_N \ll (1 - B_0)/2^{N-1} \leq 1 - B_0 \text{ due to Eq. (6.170)} \\
&\geq N \log_2[2 - 2(1 - B_0)] \\
&= N + N \log_2 B_0,
\end{aligned}$$

Since $\{A_k\}_k$ is non-decreasing in k , and since $M^{(1)}(\sigma_{(m)}) = |n_x^m| = \sqrt{A_m}$ as well as $B_0 = |n_z^0|^2$, we thus conclude that $\forall m \geq N$ the following holds

$$\frac{M^{(1)}(\sigma_{(m)})}{M^{(1)}(\rho)} \geq \frac{M^{(1)}(\sigma_{(N)})}{M^{(1)}(\rho)} = \sqrt{\frac{A_N}{A_0}} \geq \sqrt{2^{N+N \log_2(|n_z^0|^2)}} = 2^{\frac{N}{2}} \times 2^{N \log_2(|n_z^0|)}, \quad (6.173)$$

where we have used Eq. (6.172). Note that $\log_2(|n_z^0|) \leq 0$ is in general negative. Finally, to see the exponential amplification, for any given $\epsilon > 0$, which can be as small as we want, there exists an $|n_z^0|$ satisfying $0 < |\log_2 |n_z^0|| \ll \epsilon/N$, from which we have $N \log_2 |n_z^0| \geq -\epsilon$. Put this together, we have,

$$\frac{M^{(1)}(\sigma_{(m)})}{M^{(1)}(\rho)} \geq 2^{\frac{N}{2} - \epsilon} \quad \forall m \geq N. \quad (6.174)$$

To complete the proof, we note that $n = 2^N$, where n is the number of states used in N layers of the protocol. Hence, there exists an initial state that has a vanishingly small amount of coherence [Eq. (6.166)], from which a state with unboundedly more coherence can be created using this concatenation protocol. $\square$

Numerically, it can be seen in Fig 6.4 that even for states with $0 \leq |n_z^0| < 1/2$, given enough steps, an exponential increase in local coherence can be achieved. We leave a mathematical proof of this fact for future work.

6.6.5 Proof of Result 6.3.6

To prove Result 6.3.6, the following Lemma is first proved. It states that, given some operator P , the sum over the absolute value of the matrix elements of any diagonal of length k , in any basis, is upper-bounded by the k th Ky-Fan norm of P .

Lemma 6.6.2. *Let P be a operator acting on finite-dimensional Hilbert space, where $\{|\phi_i\rangle\}_{i=0}^m$ and $\{|\psi_j\rangle\}_{j=0}^n$ are orthonormal bases on the input and output spaces of P , respectively. Then we have*

$$\sum_n |\langle \psi_{n+j} | P | \phi_n \rangle| \leq \|P\|_{k\text{-KF}} \quad \forall j. \quad (6.175)$$

Proof. Firstly, we write $\langle \psi_r | P | \phi_s \rangle = |\langle \psi_r | P | \phi_s \rangle| e^{i\theta_{r,s}} \ \forall r, s$. Now, consider the operator

$$X := \sum_n e^{i\theta_{n+j,n}} |\psi_{n+j}\rangle \langle \phi_n|. \quad (6.176)$$

It can then be seen that

$$\text{tr}(X^\dagger P) = \sum_n |\langle \psi_{n+j} | P | \phi_n \rangle|. \quad (6.177)$$

It is noted that $\text{tr}(X^\dagger A) = |\text{tr}(X^\dagger A)|$, as it is non-negative. Now, the k th Ky-fan norm [140] can be written as

$$\|P\|_{k\text{-KF}} = \sup_Z \{|\text{tr}(Z^\dagger P)| : \|Z\|_{(k)}^* \leq 1\}, \quad (6.178)$$

where $\|Z\|_{(k)}^*$ is the dual of the k th Ky-Fan norm [41] and Z is an operator on the Hilbert space. By definition, the dual of the k th Ky-Fan norm is

$$\|Q\|_{(k)}^* := \sup_Y \{|\text{tr}(Y^\dagger Q)| : \|Y\|_{k\text{-KF}} \leq 1\}, \quad (6.179)$$

where Y is an operator on the Hilbert space. To write the k th Ky-Fan norm in this way, we have used the fact that taking the dual of a norm's dual returns the norm [41].

Using the above definitions, it can be stated that

$$\sum_n |\langle \psi_{n+j} | P | \phi_n \rangle| = |\text{tr}(X^\dagger P)| \leq \|P\|_{k\text{-KF}} \quad \text{if} \quad \|X\|_{(k)}^* \leq 1. \quad (6.180)$$

To see that $\|X\|_{(k)}^* \leq 1$, we explicitly evaluate it. Firstly, one can find the eigenvalues of $\sqrt{X^\dagger X}$, as these are the singular values of X . From this, it can be seen that X has k singular values that are one, with the remaining being zero.

Now, let $\|X\|_{(k)}^* = |\text{tr}[(Y^*)^\dagger X]|$, such that Y^* is the operator satisfying Eq. (6.179). The von Neumann's trace inequality [143] can then be employed to state that

$$\begin{aligned} \|X\|_{(k)}^* &= |\text{tr}[(Y^*)^\dagger X]| \leq \sum_{i=0}^{\min\{n,m\}} \mu_i(X) \mu_i(Y^*) \\ &= \sum_{i=0}^{k-1} \mu_i(Y^*) =: \|Y^*\|_{k\text{-KF}} \leq 1, \end{aligned} \quad (6.181)$$

where, for an operator C , the set $\{\mu_i(C)\}_{i=0}^{\min\{n,m\}}$ is the singular values of C ordered in decreasing order, such that $\mu_i(C) \geq \mu_i(C) \forall i$. To go from the first to second line, we have used the fact that all non-zero singular values of X are one. We have then used the definition of the k th Ky-fan norm: it is the sum over the k largest singular values. The final inequality comes from the definition of the dual norm of the k th Ky-Fan norm, where Y was restricted to have $\|Y\|_{k\text{-KF}} \leq 1$ by definition. Hence, $\|X\|_{(k)}^* \leq 1$, meaning Eq. (6.180) hold, completing the proof. $\square$

We now restate Result 6.3.6 before we begin the proof.

Result 6.3.6 Let $d^{(j)} := \dim(\mathcal{V}_{\text{in}}^{(j)})$. For any finite-dimensional state ρ , we have

$$\Delta M_A^{(j)} \leq \|(\rho_A \otimes \rho_B)^{(j)}\|_{d^{(j)}\text{-KF}} - \|\rho^{(j)}\|_1, \quad (6.182)$$

where $\|P\|_{\alpha\text{-KF}}$ is the α -Ky-Fan norm of the operator P , which is the sum of the α largest singular values of P .

Proof. The j th bipartite mode subspace, $\mathcal{B}_{AB}^{(j)}$, has $d^{(j)} := \dim(\mathcal{V}_{\text{in}}^{(j)})$ basis operators whose coefficients will contribute to the local mode upon performing the partial trace. The rough goal of coherence concentration is to maximise the value of these coefficients under conjugation by an allowed unitary. It is noted that each basis operator in $\mathcal{V}_{\text{in}}^{(j)}$ is in a distinct row and column along a diagonal of the j th bipartite mode operator.

Now, $M^{(j)}(\sigma_A) := \|\sigma_A^{(j)}\|_1$ is the sum of the absolute values of the matrix elements of σ_A in the j th mode. Each matrix element of $\sigma_A^{(j)}$ is then a sum over some subset of the coefficients of the basis states in

$\mathcal{V}_{\text{in}}^{(j)}$ for the j th bipartite mode $\sigma_{AB}^{(j)}$. Hence, using the triangle inequality, it can be seen that

$$M^{(j)}(\sigma_A) \leq \sum_{n,m} |\langle n+j, m | \sigma_{AB}^{(j)} | n, m \rangle|, \quad (6.183)$$

where $\langle n+j, m | \sigma_{AB}^{(j)} | n, m \rangle$ are the $d^{(j)}$ coefficients of $\sigma_{AB}^{(j)}$ with respect to the basis operators in $\mathcal{V}_{\text{in}}^{(j)}$. This upper bound is then a sum over the absolute values of the matrix elements of a diagonal of the j th bipartite mode operator, $\sigma_{AB}^{(j)}$, in the eigenbasis of L_{AB} . Lemma 6.6.2 can then be employed to imply that

$$\sum_{n,m} |\langle n+j, m | \sigma_{AB}^{(j)} | n, m \rangle| \leq \|\sigma_{AB}^{(j)}\|_{d^{(j)}\text{-KF}} = \|(\rho_A \otimes \rho_B)^{(j)}\|_{d^{(j)}\text{-KF}}, \quad (6.184)$$

as the Ky-Fan norm is unitarily invariant. Taking away $M^{(j)}(\rho_A)$ from both sides completes the proof. $\square$

6.6.6 Proof of Result 6.3.7

Here, we present a proof of Result 6.3.7, which is first stated for clarity.

Result 6.3.7 Let $d^{(j,c)} := \dim(\mathcal{V}_{\text{in}}^{(j,c)})$. For any d -dimensional state ρ , we have

$$\Delta M_A^{(j)} \leq \sum_{c=0}^{2d-2-j} \|\Pi_{c+j}(\rho \otimes \rho)^{(j)}\Pi_c\|_{d^{(j,c)}\text{-KF}} - \|\rho^{(j)}\|_1, \quad (6.185)$$

Proof. To prove this, one firstly needs to show that each LR-D subspace transforms independently. From here, the proof of Result 6.3.6 can then be applied to each individual space. To see the independence of each LR-D subspace, recall that an allowed unitary will apply independent unitaries within each degenerate subspace of the operator defining our chosen basis. In the concentration scenario, a global allowed unitarity V_{AB} therefore acts as an independent unitary within each degenerate space of the operator $L_{AB} = \sum_c \lambda_c \Pi_c$ with spectrum $\{\lambda_c\}_c$. More precisely, for each c , there exist unitary operators V_c such that

$$V_{AB} = \sum_{c=0}^{2d-2} V_c \Pi_c, \quad (6.186)$$

where the projector Π_c is given by

$$\Pi_c = \sum_{n+m=k+j=c} |nm\rangle \langle kl|. \quad (6.187)$$

A j th mode operator, as seen in the main text, is some operator supported in the j th mode subspace,

$$\sigma_{AB}^{(j)} \in \mathcal{B}_{AB}^{(j)} = \text{span}\{|n+k, m+j-k\rangle \langle n, m| : \forall n, m, k\}, \quad (6.188)$$

such that

$$\sigma_{AB}^{(j)} = \sum_k \sum_{n,m} q_{n+k, m+j-k, n, m} |n+k, m+j-k\rangle \langle n, m| = \sum_{n, \alpha, \beta} \tilde{q}_{n+j, n, \alpha, \beta} |n+j, \alpha\rangle \langle n, \beta|, \quad (6.189)$$

where we have performed a relabelling to reflect the degeneracies in L_{AB} , using the dummy indices α and β to label the degeneracies. From here, it can be seen that

$$\begin{aligned} V_{AB} \sigma_{AB}^{(j)} V_{AB}^\dagger &= \left(\sum_{c=0}^{2d-2} V_c \Pi_c \right) \sigma_{AB}^{(j)} \left(\sum_{e=0}^{2d-2} \Pi_e V_e^\dagger \right) = \sum_{c,e=0}^{2d-2} \sum_{n, \alpha, \beta} \tilde{q}_{n+j, n, \alpha, \beta} V_c \Pi_c |n+j, \alpha\rangle \langle n, \beta| \Pi_e V_e^\dagger \\ &= \sum_{n, \alpha, \beta} \tilde{q}_{n+j, n, \alpha, \beta} V_{n+j} |n+j, \alpha\rangle \langle n, \beta| V_n^\dagger = \sum_n V_{n+j} \Pi_{n+j} \sigma_{AB}^{(j)} \Pi_n V_n. \end{aligned} \quad (6.190)$$

where in the first line we have used the fact that $\Pi_c |n, \alpha\rangle = \delta_{c,n} |n, \alpha\rangle \forall \alpha$, and in the second line we have used

$$\Pi_{c+j} \sigma_{AB}^{(j)} \Pi_c = \sum_{\alpha, \beta} \tilde{q}_{c+j, c, \alpha, \beta} |c+j, \alpha\rangle \langle c, \beta|. \quad (6.191)$$

As the unitary V_c maps states within the c th degenerate subspace back to the c th degenerate subspace, it can be concluded that each $\Pi_{c+j} \sigma_{AB}^{(j)} \Pi_c$, that is, each LR-D subspace, transforms independently under allowed unitaries V_{AB} . We note that the c th degenerate subspace of L_{AB} has dimension

$$d(c, L_{AB}) := \begin{cases} c+1 & 0 \leq c < d-1, \\ 2d-1-c & d-1 \leq c \leq 2d-2. \end{cases} \quad (6.192)$$

Hence, the $(c+j, c)$ -LR-D operator has its input and output dimensions as $d(c+j, L_{AB})$ and $d(c, L_{AB})$, respectively. Also, there are $2d-2-j$ LR-D operators within the j th global mode.

Returning to the original labelling, one can define the $(c+j, c)$ -LR-D subspace of the j th mode space as

$$\mathcal{B}_{AB}^{(j, c)} := \text{span}\{|n+k, m+j-k\rangle \langle n, m| : n+m=c\} \subset \mathcal{B}_{AB}^{(j)}. \quad (6.193)$$

As before, this can be split into two subspaces, one where the components make a contribution to the local mode and where they do not

$$\begin{aligned} \mathcal{V}_{\text{in}}^{(j, c)} &:= \text{span}\{|n+j, m\rangle \langle n, m| : n+m=c\} \subseteq \mathcal{B}_{AB}^{(j, c)}, \\ \mathcal{V}_{\text{out}}^{(j, c)} &:= \text{span}\{|n+k, m+j-k\rangle \langle n, m| : n+m=c, k \neq j\} \subseteq \mathcal{B}_{AB}^{(j, c)}, \end{aligned} \quad (6.194)$$

where the aim of mode concentration is still to maximise the coefficients with respect to the basis operators in $\mathcal{V}_{\text{in}}^{(j, c)}$. Given that each of the LR-D subspaces evolve unitarily, the proof of Result 6.3.6 can then be applied to each LD-R subspace individually. $\square$

General Unitary Resource Enhancement: Cooling a Qubit with n others

7

7.1 Introduction

When performed under allowed operations, resource enhancement means a relocation of resources from one subsystem to another. This is due to allowed operations being resource non-generating, meaning any local increase in resource must mean resources being moved from elsewhere. If one instead considers operations outside of the allowed set, then resource enhancement can include both the relocation of existing resource, and the generation of new resources. Although, considering such operations places one firmly outside the remit of resource theories, since these theories are founded on the requirement that only resource non-generating operations are allowed: if a property of an object can be increased under operations accessible within an agent's physical limitations, then that property is not a resource to that agent.

However, in reality, these properties remain useful for performing operational tasks even under physical limitations that permit their generation. Moreover, the generation of these properties will not necessarily be a trivial task just because theoretically its possible. On the other hand, the restriction to resource non-generating operations might have been artificial, introduced only to allow resource quantifiers to be identified via the resource-theory framework, and not to reflect real physical limitations experienced in the lab. Hence, whilst the property is not technically considered a resource, it remains important to understand how such an agent could increase the amount of it they possess.

Here, we consider the specific resource enhancement tasks of increasing the athermality of a qubit given access to n other qubits and general unitary operations. Under this set of operations, athermality can be generated, and, as mentioned above, it is therefore not technically a resource. However, given a universal quantum processor — which is capable of applying any unitary operation — the creation of states with athermality remains desirable and non-trivial to achieve. Therefore, we still refer to athermality as a resource, and consider optimally enhancing this resource.

7.1.1 Motivation

The task of increasing the athermality of a quantum system using another [30, 32, 33, 35, 38, 144–146] is a foundational point of inquiry in quantum thermodynamics [70]. This resource enhancement problem has been studied in a multitude of settings — including both closed and open system dynamics — and has inspired many practical computational subroutines, such as algorithmic cooling [27–29, 31, 147, 148] and autonomous refrigerators [37, 149, 150].

Here, the task of *cooling* a thermal qubit via a unitary interaction with n other thermal qubits is investigated. Cooling, a subtask of increasing athermality, aims to specifically decrease the temperature of some target system, τ_S . To achieve this, we consider performing a joint unitary interaction on τ_S and the n other additional qubits, which we will refer to as the *machine* τ_M . This unitary cooling scenario has previously been studied in the context of state preparation for quantum computers [27–30], and can be considered a sub-routine of algorithmic cooling. Typically, these prior works have assumed that each of the n thermal qubits in the system and machine are identical, meaning they are each governed by the same Hamiltonian,

$$\tau_S \otimes \tau_M = \bigotimes_{i=1}^{n+1} \tau, \text{ where } \tau = \frac{e^{-\beta H}}{\mathcal{Z}}, \mathcal{Z} = \text{tr}[e^{-\beta H}], \beta = \frac{1}{k_B T}, \quad (7.1)$$

where H is some qubit Hamiltonian, which is identical for each qubit, T is the temperature and k_B is the Boltzmann constant. In addition, they have allowed the thermal qubits to be refreshed between successive rounds of cooling via unitaries [31–33], with asymptotic bounds on the cooling found in this case. Alternately, machines of a fixed size (meaning no qubits refreshed) have been considered, but they have either been asymptotically large [34–36], very small [37, 38] and/or used identical qubits [27–30].

Unitary cooling using machines of an arbitrary, but fixed size, with non-identical qubits, has therefore not been explored. This is despite this being the most general type of fixed size qubit machine that could be considered.

We here consider such finite sized machines of n non-identical qubits, meaning we allow each qubit to be governed by a different Hamiltonian,

$$\tau_S \otimes \tau_M = \tau_S \otimes \tau_{M_1} \otimes \tau_{M_2} \otimes \tau_{M_3} \otimes \dots \otimes \tau_{M_n}, \text{ where } \tau_S = \frac{e^{-\beta_S H_S}}{\mathcal{Z}_S}, \tau_{M_i} = \frac{e^{-\beta_{M_i} H_{M_i}}}{\mathcal{Z}_{M_i}}. \quad (7.2)$$

Specifically, we allow the Hamiltonian of each qubit to have a different energy gap — effectively corresponding to a different temperature (see Section 7.2.1). This, firstly, enables an understanding of how the energetic structure of the n thermal qubits impacts the cooling scenario. By removing the ability for the machine qubits to re-thermalise, we further isolate the impact of the energetic structure on the machines ability to cool. Secondly, this brings the unitary cooling scenario closer to practical reality given that physical qubits are unlikely to be identical in real hardware, whether unintentionally or by design. In general, by considering cooling protocols in this scenario i.e., unitary cooling with finite size qubit machines and no re-thermalising, we ensure that they can be physically implement as a single quantum circuit. See Fig. 7.1 for a schematic of the system and machine being considered.

However, heuristically speaking, this is a solved problem — even in the case of finite sized machines with non-identical qubits. One only needs to order the eigenvalues of the state of the combined system and machine in non-increasing order. The optimal ground state population that can be reached in the system — the singular parameter dictating the temperature of a qubit — is then just the sum of the largest half of these eigenvalues. To physically achieve the cooling, one just needs to perform this ordering via implementing an appropriate permutation on the system and machine eigenvalues. This has long been considered part of the quantum thermodynamics folklore [70]; although, little attention has ever been paid to the details of the dynamics needed to implement this cooling. Namely, what form does this ordering take and how does this depend on the energetic structure of the machine? This is despite the potential for vast differences in their practical application. For instance, a given set of two level permutations could simplify down to a single full swap between the system and one of the n machine qubits. Alternatively, it could require an operation controlled on many machine qubits. In the work presented in this chapter we have aimed to address this gap by presenting a framework that not only identifies the necessary permutation for optimal cooling, from which the machine's ability to cool can be quantified, but also evaluates the difficulty of implementing it.

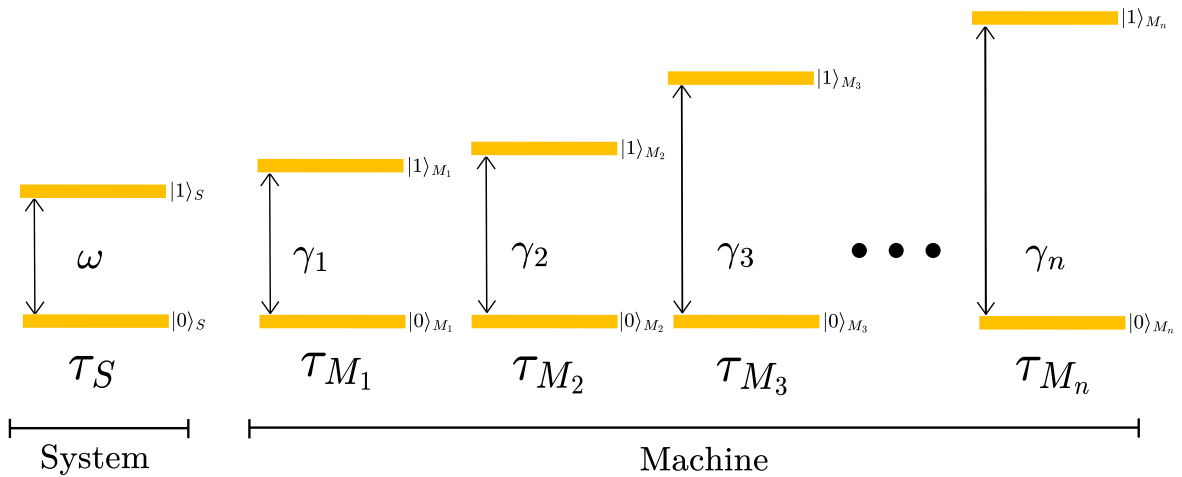


Figure 7.1: A schematic of the system and machine being considered in this unitary cooling scenario. The system consist of a single qubit governed by a Hamiltonian with energy gap ω . The machine consists of n qubits, where each of the n qubits is governed by a Hamiltonian with a different energy gap, with the i th qubit having a Hamiltonian with energy gap γ_i .

7.1.2 Structure

This chapter is organised as follows: firstly, in Section 7.2 a simple three qubit example is presented, which exposes the main insights and motivations behind this work. In Section 7.3 a set of inequalities is introduced that fully characterise the unitary cooling scenario and the optimal amount by which an agent can cool. From this, a general bound is placed on the agents ability to cool in Section 7.4. In Section 7.5 the notion of machine reducibility is introduced, which categories which cooling scenarios can be simplified. A detailed four qubit example is then given in Section 7.6, where all our introduced tools are demonstrated. The cooling protocol is then extended beyond thermal qubits to arbitrary states in Section 7.7, where its adaption makes it into a general purity enhancement protocol. Finally, in Section 7.8, cooling in our framework is considered when our dynamics are restricted by some symmetry.

This work was done in collaboration with Jake Xuereb, Alberto Rolandi, Jinming He, Marcus Huber, and Pharnam Bakhshinezhad. All resulted presented in this section I had a significant contribution to, either mathematically, conceptually or both. The result presented here, and more, contribute to the manuscript [2].

7.2 Motivating Example

7.2.1 What does it mean to cool a qubit?

Consider a qubit, $\rho \in \mathcal{D}(\mathcal{H}^2)$, with Hamiltonian $H = E_g |g\rangle\langle g| + E_e |e\rangle\langle e|$. If ρ is diagonal with respect to the energy eigenbasis — the eigenbasis of H — then τ can always have a temperature associated to it via the Gibbs state,

$$\tau = \frac{|g\rangle\langle g| + e^{-\beta(E_e - E_g)} |e\rangle\langle e|}{1 + e^{-\beta(E_e - E_g)}}, \quad \beta = \frac{1}{k_B T}, \quad (7.3)$$

To achieve this, the ground state population of ρ is equated to that of τ , from which a temperature can be found. We note that we allow this temperature to be negative, such that states with a population inversion can also written in this way *. Due to all qubits being able to be written in this form, the goal of cooling can be rephrased as increasing the ground state population of the qubit, as this effectively lowers T when H is fixed. This can also be seen to cause an increasing in the purity of the qubit. Moreover, consider two qubits with Hamiltonians H and H' , with energy gaps $E_e - E_g$ and $E'_e - E'_g$, respectively, such that $E'_e - E'_g > E_e - E_g$. At a fixed temperature, the ground state population of the qubit with Hamiltonian H' will be larger than the ground state population of the qubit with Hamiltonian H . Hence, when considered with respect to the energetic structure of the H governed qubit, the H' governed qubit effectively appears colder. We therefore refer to qubits with larger energy gaps as being “effectively colder”, with this alluding to an “effective temperature” with respect to the qubit with the smallest energy gap. In addition, and without loss of generality, we take $\hbar = 1$ and fix the computational basis to be $|g\rangle\langle g| = |0\rangle\langle 0|$, $|e\rangle\langle e| = |1\rangle\langle 1|$. We denote the partition function of τ as $\mathcal{Z} = \text{tr}[e^{-\beta H}]$.

7.2.2 An $n = 2$ qubit machine

Before introducing the general case, consider an example of three qubits, all with $\beta = 1$: a system qubit with energy gap ω , $\tau_S = \frac{1}{1+e^\omega}(|0\rangle\langle 0| + e^{-\omega} |1\rangle\langle 1|)$, and two machine qubits, $\tau_{M_1} = \frac{1}{1+e^{-\gamma_1}}(|0\rangle\langle 0| + e^{-\gamma_1} |1\rangle\langle 1|)$ and $\tau_{M_2} = \frac{1}{1+e^{-\gamma_2}}(|0\rangle\langle 0| + e^{-\gamma_2} |1\rangle\langle 1|)$ with energy gaps γ_1 and γ_2 respectively. We assume that $\omega \leq \gamma_1 \leq \gamma_2$, such that M_2 could be effectively colder than M_1 that could be effectively colder than S . Such an ordering is always possible if one aims to cool the hottest qubit. We now seek to understand the following two points:

- By how much can S be cooled via a joint unitary on S, M_1 , and M_2 for different M_1 and M_2 .
- What operations are required to achieve this optimal cooling.

* Once a temperature has been assigned to a qubit in this way, to give the temperature physical meaning, one assumes access to some arbitrarily large heat bath at this temperature from which Gibbs states at the temperature can be freely extracted without effecting the bath. This is a typically used notion of a bath in quantum thermodynamics and allows the temperature to be a property of some macroscopic object, which is a well defined notion.

The state of the system and the machine can be expressed as a diagonal matrix in the joint energy eigenbasis as

$$\tau_S \otimes \tau_M = \sum_{i_S, i_{M_1}, i_{M_2} \in \{0,1\}} \frac{e^{-(i_S \omega + i_{M_1} \gamma_1 + i_{M_2} \gamma_2)}}{\mathcal{Z}_S \mathcal{Z}_M} |i_S i_{M_1} i_{M_2}\rangle \langle i_S i_{M_1} i_{M_2}| \quad (7.4)$$

$$= \frac{1}{\mathcal{Z}_S \mathcal{Z}_M} \text{diag} \left(\underbrace{1, e^{-\gamma_2}, e^{-\gamma_1}, e^{-(\gamma_1 + \gamma_2)}}_{|0_S i_M\rangle : i_M \in \{0,1\}^2}, \underbrace{e^{-\omega}, e^{-(\omega + \gamma_2)}, e^{-(\omega + \gamma_1)}, e^{-(\omega + \gamma_1 + \gamma_2)}}_{|1_S i_M\rangle : i_M \in \{0,1\}^2} \right), \quad (7.5)$$

where $\mathcal{Z}_S = (1 + e^{-\omega})$ and $\mathcal{Z}_M = (1 + e^{-\gamma_1})(1 + e^{-\gamma_2})$. It proves useful here to adopt a bitwise product to denote the coefficients of this matrix. Specifically, we introduce the object $\Gamma = (\gamma_1, \gamma_2)$, which we term the *gap vector*, and define its product with the 2-bit-string $i_M = i_{M_1} i_{M_2}$ that labels a given basis state as $i_M \cdot \Gamma = i_{M_1} \gamma_1 + i_{M_2} \gamma_2$. It is noted that these coefficients of $\tau_S \otimes \tau_M$ are the eigenvalues of the state. Any unitary operation $U(\tau_S \otimes \tau_M)U^\dagger$ must therefore preserve these coefficients, as the spectrum of a state remains unchanged under unitary conjugation. We will refer to the coefficient currently associated to a given basis state as its *population*. With this, we are in a position to set the scene.

Initially, the ground state population of S is $p_0 = \sum_{i_M} \frac{e^{-i_M \cdot \Gamma}}{\mathcal{Z}_M}$, with it in general being obtained by summing over the populations associated to the basis states $|0_S i_M\rangle \langle 0_S i_M| : i_M \in \{0,1\}^{\times 2}$. The initial excited state population of S is then $1 - p_0 = \sum_{i_M} e^{-(\omega + i_M \cdot \Gamma)}$, similarly found in general by summing over the population of $|1_S i_M\rangle \langle 1_S i_M| : i_M \in \{0,1\}^{\times 2}$. When optimal cooling S under a joint unitary on S, M_1 and M_2 , our aim is therefore to maximise the sum over the populations of $|0_S i_M\rangle \langle 0_S i_M| : i_M \in \{0,1\}^{\times 2}$.

To have a notion of temperature, a qubit must be diagonal with respect to its Hamiltonian, as discussed above. As we are considering cooling — lowering the qubits temperature — we therefore require that our output system state is diagonal in the energy eigenbasis, such that a temperature can be assigned to it. We place no such restrictions on coherences being created within the machine. However, it can be shown that when aiming to optimally increase the ground state population of the system qubit, any coherences in the energy eigenbasis of the output system and machine state will provide no advantage (see Section 7.10.1 for a proof). We can therefore assume without loss of generality that the output state is diagonal in the energy eigenbasis. Since $\tau_S \otimes \tau_M$ is initially diagonal in the energy eigenbasis, and the output state can be optimally cooled whilst diagonal in the energy eigenbasis, it is therefore sufficient to consider the subset of unitaries that are *permutations* of the energy eigenbasis when performing optimal cooling.

To achieve cooling with these permutations, one must ensure that the populations permuted to the basis states $|0_S i_M\rangle \langle 0_S i_M| : i_M \in \{0,1\}^{\times 2}$ lead to their sum being greater than p_0 . To achieve *optimal* cooling with these permutations, one must ensure that *all* the largest populations are permuted to these basis states. Hence, the optimal ground state population of S that can be achieved is the sum over the four largest populations of $\tau_S \otimes \tau_M$ in the energy eigenbasis. From here, one can determine by how much S can be cooled via a joint unitary interaction on S, M_1 and M_2 . It remains to determine what dynamics are needed to perform this optimal cooling. Specifically, we seek a permutation that assigns the four largest populations to the basis states $|0_S i_M\rangle \langle 0_S i_M| : i_M \in \{0,1\}^{\times n}$.

To do this, we first note that all permutations can be broken down into a product of two-level permutations (TLP), which are permutations that permute at most two populations at a time i.e., they swap two populations. Hence, an optimal cooling unitary can be characterised by some set of TLPs. Moreover, it can easily be seen that any TLP of form $|0_S i_M\rangle \leftrightarrow |0_S j_M\rangle$ or $|1_S i_M\rangle \leftrightarrow |1_S j_M\rangle$ for all i_M, j_M leads to no change in the populations of S , and hence are unable to perform any cooling. Only TLPs of the form $|0_S i_M\rangle \leftrightarrow |1_S j_M\rangle$ are therefore able to cool. However, not even all TLP of this form facilitate cooling:

Definition 7.2.1. (Cooling TLP) A TLP of the form $|0_S i_M\rangle \leftrightarrow |1_S j_M\rangle$ is cooling if and only if

$$e^{-i_M \cdot \Gamma} < e^{-(\omega + j_M \cdot \Gamma)}, \quad (7.6)$$

as the new population associated to the basis state $|0_S i_M\rangle\langle 0_S i_M|$ will be larger, meaning the ground state population of S will be larger.

For a given population of $|0_S i_M\rangle\langle 0_S i_M|$, there will in general be many TLPs involving it that are cooling. Although, not all cooling TLPs are necessary for optimal cooling; a TLP could be cooling whilst also moving one of the four largest populations away from $|0_S i_M\rangle\langle 0_S i_M| : i_M \in \{0, 1\}^{\times 2}$. A possible strategy to correctly order the four largest populations is to consider iteratively permuting the largest population in $|1_S j_M\rangle\langle 1_S j_M| : i_M \in \{0, 1\}^{\times 2}$ with the smallest population in $|0_S i_M\rangle\langle 0_S i_M| : i_M \in \{0, 1\}^{\times 2}$. We term such a strategy the *Eigenvalue Cooling Algorithm*, and it is detailed in Proposition 7.2.1 (in the literature, cooling a quantum system via a unitary has typically consisted of just this step, as mentioned in the introduction of this chapter).

We now implement the Eigenvalue Cooling Algorithm using the assumption of a non-decreasing gap structure, $\omega \leq \gamma_1 \leq \gamma_2$. The set of populations of $|0_S i_M\rangle\langle 0_S i_M| : i_M \in \{0, 1\}^{\times 2}$ ordered in non-increasing order is then

$$1 \geq e^{-\gamma_1} \geq e^{-\gamma_2} \geq e^{-(\gamma_1 + \gamma_2)}. \quad (7.7)$$

Given the $|1_S i_M\rangle\langle 1_S i_M|$ populations are the $|0_S i_M\rangle\langle 0_S i_M|$ populations multiplied by the constant $e^{-\omega}$, they too are ordered in the same way. The TLP that exchanges the largest and smallest populations in each of these sets is therefore cooling if $e^{-(\gamma_1 + \gamma_2)} < e^{-\omega}$, as per Definition 7.2.1. This can be seen to hold if $\omega < \gamma_1 + \gamma_2$, which is always true due to the non-decreasing gap structure. Hence, the TLP $|0_S 11\rangle \leftrightarrow |1_S 00\rangle$ is always cooling. Notably, its also the single TLP that leads to the largest amount of cooling.

One can then check the next largest in the set of populations of $|1_S j_M\rangle\langle 1_S j_M|$ against the next smallest in the set of populations of $|0_S i_M\rangle\langle 0_S i_M|$. The associated TLP for this is cooling if $e^{-\gamma_2} < e^{-(\omega + \gamma_1)}$, which holds if $\omega < \gamma_2 - \gamma_1$. This will not always hold, and will depend on the energetic structure of the machine. If it does not hold, no other TLP will be cooling either, due to the ordering of the populations. If it does hold, one can then consider if $e^{-\gamma_1} < e^{-(\omega + \gamma_2)}$, to see if exchanging the next largest and smallest populations is cooling. This will hold if $\omega < \gamma_1 - \gamma_2$, and hence never holds, again due to the non-decreasing gap structure.

From here, one can see that $|0_S 00\rangle\langle 0_S 00|$ and $|0_S 01\rangle\langle 0_S 01|$ already hold two of the four largest populations of $|0_S i_M\rangle\langle 0_S i_M| : i_M \in \{0, 1\}^{\times 2}$. Moreover, $|1_S 00\rangle\langle 1_S 00|$ holds one of the four largest, and the TLP $|0_S 11\rangle \leftrightarrow |1_S 00\rangle$ always ensures its correct placement. This leaves just one final population, with two possible cases:

- (a) If $\omega \geq \gamma_2 - \gamma_1$ then $|0_S 01\rangle\langle 0_S 01|$ holds the fourth largest population already
- (b) If $\omega < \gamma_2 - \gamma_1$ then $|1_S 10\rangle\langle 1_S 10|$ holds the fourth largest population and the TLP $|0_S 10\rangle \leftrightarrow |1_S 01\rangle$ is needed to ensure its correct placement.

Before considering each case individually, we give an example of a TLP that is cooling, but not necessary for optimal cooling; $|0_S 10\rangle \leftrightarrow |1_S 00\rangle$ is cooling, as $\omega < \gamma_1$, but it swaps one of the four largest populations away from $|0_S i_M\rangle\langle 0_S i_M| : i_M \in \{0, 1\}^{\times 2}$.

Now, in case a) the only TLP that must be applied is $|0_S 11\rangle \leftrightarrow |1_S 00\rangle$. The unitary

$$S_{011,100} = |011\rangle\langle 100| + |100\rangle\langle 011| + \mathbb{I}_{\text{Rest}}, \quad \mathbb{I}_{\text{Rest}} = \sum_{k \in \{0,1\}^3} |k\rangle\langle k| - |011\rangle\langle 011| - |100\rangle\langle 100|, \quad (7.8)$$

therefore optimally cools S . The final ground state population of the system, $p_0^{(a)}$, is then

$$p_0^{(a)} = \frac{1}{\mathcal{Z}_S \mathcal{Z}_M} (1 + e^{-\gamma_2} + e^{-\gamma_2} + e^{-\omega}). \quad (7.9)$$

It can be seen that $p_0^{(a)}$ is larger than the initial ground state population of M_2 — the effective coldest

qubit — if $p_0^{(a)} > 1/(1 + e^{-\gamma_2})$. This holds if $\omega > \gamma_2 - \gamma_1$, which is the criteria for scenario a), and hence is always true here. Therefore, in case a) it is possible to cool below the lowest effective temperature of all the qubits i.e., increase the ground state population below that of the machine qubit governed by the Hamiltonian with the largest energy gap.

In case b) both $|0_S 11\rangle \leftrightarrow |1_S 00\rangle$ and $|0_S 01\rangle \leftrightarrow |1_S 10\rangle$ must be implemented. With both conditions fulfilled, i_{M_1} becomes a free index such that any unitary that performs the TLPs

$$|0_S i_{M_1} 1\rangle \leftrightarrow |1_S i_{M_1} 0\rangle : i_{M_2} \in \{0, 1\}, \quad (7.10)$$

will optimally cool S . A full swap between the subsystems S and M_2 , for exmaple, will achieve this. The final ground state population in this case, $p_0^{(b)}$, is therefore implicitly that of M_2 i.e., $p_0^{(b)} = 1/(1 + e^{-\beta\gamma_2})$.

From here, it can seen that in case b), where the energetic gap of S is smaller than the difference between the energy gaps of the machine qubits, the state in subsystem M_1 does not contribute to the optimal cooling. The dynamics for performing optimal cooling is a purely bipartite cooling scenario where a qubit in the machine is left unused. On the other hand, in case a), where the energetic gap of S is larger than the difference between the energy gaps of the machine qubits, cooling beyond the machine qubit with the lowest effective temperature is possible. Although, given access to the gate set $\mathcal{G} = \{\mathbf{X}, \text{CNOT}, \text{TOFFOLI}\}$, as defined in Definition 7.2.2, the dynamics for performing this optimal cooling are necessarily tripartite due to the optimal cooling permutation being odd. See Observation 7.2.2 and Section 7.10.2 for more details. Considering such a gate set is natural, as it is universal for revisable classical computing on three bits [151].

In this example, it can be seen that even cooling a single qubit with access to two other qubits seems to lead to an intricate set of scenarios which, whilst complex, is seemingly captured by a simple set of relationships between the energy gaps of the system and machine. In the next section, we will generalise this to the n -qubit machine case.

Definition 7.2.2 (Gate Set $\mathcal{G}$). Let $\mathcal{G} = \{\mathbf{X}, \text{CNOT}, \text{TOFFOLI}\}$ be a gate set, where

$$\begin{aligned} \mathbf{X} &= |0\rangle\langle 1| + |1\rangle\langle 0| \in \mathcal{L}(\mathcal{H}^2) \\ \text{CNOT} &= |0\rangle\langle 0|_A \otimes \mathbb{I}_B + |1\rangle\langle 1|_A \otimes \mathbf{X}_B \in \mathcal{L}(\mathcal{H}_A^2 \otimes \mathcal{H}_B^2) \\ \text{TOFFOLI} &= (\mathbb{I}_{AB} - |11\rangle\langle 11|_{AB}) \otimes \mathbb{I}_C + |11\rangle\langle 11|_{AB} \otimes \mathbf{X}_C \in \mathcal{L}(\mathcal{H}_A^2 \otimes \mathcal{H}_B^2 \otimes \mathcal{H}_C^2). \end{aligned} \quad (7.11)$$

Proposition 7.2.1 (Eigenvalue Cooling Algorithm). Given that the initial state of the system and machine is diagonal in the energy eigenbasis, all the populations can be collected into two vectors $\mathbf{q}_0$ and $\mathbf{q}_1$ such that

$$p_0 = \sum_{i=1}^{2^n} q_i, \quad q_i \in \mathbf{q}_0, \quad p_1 = 1 - p_0 = \sum_{i=1}^{2^n} l_i, \quad l_i \in \mathbf{q}_1, \quad (7.12)$$

where p_0 and p_1 are the initial ground and excited state populations of the system qubit. A heuristic way to perform optimal cooling would be to first order $\mathbf{q}_0$ and $\mathbf{q}_1$ in non-decreasing order, giving $\mathbf{q}_0^\downarrow$ and $\mathbf{q}_1^\downarrow$ respectively. One can then compare the first component of $\mathbf{q}_1^\downarrow$ with the final component of $\mathbf{q}_0^\downarrow$ and perform a permutation between the two levels if the first component of $\mathbf{q}_1^\downarrow$ is greater than the final component of $\mathbf{q}_0^\downarrow$. If valid, this will perform the optimal exchange for increasing the ground state population of the system qubit in a single two level permutation. Next, one compares the second component of $\mathbf{q}_1^\downarrow$ with the second to last component of $\mathbf{q}_0^\downarrow$ and performs a permutation between the two levels if the second component of $\mathbf{q}_1^\downarrow$ is greater than the second to last component of $\mathbf{q}_0^\downarrow$, and so on. Each comparison of components will lead to a two-level permutation that represents the new optimal energy level exchange for increasing the ground state population (given that the previous

permutations have been performed). If one reaches the k th component of $\mathbf{q}_1^\downarrow$ and it is not greater than the $(2^n - k)$ th component of $\mathbf{q}_0^\downarrow$, one has successfully moved the populations such that the ground state population of the system qubit is maximised and, hence, they have performed optimal cooling via unitary dynamics. See Fig. 7.2 for a pictorial depiction of the Eigenvalue Cooling Algorithm

In total, one swaps the k largest elements of $\mathbf{p}_1$ with the k smallest elements of $\mathbf{p}_0$, where the size of k is dependent on the system and machine being used. Note, once the k elements that must be swapped are known, it does not matter for obtaining the largest attainable ground state population which of the k elements of $\mathbf{q}_0$ are swapped for the k elements of $\mathbf{q}_1$. All that is important is that all k elements are swapped.

Observation 7.2.2. When considering a three qubit Hilbert space, the set of all permutations on the computational basis states are decomposable into the gate set $\mathcal{G} = \{\mathbf{X}, \text{CNOT}, \text{TOFFOLI}\}$. A permutation is said to be *even* if the minimum number of TLPs it can be decomposed into is even, and vice versa if odd. Using only the single and bipartite gates, $\mathbf{X}$ and CNOT respectively, only even permutations can be implemented on the three qubit space [151]. In fact, not even all the even permutations can be implemented using only these bipartite gates. The tripartite TOFFOLI gate, on the other hand, is able to implement odd permutations. Specifically, in conjunction with $\mathbf{X}$, it is able to implement all individual TLPs on the three qubit space. As a result, using $\mathcal{G}$ all permutations on the three qubit space can be implemented.

Given the optimal cooling unitary $S_{011,100}$, detailed in Eq. 7.8, is composed of a single TLP, it is an odd permutation. Hence, any decomposition of $S_{011,100}$ into gates in $\mathcal{G}$ must include at least one TOFFOLI gate, and it is therefore necessarily a tripartite unitary. A potential strategy one could consider to make the optimal cooling in scenario (a) achievable by bipartite interactions is to add an additional unnecessary TLP e.g., $|0_S 00\rangle \leftrightarrow |0_S 10\rangle$, to make the applied permutation even. Once the permutation is even, it may be decomposable in terms of only $\mathbf{X}$ and CNOT gates. However, we show in Section 7.10.2 that this is not possible in this three qubit case, and that when using the gate set $\mathcal{G}$, some form of tripartite interaction is genuinely necessary to perform optimal cooling.

A well known separation of quantum computing and reversible classical computing is the existence of a universal gate set with bipartite gates rather than at least tripartite gates [152]. Hence, given access to the gate set $\{\mathbf{X}, \mathbf{H}, \mathbf{T}, \text{CNOT}\}$ — a universal gate set for quantum computing — one could perform optimal cooling in scenario (a) using only bipartite interactions. This is in comparison to using a classical reversible computing gate set, such as $\mathcal{G}$, where a tripartite interaction is necessary for performing optimal cooling. Therefore, a reduction in the necessary number of systems that need to be simultaneously interacted with to perform optimal cooling in this $n = 2$ qubit machine scenario is a thermodynamic consequence of this separation between quantum computing and reversible classical computing.

7.3 Framework

Armed with our bit-string notation, we now investigate the general case of an n -qubit machine, again aiming to identify which TLPs are necessary for performing optimal cooling, and by how much the system will be cooled if these TLPs are implemented. We consider a qubit with an energetic gap of ω and a temperature β_S as our system S , and n -qubits at temperature β_M with an energetic gap structure $\Gamma = (\gamma_1, \gamma_2, \dots, \gamma_n)$, such that $\gamma_i \leq \gamma_{i+1} \forall i \in \{1, N\}$, as the machine M . The system and machine are therefore initially in the state

$$\tau_S \otimes \tau_M = \sum_{\substack{i_S \in \{0,1\} \\ i_M \in \{0,1\}^{\times n}}} \frac{e^{-(\beta_S i_S \omega + \beta_M i_M \cdot \Gamma)}}{\mathcal{Z}_S \mathcal{Z}_M} |i_S i_M\rangle \langle i_S i_M|, \quad (7.13)$$

where $\mathcal{Z}_S = (1 + e^{-\beta_S \omega})$ and $\mathcal{Z}_M = \prod_{j=1}^n (1 + e^{-\beta_M \gamma_j})$, such that each population in the SM system can be identified by an $(n+1)$ -bit-string $i_S i_M \in \{0,1\}^{\times (n+1)}$.

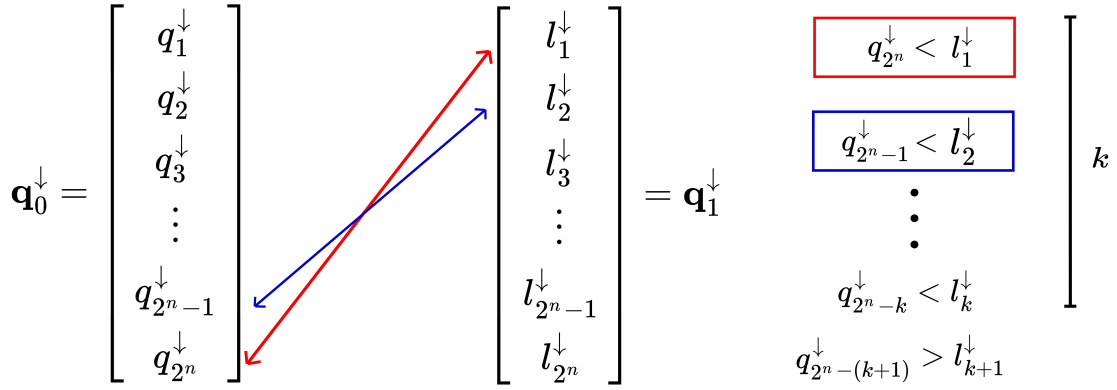


Figure 7.2: A diagram of the Eigenvalue Cooling Algorithm. The vectors of populations are first placed in non-decreasing order. The final component of the ground state population vector is compared to the first component of the excited state population vector and a swap between the levels is performed if the former is larger. This process is repeated until a point is reached where the component from the ground state population vector is larger than the component from the excited state population vector. Given at each stage the largest remaining component from the ground state population vector is swapped with the smallest remaining component of the excited state population vector, it represents the optimal (in terms of increasing the system ground state population) two-level permutation that can be performed.

Firstly, we identify the necessary set of TLP that must be performed to achieve optimal cooling. As seen in Section 7.2.2, any TLP $|0_S i_M\rangle \leftrightarrow |1_S j_M\rangle$ on $\tau_S \otimes \tau_M$ that an agent performs is cooling on S if their associated populations satisfy

$$e^{-\beta_M i_M \cdot \Gamma} < e^{-(\beta_S \omega + \beta_M j_M \cdot \Gamma)}. \quad (7.14)$$

This implies that any pair of energy eigenkets in the SM Hilbert space that when permuted cool S must satisfy

$$\frac{T_M}{T_S} \omega < (i_M - j_M) \cdot \Gamma. \quad (7.15)$$

To find the TLP's necessary for performing optimal cooling one can implement the Eigenvalue Cooling Algorithm detailed in Proposition 7.2.1, which iteratively swaps the largest populations $|1_S j_M\rangle$ with the smallest populations $|0_S i_M\rangle$. Conveniently, this always outputs TLPs of the following form:

Lemma 7.3.1. *The TLPs output from the Eigenvalue Cooling Algorithm are always of the form*

$$|0_S i_M\rangle \leftrightarrow |1_S (i_M \oplus 1)\rangle, \quad (7.16)$$

where $\oplus 1$ is bit-wise addition modulo 2 and $i_M \in \{0, 1\}^{\times n}$ is an n -bit string labelling a basis state of the machine M .

Proof Sketch. This can be proved by directly applying the Eigenvalue Cooling Algorithm to $\tau_S \otimes \tau_M$. It is shown that the vector containing the populations of the global state that contribute to the ground state population of the system, $\mathbf{q}_0$, and the vector that contains that global populations that do not contribute to the ground state population, $\mathbf{q}_1$, are equivalent up to the constant $e^{-\beta_S \omega}$. This means one can focus on just one vector. The populations in this vector are then converted to a vector of energies that generate them. One can then see from here that if $i_M \cdot \Gamma$ gives the k -largest energy of the machine, then $(i_M \oplus 1) \cdot \Gamma$ gives the $(2^n - k)$ -largest energy. Converting this notion back to the populations and once again considering both vectors, it says that if i_M labels the k th largest population in $\mathbf{q}_1$, it also labels the k th smallest population in $\mathbf{q}_0$. These are exactly the TLPs that the Eigenvalue Cooling Algorithm performs. See Section 7.10.3 for a full proof $\square$.

Using Eq. (7.15) and Lemma 7.3.1, a given TLP is therefore necessary for optimal cooling if

$$\frac{T_M}{T_S} \omega < (i_M - (i_M \oplus 1)) \cdot \Gamma, \quad (7.17)$$

which, given that

$$(i_M \oplus 1) \cdot \Gamma = \sum_{l=1}^n (i_l \oplus 1) \gamma_l = \sum_{l=1}^n \gamma_l - \sum_l i_l \gamma_l = E_{\text{Max}} - (i_M \cdot \Gamma) \quad (7.18)$$

where $E_{\text{Max}} = \sum_{l=1}^n \gamma_l$, simplifies to

$$\frac{1}{2} \left(\frac{T_M}{T_S} \omega + E_{\text{Max}} \right) < i_M \cdot \Gamma. \quad (7.19)$$

This inequality therefore detects whether the population initially associated to a basis state must be exchanged to optimally cool S . The full set of populations that must be exchanged to perform optimal cooling can then be summarised in the following result.

Result 7.3.2. *Let*

$$\mathcal{S} = \left\{ i_M : i_M \in \{0, 1\}^{\times n}, \frac{1}{2} \left(\frac{T_M}{T_S} \omega + E_{\text{Max}} \right) < i_M \cdot \Gamma \right\}. \quad (7.20)$$

To perform optimal cooling of the system S using the machine M , the populations of the following basis states must be permuted

$$\{|0_S i_M\rangle : i_M \in \mathcal{S}\} \leftrightarrow \{|1_S (i_M \oplus 1)\rangle : i_M \in \mathcal{S}\}, \quad (7.21)$$

meaning each population associated to a basis states in the left set must be permuted with a population associated to a basis states in the right set, such that the total population associated to the basis states in each set has been exchanged.

Proof. Here, we collect all n -bit-strings that satisfy Eq. (7.19) into a set $\mathcal{S}$. These are exactly the bit-strings labelling the basis states of M who's initial populations must be swapped in order to perform optimal cooling. As shown in Lemma 7.3.1, the bit-string labelling basis states of M whos initial populations these must be swapped with, is given by $\mathcal{S} \oplus 1 = \{i_M \oplus 1 : i_M \in \mathcal{S}\}$. Hence, performing the TLPs $|0_S i_M\rangle \leftrightarrow |1_S (i_M \oplus 1)\rangle : i_M \in \mathcal{S}$ is one collection of TLPs that would perform optimal cooling. As mentioned in Proposition 7.2.1, any collection of TLPs that exchanged all basis states in $\mathcal{S}$, with basis states in $\mathcal{S} \oplus 1$, will perform optimal cooling. The specific TLPs implement do not matter, only that all necessary populations are exchanged. $\square$

Result 7.3.2 details what TLPs must be implemented to perform optimal cooling by comparing the energy of levels in the machine, given by $i_M \cdot \Gamma$, to a threshold value of $\frac{1}{2} \left(\frac{T_M}{T_S} \omega + E_{\text{Max}} \right)$. Whilst giving the same set of TLPs as the Eigenvalue Cooling Algorithm, this result provides an insight to the physics of why a given TLP must be performed. Every energy level in the machine is labelled by some $i_M \cdot \Gamma$ for $i_M \in \{0, 1\}^{\times n}$, with the largest energy level being $E_{\text{Max}} = 1^n \cdot \Gamma$ and the smallest $0 = 0^n \cdot \Gamma$. As $i_M \cdot \Gamma + (i_M \oplus 1) \cdot \Gamma = E_{\text{Max}}$, it can be seen that if $i_M \cdot \Gamma$ is above $E_{\text{Max}}/2$ then $(i_M \oplus 1) \cdot \Gamma$ is below $E_{\text{Max}}/2$. From this, it can be deduced that $E_{\text{Max}}/2$ is the median of the energies in the machine. Result 7.3.2 can then be interpreted as saying that if the energy $i_M \cdot \Gamma$ is above the shifted median value $\frac{1}{2} \left(\frac{T_M}{T_S} \omega + E_{\text{Max}} \right)$ then there exists at least one energy $j_M \cdot \Gamma$ below that shifted median such that the TLP $|0_S i_M\rangle \leftrightarrow |1_S j_M\rangle$ will cool S .

We now briefly consider the number of different unitaries able to implement the collection of TLPs characterised by $\mathcal{S}$. We denote the set of energy levels to be swapped as $\mathfrak{A} = \{|0_S i_M\rangle : i_M \in \mathcal{S}\}$ and $\mathfrak{B} = \{|1_S (i_M \oplus 1)\rangle : i_M \in \mathcal{S}\}$. It can be seen that we have the freedom to pair each $|0_S i_M\rangle \in \mathfrak{A}$ with any $|1_S j_M\rangle \in \mathfrak{B}$ via a TLP and obtain the same increase in ground state population, Δp_0^* . There are therefore at least $|\mathcal{S}|!$ unitaries of the form $U_{\text{cool}} = |A\rangle\langle B| + |B\rangle\langle A| + \mathbb{1}_{\text{Rest}}$ that can cool S to the same temperature. This is because we have two lists of size $|\mathcal{S}|$. Starting with one element in $|A\rangle$ we have

$|S|$ choices to pair it with in $|B\rangle$, whereas the next element has $|S| - 1$ possibilities and so on giving $|S| \times |S| - 1 \times |S| - 2 \times \dots \times 1 = |S|!$ possible pairings in general. Other unitaries may be considered that cool optimally but also act on levels outside $|A\rangle$ without any effect on the cooling, meaning $|S|$ is a lower bound on the number of unitaries.

From Result 7.3.2 the TLPs that must be performed to achieve optimal cooling can be found. From this, we can now calculate the optimal achievable cooling of S . The figure of merit we are interested in is the optimal change in ground state population of S , $\Delta p_0^* = p'_0 - p_0$, where p'_0 is the ground state population after applying the optimal cooling unitary. To investigate Δp_0^* , we begin by noting that the ground state population after optimal cooling can be expressed as a sum of populations of unchanged energy levels and swapped energy levels

$$p'_0 = \underbrace{\sum_{i_M \in \{0,1\}^{\times n} \setminus S} \frac{e^{-\beta_M i_M \cdot \Gamma}}{\mathcal{Z}_S \mathcal{Z}_M}}_{\text{unchanged}} + \underbrace{\sum_{j_M \in S} \frac{e^{-\beta_S \omega - \beta_M (j_M \oplus 1) \cdot \Gamma}}{\mathcal{Z}_S \mathcal{Z}_M}}_{\text{swapped}}, \quad (7.22)$$

where S is a set of n -bit strings characterising the TLPs necessary for performing optimal cooling, as given in Result 7.3.2. From here, the following Corollary can be used to find Δp_0^* :

Result 7.3.3. The optimal change in the ground state population of S , denoted Δp_0^* , is given by

$$\Delta p_0^* = \sum_{j_M \in S} \frac{e^{-\beta_S \omega - \beta_M (E_{\text{Max}} - j_M \cdot \Gamma)} - e^{-\beta_M j_M \cdot \Gamma}}{\mathcal{Z}_S \mathcal{Z}_M}, \quad (7.23)$$

where the set of TLPs necessary for optimal cooling is characterised by the set of bit-string S ,

Proof. The initial ground state population can be expressed as $p_0 = 1/\mathcal{Z}_S$, allowing us to express the difference in populations as

$$\begin{aligned} \Delta p_0^* &= \frac{1}{\mathcal{Z}_S \mathcal{Z}_M} \left(\sum_{i_M \in \{0,1\}^{\times n} \setminus S} e^{-\beta_M i_M \cdot \Gamma} + \sum_{j_M \in S} e^{-\beta_S \omega - \beta_M (j_M \oplus 1) \cdot \Gamma} \right) - \frac{1}{\mathcal{Z}_S} \\ &= \frac{1}{\mathcal{Z}_S \mathcal{Z}_M} \left(\sum_{i_M \in \{0,1\}^{\times n} \setminus S} e^{-\beta_M i_M \cdot \Gamma} + \sum_{j_M \in S} e^{-\beta_S \omega - \beta_M (j_M \oplus 1) \cdot \Gamma} - \mathcal{Z}_M \right). \end{aligned} \quad (7.24)$$

We can then express the machine's partition function using the bitwise notation as

$$\begin{aligned} \mathcal{Z}_M &= \prod_{j=1}^n (1 + e^{-\beta_M \gamma_j}) \\ &= \sum_{l_M \in \{0,1\}^{\times n}} e^{-\beta_M l_M \cdot \Gamma} \\ &= \sum_{l_M \in \{0,1\}^{\times n} \setminus S} e^{-\beta_M l_M \cdot \Gamma} + \sum_{l_M \in S} e^{-\beta_M l_M \cdot \Gamma}, \end{aligned} \quad (7.25)$$

where we have split the partition function into a contribution from S and $\{0,1\}^{\times n} \setminus S$. From this reorganisation of the partition function it can be seen that

$$\Delta p_0^* = \frac{1}{\mathcal{Z}_S \mathcal{Z}_M} \left(\sum_{j_M \in S} e^{-\beta_S \omega - \beta_M (j_M \oplus 1) \cdot \Gamma} - \sum_{l_M \in S} e^{-\beta_M l_M \cdot \Gamma} \right). \quad (7.26)$$

Recalling that $(j_M \oplus 1) \cdot \Gamma = E_{\text{Max}} - j_M \cdot \Gamma$, we obtain

$$\begin{aligned} \Delta p_0^* &= \frac{1}{\mathcal{Z}_S \mathcal{Z}_M} \left(\sum_{j_M \in S} e^{-\beta_S \omega - \beta_M (E_{\text{Max}} - j_M \cdot \Gamma)} - \sum_{l_M \in S} e^{-\beta_M l_M \cdot \Gamma} \right) \\ &= \frac{1}{\mathcal{Z}_S \mathcal{Z}_M} \left(\sum_{j_M \in S} \left[e^{-\beta_S \omega - \beta_M (E_{\text{Max}} - j_M \cdot \Gamma)} - e^{-\beta_M j_M \cdot \Gamma} \right] \right), \end{aligned} \quad (7.27)$$

completing the proof. $\square$

7.4 Bounds

If S is not completely known, but instead a subset of S is known, a lower bound can at least be placed on Δp_0^* . For example, if $(T_M/T_S)\omega \leq E_{\text{Max}}$ then a simple lower bound can be found by noting that $1^n \in S$. This corresponds to performing the TLP $|0_S 1^n\rangle \leftrightarrow |1_S 0^n\rangle$ which is the single TLP that causes the largest change in ground state population. The new ground state population after this will be

$$\tilde{p}'_0 = p_0 + \frac{e^{-\beta_S \omega} - e^{-\beta_M E_{\text{Max}}}}{\mathcal{Z}_S \mathcal{Z}_M}. \quad (7.28)$$

From here, the following lower bound on the optimal change in the ground state population of S can easily be found

$$\Delta p_0^* \geq \frac{e^{-\beta_S \omega} - e^{-\beta_M E_{\text{Max}}}}{\mathcal{Z}_S \mathcal{Z}_M}. \quad (7.29)$$

This contribution is due to what is known as the *virtual qubit subspace swap* in the literature [32, 33, 153].

More generally, if one considers the more restrictive assumption that $(T_M/T_S)\omega \leq \gamma_1$, then the following tighter lower bound can be found.

Result 7.4.1. *If $(T_M/T_S)\omega \leq \gamma_1$, then the optimal change in the ground state population of S , Δp_0^* , is lower bounded by*

$$\Delta p_0^* \geq \frac{1}{\mathcal{Z}_S \mathcal{Z}_M^{\tilde{k}:n}} (e^{-\beta_S \omega} - e^{-\beta_M \sum_k^n \gamma_i}) + \frac{1}{\mathcal{Z}_S \mathcal{Z}_M} (e^{-\beta_M \sum_k^n \gamma_i} - e^{-\beta_S \omega - \beta_M (E_{\text{Max}} - \sum_k^n \gamma_i)}), \quad (7.30)$$

where $\mathcal{Z}_M^{\tilde{k}:n} = \prod_{i=\tilde{k}}^n (1 + e^{-\beta_M \gamma_i})$ is the partition function of the $n - \tilde{k}$ machine qubits with the largest energy gaps.

Proof Sketch. Under the assumption that $(T_M/T_S)\omega \leq \gamma_1$, it can be seen that there exists a bit-string above which, in terms of lexicographic order, all strings are in S — the set that should be swapped. Given we know that at least the basis states corresponding to these states should be permuted, one can calculate a lower bound on Δp_0^* by finding the change in ground-state population from permuting just these basis states. See Section 7.10.4 for a full proof. $\square$

We assess the quality of the bound in Result 7.4.1 by comparing it to the exact optimal change for degenerate, linear and exponential gap vectors Γ i.e., the qubits energy gaps are a fixed size, increasing linearly, or increase exponentially respectively with each additional qubit. The comparisons are plotted in Fig. 7.3 and we see that, whilst the lower bound performs well for machines with energetic structures that grow quickly (linear, exponential), especially for larger n , it performs poorly for machines with fixed gaps (degenerate). This is because Result 7.4.1 captures a fixed set of strings corresponding to TLPs that in the linear and exponential cases capture the TLPs most significantly contributing to the cooling, even as n grows. However, in the degenerate case we see that this fixed set of TLPs contributes less and less to the growing number of necessary TLPs for performing optimal cooling as n grows.

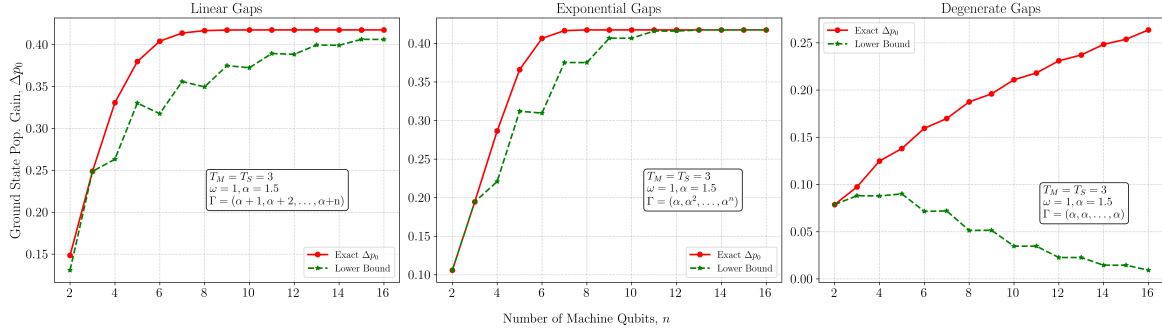


Figure 7.3: In this figure we compare the change in ground-state population of S , Δp_0^* , induced by the cooling protocol for different machine energy gap structures as the number of qubits in the machine n increases. In green, we compare this exact value to the lower bound given in Result 7.4.1.

7.5 Reducibility of Machines

Within the model presented in this chapter it is perhaps natural to think that increasing the number of qubits in the machine is always helpful. And indeed it is, as increasing the number of qubits in the machine increases E_{Max} . However, one can ask whether it is only the increased energy that improves the cooling, or also the extra dimensions in the Hilbert space. In other words, we must investigate whether a cooling operation involving S and an n -qubit machine M is ever reducible to an operation involving just some $k < n$ coldest qubits of the machine. In which case, the energetic gap structure Γ extended to n qubits appears dimensionally *wasteful* and could be reduced to a machine with fewer qubits involving only a subset of Γ . Here, we first investigate the extreme cases of $(n - 1)$ -reducibility and irreducibility, before discussing the general case of k -reducibility.

7.5.1 $(n - 1)$ -Reducibility

To begin this investigation, we look at the most egregiously wasteful kind of machine, an n qubit machine that is reducible to a bipartite interaction involving the effectively coldest qubit in the machine and S . We term such a machine $(n - 1)$ -reducible, meaning that the effectively warmest $(n - 1)$ qubits do not participate in the cooling process such that the optimal cooling unitary acts as the identity on all $(n - 1)$ qubits. Mathematically, this means that the set of energy levels to be exchanged can be characterised by $S = \{0_S i_M^{n-1} 1 : i_M^{n-1} \in \{0, 1\}^{\times(n-1)}\}$, such that there is no dependence on the $(n - 1)$ -hottest qubits. Using Result 7.3.2, to have this set of necessary TLPs it must be the case that

$$\frac{1}{2} \left(\frac{T_M}{T_S} \omega + E_{\text{Max}} \right) < (i_M^{n-1} 1) \cdot \Gamma \quad \forall i_M^{n-1} \in \{0, 1\}^{\times(n-1)}. \quad (7.31)$$

Most stringently, this must also be valid for $i_M^{n-1} = 0^{n-1}$ where only the qubit with gap γ_n contributes to cooling. Considering this leads to the following necessary and sufficient condition for a machine to be $(n - 1)$ -reducible.

Result 7.5.1. A necessary and sufficient condition for an n -qubit machine with gap vector $\Gamma = (\gamma_1, \gamma_2, \dots, \gamma_n)$, at temperature T_M , to be $(n - 1)$ -reducible when cooling a qubit with gap ω , at temperature T_S , is

$$\frac{1}{2} \left(\frac{T_M}{T_S} \omega + E_{\text{Max}} \right) < \gamma_n. \quad (7.32)$$

Proof. The condition is found by inputting $i_M^{n-1} = 0^{n-1}$ into Eq. (7.31). As Eq. (7.31) must hold for all bit-strings for the machine to be $(n - 1)$ -reducible, this is a necessary condition. Moreover, by noting that if Eq. (7.31) holds for $i_M = 0^{n-1}$, then it will hold for all other bit-strings $i_M \in \{0, 1\}^{\times(n-1)}$, the condition becomes sufficient. $\square$

From Result 7.5.1, it can be seen that if the functional form of Γ grows steeply enough with n then all that matters for cooling is the energetic gap of the largest qubit, and rest of the machine is *useless*. Interestingly, we have already encountered this situation in our motivating example in Section 7.2.2. Here, only one of the two qubits in the $n = 2$ qubit machine was involved in the cooling operation when $\omega < \gamma_2 - \gamma_1$ (scenario b). A quick calculation shows this is precisely the $(n - 1)$ -reducibility condition given in Result 7.5.1 for $T_M = T_S$ and a 2 qubit machine.

7.5.1.1 Exponential Machines

Whilst the condition in Result 7.5.1 is quite restrictive, it arises frequently in one of the machine examples we have considered, namely exponential machines. An exponential n -qubit machine is one whose gap structure grows exponentially with n such that $\Gamma = (\gamma, \gamma^2, \dots, \gamma^n)$. To consider the reducibility of such machines, first recall that the sum of a finite geometric series is $\sum_{i=0}^n ar^i = a \left(\frac{1-r^{n+1}}{1-r} \right)$ for $r \neq 1$. We can use this to express the sum of gaps in the exponential machine as $E_{\text{Max}} = \gamma \left(\frac{1-\gamma^n}{1-\gamma} \right)$. Inputting this into the $(n - 1)$ reducibility condition in Result 7.5.1 we find

$$\frac{1}{2} \left(\frac{T_M}{T_S} \omega + \gamma \left(\frac{1-\gamma^n}{1-\gamma} \right) \right) < \gamma^n \quad (7.33)$$

$$\frac{T_M}{T_S} \omega < \frac{\gamma(2\gamma^n - 2\gamma^{n-1} - 1)}{1-\gamma}. \quad (7.34)$$

Since $1 < \frac{(2\gamma^{n-1} - 2\gamma^{n-2} - 1)}{1-\gamma}$ if $1 < \gamma$, the $(n - 1)$ reducibility condition holds if $\frac{T_M}{T_S} \omega < \gamma$ and $1 < \gamma$. Hence, these are sufficient conditions for an exponential machines to be $(n - 1)$ -reducible.

7.5.2 Irreducible Machines

We now investigate the opposite extreme, where all machine qubits are used in performing optimal cooling, with such a machine being referred to as irreducible. This is arguably the most important reducibility criterion because it teaches us how to design *useful* machines where adding more qubits always leads to more potential for cooling.

An irreducible machine must use every qubit for cooling, even its effectively warmest qubit, implying that i_1 cannot be a free index. Hence, there must exist at least one $l^{n-1} \in \{0, 1\}^{\times n-1}$ such that $|0_S 1 l^{n-1}\rangle \in \mathcal{S}$ whilst $|0_S 0 l^{n-1}\rangle \notin \mathcal{S}$. If not true, then the joint action of all these permutations renders the first qubit of the machine useless, making the machine reducible. The first condition $|0_S 1 l^{n-1}\rangle \in \mathcal{S}$ imposes the upper bound $\frac{1}{2} \left(\frac{T_M}{T_S} \omega + E_{\text{Max}} \right) < 1 l^{n-1} \cdot \Gamma$, whilst the second condition $|0_S 0 l^{n-1}\rangle \notin \mathcal{S}$ imposes the lower bound $0 l^{n-1} \cdot \Gamma \leq \frac{1}{2} \left(\frac{T_M}{T_S} \omega + E_{\text{Max}} \right)$. Taken together, they create the following necessary and sufficient condition for a machine to be irreducible.

Result 7.5.2. A necessary and sufficient condition for an n -qubit machine with gap vector $\Gamma = (\gamma_1, \gamma_2, \dots, \gamma_n)$, at temperature T_M , to be irreducible when cooling a qubit with gap ω , at temperature T_S , is for there to exist at least one $l^{n-1} \in \{0, 1\}^{\times (n-1)}$ such that

$$0 \leq \frac{1}{2} \left(\frac{T_M}{T_S} \omega + E_{\text{Max}} \right) - l^{n-1} \cdot \Gamma_{2:n} < \gamma_1, \quad (7.35)$$

where $\Gamma_{k:n} = (\gamma_k, \gamma_{k+1}, \dots, \gamma_n)$.

Proof. It is first noted that $1 l^{n-1} \cdot \Gamma = \gamma_1 + l^{n-1} \cdot \Gamma_{2:n}$ whilst $0 l^{n-1} \cdot \Gamma = l^{n-1} \cdot \Gamma_{2:n}$. Subtracting $l^{n-1} \cdot \Gamma_{2:n}$ from both sides gives the condition. The condition is necessary as if it does not hold then the machine is 1-reducible and hence not irreducible. It is sufficient as if the machine is not at least 1-reducible, then it is irreducible. $\square$

Intuitively, one can understand this inequality to mean that there is at least one energy level l^{n-1} of the

$n - 1$ coldest machine qubits which is only large enough to optimally cool if γ_1 is included – exchanging l^{n-1} without a contribution from the effectively warmest qubit would not cool S .

7.5.2.1 Degenerate Machines

Using Result 7.5.2, we can now prove that degenerate machines, such as those typically used in algorithmic and dynamical cooling [28, 30], are irreducible. Hence, the cooling potential of these machines always increases as the number of qubits in the machine increases. For an n -qubit degenerate machine, $\Gamma = \underbrace{(\gamma, \gamma, \dots, \gamma)}_n$, Result 7.5.2 takes the form

$$0 \leq \frac{1}{2} \left(\frac{T_M}{T_S} \omega + n\gamma - 2t\gamma \right) < \gamma, \quad (7.36)$$

as $E_{\text{Max}} = n\gamma$, and $l^{n-1} \cdot \Gamma_{2:n} = t\gamma$ for some $t \in [0, n - 1]$. Simplifying, we arrive at

$$-(n - 2t)\gamma \leq \frac{T_M}{T_S} \omega \leq 2\gamma - (n - 2t)\gamma. \quad (7.37)$$

To show that degenerate machines are irreducible, one must always be able to find a t such that the above condition is satisfied. To see this holds, consider the integers $n/2 \leq t \leq n - 1$ gives the following set of conditions

$$\begin{aligned} t = \frac{n}{2} : 0 &\leq \frac{T_M}{T_S} \omega \leq 2\gamma \\ t = \frac{n}{2} + 1 : 2\gamma &\leq \frac{T_M}{T_S} \omega \leq 4\gamma \\ t = \frac{n}{2} + 2 : 4\gamma &\leq \frac{T_M}{T_S} \omega \leq 6\gamma \\ &\vdots \\ t = n - 1 : (n - 2)\gamma &\leq \frac{T_M}{T_S} \omega \leq n\gamma. \end{aligned} \quad (7.38)$$

Hence, each condition sets an interval on the number line $[0, n\gamma]$ such that the whole number line is captured. Therefore, if $0 < (T_M/T_S) \omega \leq n\gamma$, then a t can always be found such that one of these conditions is satisfied. This lower bound on $(T_M/T_S) \omega$ always holds by definition, as $\omega, T_M, T_S \geq 0$. Hence, for degenerate machines satisfying $T_M/T_S \omega \leq n\gamma$, an $l^{n-1} \in \{0, 1\}^{\times(n-1)}$ can be found such that Result 7.5.2 is satisfied, meaning that such degenerate machines are irreducible.

7.5.3 k -Reducibility

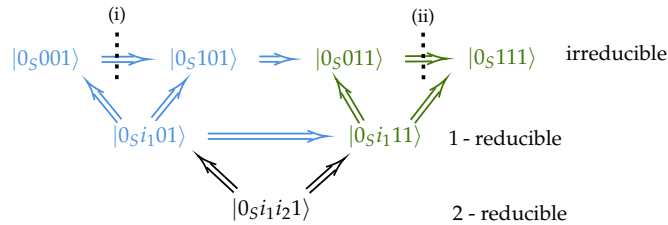


Figure 7.4: Diagram of reducibility inequalities for a 3 qubit machine showing the partial order in the inequalities, where $|0_S i_M\rangle \implies |0_S j_M\rangle$ means that if $|0_S i_M\rangle \in S$, then so is $|0_S j_M\rangle \in S$. Although, the reverse implication does not necessarily hold.

We now consider the general case of k -Reducibility, and demonstrate the complexity of finding a condition for k -Reducibility by investigating the $n = 3$ qubit machine case.

For a machine to be k -reducible, one must examine under which condition all the energy levels $|0_S i_M^k 1_M^{n-k}\rangle : i_M^k \in \{0, 1\}^{\times k}$ need to be exchanged. This situation is not straightforward, as one must

consider $|0_S i_M^k l_M^{n-k}\rangle$ for every possible choice of $l^{n-k} \in \{0,1\}^{\times n-k}$. To see the difficulty in this, consider the $n = 3$ qubit case visualised in Fig 7.4. This has 3 levels corresponding to the machine being 2-reducible, meaning the optimal cooling interaction involves only the effectively coldest qubit in the machine, 1-reducible, meaning the optimal cooling involves the effectively coldest two qubits in M , and irreducible, meaning it involves all the qubits in M . If the machine is 2-reducible, that is $|0_S i_1 i_2 1\rangle$ satisfies Eq. (7.19) for any $i_1, i_2 \in \{0,1\}$, then every energy level in the hierarchy is also a member of S and must be exchanged. Ascend one level of the hierarchy we find two scenarios, only one of which leads to 1-reducibility. It must be the case that $|0_S i_1 11\rangle \in S$ whilst $|0_S i_1 01\rangle \notin S$ for $i_1 \in \{0,1\}$. If $|0_S i_1 01\rangle \in S$ then it implies that $|0_S i_1 11\rangle \in S$ also, leading again to the condition for 2-reducibility. At the final level, two scenarios lead to irreducibility. Firstly, in situation (i) in Fig 7.4 we have $|0_S 001\rangle \notin S$ and $|0_S 101\rangle \in S$, with this second condition implying that $|0_S 011\rangle, |0_S 111\rangle \in S$ also. Whilst this gives $|0_S i_1 11\rangle \in S$ for $i_1 \in \{0,1\}$, as we have $|0_S 101\rangle \in S$ but $|0_S 001\rangle \notin S$, meaning Result 7.5.2 is satisfied for $l^{(2)} = 01$, and hence the machine is irreducible. Secondly, in situation (ii) in Fig 7.4, the machine is irreducible if $|0_S 011\rangle \notin S$ and $|0_S 111\rangle \in S$ as Result 7.5.2 is satisfied for $l^{(2)} = 11$. This begins to elucidate the complex hierarchical structure of the energetic inequalities we have explored in this manuscript, and indicates the difficulty of finding a simple condition to capture k -reducibility

7.6 $n = 3$ Qubit Machine Example

Here, we will demonstrate all of the tools introduced in this chapter so far via the case of an $n = 3$ qubit machine, $\Gamma = (\gamma_1, \gamma_2, \gamma_3)$, being used to cool a thermal qubit S with gap ω , such that $\omega \leq \gamma_1 \leq \gamma_2 \leq \gamma_3$. We set $\beta_S = 1$ and $\beta_M = 1$ for simplicity. We will use our tools to assess: which TLPs should be performed for optimal cooling, what the optimal achievable ground state population of S is, which machines are irreducible or reducible, and finally, to evaluate a lower bound on optimal achievable ground state population[†]. Note, this section can be skipped without any detriment to the reader in understanding future sections.

The population vector for the ground and excited state of the system, p_0 and p_1 respectively, are

$$\begin{aligned} p_0 &= \frac{1}{\mathcal{Z}_S \mathcal{Z}_M} [1, e^{-\gamma_3}, e^{-\gamma_2}, e^{-(\gamma_2+\gamma_3)}, e^{-\gamma_1}, e^{-(\gamma_1+\gamma_3)}, e^{-(\gamma_1+\gamma_2)}, e^{-(\gamma_1+\gamma_2+\gamma_3)}], \\ p_1 &= \frac{1}{\mathcal{Z}_S \mathcal{Z}_M} [e^{-\omega}, e^{-(\omega+\gamma_3)}, e^{-(\omega+\gamma_2)}, e^{-(\omega+\gamma_2+\gamma_3)}, e^{-(\omega+\gamma_1)}, e^{-(\omega+\gamma_1+\gamma_3)}, e^{-(\omega+\gamma_1+\gamma_2)}, e^{-(\omega+\gamma_1+\gamma_2+\gamma_3)}], \end{aligned} \quad (7.39)$$

where $\mathcal{Z}_S = (1 + e^{-\omega})$ and $\mathcal{Z}_M = (1 + e^{-\gamma_1})(1 + e^{-\gamma_2})(1 + e^{-\gamma_3})$. Using the Eigenvalue Cooling Algorithm, detailed in Proposition 7.2.1, the TLPs necessary for performing optimal cooling can be found by comparing elements of p_0 and p_1 . Alternatively, using Result 7.3.2 the necessary TLPs can be calculated using only Γ . In this case, Result 7.3.2 takes the form

$$\frac{1}{2} (\omega + \gamma_1 + \gamma_2 + \gamma_3) < i_1 \gamma_1 + i_2 \gamma_2 + i_3 \gamma_3, \quad (7.40)$$

with the two level swap $|0_S i_1 i_2 i_3\rangle \leftrightarrow |1_S (i_1 \oplus 1)(i_2 \oplus 1)(i_3 \oplus 1)\rangle$ being a method for optimally cooling the system if Eq. (7.40) is satisfied for a given $i_M = i_1 i_2 i_3$ where $i_1, i_2, i_3 \in \{0,1\}$. Specifically, the following

[†] Whilst unnecessary given that the optimal ground state populations can be analytically calculated easily in this case, this will be used to demonstrate Result 7.4.1)

two level swaps can be performed to achieve optimal cooling if the related conditions are satisfied

$$|0_S000\rangle \longleftrightarrow |1_S111\rangle \text{ if } \omega < -\gamma_1 - \gamma_2 - \gamma_3, \quad (7.41)$$

$$|0_S100\rangle \longleftrightarrow |1_S011\rangle \text{ if } \omega < \gamma_1 - \gamma_2 - \gamma_3, \quad (7.42)$$

$$|0_S010\rangle \longleftrightarrow |1_S010\rangle \text{ if } \omega < -\gamma_1 + \gamma_2 - \gamma_3, \quad (7.43)$$

$$|0_S110\rangle \longleftrightarrow |1_S001\rangle \text{ if } \omega < \gamma_1 + \gamma_2 - \gamma_3, \quad (7.44)$$

$$|0_S001\rangle \longleftrightarrow |1_S110\rangle \text{ if } \omega < -\gamma_1 - \gamma_2 + \gamma_3, \quad (7.45)$$

$$|0_S101\rangle \longleftrightarrow |1_S010\rangle \text{ if } \omega < \gamma_1 - \gamma_2 + \gamma_3, \quad (7.46)$$

$$|0_S011\rangle \longleftrightarrow |1_S100\rangle \text{ if } \omega < -\gamma_1 + \gamma_2 + \gamma_3, \quad (7.47)$$

$$|0_S111\rangle \longleftrightarrow |1_S000\rangle \text{ if } \omega < \gamma_1 + \gamma_2 + \gamma_3. \quad (7.48)$$

Due to the assumed gap structure of $\omega \leq \gamma_1 \leq \gamma_2 \leq \gamma_3$, it can be seen that 7.48, 7.47 and 7.46 are always satisfied. This therefore means that their associated two level swaps always cool the system. Note, from the perspective of the Eigenvalue Cooling Algorithm, these conditions always being satisfied are equivalent to the fact that the following conditions on the populations from p_0 and p_1 are always satisfied

$$\omega < \gamma_1 - \gamma_2 + \gamma_3 \iff e^{-(\omega+\gamma_2)} > e^{-(\gamma_1+\gamma_3)}, \quad (7.49)$$

$$\omega < -\gamma_1 + \gamma_2 + \gamma_3 \iff e^{-(\omega+\gamma_1)} > e^{-(\gamma_2+\gamma_3)}, \quad (7.50)$$

$$\omega < \gamma_1 + \gamma_2 + \gamma_3 \iff e^{-\omega} > e^{-(\gamma_1+\gamma_2+\gamma_3)}. \quad (7.51)$$

Meanwhile, 7.41, 7.42 and 7.43 are never satisfied, meaning their associated two level swaps never cool the system. As above, these conditions directly correspond to conditions on the populations from p_0 and p_1 that are never satisfied.

This leaves two possible conditions, namely $\omega < \gamma_1 + \gamma_2 - \gamma_3$ (7.44) and $\omega < -\gamma_1 - \gamma_2 + \gamma_3$ (7.45) that one must consider. Given these, there are four possible cases to consider:

Case 1: 7.44 and 7.45 are satisfied.

It can easily be seen that this is never possible. This is due to $\omega \geq 0$ and the right hand sides of 7.44 and 7.45 being the negative of each other.

Case 2: 7.44 is satisfied and 7.45 is not.

In addition to the three TLPs that should always be performed, the TLP

$$|0_S110\rangle \longleftrightarrow |1_S001\rangle \text{ as } \omega < \gamma_1 + \gamma_2 - \gamma_3 \iff e^{-(\omega+\gamma_3)} > e^{-(\gamma_1+\gamma_2)} \quad (7.52)$$

is also needed for optimal cooling in this case. If all TLPs are performed then the final ground state population in this case is

$$p_0^*(2) = \frac{1 + e^{-\gamma_1} + e^{-\gamma_2} + e^{-\gamma_3}}{\mathcal{Z}_M}, \quad (7.53)$$

where

$$\omega < \gamma_1 + \gamma_2 - \gamma_3 \implies p_0^*(2) > \frac{1}{1 + e^{-\gamma_3}} \quad (7.54)$$

meaning that in this case one cools the system further than the lowest effective temperature of the machine qubits. Note, we have used the notion $p_0^*(i)$ to denote the optimal ground state population of the system in case i .

Case 3: 7.45 is satisfied and 7.44 is not.

In addition to the three TLPs that should always be performed, the TLP

$$|0_S001\rangle \longleftrightarrow |1_S110\rangle \text{ as } \omega < -\gamma_1 - \gamma_2 + \gamma_3 \iff e^{-(\omega+\gamma_1+\gamma_2)} > e^{-\gamma_3} \quad (7.55)$$

is also needed for optimal cooling in this case. If all TLPs are performed then the final ground state population in this case is

$$p_0^*(3) = \frac{1}{1 + e^{-\gamma_3}}, \quad (7.56)$$

meaning the optimal ground state population of the system that can achieve is that of the effectively coldest qubit. Hence, the optimal unitary for cooling in this case is simply a SWAP between the system and the machine qubit with the lowest effective temperature.

Case 4: 7.44 and 7.45 are not satisfied.

This leads to no additional TLPs beyond the three always performed TLPs. If all three TLPs are performed then the final ground state population is

$$p_0^*(4) = \frac{1 + e^{-\omega} + e^{-\gamma_1} + e^{-\gamma_2} + e^{-\gamma_3} + e^{-(\omega+\gamma_2)} + e^{-(\omega+\gamma_3)} + e^{-(\gamma_1+\gamma_2)}}{\mathcal{Z}_S \mathcal{Z}_M}. \quad (7.57)$$

From here, it can be seen that

$$\omega > -\gamma_1 - \gamma_2 + \gamma_3 \implies p_0^*(3) > \frac{1}{1 + e^{-\gamma_3}}, \quad (7.58)$$

where the first condition comes from 7.45 not being satisfied. Hence, one cannot cool past the lowest effective temperature of the machine qubits.

In Section 7.5, the notion of machine reducibility was discussed. It can be seen that Case 3 fits the definition of an $(n - 1)$ -reducible machine (Section. 7.5.1) given that two of the three machine qubits are not involved in the cooling process — the final ground state populations does not depend on γ_1 or γ_2 . The condition for $(n - 1)$ -reducibility given in Result 7.5.1 in this case takes the form

$$\frac{1}{2}(\omega + \gamma_1 + \gamma_2 + \gamma_3) < \gamma_3, \quad (7.59)$$

which can equivalently be expressed as

$$\omega < -\gamma_1 - \gamma_2 + \gamma_3. \quad (7.60)$$

Hence, 7.45 is exactly the condition for $(n - 1)$ -reducibility when $n = 3$. It being satisfied therefore immediately tells us that the optimal cooling that can be achieved is to the effective temperature of the effectively coldest machine qubit, and that a full swap between the effectively coldest machine qubit and system is an optimal cooling unitary.

We now consider whether the machines in case 2 and 4 are irreducible (Section. 7.5.2) meaning all qubits in the machine contribute to the optimal cooling. In Result. 7.5.2, a condition for a machine to be irreducible is given, which in this case ask if there exists an $i_2, i_3 \in \{0, 1\}$ such that

$$0 \leq \frac{1}{2}(\omega + \gamma_1 + \gamma_2 + \gamma_3) - (i_2\gamma_2 + i_3\gamma_3) < \gamma_1. \quad (7.61)$$

We first consider case 2, where 7.44 is satisfied and 7.45 is not. For $i_2 = 1$ and $i_3 = 0$, Eq. (7.61) becomes

$$-\gamma_1 + \gamma_2 - \gamma_3 < \omega < \gamma_1 + \gamma_2 - \gamma_3. \quad (7.62)$$

Given 7.44 is satisfied the right hand side of this equation is satisfied. Given the non-decreasing energy structure of the machine, the left hand side is always satisfied. Hence, the machine in case 2 is irreducible.

We then consider case 4, where 7.44 is satisfied and 7.45 is not. For $i_2 = 0$ and $i_3 = 1$ Eq. (7.61) becomes

$$-\gamma_1 - \gamma_2 + \gamma_3 < \omega < \gamma_1 - \gamma_2 + \gamma_3. \quad (7.63)$$

Given the non-decreasing energy structure of the machine the right hand side is always statised. Then, given 7.45 is not satisfied, the left hand side is satisfied. Therefore, the machine in case 4 is also irreducible. Given that the optimal achievable ground state populations in case 2 and case 4, $p_0^*(2)$ and $p_0^*(4)$ respectively, depend on all elements in Γ , intuitively one would assume the machines are irreducible. However, the above calculations confirm this.

Lastly, one can lower bound the optimal achievable change in population of the system, Δp_0^* , using Result 7.4.1, which, in this case gives

$$\Delta p_0^* \geq \frac{1}{\mathcal{Z}_S(1 + e^{-\gamma_3})}(e^{-\omega} - e^{-\gamma_3}) + \frac{1}{\mathcal{Z}_S \mathcal{Z}_M}(e^{-\gamma_3} - e^{-(\omega + \gamma_1 + \gamma_2)}). \quad (7.64)$$

where it is noted that the second term is negative if 7.45 is satisfied.

7.7 Generalisation to Arbitrary Qubit States

Throughout this chapter we have focused on the task of cooling a qubit, which was shown to be equivalent to increasing the probability of being in the ground state of the system Hamiltonian. However, given we allow the application of arbitrary unitaries, once we have increased the probability of being in the ground state we could arbitrarily change the basis of the state. As cooling is only applicable when considering the energy eigenbasis, and increasing the purity of a qubit just means increase its largest eigenvalue, it would be more appropriate to view the protocol detailed here as a *purity enhancement* protocol[‡]. In the same vein, our protocol can easily be adapted to perform purity enhancement of an arbitrary system qubit given access to a machine of arbitrary qubits i.e., it can be applied more generally than just to thermal states. This can be achieved given that all qubit states can be considered to be a thermal state with respect to some Hamiltonian. Specifically, given some state $\rho \in \mathcal{D}(\mathcal{H}^2)$, with spectral decomposition

$$\rho = \lambda_0 |\lambda_0\rangle\langle\lambda_0| + (1 - \lambda_0) |\lambda_1\rangle\langle\lambda_1|, \quad \lambda_0 \geq 2^{-1}, \quad (7.65)$$

an effective Hamiltonian can be defined as

$$H^{\text{eff}} = \gamma |\lambda_1\rangle\langle\lambda_1|, \quad \gamma = -\frac{1}{\beta} \ln \left(\frac{1}{\lambda_0} - 1 \right), \quad (7.66)$$

where β is some effective temperature that can be chosen to tune the gap. Hence, given some arbitrary set of states

$$\rho_S \otimes \rho_M = \rho_S \otimes \rho_{M_1} \otimes \rho_{M_2} \otimes \rho_{M_3} \otimes \dots \otimes \rho_{M_n}, \quad (7.67)$$

one can define an effective Hamiltonian for the system and each machine qubit such that

$$\rho_S = \frac{e^{-\beta_S H_S^{\text{eff}}}}{Z_S} \quad \text{and} \quad \rho_{M_i} = \frac{e^{-\beta_M H_i^{\text{eff}}}}{Z_i} \quad \forall i, \quad (7.68)$$

with total effective Hamiltonian

$$H_T^{\text{eff}} = H_S^{\text{eff}} + \sum_{i=1}^n H_i^{\text{eff}}. \quad (7.69)$$

From this, an effective system gap, ω^{eff} , and effect gap vector, Γ^{eff} , can be defined. The qubits in the machine can be relabelled such that the gaps in the gap vector are in non-decreasing order. A β_S can then always be chosen for any $\rho \in \mathcal{D}(\mathcal{H}^2)$ such that $\omega^{\text{eff}} \leq \gamma_1^{\text{eff}} : \gamma_1^{\text{eff}} \in \Gamma^{\text{eff}}$. The framework is therefore the same as presented above, meaning that the same protocol can be performed. Although, TLPs of the joint eigenbasis of the system and machine are now considered, rather than TLPs of the energy eigenbasis e.g., TLPs of the form

$$|\lambda_0^S \lambda_{i_1}^1 \lambda_{i_2}^2 \dots \lambda_{i_n}^{M_n}\rangle \leftrightarrow |\lambda_1^S \lambda_{j_1}^{M_1} \lambda_{j_2}^{M_2} \dots \lambda_{j_n}^{M_n}\rangle, \quad (7.70)$$

[‡] If our restriction to unitaries that are permutations of the energy eigenbasis was a physical restriction, rather than just a result of them being sufficient for cooling, our protocol would be strictly cooling

where $\{|\lambda_0^S\rangle\langle\lambda_0^S|, |\lambda_1^S\rangle\langle\lambda_1^S|\}$ and $\{|\lambda_0^{M_k}\rangle\langle\lambda_0^{M_k}|, |\lambda_1^{M_k}\rangle\langle\lambda_1^{M_k}|\}$ are the eigenbasis of the system (the chosen qubit being purified) and the k th machine qubit respectively. Therefore, $i_M = i_1 i_2 \dots i_n$ and $j = j_1 j_2 \dots j_n$ are bit-strings labelling the eigenstates of H_T which are tensor products of the local eigenstates. These bit-string are then used in determining which TLPs should be performed to purify (cool) the system (using the inequalities derived above). The purifying unitary generated by our protocol will then be in this joint eigenbasis, but can of course be decomposed into any basis once it has been found. We emphasize that the local eigenbasis of each qubit can be different.

This leaves open the question of whether this is the optimal protocol for increasing the purity of the system qubit. To see that it is, we first note the similarities between the informational non-equilibrium concentration problem detailed in Section 6.2, and the problem stated here. There, we aimed to optimally increase the maximum eigenvalue[§] of some state given access to two copies of said state and global unitary interactions. Here, we are aiming to optimally increase the maximum eigenvalue — and hence purity — of a qubit state given access to some machine and global unitary interactions. In this knowledge, we can employ the fact that

$$\begin{aligned} \|\sigma_S\|_\infty &= \max_{V_{SM}} \left\| \text{tr}_M [V_{SM}(\rho_S \otimes \rho_M) V_{SM}^\dagger] \right\|_\infty \\ &= \|\rho_S \otimes \rho_M\|_{2^n - k_Y}, \end{aligned} \quad (7.71)$$

where $\dim(\rho_M) = 2^n$, which was proved in Section 6.5.1. Using this, we can see that under global unitaries the maximum value that the largest eigenvalue of the system can take is the sum over the 2^n largest eigenvalues of $\rho_S \otimes \rho_M$. This is exactly what the above algorithm will achieve, applying TLPs to assign the 2^n largest eigenvalues to the 2^n basis states of the form $|\lambda_0^S \lambda_{i_1}^1 \lambda_{i_2}^2 \dots \lambda_n^{M_n}\rangle : i_M \in \{0, 1\}^{\times n}$ i.e., the 2^n basis states with λ_0^S as its first label. Hence, this effective cooling protocol will optimally increase the maximum eigenvalue of the system qubit, and therefore perform optimal purity enhancement on the system qubit.

Lastly, using the above, it can be stated that the optimal change in the maximum eigenvalue of the system, $\Delta\lambda_0^*$, is given by

$$\Delta\lambda_0^* = \|\rho_S \otimes \rho_M\|_{2^n - k_Y} - \|\rho_S\|_\infty, \quad (7.72)$$

when ρ_M is an n -qubit machine. From here, it can be easily shown that the change in purity of the system from this change in the maximum eigenvalue, $\Delta\text{Pur}(S) := \text{tr}[\sigma_S^2] - \text{tr}[\rho_S^2]$, is given by

$$\Delta\text{Pur}(S) = 2\Delta\lambda_0^* [2\|\rho_S\|_\infty - 1] + 2(\Delta\lambda_0^*)^2. \quad (7.73)$$

All of the above result for cooling still apply to the effect cooling protocol presented here. One can therefore, for example, apply the notion of reducibility to optimal purity enhancement, removing qubits that cannot contribute to the purity enhancement.

7.8 Cooling With Symmetries

In the above, we have assumed access to any unitary when cooling the system (although ultimately we have argued that only permutations in the energy eigenbasis are necessary). Here, returning from the notion of purity enhancement to cooling, we now consider restricting the dynamics due to the existence of some symmetry and assess its effect on our ability to optimally cool the system.

Given some observable Q , we will now consider unitaries U such that $[U, Q] = 0$. Under such dynamics Q is said to be symmetric as the commutator implies that $Q = UQU^\dagger$. Physically, this means that under evolution via U the probabilities of getting a given measurement outcome of Q will remain a constant. As a consequence, the expectation value of Q will also remain constant. Hence, this symmetry in the dynamics is said to have led to the conservation of Q , with this being an example of Noether Theorem [154].

[§] In the context we looked to maximally increase the max relative entropy between the state and the maximally mixed state, but under the hood this just means maximising the maximum eigenvalue

7.8.1 Energy Conservation

Firstly, we will consider $H_T = H_S + H_M$, where

$$H_T = \sum_{j_S \in \{0,1\}} \sum_{i_M \in \{0,1\}^n} (j_S \cdot \omega + i_M \cdot \Gamma) |j_S \cdot \omega + i_M \cdot \Gamma\rangle \langle j_S \cdot \omega + i_M \cdot \Gamma|, \quad (7.74)$$

such that H_S, H_M are the Hamiltonians of the system and machine respectively, so that H_T is the total Hamiltonian. The symmetry in the dynamics U , i.e., $[U, H_T] = 0$, then leads to the total energy of the system and machine being conserved. We note, this now places us back within the resource theory of athermality [17, 19], as we are considering only resource non-generating operations.

A unitary U commutes with H_T if it acts an independent unitary within each degenerate subspace of H_T ,

$$U = \bigoplus_c U_c, \quad U_c U_c^\dagger = \mathbb{I} \quad (7.75)$$

such that

$$U_c = \sum_{j_S \in \{0,1\}} \sum_{i_M \in \{0,1\}^n} u_{i_S, i_M} |j_S \cdot \omega + i_M \cdot \Gamma\rangle \langle j_S \cdot \omega + i_M \cdot \Gamma| : j_S \cdot \omega + i_M \cdot \Gamma = E_c, \quad (7.76)$$

where E_c is the energy of the c th degenerate subspace and u_{i_S, i_M} is some matrix coefficient of U_c in the eigenbasis of H_T . Applying the same logic as before, but now to each degenerate subspace, it can be concluded that optimal cooling of the system can be achieved by applying permutations of the energy eigenbasis within each degenerate subspace of H_T . As before, a permutation can only change the ground state population of the system qubit if it is of the form

$$|0_S i_M\rangle \leftrightarrow |1_S k_M\rangle : i_M, k_M \in \{0,1\}^{\times n}. \quad (7.77)$$

Hence, to achieve any cooling there must exist at least one $i_M, k_M \in \{0,1\}^{\times n}$ such $i_M \cdot \Gamma = \omega + k_M \cdot \Gamma$, as this means there will exist a degenerate subspace of H_T that can facilitate such a TLP. As before, this TLP will be cooling if

$$\frac{T_M}{T_S} \omega < (i_M - k_M) \cdot \Gamma. \quad (7.78)$$

Although, due to the above equality, this will be satisfied if $T_S > T_M$. This result is expected, as the thermal state of H_T is

$$\tau_{SM} = \frac{1}{\mathcal{Z}_{SM}} e^{-\beta H_T} = \frac{e^{-\beta H_S}}{\mathcal{Z}_S} \otimes \frac{e^{-\beta H_M}}{\mathcal{Z}_M}, \quad (7.79)$$

such that the system and machine are locally thermal, each at the same temperature β . As energy conserving dynamics act invariantly on the thermal state i.e., $U \tau_{SM} U^\dagger = \tau_{SM}$ if $[U, H_T] = 0$, no cooling of the system can be achieved if $T_M = T_S$ (as no populations can be moved). However, if the system and machine are at different temperatures, they are not initially in a thermal state of H_T . Hence, the initial state is not invariant under energy conserving unitaries and there exists the possibility to cool. This can be summarised as follows:

Observation 7.8.1. *If cooling a system and machine using a unitary U , where $[U, H_T] = 0$ with $H_T = H_S + H_M$ the total Hamiltonian of the system and machine, the system can only be cooled if*

- (a) H_T has degeneracies facilitating TLP of the form $|0_S i_M\rangle \leftrightarrow |1_S k_M\rangle : i_M, k_M \in \{0,1\}^{\times n}$, meaning there exists at least one $i_M, k_M \in \{0,1\}^{\times n}$ such $i_M \cdot \Gamma = \omega + k_M \cdot \Gamma$.
- (b) The initial (real) temperature of the machine is less than the initial temperature of the system, $T_M < T_S$.

The change in ground state temperature that can be achieved under energy conserving unitaries will then depend on both the dimension of the degenerate subspaces that can facilitate permutations of the form of Eq. (7.77) — as this dictates the number of TLPs that can be performed within the given

degenerate subspace — and the temperature difference between the system and machine. To see this, it is noted that the change in the ground state population from performing the TLP $|0_S i_M\rangle \leftrightarrow |1_S k_M\rangle$, within the E_c th degenerate subspace, is given by

$$\begin{aligned}\Delta p_c &= \frac{e^{-\beta_S \omega - \beta_M k_M \cdot \Gamma} - e^{-\beta_M i_M \cdot \Gamma}}{\mathcal{Z}_S \mathcal{Z}_M} \\ &= \frac{e^{-\beta_M E_c} (e^{\omega(\beta_M - \beta_S)} - 1)}{\mathcal{Z}_S \mathcal{Z}_M},\end{aligned}\quad (7.80)$$

where we have used the fact that $i_M \cdot \Gamma = \omega + k_M \cdot \Gamma = E_c$. Hence, it can be seen that the larger the disparity between T_S and T_M , the larger $\beta_M - \beta_S$, and the larger the possible change in ground state population.

To solidify the notions introduced here, we will now consider energy conserving unitaries in the motivating example of the two qubit machine considered in Section 7.2[¶]. Given the assumption of the gap vector being ordered in non-increasing order, $\omega \leq \gamma_1 \leq \gamma_2$, there are two possible ways in which H_T can have degeneracies that facilitate TLPs of the form of Eq. (7.77): (1) $\omega + \gamma_1 = \gamma_2$, and (2) $\omega = \gamma_1 = \gamma_2 = \gamma$.

In case (1), there is a single degenerate subspace: $\text{span}\{|0_S 01\rangle, |1_S 10\rangle\}$. Hence, the TLP $|0_S 01\rangle \leftrightarrow |1_S 10\rangle$ is all that can be performed, leading to an optimal change in ground state population of the system of

$$\Delta p^*(1) = \frac{e^{-\beta_M \gamma_2} (e^{\omega(\beta_M - \beta_S)} - 1)}{\mathcal{Z}_S \mathcal{Z}_M}, \quad (7.81)$$

where again the (i) is used to denote that this is the optimal achievable ground state population in case (i). In case (2), there are two degenerate subspaces:

$$\begin{aligned}\text{Span}\{|0_S 01\rangle, |0_S 10\rangle, |1_S 00\rangle\}, \\ \text{Span}\{|0_S 11\rangle, |1_S 01\rangle, |1_S 10\rangle\}.\end{aligned}\quad (7.82)$$

In the first subspace, either the TLP $|0_S 01\rangle \leftrightarrow |1_S 00\rangle$ or $|0_S 10\rangle \leftrightarrow |1_S 00\rangle$ must be performed. An equivalent choice of TLPs then exists for the other degenerate subspace. Taken together, they lead to an optimal change in ground state population of

$$\Delta p^*(2) = \frac{(e^{-\beta_M \omega} + e^{-2\beta_M \omega}) (e^{\omega(\beta_M - \beta_S)} - 1)}{\mathcal{Z}_S \mathcal{Z}_M}. \quad (7.83)$$

7.8.2 Additional Conserved Quantities

In addition to energy, one can also consider other observables that must be conserved under the cooling dynamics. For example, one can consider symmetry with respect to

$$J_Z = \sigma_Z^S \otimes \mathbb{I}_M + \mathbb{I}_S \otimes \sum_{i=1}^n \sigma_Z^i, \quad \sigma_Z^i = \mathbb{I}_1 \otimes \mathbb{I}_2 \dots \mathbb{I}_{i-1} \otimes \sigma_Z \otimes \mathbb{I}_{i+2} \dots \mathbb{I}_n, \quad (7.84)$$

which is the total spin observable (with constants set to 1). As before, the unitaries that conserve total spin are those that commute with J_Z , $[U, J_Z] = 0$, and hence they act invariantly within the degenerate subspaces of J_Z . If both H and J_Z are conserved, one therefore needs to consider unitaries U such that $[U, H] = [U, J_Z] = 0$. Such a condition can be very restrictive, allowing arbitrary unitaries to act only within the joint degenerate eigenspaces of H and J_Z . Although, if one assumes that the energy eigenvectors and spin eigenvectors align, such that $[H, J_Z] = 0$, the problem becomes more tractable. For example, in the case of the two-qubit motivating example, the total spin operator in the joint eigenbasis under this assumption is

$$J_Z = \text{diag}(3, 1, 1, -1, 1, -1, -1, -3). \quad (7.85)$$

[¶] We note that this is a very similar paradigm to that discussed in [37].

Hence, it has the same degenerate subspaces as case (2) detailed above. Therefore, in case (2) both H_T and J_Z can be conserved whilst performing cooling. However, in case (1) there is no overlapping degenerate subspaces between J_Z and H_T , meaning, in this case, that it is not possible to perform any cooling whilst conserving both quantities.

7.9 Conclusion

In this chapter we have built a comprehensive model to understand how to optimally cool a thermal qubit using n -other thermal qubits and global unitary interactions. Such a task has been considered in many different contexts before, but we here go beyond these prior works by allowing each qubit to be governed by a different Hamiltonian. This development was made possible by the introduction of the gap vector, and the resulting bit-string notion. From this, an understanding of the TLPs necessary for performing optimal cooling could be found, from which the optimal change in ground state population could be cleanly characterised. It also clearly led to the notion of reducibility, allowing for an understanding of when certain qubits in the machine are useless.

These results offer an alternative perspective on the well studied problem of cooling via unitary dynamics; we have taken the folkloric idea that cooling a quantum system given access to another consists of only reordering the eigenvalues of the joint state in descending order — as summarised in the Eigenvalue Cooling Algorithm in Proposition 7.2.1 — and examined the structure of this sorting in detail. We found that the intricacies of this rearrangement is determined by a simple set of inequalities which depend on the global temperature of the machine and its total energy E_{Max} , and the temperature and energy gap of the system. This has elucidated the physics behind the Eigenvalue Cooling Algorithm, showing that energy levels in the machine are only useful for cooling if they are above some shift median value. Moreover, understanding these inequalities is powerful as it tells one how to cool the system without them having to consider the whole system-machine density operator, they can simply check whether a given energy level of the machine is above the threshold value $\frac{1}{2}(T_M\omega/T_S + E_{\text{Max}})$ and know whether a TLP should be performed.

We were also able to show that these inequalities are sensitive to what number of qubits in the machine are contributing to an optimal cooling interaction with S . Here, we found that machines whose gaps increase exponentially are generally fully reducible to an interaction involving only their coldest qubit ($(n - 1)$ -reducible). And, we were able to verify that degenerate machines are generally irreducible, that is all their qubits contribute to optimally cooling S . This irreducibility criterion should prove especially helpful when designing quantum thermal machines for cooling quantum systems as it allows one to ensure that the whole machine they have constructed will be contribute to the cooling.

We then showed that whilst we presented the above protocol in the context of cooling, it is better thought of as a purity enhancement protocol, with all results still applying — under slight modifications — if one considers increasing the purity of some arbitrary state given access to a machine of arbitrary states. The core contribution of this work is therefore an algorithm for generating the unitary dynamics that optimally increase the purity of some target qubit, given some number of initially uncorrelated qubits. To find this unitary via this protocol, however, one would need a complete description of each state in the system and machine. Lastly, we considered cooling via unitaries constrained by some symmetry.

In this work, we chose to focus on a simple set up: a machine consisting of thermal qubits in a separable state, which in turn is in a separable state with the qubit system to be cooled. An interesting direction to further this work would be to generalise the framework to more complicated machines. For example, those featuring higher-dimensional systems such as qudits, or those with initial correlations either within the machine, or between the system and machine. A further interesting extension would be to consider cooling under dynamics with multiple conserved quantities (greater than 2), as was briefly explored in [155]. Whilst it was shown in Section 7.8.2 that having two conserved quantities can be restrictive enough to remove any ability to cool, this was a system with only three qubits. In larger systems it could still be possible to perform cooling despite such added restriction. In addition, it would itself be interesting to characterise in what situations symmetric dynamics become so restrictive that cooling cannot occur. Moreover, it would be timely and interesting to also consider situations in which the conserved quantities do not commute. Considering non-commuting conserved quantities in the

process of thermalisation has already proved interesting [156], it would now be pertinent to consider their role in opposite task: cooling.

7.10 Additional Material

7.10.1 Final State Coherences Cannot Aid in Cooling

Here, we show that given a state initially diagonal in the joint energy eigenbasis, coherences in the final state cannot aid in cooling the system qubit — where cooling means increasing the ground state population. We note that the calculation performed here is very similar to the technique used to prove Result 6.2.1 in Section 6.5.1. In fact, one could perform this calculation using results derived in that proof. We give the full calculation here regardless for clarity. Moreover, we perform this calculation for n -qubit machines to be as general as possible, but in doing so use notion introduced after this section is refereed to in the main text. See Section 7.3 for the details of the notation used if it is unclear.

Firstly, we note that the ground state population of the system qubit, p_0 , for some arbitrary state in the system and machine Hilbert space σ_{SM} is given by

$$p_0 = \text{tr}[\Pi_g \sigma_{SM}], \quad \Pi_g = \sum_{i_M \in \{0,1\}^{\times n}} |0_S i_M\rangle \langle 0_S i_M|, \quad (7.86)$$

such that Π_g is a rank $d_M = \dim(\text{tr}_S[\sigma_{SM}])$ projector i.e., the rank of the projector is the dimension of the machine Hilbert space. Given an initial system and machine state $\tau_S \otimes \tau_M$ cooled via a unitary U_{SM} , the ground state is therefore given by

$$\begin{aligned} p'_0 &= \text{tr}[\Pi_g U_{SM}(\tau_S \otimes \tau_M) U_{SM}^\dagger] \\ &= \text{tr}[U_{SM}^\dagger \Pi_g U_{SM}(\tau_S \otimes \tau_M)] \\ &\leq \max_{\Pi_{d_M}} \text{tr}[\Pi_{d_M}(\tau_S \otimes \tau_M)], \end{aligned} \quad (7.87)$$

where we have used the fact that $U_{SM}^\dagger \Pi_g U_{SM}$ is still a rank d_M projector and in the final line the maximisation is over all rank- d_M projectors, leading to the equality becoming a bound. We now explicitly write the initial system and machine state as

$$\tau_S \otimes \tau_M = \sum_{j_S \in \{0,1\}} \sum_{i_M \in \{0,1\}^{\times n}} \lambda_{j_S, i_M} |j_S i_M\rangle \langle j_S i_M|, \quad (7.88)$$

where we have set $\lambda_{j_S, i_M} = e^{-\beta_S(j_S \cdot \omega) - \beta_M(i_M \cdot \Gamma)}$ for brevity. Using this, an upper bound on the output ground state population is then given by

$$p'_0 \leq \max_{\Pi_{d_M}} \sum_{j_S \in \{0,1\}} \sum_{i_M \in \{0,1\}^{\times n}} \lambda_{j_S, i_M} \langle j_S i_M | \Pi_{d_M} | j_S i_M \rangle. \quad (7.89)$$

By noting that

$$\sum_{j_S \in \{0,1\}} \sum_{i_M \in \{0,1\}^{\times n}} \langle j_S i_M | \Pi_{d_M} | j_S i_M \rangle = d_M \quad \text{and} \quad \langle j_S i_M | \Pi_{d_M} | j_S i_M \rangle \leq 1 \quad \forall j_S, i_M, \quad (7.90)$$

it can be seen that this sum is maximised when Π_{d_M} is the projector onto the largest d_M eigenvectors of $\tau_S \otimes \tau_M$.

Therefore, if one applied the unitary U_{SM} that permutes the energy eigenstates such that the largest d_M eigenvalues are associated to the basis states $|0_S i_M\rangle : i_M \in \{0,1\}^{\times n}$ then this upper bound becomes saturated, and the inequality becomes an equality. Hence, the ground state population of the system can always be maximised by just permuting the energy eigenbasis. Such permutations create no coherences in the energy eigenbasis of the output system and machine state. Given optimality is still achieved, this shows that coherences do not provide any advantage when aiming to optimally increasing the ground

state population of the system. When investigating optimal cooling via unitary dynamics, it is therefore sufficient to consider the subset of unitaries that are permutations of the energy eigenbasis.

7.10.2 Proof That Optimal Cooling in $n = 2$ Machine Can Require a TOFFOLI Gate

Result 7.10.1. *Given a two qubit machine such that $\omega \geq \gamma_2 - \gamma_1$, and access to $\mathcal{G} = \{\mathbf{X}, \text{CNOT}, \text{TOFFOLI}\}$, a TOFFOLI gate is necessary for performing optimal cooling.*

Proof. In Section 7.2.2 and Observation 7.2.2 it was argued that to implement the unitary $S_{011,100}$ — the unitary necessary to perform optimal cooling — a TOFFOLI gate is needed. This was based off the notion of odd and even permutations. Here, we show that in general there exists no transformation using just $\mathbf{X}$ and CNOT able to perform this optimal cooling, meaning a TOFFOLI (or some tripartite) gate is necessary.

Given access to the restricted gate set $\tilde{\mathcal{G}} = \{\mathbf{X}, \text{CNOT}\}$ and a single ancilla bit, all implementable operations on $x \in \{0, 1\}^{\times n}$ are affine transformations [157],

$$f(x) = Ax \oplus b, \quad (7.91)$$

where $A \in \mathbb{F}_2^{2^n \times 2^n}$ and $b \in \mathbb{F}_2^{2^n \times 1}$. From this definition of an affine transformation, it can be seen that if

$$x \oplus y = z \oplus q, \text{ where } x, y, z, q \in \{0, 1\}^{\times n}, \quad (7.92)$$

then

$$f(x) \oplus f(y) = f(z) \oplus f(q). \quad (7.93)$$

In the 2-qubit machine case, where $n = 3$, one can consider

$$\begin{aligned} 0_s 11 \oplus 0_i 1 i_2 &= 0(1 \oplus i_1)(1 \oplus i_2) \\ 0_s 00 \oplus 0(1 \oplus i_1)(1 \oplus i_2) &= 0(1 \oplus i_1)(1 \oplus i_2) \end{aligned} \quad (7.94)$$

where $i_1, i_2 \in \{0, 1\}^{\times 1}$ but not $i_1 = i_2 = 0$, such that the XOR product of these pairs of bit-string are equal, but they are not the same bit-strings. The left hand-side of Eq. (7.94) then contains all four of the three-bit-strings with a 0 in S .

In scenario a) in Section 7.2.2, it was seen that the TLP $|0_s 11\rangle \leftrightarrow |1_s 00\rangle$ was needed to perform optimal cooling i.e. $f(0_s 11) = 1_s 00$ and $f(1_s 00) = 0_s 11$ is needed. However, permutations within $|0_s i_M\rangle\langle 0_s i_M| : i_M = i_1 i_2 \in \{0, 1\}^{\times 2}$ and $|1_s i_M\rangle\langle 1_s i_M| : i_M = i_1 i_2 \in \{0, 1\}^{\times 2}$ do not effect the final ground state population of the system, and hence they can be freely performed. One can therefore be more general, and say that to perform optimal cooling there must be only one TLP of the form $|0_s i_M\rangle \leftrightarrow |1_s j_M\rangle$, for some $i_M, j_M \in \{0, 1\}^{\times 2}$. Put alternatively, only one TLP between a bit-string with a 0 in S and a bit-string with a 1 in S can be performed. If any more TLP of this form is performed, some of the four largest populations will be moved away from $|0_s i_M\rangle\langle 0_s i_M|$ and the cooling will cease to be optimal. In terms of the affine functions, this is $f(0_s i_M) = 1_s j_M$ and $f(1_s j_M) = 0_s i_M$. Given there can only be one permutation of this type, all other strings output from $f(\cdot)$ with a 0 in S must output a bit-string with a 0 in S . Hence,

$$f(0_s 11) \oplus f(0_i 1 i_2) \neq f(0_s 00) \oplus f(0(1 \oplus i_1)(1 \oplus i_2)), \quad (7.95)$$

as either the left-hand side has a 1 in S after the addition, or the right hand side, but not both. Hence, as this can never be equal there exists no affine transformation to implement such an operation, meaning it cannot be achieved with $\tilde{\mathcal{G}}$ and single qubit ancilla alone. In our case, we restrict even further by not allowing a single qubit ancilla. Therefore, optimal cooling requires a TOFFOLI gate. $\square$

7.10.3 Proof of Lemma 7.3.1

Lemma 7.3.1 *The TLPs output from the Eigenvalue Cooling Algorithm are always of the form*

$$|0_S i_M\rangle \leftrightarrow |1_S(i_M \oplus 1)\rangle, \quad (7.96)$$

where $\oplus 1$ is a bit-wise addition modulo 2 and $i_M \in \{0, 1\}^{\times n}$ is an n -bit string labelling a basis state of the machine M .

Proof. To prove this, we apply the Eigenvalue Cooling Algorithm detailed in Proposition 7.2.1 to the input state $\tau_S \otimes \tau_M$ given in Eq. (7.13). To do this, we first note that each element of $\mathbf{q}_0$ and $\mathbf{q}_1$ can be identified with an n -bit-string, i_M , as

$$\mathbf{q}_0 = \left\{ \frac{e^{-\beta_M i_M \cdot \Gamma}}{\mathcal{Z}_S \mathcal{Z}_M} : i_M \in \{0, 1\}^{\times n} \right\}, \quad \mathbf{q}_1 = \left\{ \frac{e^{-\beta_S \omega - \beta_M i_M \cdot \Gamma}}{\mathcal{Z}_S \mathcal{Z}_M} : i_M \in \{0, 1\}^{\times n} \right\}. \quad (7.97)$$

Therefore, the k elements of $\mathbf{q}_0$ that must be permuted can be identified with a set of n -bit-strings, as can the k elements of $\mathbf{q}_1$ that they must be permuted with. Moreover, from the above definitions it can easily be seen that $\mathbf{q}_1 = e^{-\beta_S \omega} \mathbf{p}_0$, and hence that $\mathbf{q}_1^\downarrow = e^{-\beta_S \omega} \mathbf{p}_0^\downarrow$ — multiplying each element of an ordered list by a positive constant does not change the ordering. Therefore, finding the set of n -bit-strings that identify the k largest elements of $\mathbf{q}_1$ is equivalent to finding the set of n -bit-strings that identify the k largest elements of $\mathbf{q}_0$. One can therefore consider only $\mathbf{q}_0$ and aim to find the set of n -bit-strings that identify the smallest k elements of $\mathbf{q}_0$, denoted by $\mathbb{T}$, and the set of n -bit-strings that identify the largest k elements $\mathbf{q}_0$, denoted by $\overline{\mathbb{T}}$. From the above reasoning, the bit-strings that identify the k largest elements of $\mathbf{q}_0$, $\overline{\mathbb{T}}$, will also then identify the k largest elements of $\mathbf{q}_1$. We can then consider the following Lemma, which states that if the bit-string i_M is in the set of n -bit-strings that identify one of the k smallest populations of $\mathbf{q}_0$, then its conjugate $i_M \oplus 1$ is in the set of n -bit-strings that identify one of the k largest populations of $\mathbf{q}_1$.

Lemma 7.10.2. *If $i_M \in \mathbb{T}$, then $i_M \oplus 1 \in \overline{\mathbb{T}}$ where $i_M \in \{0, 1\}^n$.*

Proof. To begin, we first convert the vector of populations to a vector of energies

$$\mathbf{E}_0 = -\frac{1}{\beta_S} \ln(\mathcal{Z}_S \mathcal{Z}_M \mathbf{q}_0) = \{i_M \cdot \Gamma : i_M \in \{0, 1\}^n\}. \quad (7.98)$$

The constant $-1/\beta_S \ln(\mathcal{Z}_S \mathcal{Z}_M)$ does not effect the ordering. Then, given that $1/\beta_S$ is a positive constant and $\ln(a) \leq \ln(b)$ if $a \leq b$, $-\mathbf{E}_0$ has the same ordering as $\mathbf{q}_0^\downarrow$; the negative then flips the ordering. Therefore, the set of n -bit-strings that identify the k smallest values of $\mathbf{q}_0$ identify the k -largest values of $\mathbf{E}_0$, and vice versa for the k largest values of $\mathbf{q}_0$. Hence, $\mathbb{T}$ now identifies the largest k values of $\mathbf{E}_0$, and $\overline{\mathbb{T}}$ the k largest values of $\mathbf{E}_1$ (defined as in Eq. (7.98) but using $\mathbf{q}_1$).

Next, we note that

$$i_M \cdot \Gamma + (i_M \oplus 1) \cdot \Gamma = E_{\text{Max}}, \quad \text{where} \quad E_{\text{Max}} = \sum_{i=1}^n \gamma_i, \quad (7.99)$$

meaning that the component of $\mathbf{E}_0$ identified by the bit-string i_M is paired to the component of $\mathbf{E}_0$ identified by the bit-string $(i_M \oplus 1)$. Evidently, the larger $i_M \cdot \Gamma$, the smaller $(i_M \oplus 1) \cdot \Gamma$ and vice versa. From here, it can already be seen that if a n -bit-string leads to one of the k largest values of $\mathbf{E}_0$, its conjugate will lead to one of the k smallest. More formally, one can define

$$\begin{aligned} E_{\text{sup}} &= \min_{i_M \in \mathbb{S}} i_M \cdot \Gamma = i_{\text{sup}} \cdot \Gamma, \\ E_{\text{inf}} &= \max_{i_M \in \{0, 1\}^n \setminus \mathbb{S}} i_M \cdot \Gamma = i_{\text{inf}} \cdot \Gamma, \end{aligned} \quad (7.100)$$

such that E_{sup} is the smallest component of $\mathbf{E}_0$ of the k largest components (acting as a kind of supremum), and E_{inf} is the largest component of $\mathbf{E}_0$ not in the k largest (acting as a kind of infimum), giving $E_{\text{sup}} \geq E_{\text{inf}}$. Therefore,

$$E_{\text{Max}} - E_{\text{inf}} \geq E_{\text{Max}} - E_{\text{sup}} \implies (i_{\text{inf}} \oplus 1) \cdot \Gamma \geq (i_{\text{sup}} \oplus 1) \cdot \Gamma. \quad (7.101)$$

All other n -bit-strings in $\mathbb{T}$ give components of $\mathbf{E}_0$ larger than E_{sup} and hence their conjugates lead to components smaller than $E_{\text{Max}} - E_{\text{sup}}$. Thus far, we have shown that if i_M is an n -bit-string that represents one of the k largest energies of $\mathbf{E}_0$, $(i_M \oplus 1)$ is an n -bit-string that represents one of the k smallest energies of $\mathbf{E}_0$.

Now, given that the ordering of $\mathbf{E}_0$ is that of $\mathbf{p}_0$ reversed, if i_M is an n -bit-string that represents one of the k largest energies of $\mathbf{E}_0$, it is an n -bit-string that represents one of the k smallest populations of $\mathbf{p}_0$. Hence, $(i_M \oplus 1)$ is an n -bit-string that represents one of the k largest populations of $\mathbf{p}_0$. To complete the proof, one utilises the fact that the ordering of $\mathbf{p}_0$ in non-decreasing order is the same as that of $\mathbf{p}_1$. Hence, the n -bit-strings that identify the k largest populations of $\mathbf{p}_0$ also identify the k largest populations of $\mathbf{p}_1$. Finally, given they are the same set of bit-strings, any relationship between the n -bit strings identifying the k largest and smallest energies of $\mathbf{E}_0$ must also hold for those n -bit-strings representing the k largest and smallest populations of $\mathbf{p}_0$ and $\mathbf{p}_1$. They then must hold between the n -bit strings identifying the k smallest populations of $\mathbf{p}_0$ and k largest populations of $\mathbf{p}_1$ for the same reason. Using the relationship proved between the n -bit strings representing the k largest and smallest energies of $\mathbf{E}_0$, it can therefore be concluded that if $i_M \in \mathbb{S}$ then $(i_M \oplus 1) \in \bar{\mathbb{S}}$. Interestingly, one can go further and note that if the n -bit-string i'_M identifies the k' largest component of $\mathbf{p}_0$, then $(i'_M \oplus 1)$ identifies the k' smallest component (the $2^n - k'$ largest) of $\mathbf{p}_1$. $\square$

Returning now to the Eigenvalue Cooling Algorithm, Lemma 7.10.2 states that if $\mathbb{T}$ represents the set of n -bit-strings that identify the smallest components of $\mathbf{q}_0$, then the set of n -bit-strings that identify the largest components of $\mathbf{q}_1$, $\bar{\mathbb{T}}$, are given by $\bar{\mathbb{T}} = \{i_M \oplus 1 : i_M \in \mathbb{S}\}$. Given that

$$\min_{j_M \in \bar{\mathbb{S}}} j_M \cdot \Gamma > \max_{i_M \in \mathbb{S}} i_M \cdot \Gamma, \quad (7.102)$$

the smallest component of $\mathbf{q}_1$ to be swapped is still larger than the largest component of $\mathbf{q}_0$ to be swapped (where this ceases to be true defines the end of the Eigenvalue Cooling Algorithm i.e. what is the value of k). Which of these k components are paired with each other to be exchanged does not matter — they all lead to positive changes in the ground state population of the system. Therefore, without loss of generality, one can compare the population of $\mathbf{p}_0$ identified by the n -bit-string i_M with the population of $\mathbf{p}_1$ identified by the n -bit-string $(i_M \oplus 1)$ to see if the population identified by i_M should be swapped. Given that if the n -bit-string $(i'_M \oplus 1)$ identifies the k' largest component of $\mathbf{p}_1$, then i'_M identifies the k' smallest component of $\mathbf{p}_0$ (the $2^n - k'$ largest component), the strategy of swapping opposite Hamming weight pairs therefore exactly outputs the same set of TLPs as performing the Eigenvalue Cooling Algorithm, in which at each stage the optimal TLP to increase the ground state population of the system was performed. $\square$

7.10.4 Proof of Result 7.4.1

Result 7.4.1 If $(T_M/T_S)\omega \leq \gamma_1$, then the optimal change in the ground state population of S , Δp_0^* , is lower bounded by

$$\Delta p_0^* \geq \frac{1}{\mathcal{Z}_S \mathcal{Z}_M^{\bar{k}:n}} (e^{-\beta_S \omega} - e^{-\beta_M \Sigma_k^n \gamma_i}) + \frac{1}{\mathcal{Z}_S \mathcal{Z}_M} (e^{-\beta_M \Sigma_k^n \gamma_i} - e^{-\beta_S \omega - \beta_M (E_{\text{max}} - \Sigma_k^n \gamma_i)}), \quad (7.103)$$

where $\mathcal{Z}_M^{\bar{k}:n} = \prod_{i=\bar{k}}^n (1 + e^{-\beta_M \gamma_i})$ is the partition function of the $n - \bar{k}$ machine qubits with the largest energy gaps.

Proof. If $(T_M/T_S)\omega \leq \gamma_1$, then it can be seen that all bit-strings above and including $i_M = 10^{n-k-1}1^k$, where $k = n/2$ if n is even and $k = (n-1)/2$ if n is odd, are guaranteed to satisfy Eq. (7.19) for all Γ i.e.,

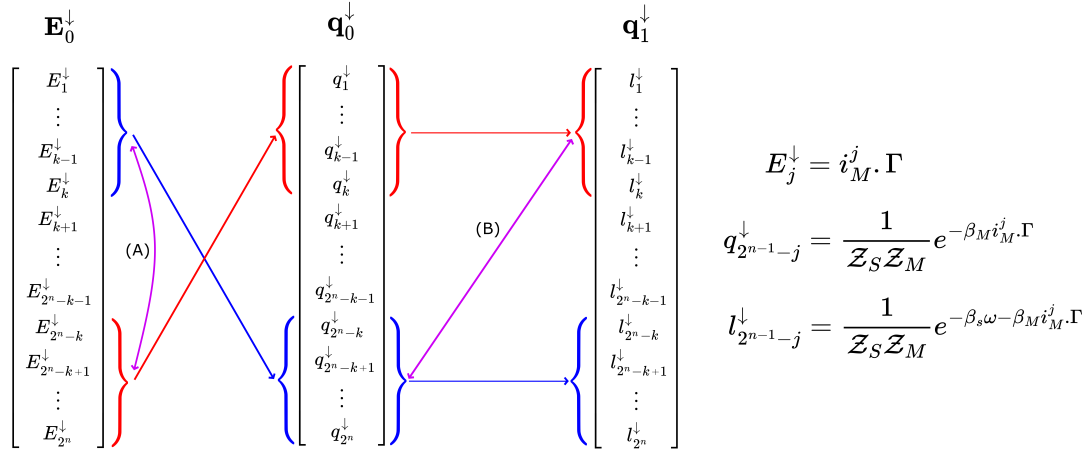


Figure 7.5: A figure detailing the proof strategy used in Lemma 7.10.2. One first considers a vector of energies, $\mathbf{E}_0$, ordered in non-decreasing order and generated from the ordered population vectors. Firstly, it can be shown that if i_M identifies the k th largest energy in $\mathbf{E}_0$, then $(i_M \oplus 1)$ identifies the $(2^n - k)$ largest energy in $\mathbf{E}_0$. Then, given that $\mathbf{E}_0$ and $\mathbf{p}_0$ have reverse orderings, if i_M identifies the k th largest energy in $\mathbf{E}_0$, it identifies the $(2^n - k)$ th largest population in $\mathbf{p}_0$. Next, as $\mathbf{p}_0$ and $\mathbf{p}_1$ have the same ordering, it can be concluded that if i_M identifies the $(2^n - k)$ th largest population in $\mathbf{p}_0$, it identifies the $(2^n - k)$ th largest population in $\mathbf{p}_1$ as well. Hence, the relationships found between the n -bit-strings identifying the k largest and smallest energies of $\mathbf{E}_0$ also apply to the n -bit-strings identifying the k smallest populations of $\mathbf{p}_0$ and the k largest populations of $\mathbf{p}_1$. Red and blue lines represent components of the respective vectors which are identified by the same n -bit-strings. The (A) purple line represents the relationship proved between the sets of n -bit strings identifying the k largest and smallest energies of $\mathbf{E}_0$ that can then be used to infer the relationship between the sets of n -bit strings identifying the k smallest populations of $\mathbf{p}_0$ and k largest populations of $\mathbf{p}_1$, represented by the (B) purple line.

they are in $\mathcal{S}$. To see this, inputting i_M into Eq. (7.19) says that the population of i_M should be swapped with $i_M \oplus 1$ if

$$\frac{T_M}{T_S} \omega < \sum_{j=\tilde{k}}^n \gamma_j - \sum_{j=2}^{\tilde{k}-1} \gamma_j + \gamma_1, \quad \tilde{k} = \begin{cases} \frac{n}{2} + 1, & n \text{ even} \\ \frac{n}{2} + \frac{3}{2}, & n \text{ odd} \end{cases} \quad (7.104)$$

Consider now the assumption that Γ is ordered in non-decreasing order, so that $\gamma_i \leq \gamma_j$ if $i \leq j \forall i, j$. From this it can be seen that

$$\gamma_1 \leq \sum_{j=\tilde{k}}^n \gamma_j - \sum_{j=2}^{\tilde{k}-1} \gamma_j + \gamma_1. \quad (7.105)$$

As we have also assumed that $(T_M/T_S)\omega \leq \gamma_1$, Eq. (7.104) must be satisfied. All bit-strings above i_M then lead to an Eq. (7.19) with a right hand side larger than Eq. (7.104), as for all other bit-strings above i_M parts of the negative sum in Eq. (7.104) are turned positive and hence it can only be larger. Therefore, under the assumption that $(T_M/T_S)\omega \leq \gamma_1$, at least all the bit-strings above and including i_M should be swapped.

To find a lower bound using Eq. (7.27) and this set of bit-strings, one can first consider performing all TLPs from $|0^{n-k}1^k\rangle \rightarrow |1^n\rangle$, and then removing the contribution arising from the inclusion of $|0^{n-k}1^k\rangle$. Initially, one therefore has

$$\begin{aligned} \Delta p_0^{\text{all}} &= \frac{1}{Z_S Z_M} \left(\sum_{i_M \in \{0,1\}^{\times(n-k)}} e^{-\beta_S \omega - \beta_M (E_{\text{Max}} - i_M^{n-k} 1^k \cdot \Gamma)} - e^{-\beta_M i_M^{n-k} 1^k \cdot \Gamma} \right), \\ &= \frac{1}{Z_S Z_M} \left(e^{-\beta_S \omega} \sum_{i_M \in \{0,1\}^{\times(n-k)}} e^{-\beta_M (i_M^{n-k} 1^k \oplus 1) \cdot \Gamma} - \sum_{i_M \in \{0,1\}^{\times(n-k)}} e^{-\beta_M i_M^{n-k} 1^k \cdot \Gamma} \right), \end{aligned} \quad (7.106)$$

where we have used the fact that $i_M \cdot \Gamma + (i_M \oplus 1) \cdot \Gamma = E_{\text{Max}}$. We now introduce the notation

$\Gamma_{1:l} = (\gamma_1, \gamma_2, \dots, \gamma_l)$ to select parts of Γ . Using this, the above expression becomes

$$\begin{aligned}\Delta p_0^{\text{all}} &= \frac{1}{\mathcal{Z}_S \mathcal{Z}_M} \left(e^{-\beta_S \omega} \sum_{i_M \in \{0,1\}^{\times(n-k)}} e^{-\beta_M (i_M^{n-k} \oplus 1) \cdot \Gamma_{1:n-k}} - e^{-\beta_M \sum_{l=\bar{k}}^n \gamma_l} \sum_{i_M \in \{0,1\}^{\times(n-k)}} e^{-\beta_M i_M^{n-k} \cdot \Gamma_{1:n-k}} \right), \\ &= \frac{1}{\mathcal{Z}_S \mathcal{Z}_M} (e^{-\beta_S \omega} - e^{-\beta_M \sum_{l=\bar{k}}^n \gamma_l}) \sum_{i_M \in \{0,1\}^{\times(n-k)}} e^{-\beta_M (i_M^{n-k} \cdot \Gamma_{1:n-k})},\end{aligned}\quad (7.107)$$

where the constants from the fixed 1s have been factorised out and, the fact that the sets $\{i_M \oplus 1 : i_M \in \{0,1\}^{\times n}\}$ and $\{0,1\}^{\times n}$ are equivalent means that

$$\sum_{i_M \in \{0,1\}^{\times(n-k)}} e^{-\beta_M (i_M^{n-k} \oplus 1) \cdot \Gamma_{1:n-k}} = \sum_{i_M \in \{0,1\}^{\times(n-k)}} e^{-\beta_M (i_M^{n-k} \cdot \Gamma_{1:n-k})}. \quad (7.108)$$

From here, it is then noted that this is the partition function over the $(n-k)$ hottest qubits and hence Δp_0^{all} can be written as

$$\begin{aligned}\Delta p_0^{\text{all}} &= \frac{1}{\mathcal{Z}_S \mathcal{Z}_M} (e^{-\beta_S \omega} - e^{-\beta_M \sum_{l=\bar{k}}^n \gamma_l}) \prod_{j=1}^{\bar{k}-1} (1 + e^{-\beta_M \gamma_j}), \\ &= \frac{1}{\mathcal{Z}_S \prod_{j=\bar{k}}^n (1 + e^{-\beta_M \gamma_j})} (e^{-\beta_S \omega} - e^{-\beta_M \sum_{l=\bar{k}}^n \gamma_l}), \\ &= \frac{1}{\mathcal{Z}_S \mathcal{Z}_M^{\bar{k}:n}} (e^{-\beta_S \omega} - e^{-\beta_M \sum_{k:n}^n \gamma_i}), \quad \mathcal{Z}_M^{\bar{k}:n} = \prod_{j=\bar{k}}^n (1 + e^{-\beta_M \gamma_j})\end{aligned}\quad (7.109)$$

One now needs to take into account the change in population included in Eq. (7.109) from including $i_M = 0^{n-k} 1^k$ in S . Performing this swap would give a change in population of

$$\Delta p_0^{\text{corr}} = \frac{1}{\mathcal{Z}_S \mathcal{Z}_M} (e^{-\beta_S \omega - \beta_M (E_{\max} - \sum_k^n \gamma_i)} - e^{-\beta_M \sum_k^n \gamma_i}). \quad (7.110)$$

A lower bound on Δp_0^* in this case can then be found as

$$\begin{aligned}\Delta p_0^* &\geq \Delta p_0^{\text{all}} - \Delta p_0^{\text{corr}}, \\ &\geq \frac{1}{\mathcal{Z}_S \mathcal{Z}_M^{\bar{k}}} (e^{-\beta_S \omega} - e^{-\beta_M \sum_k^n \gamma_i}) + \frac{1}{\mathcal{Z}_S \mathcal{Z}_M} (e^{-\beta_M \sum_{k:n}^n \gamma_i} - e^{-\beta_S \omega - \beta_M (E_{\max} - \sum_k^n \gamma_i)}),\end{aligned}\quad (7.111)$$

completing the proof. $\square$

RESULTS PART 2 — EXCLUSION

Operational Interpretation of the Choi Rank Through k-State Exclusion

8

8.1 Introduction

States give us only a snapshot in time. To model how systems evolve, interact with other systems, and respond to external stimuli, it is essential to understand dynamics. Such understanding then enables the prediction of a system's future state, facilitates the design and implementation of controls for on-demand manipulation, and allows for the characterisation of a system's response to external influences, such as noise.

In closed quantum systems, dynamics is modelled by unitary operators; the evolution of a state ρ to ρ' is given by $\rho' = U\rho U^\dagger$ for some unitary U . A more general notion of quantum dynamics is captured by *quantum channels*, or simply *channels*, which are *completely-positive trace-preserving* (CPTP) linear maps (we will here give a small overview of CPTP maps, see Section 2.4 for more details). Operationally, channels can be thought of as modelling the dynamics of open quantum systems, where a system evolves whilst interacting with an environment. Any channel $\mathcal{N}$ acting on a system S has a *Stinespring dilation* [158] given by $\mathcal{N}(\rho) = \text{tr}_E[U_{SE}(\rho_S \otimes \tau_E)U_{SE}^\dagger]$, where S, E represent the system and environment respectively, τ_E is some environment state, and U_{SE} is a unitary acting on SE . All quantum dynamics can therefore be modelled as unitary with respect to some higher dimensional space, and can be described by a single state (τ_E) and unitary (U_{SE}). Whilst the Stinespring dilation gives a physically motivated description of quantum dynamics, a more mathematically motivated description is given through a *Kraus decomposition* [39]*. For a channel $\mathcal{N}$ there always exists a set of M operators $\{K_x\}_{x=1}^M$, where $\sum_{x=1}^M K_x^\dagger K_x = \mathbb{I}$, such that the action of $\mathcal{N}$ on a state ρ is given by $\mathcal{N}(\rho) = \sum_{x=1}^M K_x \rho K_x^\dagger$. The Kraus decomposition is useful when applying a channel since one only needs to consider the input state and not the environment.

A given channel can have infinitely many Stinespring dilations and Kraus representations – all sets of operators that are unitarily equivalent to $\{K_x\}_{x=1}^M$ define the same channel, for example. However, and perhaps surprisingly, a channel can be *uniquely* described through its action on a single quantum state. The *Choi-Jamiołkowski isomorphism* [159, 160] is a linear mapping between quantum channels and bipartite quantum states. For a channel $\mathcal{N}_A$ acting on a system A with dimension d , its *Choi-state* $\mathcal{J}_{AB}^\mathcal{N}$ is a bipartite state in AB (where B also has dimension d) defined by

$$\mathcal{J}_{AB}^\mathcal{N} = (\mathcal{N}_A \otimes \mathcal{I}_B)(|\Phi^+\rangle\langle\Phi^+|_{AB}), \quad (8.1)$$

where $|\Phi^+\rangle_{AB} := \sum_{i=0}^{d-1} |ii\rangle_{AB} / \sqrt{d}$ is a maximally entangled state in AB ($|i\rangle_A$ and $|i\rangle_B$ are elements of fixed orthonormal bases of A, B , respectively), and $\mathcal{I}_B$ is an identity channel acting on B . Subscripts will be used to explicitly denote the corresponding (sub-)systems if needed. From the Choi-state, the action of $\mathcal{N}_A$ on a state ρ can be recovered as $\mathcal{N}_A(\rho) = d \text{tr}_B[(\mathbb{I}_A \otimes \rho^t) \mathcal{J}_{AB}^\mathcal{N}]$, where $(\cdot)^t$ is the transpose operation in the fixed basis of $|\Phi^+\rangle_{AB}$. Hence, remarkably, one can fully characterise the action of $\mathcal{N}$ on arbitrary states by a single action on $|\Phi^+\rangle_{AB}$.

Choi-states have proved to be one of the most powerful tools for understanding and characterising channels, both analytically and numerically. For instance, a linear map $\mathcal{N}_A$ acting on A is a channel if and only if its Choi-state is positive semi-definite, $\mathcal{J}_{AB}^\mathcal{N} \geq 0$, and maximally mixed in the B subspace, $\text{tr}_A[\mathcal{J}_{AB}^\mathcal{N}] = \mathbb{I}_B/d$ [159, 160]. By applying further operations to Choi-states, channels can be categorised into relevant subsets—a task often challenging when relying on Stinespring dilation or Kraus decomposition [23, 102, 161, 162]. For example, a channel is entanglement-breaking [163] if its Choi-state is separable; it is steering-breaking if its Choi-state is steerable [164]. Also, non-signalling conditions can be described by simple formulae of Choi-states [165–171]. Generally, Choi-states allow problems concerning quantum dynamics to be reformulated as numerically feasible convex optimisations [61, 172], making

* Note, other methods of modelling the dynamics of open quantum systems beyond quantum channels does exist, one could also consider microscopically derived master equations or the Von Neumann equation, for example.

† In general $|\Phi\rangle_{AB}$ can be any full-Schmidt-rank pure state.

them efficiently solvable. To date, Choi-states are widely recognised as an essential tool in characterising and quantifying dynamical quantum signatures—they provide an alternative, often simpler, approach to dealing with channels [5, 23, 26, 49, 61, 100, 102, 103, 105, 162–183].

8.1.1 Motivation

An important property of the Choi-state is its rank, r_c^N , termed *Choi-rank*. This is a key characterising property that provides insight into the structure of quantum dynamics; it has been shown to place mathematical bounds on the channel's description. For example, it is one, $r_c^N = 1$, if and only if the channel is unitary [184]; it serves as a lower bound on the number of operators needed in the Kraus decomposition of the channel, $r_c^N \leq M$ [39]; it equals the minimum dimension of the environment of a channel's Stinespring dilation [185]; and, when considering mixed unitary channels, is used to bound the number of unitaries needed to define the channel [184].

To date, the Choi-rank has been used solely as a mathematical tool, lacking a clear operational interpretation. In this chapter, we provide the Choi-rank with an operational interpretation, further cementing its importance as a characterising property. To this end, we introduce the task of *entanglement-assisted sub-channel exclusion*, phrasing them as communication tasks that resemble super-dense coding [69], but where state exclusion [64–66] replaces state discrimination [59] at the culmination of the protocol. We show that the Choi-rank establishes a tight, fundamental upper bound on one's ability to succeed in a task of this nature. Hence, this bound can be used to provide the Choi-rank with a physical and operational meaning.

Informally, in the task of state exclusion, a referee gives a player a state from a set of N possible predetermined states. The player then performs a measurement on the state and aims to exclude a set of k states that were *not* given to them. This leaves the player with $N - k$ possible states that they were sent. In comparison, if performing the task of state discrimination, after the measurement the player would aim to declare a single state that they *were* sent. For some sets of states, the player can perform *conclusive* state exclusion — excluding k states with unit probability — even when they can say nothing deterministically about what state they *do have*. Exclusionary information [186] — knowledge about what state the player does not have — can, therefore, be the only certain knowledge about the state that it is possible for the player to obtain. Hence, when using states to encode messages for communication (as discussed in Section 3.3), one may be able to deterministically say what message was not encoded, whilst only being able to probabilistically say what message was encoded. The importance of exclusionary information has already been demonstrated in the foundations of quantum theory [64, 65, 187] and in the quantification of quantum resources [67, 186, 188]. Here, we build on its ongoing significance in quantum information theory [66, 189–195] by establishing a connection between exclusionary information and the Choi-state.

8.1.2 Structure

Firstly, the structure of a k -state exclusion tasks is detailed, before the notions of weak and strong k -state exclusion are introduced and defined. Following this, a novel necessary condition for performing k -state exclusion on a set of mixed states is introduced. Then, an operational task is introduced, which can be seen as a generalisation of superdense coding, and for which the Choi-rank is shown to bound an agent's ability to succeed in the task. This is done using our novel necessary condition for k -state exclusion. The operational task is then considered for several examples. The result presented here are published in the manuscript [4].

8.2 K-state Exclusion

The notion of k -state exclusion was introduced in detail in Section 3.2. We here give another quick overview in order to keep this chapter self contained, giving enough for the reader to be able to understand our novel results in this area.

Formally, in a state exclusion task [64–66], the referee gives a player a state from the ensemble

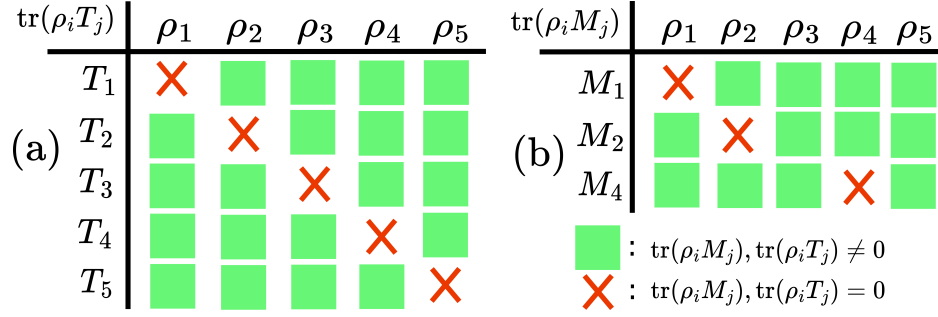


Figure 8.1: A graphical depiction of weak versus strong state exclusion. (a) Strong state exclusion: each outcome of the POVM $\{T_a\}_{a=1}^5$ excludes a different state from the set $\{\rho_x\}_{x=1}^5$. (b) Weak state exclusion: each outcome of the POVM $\{M_a\}_{a \in \{1,2,4\}}$ excludes one state from the set $\{\rho_x\}_{x=1}^5$. However, only some states in $\{\rho_x\}_{x=1}^5$ are ever excluded. There exists no POVM element in $\{M_a\}_{a \in \{1,2,4\}}$ to exclude ρ_3 or ρ_5 .

$(p_x, \rho_x)_x$, with the ensemble known to both referee and player. If the ensemble contains N states, the player therefore ends up with a state from the set $\{\rho_x\}_{x=1}^N$, where they get the state ρ_x with probability p_x . The player then performs a general N -outcome measurement described by a POVM $\{T_a\}_{a=1}^N$, such that $T_a \geq 0 \forall a$ and $\sum_{a=1}^N T_a = \mathbb{I}$, on the state, and outputs a label $g \in \{1, \dots, N\}$. They win if $g \neq x$ and fail if $g = x$. Namely, the player wins if they successfully exclude the state by outputting a label that was not associated to the sent state; they fail if they output the label associated to the sent state.

If the player outputs a single label g such that $g \neq x$ with certainty, this is *conclusive 1-state exclusion*. This occurs if the player is able to find a POVM such that

$$\text{tr}[T_x \rho_x] = 0 \quad \forall x \in \{1, \dots, N\}. \quad (8.2)$$

If the player gets the measurement outcome associated to T_g , they output g knowing with certainty the referee could not have sent ρ_g .

If the player outputs a set of k labels, $\{g_i\}_{i=1}^k$, such that $x \notin \{g_i\}_{i=1}^k$ with certainty, this is *conclusive k -state exclusion*. There are $\binom{N}{k}$ different sets of k labels the player could exclude, corresponding to all the different subsets of $\{1, \dots, N\}$ of length k . Therefore, when performing k -state exclusion, the player aims to find a POVM with $\binom{N}{k}$ elements such that each measurement outcome allows the player to exclude a subset of states from $\{\rho_x\}_{x=1}^N$ of length k .

The task of state exclusion is reminiscent of state discrimination, where the player instead tries to output a label g such that $g = x$. It can be seen that conclusive state discrimination, where a player outputs a label $g = x$ with certainty, is a special case of conclusive k -state exclusion[‡] where $k = N - 1$. Outputting $N - 1$ labels of states that were definitely not sent is equal to outputting one label of the state that definitely was sent. It is a well-known result that conclusive state discrimination, and hence conclusive $(N - 1)$ -state exclusion, is only possible if all states in $\{\rho_x\}_{x=1}^N$ are orthogonal, as detailed in Theorem 3.1.2 and Theorem 3.2.2.

Here, we will also consider the closely related task of sub-channel exclusion, as detailed in Section 3.2.4. Specifically, the special case that all sub-channels are a probabilistically applied CPTP map will be considered.

8.2.1 Weak and Strong State Exclusion

Whilst the notion of state exclusion is widely understood, some of the nuances in the definition are not agreed upon. In addition to Eq. (8.2) being a condition for conclusive state exclusion, the following additional condition,

$$\sum_{x=1}^N \text{tr}[T_a \rho_x] \neq 0 \quad \forall a \in \{1, \dots, N\}, \quad (8.3)$$

[‡] strong conclusive k -state exclusion

has also been implicitly or explicitly enforced on occasion [66, 189], while on other occasions it has not [190]. This additional condition ensures that all outcomes of the POVM $\{T_a\}_{a=1}^N$ have some probability of occurring. By enforcing both Eq. (8.2) and Eq. (8.3), conclusive 1-state exclusion on the set $\{\rho_x\}_{x=1}^N$ is defined to be the existence of an N element POVM where each element excludes a different state from $\{\rho_x\}_{x=1}^N$ with certainty, as seen in Fig. 8.1 (a). We define this to be *strong state exclusion*.

On the other hand, by only enforcing Eq. (8.2), conclusive 1-state exclusion on $\{\rho_x\}_{x=1}^N$ is defined to be the existence of a POVM with L non-zero elements, where $L \leq N$, such that each conclusively exclude a different state from a subset of $\{\rho_x\}_{x=1}^N$ of size L , as seen in Fig. 8.1 (b). We define this to be *weak state exclusion*.

Definition 8.2.1 (Strong State Exclusion). *Given a set of states $\{\rho_x\}_{x=1}^N$, strong conclusive 1-state exclusion is possible if there exists a POVM $T = \{T_a\}_{a=1}^N$ such that*

$$\text{tr}[T_x \rho_x] = 0 \quad \forall x \in \{1, \dots, N\} \quad \text{and} \quad \sum_{x=1}^N \text{tr}[T_a \rho_x] \neq 0 \quad \forall a \in \{1, \dots, N\}. \quad (8.4)$$

Definition 8.2.2 (Weak State Exclusion). *Given a set of states $\{\rho_x\}_{x=1}^N$, weak conclusive 1-state exclusion is possible if there exists a POVM $T = \{T_a\}_{a=1}^N$ such that*

$$\text{tr}[T_x \rho_x] = 0 \quad \forall x \in \{1, \dots, N\}. \quad (8.5)$$

When extended to k -state exclusion, strong exclusion means that there exists a POVM that can exclude all possible sub-sets of $\{\rho_x\}_{x=1}^N$ of length k . Weak exclusion then means that there exists a POVM that can only exclude *some subsets* of $\{\rho_x\}_{x=1}^N$ of length k .

It is clearly the case from the above definitions that weak state exclusion is a requisite for strong state exclusion — justifying their respective names. Moreover, if one is able to perform strong state exclusion, they can trivially convert this into weak state exclusion via classical post-processing of the measurement outcomes. It can also be seen that if any of the states in $\{\rho_x\}_{x=1}^N$ are full-rank, then strong state exclusion is never possible. This is due to $\text{tr}[T_g \rho_x] = 0$ if and only if $T_g = 0$ when ρ_x is full-rank.

Previously, our proposed definition of weak state exclusion has been used as the general definition of state exclusion [190]. However, this definition has attracted (indirect) criticism for trivialising the problem of state exclusion [189], as if a player can perform conclusive 1-state exclusion on any two states $\{\rho_1, \rho_2\} \subseteq \{\rho_x\}_{x=1}^N$ using the two-element POVM $\{M_1, M_2\}$, then by definition 1-state exclusion could trivially be performed on the whole set $\{\rho_x\}_{x=1}^N$ by considering the N element POVM

$$\{T_1 = M_1, T_2 = M_2, T_3 = 0, \dots, T_N = 0\}. \quad (8.6)$$

Each measurement outcome would exclude one state, but some states would never be excluded. Whilst such an example is indeed trivial, there exist plenty of intermediate scenarios between this and strong state exclusion that could prove useful in operationally motivated tasks. For example, consider a task where player A is trying to communicate to player B which of N wires can be cut to diffuse a bomb [186]. Player A can aim to send exclusionary information to player B that says “do not cut wire α or wire β .” They do this by encoding exclusionary information into some quantum state and sending it to player B. Player B then measures a POVM that performs conclusive 2-state exclusion, allowing them to say with certainty that they do not have states associated to the label α or β and hence those are the wires not to cut. To succeed, player B must output a list of wires not to cut; he does not need to be able to exclude all possible subsets of wires of length 2. Hence, weak k -state exclusion would still prove useful in the task of preventing accidental detonation during defusal.

8.2.2 Necessary condition for k -state exclusion

It has previously been shown that all k -state exclusion tasks can be recast as 1-state exclusion tasks by reformulating the set $\{\rho_x\}_{x=1}^N$ (see Appendix I of Ref. [66] or Section 3.2.2). Conceptually, this means

that all k -state exclusion tasks are mathematically equivalent to a 1-state exclusion task — solving one task means you have solved the other. Consequently, all state exclusion tasks can be studied within the 1-state exclusion framework through this “equivalent” task. This has led to a consensus that only the task of 1-state exclusion needed to be studied, and hence, all feasibility conditions in the literature for both weak and strong state exclusion tasks have been for conclusive 1-state exclusion [64, 66, 189–192]. However, when using the reformulation method for accessing k -state exclusion tasks, the size of the reformulated sets can get very large for particular values of N and k , potentially making the equivalent task computationally infeasible to access. In addition, scenarios may exist where one wants to consider the original task rather than the equivalent task; this may happen, for instance, if the set of states upon which exclusion is being performed holds some operational significance. By reformulating the set into the equivalent task, the states in the set will differ, potentially significantly, from the physical states considered in the original task. Hence, a condition for k -state exclusion that is dependent only on the original set $\{\rho_x\}_{x=1}^N$ is of value — it by-passes the need to consider the equivalent task. Here, a necessary condition of this form is presented.

Lemma 8.2.1. *A referee has a set of N , d -dimensional quantum states, $\{\rho_x\}_{x=1}^N$. A necessary condition for the existence of a POVM such that the player can perform conclusive strong or weak k -state exclusion is*

$$\sum_{x=1}^N \Pi_x \leq (N - k)\mathbb{I}, \quad (8.7)$$

where Π_x is the projector onto the support of ρ_x for all x .

Proof. A referee has a set of states $\{\rho_x\}_{x=1}^N$. Let $\mathcal{Y}_{(N,k)}$ be the set of all subsets of length k of the set $\{1, \dots, N\}$ [66]. During each round of the task, the referee randomly generates a label x and sends the state ρ_x to the player. The player measures a POVM on the state ρ_x and aims to output a set of k labels $Y \in \mathcal{Y}_{(N,k)}$ such that $x \notin Y$. Such a measurement will be a POVM with $\binom{N}{k}$ elements, denoted by $\mathbf{S} := \{S_Y\}_{Y \in \mathcal{Y}_{(N,k)}}$. The player is able to perform conclusive k -state exclusion if for all $Y \in \mathcal{Y}_{(N,k)}$ there exists an S_Y such that $\text{tr}[S_Y \rho_y] = 0 \forall y \in Y$. If the player gets the measurement outcome associated to S_Y , they can output the set Y knowing with certainty the referee could not have sent any of the states in the set $\{\rho_y\}_{y \in Y}$. By defining the operator $R_Y := \sum_{y \in Y} \rho_y$, the conclusive k -state exclusion task can then be succinctly expressed as

$$\text{tr}[S_Y R_Y] = 0 \quad \forall Y \in \mathcal{Y}_{(N,k)}. \quad (8.8)$$

Letting $L := \binom{N}{k}$, one can order $\mathbf{S}$'s elements and write $\mathbf{S} := \{S_l\}_{l=1}^L$. Similarly, we also order the operators R_Y 's by the same label and write $\{R_l\}_{l=1}^L$. Now, it can be seen that a given $x \in \{1, \dots, N\}$ will appear in $\binom{N-1}{k-1} = Lk/N$ many subsets in $\mathcal{Y}_{(N,k)}$. To see this, consider a set of N elements and fix one to appear in all subsets. For subsets of size k , one then needs to find the number of different subsets of the remaining $N - 1$ elements of length $k - 1$ there are, as adding the fixed element into each of these subsets will then give all the subsets of length k of the full set containing the fixed element. There are $\binom{N-1}{k-1}$ such subsets.

Returning to the proof, this means that, for each state ρ_x , there are Lk/N many operators R_l 's that contain it. For each $x \in \{1, \dots, N\}$, let X_x denote the set of all labels l corresponding to these R_l 's. Each X_x thus contain Lk/N many labels, and we have $\text{tr}[S_l \rho_x] = 0 \forall l \in X_x$. Equation (8.8) thus implies

$$\text{tr} \left[\rho_x \left(\sum_{l \in X_x} S_l \right) \right] = 0 \quad \forall x \in \{1, \dots, N\}. \quad (8.9)$$

We then use the following Lemma, proved in Section 8.6.1.

$$\begin{array}{c}
\text{Original Set} \\
\overline{\{\rho_1, \rho_2, \rho_3, \rho_4\}} \rightarrow \underbrace{\{\rho_1 + \rho_2, \rho_1 + \rho_3, \rho_1 + \rho_4, \rho_2 + \rho_3, \rho_2 + \rho_4, \rho_3 + \rho_4\}}_{K=2 - \text{Exclusion Set}}
\end{array}$$

$$\begin{array}{c}
\left. \begin{array}{l}
\text{tr}[S_{Y_1}(\rho_1 + \rho_2)] = 0 \\
\text{tr}[S_{Y_2}(\rho_1 + \rho_3)] = 0 \\
\text{tr}[S_{Y_3}(\rho_1 + \rho_4)] = 0 \\
\text{tr}[S_{Y_4}(\rho_2 + \rho_3)] = 0 \\
\text{tr}[S_{Y_5}(\rho_2 + \rho_4)] = 0 \\
\text{tr}[S_{Y_6}(\rho_3 + \rho_4)] = 0
\end{array} \right\} \Rightarrow \left\{ \begin{array}{l}
\text{tr}[\rho_1(S_{Y_1} + S_{Y_2} + S_{Y_3})] = 0 \Rightarrow \Pi_{\rho_1} \leq \mathbb{I} - (S_{Y_1} + S_{Y_2} + S_{Y_3}) \\
\text{tr}[\rho_2(S_{Y_1} + S_{Y_4} + S_{Y_5})] = 0 \Rightarrow \Pi_{\rho_2} \leq \mathbb{I} - (S_{Y_1} + S_{Y_4} + S_{Y_5}) \\
\text{tr}[\rho_3(S_{Y_2} + S_{Y_4} + S_{Y_6})] = 0 \Rightarrow \Pi_{\rho_3} \leq \mathbb{I} - (S_{Y_2} + S_{Y_4} + S_{Y_6}) \\
\text{tr}[\rho_4(S_{Y_3} + S_{Y_4} + S_{Y_6})] = 0 \Rightarrow \Pi_{\rho_4} \leq \mathbb{I} - (S_{Y_3} + S_{Y_4} + S_{Y_6})
\end{array} \right\} \Rightarrow \sum_{x=1}^4 \Pi_{\rho_x} \leq (N-k)\mathbb{I} = 2\mathbb{I} \quad (D)
\end{array}$$

(A) (B) (C)

Figure 8.2: An example and a graphical depiction of the proof of Lemma 8.2.1 for performing $k = 2$ exclusion on $N = 4$ states. Firstly, the set of states upon which exclusion is to be performed, $\{\rho_1, \rho_2, \rho_3, \rho_4\}$, is reformulated into a set of all subsets of $\{\rho_1, \rho_2, \rho_3, \rho_4\}$ of length two. In (A) the definition for conclusive weak state exclusion, detailed in Definition 8.2.2, is assumed to hold for the POVM $\mathbf{S} := \{S_Y\}_{Y \in \mathcal{Y}_{(4,2)}}$. In (B), these conditions are rewritten to factorise out the states. From each of these four conditions — one for each of the four input states — Lemma 8.2.2 is used to create the conditions in (C). Finally, these conditions are summed over in (D) to give the final expression.

Lemma 8.2.2. Let $\sigma \geq 0$ and $0 \leq Q \leq \mathbb{I}$ be operators such that $\text{tr}[\sigma Q] = 0$, then

$$\Pi_\sigma \leq \mathbb{I} - Q, \quad (8.10)$$

where Π_σ is the projector onto the support of σ .

Employing Lemma 8.2.2, the above N equations implies that

$$\Pi_x \leq \mathbb{I} - \sum_{l \in X_x} S_l \quad \forall x \in \{1, \dots, N\}, \quad (8.11)$$

where Π_x is the projector onto the support of the state ρ_x . Summing over the N individual conditions in Eq. (8.11) gives

$$\sum_{x=1}^N \Pi_x \leq N\mathbb{I} - \sum_{x=1}^N \sum_{l \in X_x} S_l. \quad (8.12)$$

By Eq. (8.8), for each l , there are exactly k many possible labels x 's such that $l \in X_x$. Hence, each POVM element S_l appears exactly k times, meaning that $\sum_{x=1}^N \sum_{l \in X_x} S_l = \sum_{l=1}^L k S_l = k\mathbb{I}$ and thus $\sum_{x=1}^N \Pi_x \leq (N-k)\mathbb{I}$, as desired. See Fig. 8.2 for an example and graphical depiction of this proof. $\square$

Given that every sub-channel exclusion task induces an effective state exclusion task, Lemma 8.2.1 can be applied to both state and sub-channel exclusion tasks. The proof of Lemma 8.2.1 does not enforce Eq. (8.3) as a condition, and, in fact, is provably only a necessary condition for strong state exclusion $\hat{\mathbf{S}}$. However, it is left open as to whether Lemma 8.2.1 is a sufficient condition for weak k -state exclusion. Although, if the inequality is saturated, then Lemma 8.2.1 is sufficient for strong 1-state exclusion. In this case, measuring the POVM $\{(\mathbb{I} - \Pi_x)/k\}_{x=1}^N$ would perform 1-state exclusion.

As an application of Lemma 8.2.1, consider $\{\rho_x\}_{x=1}^N$ to be a set of N orthogonal states. It follows that $\sum_{x=1}^N \Pi_x \leq \mathbb{I}$, and, hence, the largest value of k such that Lemma 8.2.1 is satisfied is $k = N - 1$. Lemma 8.2.1 therefore implies the ability to perform conclusive state discrimination on a set of N orthogonal states, as expected. This also shows that there exists a set of states for all values of N and d for which Eq. (8.7) is saturated; namely, Lemma 8.2.1 is tight. In addition, if considering rank r states of dimension d , one can always find a weak exclusion task for which Eq. (8.7) is saturated if $N = \binom{d}{r}$.

[§] Consider the set of states with projectors onto their supports of $\{|00\rangle\langle 00|, |01\rangle\langle 01|, |00\rangle\langle 00| + |10\rangle\langle 10|, |01\rangle\langle 01| + |11\rangle\langle 11|\}$. These satisfy Lemma 8.2.1 for $k = 2$, but by considering the equivalent task it can be seen that strong state exclusion is never possible as there is a full-rank state in the reformulated set.

Firstly, let $\{|i\rangle\}_{i=0}^{d-1}$ be a basis in the d -dimensional space. One can then consider $\{\rho_x\}_{x=1}^N$ such that each Π_x is a projector onto the basis elements contained in each subset of $\{|i\rangle\}_{i=0}^{d-1}$ of length r , of which there are N of them. By measuring the POVM $\{|i\rangle\langle i|\}_{i=0}^{d-1}$, conclusive weak k -state exclusion can be performed with $k = \binom{d-1}{r}$. This is as predicted by Lemma 8.2.1, as $\sum_{x=1}^N \Pi_x = \binom{d-1}{r-1} \mathbb{I}$, with $\binom{d}{r} - \binom{d-1}{r} = \binom{d-1}{r-1}$.

Lemma 8.2.1 also leads to the following corollary on the maximum value of k .

Corollary 8.2.3. *When performing conclusive state exclusion on $\{\rho_x\}_{x=1}^N$, an upper bound on the number, k , of states that can be excluded is given by*

$$k \leq N - \left(2^{D_{\max}(\omega \| \mathbb{I}/d)} \alpha \right) / d \leq N(d-1)/d, \quad (8.13)$$

where $\alpha := \text{tr}[\sum_{x=1}^N \Pi_x]$, $\omega := \sum_{x=1}^N \Pi_x / \alpha$, and $D_{\max}(\psi \| \sigma) := \log_2 \min\{\lambda \geq 1 : \psi \leq \lambda \sigma\}$, such that $\psi \in \mathcal{D}(\mathcal{H})$ and $\sigma \geq 0$, is the max relative entropy (see Definition 2.7.3 and [47]).

Proof. By comparison to Lemma 8.2.1, it can be seen that the λ used in the definition of $D_{\max}$ can be related to $N - k$ when considering the largest possible k . Rearranging and noting that ψ must be a quantum state gives the first inequality. By taking the trace of both sides of Lemma 8.2.1, the second inequality can be shown. The trace of the left-hand side is lower bounded by N , which is achieved when all ρ_x 's are rank-one projectors, i.e., all of them are pure states. In this best-case scenario where all ρ_x 's are rank-one projectors, we have $\alpha = N$, such that in general $\alpha \geq N$. It can then be seen that the second inequality always upper bounds the first given that $0 \leq D_{\max}(\psi \| \sigma) \forall \psi, \sigma$. $\square$

Corollary 8.2.3 sets a fundamental limit on the number of states that can be excluded, and, interestingly, gives the max-relative entropy a novel operational meaning in terms of state exclusion tasks. Specifically, given a set of states, the max-relative entropy establishes an upper-bound on the number of states that can be conclusively excluded. In the next section, the Choi-rank is given a novel operational interpretation using Lemma 8.2.1 — its shown to set a universal upper bound on the number of states that can be excluded in a communication task.

8.3 Operational Interpretation of The Choi-Rank

An operational interpretation of the Choi-rank of a channel $\mathcal{N}$ is now presented through an *entanglement-assisted sub-channel exclusion task*. The task is defined as a communication task between two spatially separated parties, Alice (A) and Bob (B). Alice aims to use a pre-shared entangled state to increase the amount of classical information she can send to Bob through a single use of a channel $\mathcal{N}$, as in super-dense coding [69] (see Section 3.3). It is assumed that the message Alice is sending is of the utmost importance, meaning Bob chooses to only output information about the encoded message that he is certain of.

Task 8.3.1 (Choi-rank Operational Task). *Alice and Bob share a bipartite maximally entangled state of local dimension d . Alice encodes one of N messages into her half of the state by applying one of the N unitary channels $\{U_x\}_{x=1}^N$. She then sends her half of the maximally entangled state to Bob via the channel $\mathcal{N}$. Bob performs a joint measurement and aims to output a set of k messages that he is certain Alice did not encode.*

See Fig. 8.3 for a graphical depiction of Task 8.3.1. This task aims to operationally quantify the information about Alice's encoded message (communicated via $\mathcal{N}$) that Bob can obtain with certainty. Its significance can be understood by first considering the case with $k = N - 1$ and $N = d^2$. If Bob achieves this, he can exclude all messages but one, effectively outputting a single message, l , such that $l = x$ with certainty. Hence, Bob knows with certainty what message Alice encoded. In this case, Task 8.3.1 becomes equivalent to (conclusive) super-dense coding. However, it is known that Bob can only achieve $k = N - 1$ if the states after encoding and sending are orthogonal. This is, in general, not true whenever $\mathcal{N}$ introduces noise (see, e.g., Ref. [196]). Bob can therefore instead attempt to say something

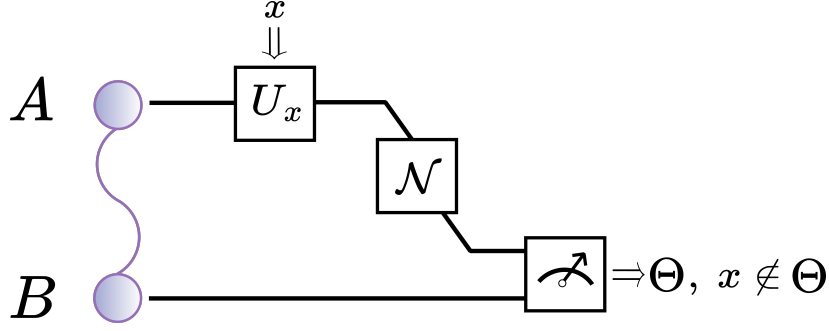


Figure 8.3: A diagram of the entanglement-assisted sub-channel exclusion task, detailed in Task 8.3.1, where x is the message Alice encodes, and Θ is the set of messages Bob outputs that will be a subset of the set of all messages.

with certainty about Alice's encoded message by outputting a set of k messages, where $k \leq N - 1$, which he is certain *does not* contain x .

We will focus on Bob's ability to maximise the value of k , measuring his success in Task 8.3.1 by the maximum number of messages that it is possible for him to exclude with certainty. This is a justified measure of success as the larger the value of k , the more Bob knows about which message Alice encoded. This culminates in Bob performing conclusive state discrimination if $k = N - 1$ and hence knowing Alice's encoded message with certainty — the aim of a communication task. The following result upper-bounds k via the Choi-rank of $\mathcal{N}$ and holds for all possible unitary-encoding and decoding (POVMs) strategies:

Result 8.3.2. *The maximum number of messages, k , that Bob can exclude in Task 8.3.1 is*

$$k \leq \frac{N(d^2 - r_c^{\mathcal{N}})}{d^2}. \quad (8.14)$$

Proof. Alice and Bob share a maximally entangled state $|\Phi^+\rangle_{AB} = \sum_{i=0}^{d-1} |ii\rangle_{AB} / \sqrt{d}$ with an equal local dimension d (here, A, B denotes Alice's and Bob's systems). If Alice encodes the message x (via the unitary channel $\mathcal{U}_{x,A} = U_x(\cdot)U_x^\dagger$ in A) and then sends her half of the state to Bob via the channel $\mathcal{N}_A$, Bob has the state

$$\begin{aligned} \rho_{AB}^{x|\mathcal{N}} &:= (\mathcal{N}_A \otimes \mathcal{I}_B) \circ (\mathcal{U}_{x,A} \otimes \mathcal{I}_B) (|\Phi^+\rangle\langle\Phi^+|_{AB}) \\ &= (\mathcal{I}_A \otimes \mathcal{U}_{x,B}^t) (\mathcal{J}_{AB}^{\mathcal{N}}). \end{aligned} \quad (8.15)$$

where $\mathcal{U}_x^t = U_x^t(\cdot)[U_x^t]^\dagger$ where $(\cdot)^t$ is the transpose in the basis defined by $|\Phi^+\rangle_{AB}$. Note that, after the channel $\mathcal{N}_A$, Bob has the whole bipartite state. From Bob's point of view, he, therefore, has a state from the set $\{\rho_{AB}^{x|\mathcal{N}}\}_{x=1}^N$. Note that all elements of this set have the same rank — the Choi-rank, $r_c^{\mathcal{N}}$, of the channel $\mathcal{N}_A$. This is due to the rank of states being invariant under unitary channels. Bob now aims to perform conclusive state exclusion on this set and hence Lemma 8.2.1 can be applied. Given all states in the set are of rank $r_c^{\mathcal{N}}$, all projectors onto the support of those states are of rank $r_c^{\mathcal{N}}$. Taking the trace of both sides of Eq. (8.7) in Lemma 8.2.1 therefore gives $Nr_c^{\mathcal{N}} \leq (N - k)d^2$. Rearranging completes the proof. $\square$

Result 8.3.2 is derived under minimal assumptions about dynamics — namely, that they are modelled by quantum channels — making it broadly applicable. Below, it is shown that there exist situations in which this bound is tight, meaning the inequality can be saturated. This tight bound on an operational task (Task 8.3.1) can then be used to provide the Choi-rank with an operational interpretation — it is the quantity that limits the number of messages that could ever be excluded in Task 8.3.1 given a fixed N and d . Put alternatively, the Choi-rank of a quantum channel limits what can be decoded with certainty about an encoded message communicated via a channel $\mathcal{N}$. See Proposition 8.3.3 for a discussion about when/if a bound can be considered an operational interpretation.

We will now consider examples of Task 8.3.1, where $\mathcal{N}$ is a depolarising channel and a dephasing channel, that demonstrate that Result 8.3.2 is tight.

Proposition 8.3.3 (when can a bound be an operational interpretation?). Consider some operation task and some quantity x . If it holds that one's ability to succeed in this task is bounded by x , e.g., $\text{Success} \leq x$, then this bound *might* provide the quantity x with an operational interpretation. However, it could be the case that the quantity x actually has no relevance to the operational task it is bounding. To see this, consider the tasks of guessing the outcome of a coin toss i.e., guessing either Heads or Tails. The probability of success in this task is upper-bounded by 1, $P(\text{Success}) \leq 1$. Now, let x be the number of arms on an average human i.e., $x = 2$. Hence, $P(\text{Success}) \leq x$, meaning that the number of arms on the average human does upper-bound the ability to succeed in this task. However, in this case, it is clear that x has no relevance to one's ability to succeed in the operational task. Hence, only upper bounding the success probability is not enough to determine that x is an operational interpretation. An additional piece of information that could add to the likelihood of x being relevant to the operational task is if the bound is achievable — if it is tight. Whilst being tight does not guarantee that x has relevance to the operational task, it does provide further evidence to suggest it does. In the above example, the bound is not tight, which suggests to the irrelevance of quantity x to the operational task.

8.3.0.1 Depolarising Channel

Firstly, we consider the scenario that Alice is encoding one of N messages into one half of a d -dimensional maximally entangled state, using the unitaries $\{U_x\}_{x=1}^N$. She then sends her half of the state to Bob via a depolarising channel, $\mathcal{D}_p^{\text{pol}}(\rho) := p\rho + (1-p)\text{tr}[\rho]\mathbb{I}/d$. Note, we assume no relationship between N and d .

In his possession, Bob will therefore have one of the following states

$$\begin{aligned}\rho_{AB}^{x|\mathcal{D}^{\text{pol}}} &:= (\mathcal{D}_p^{\text{pol}}(\cdot) \otimes \mathcal{I}_B) \circ (\mathcal{U}_{x,A} \otimes \mathcal{I}_B)(|\Phi^+\rangle\langle\Phi^+|_{AB}) \\ &= (\mathcal{I}_A \otimes \mathcal{U}_{x,B}^t)(\mathcal{J}_{AB}^{\mathcal{D}_p^{\text{pol}}}), \quad x \in \{1, 2, \dots, l\},\end{aligned}\tag{8.16}$$

where $\mathcal{U}_x = U_x(\cdot)U_x^\dagger$, such that $\mathcal{U}_x^t = U_x^\dagger(\cdot)U_x$, and $\mathcal{J}_{AB}^{\mathcal{D}_p^{\text{pol}}}$ is the Choi-state of the Depolarising channel,

$$\begin{aligned}\mathcal{J}_{AB}^{\mathcal{D}_p^{\text{pol}}} &:= (\mathcal{D}_p^{\text{pol}}(\cdot) \otimes \mathcal{I}_B)(|\Phi^+\rangle\langle\Phi^+|_{AB}) \\ &= p|\Phi^+\rangle\langle\Phi^+|_{AB} + (1-p)\mathbb{I}_{AB}/d^2.\end{aligned}\tag{8.17}$$

As $\mathbb{I}_{AB}/d^2$ is full-rank, and taking a convex mixture with a positive operator cannot lower the rank, the Choi-state of the depolarising channel must also be full-rank if $p < 1$. Hence, $r_c^N = d^2$ for all $p < 1$, and Result 8.3.2 implies that if Bob performs exclusion on this set of states, then $k \leq 0$, meaning $k = 0$ as k is a positive integer.

Now, given that all states output from a depolarising channel are full-rank, there exists no POVM element, M_i for example, such that $\text{tr}[M_i\mathcal{D}_p^{\text{pol}}(\rho)] = 0$ for all possible input states, $\rho \in \mathcal{D}(\mathcal{H})$ (other than the trivial case of $M_i = 0$). Hence, it must be the case that $k = 0$, meaning Result 8.3.2 is tight in this scenario. Physically, this means that Bob can say *nothing* with certainty about the message encoded by Alice when $\mathcal{N} = \mathcal{D}_p^{\text{pol}}(\cdot)$ if $p < 1$, regardless of N and d .

8.3.0.2 Dephasing channel

We now consider the scenario that Alice is trying to perform super-dense coding in the presence of dephasing noise: she encodes one of d^2 messages into a maximally entangled state with local dimension d using the Heisenberg-Weyl operators [10], before sending her half of the state to Bob via a *dephasing channel*, $\mathcal{D}_p^{\text{ph}}(\rho) := p\rho + (1-p)\sum_{n=0}^{d-1}|n\rangle\langle n|\rho|n\rangle\langle n|$. Bob then measures the joint state in the maximally

entangled basis generated by these Heisenberg-Weyl operators and $|\Phi^+\rangle_{AB}$.

Firstly, as a recap of Section 2.9, in a space of dimension d the *Heisenberg-Weyl operators*[10] are defined as

$$W_{a,b} := U^a V^b = \sum_{n=0}^{d-1} \Omega^{bn} |n+a\rangle\langle n|, \quad (8.18)$$

where $a, b \in \{0, 1, \dots, d-1\}$ are cyclic and,

$$U := \sum_{n=0}^{d-1} |n+1\rangle\langle n|, \quad V := \sum_{n=0}^{d-1} \Omega^n |n\rangle\langle n|, \quad \Omega := e^{\frac{2\pi i}{d}}. \quad (8.19)$$

Using the Heisenberg-Weyl operators, a maximally entangled basis in AB can be generated as

$$|\Phi_{ab}^+\rangle_{AB} := (\mathbb{I}_A \otimes W_{a,b})(|\Phi_{00}^+\rangle_{AB}) \quad \forall a, b \in \{0, 1, \dots, d-1\}, \quad (8.20)$$

where, in this notation, we have $|\Phi_{00}^+\rangle_{AB} = |\Phi^+\rangle_{AB}$.

One of the d^2 messages encoded by Alice can therefore be characterised by the tuple (a', b') , where she applies $W_{a',b'}$ locally to encode the message (a', b') . At the culmination of the protocol, Bob then measures the non-degenerate observable

$$O = \sum_{a,b=0}^{d-1} \lambda_{a,b} |\Phi_{ab}^+\rangle\langle\Phi_{ab}^+|, \quad (8.21)$$

and decodes Alice's measures as (a', b') if he gets the measurement outcomes $\lambda_{a',b'}$.

In the scenario outline above, depending on the message encoded by Alice, Bob will have one of the following d^2 -many bipartite states,

$$\begin{aligned} \rho_{AB}^{(a,b)|\mathcal{D}^{\text{ph}}} &:= (\mathcal{D}_p^{\text{ph}}(\cdot) \otimes \mathcal{I}_B) \circ (\mathcal{W}_{a,b} \otimes \mathcal{I}_B)(|\Phi^+\rangle\langle\Phi^+|_{AB}) \\ &= (\mathcal{I}_A \otimes \mathcal{W}_{a,b}^t)(\mathcal{J}_{AB}^{\mathcal{D}_p^{\text{ph}}}) \quad \forall a, b \in \{0, 1, \dots, d-1\}. \end{aligned} \quad (8.22)$$

where $\mathcal{W}_{a,b}[\cdot] := W_{a,b}[\cdot]W_{a,b}^\dagger$, such that $\mathcal{W}_{a,b}^t[\cdot] := W_{a,b}^t[\cdot](W_{a,b}^t)^\dagger$, and $\mathcal{J}_{AB}^{\mathcal{D}_p^{\text{ph}}}$ is the Choi-state of the dephasing channel,

$$\mathcal{J}_{AB}^{\mathcal{D}_p^{\text{ph}}} = \alpha |\Phi_{00}^+\rangle\langle\Phi_{00}^+|_{AB} + \frac{(1-\alpha)}{d-1} \sum_{c=1}^{d-1} |\Phi_{0c}^+\rangle\langle\Phi_{0c}^+|_{AB}, \quad (8.23)$$

where $\alpha = [p(d-1) + 1]/d$. See Section 8.6.2 for a full derivation of this Choi-state.

From here, it can be seen that the Choi-rank of the dephasing channel is d if $p < 1$, that is, $r_c^{\mathcal{N}} = d$ when $\mathcal{N} = \mathcal{D}_p^{\text{ph}}(\cdot)$ for $p < 1$. Result 8.3.2 therefore implies that k -state exclusion is possible for $k \leq d^2 - d$.

We now show that Bob can saturate this bound if he measures O . Firstly, we input the Choi-state of the dephasing channel, Eq. (8.23), into the set of states in Bob possession, Eq. (8.22), which gives

$$\begin{aligned} \rho_{AB}^{(a,b)|\mathcal{D}^{\text{ph}}} &= (\mathcal{I}_A \otimes \mathcal{W}_{a,b}^t)(\mathcal{J}_{AB}^{\mathcal{D}_p^{\text{ph}}}) \\ &= (\mathcal{I}_A \otimes \mathcal{W}_{-a,b}) \left(\alpha |\Phi_{00}^+\rangle\langle\Phi_{00}^+|_{AB} + \frac{1-\alpha}{d-1} \sum_{c=1}^{d-1} (\mathcal{I}_A \otimes \mathcal{W}_{0,c})(|\Phi_{00}^+\rangle\langle\Phi_{00}^+|_{AB}) \right) \\ &= \alpha |\Phi_{-a,b}^+\rangle\langle\Phi_{-a,b}^+|_{AB} + \frac{1-\alpha}{d-1} \sum_{c=1}^{d-1} |\Phi_{-a,b+c}^+\rangle\langle\Phi_{-a,b+c}^+|_{AB}, \end{aligned} \quad (8.24)$$

where we have used the following identities,

$$\begin{aligned} W_{a,b}^t &= \Omega^{-ab} W_{-a,b} \\ W_{a,b} W_{n,m} &= \Omega^{bn} W_{a+n,b+m} = \Omega^{bn-am} W_{n,m} W_{a,b}, \end{aligned} \quad (8.25)$$

which hold for every a, b, n, m [10].

From Eq. (8.24), it can be seen that if Alice encodes any of the messages

$$\{(a, b) : b \in \{0, 1, \dots, d-1\}\}, \quad (8.26)$$

all the possible states that could be in Bob's possession will have identical support. Namely, they will always be supported in the subspace

$$\text{span}\left\{|\Phi_{d-a,b}^+\rangle\langle\Phi_{d-a,b}^+|_{AB} : b \in \{0, 1, \dots, d-1\}\right\}, \quad (8.27)$$

where we have used the fact that the labels are cyclic mod d to set $-a = d - a$. If Alice instead encoded any of the messages $\{(l, b) : b \in \{0, 1, \dots, d-1\}\}$, where $l \neq a$, the possible states in Bob's possession are again all supported in the same subspace. However, this subspace will be orthogonal to the subspace for $l = a$. Using this fact, the Hilbert space can be split into d , d -dimensional orthogonal subspaces such that the possible states in Bob's possession for the encoded messages $(a', b') \forall a' \in \{0, 1, \dots, d-1\}$ are supported in a different subspace, whilst, for a given a' , for all $b' \in \{0, 1, \dots, d-1\}$ the states are supported in the same subspace.

Therefore, if Bob measures O and gets one of the outcomes $\{\lambda_{d-r,b} : b \in \{0, 1, \dots, d-1\}\}$, he knows for certain that Alice encoded one of the messages $\mathfrak{A}^{\text{enc}} = \{(r, b) : b \in \{0, 1, \dots, d-1\}\}$. Alternatively, this can be phrased as: Bob knows for certain that Alice did not encode a message from the set

$$\{(a, b) : a, b \in \{0, 1, \dots, d-1\}\} \setminus \mathfrak{A}^{\text{enc}}. \quad (8.28)$$

As $|\mathfrak{A}^{\text{enc}}| = d$, and there are d^2 total messages, measuring O therefore excludes $k = d^2 - d$ messages. Hence, Result 8.3.2 is saturated in the case.

Physically, this result can be understood as follows: given that Alice encodes a message (a', b') and then sends her state to Bob under a dephasing channel with $p < 1$, if Bob measures O he is able to say with certainty what the first parameter, a' , of the encoded message is — he is able to perfectly distinguish the first parameter. However, as a result of the noise, nothing can be said for certain about the second parameter. Alice is therefore able to send Bob d messages with certainty. Hence, if insisting on certainty of the decoded message, any amount of dephasing noise will remove the quantum advantage of superdense coding. In this case, Alice can only send one of d -messages to Bob when sending him a d -dimensional state — exactly what is achievable classically using a d -dimensional, as seen in Section 3.3. This is in contrast to the d^2 -messages that can be sent using a d -dimensional state in noiseless superdense coding.

8.4 Encoding via Unital Channels

A natural extension to Result 8.3.2 is to ask if it also holds for encodings via general quantum channels in Task 8.3.1, rather than just unitary channels. As a first attempt at this, we extend Result 8.3.2 to hold for all encodings via unital channels that do not change the dimension of the system.

A channel $\mathcal{E}$ is unital if $\mathcal{E}(\mathbb{I}) = \mathbb{I}$, meaning it acts invariantly on the identity, as defined in Definition 4.3.3. We will use the fact that the transpose of a unital channel is also a unital channel, proved in Section 8.6.3, and that unital channels that do not change the dimension are rank-non-decreasing, proved in Section 8.6.4.

We now consider Task 8.3.1 where Alice encodes her message into one half of the d -dimensional maximally entangled state using one of a set of unital channels $\{\mathcal{E}_x\}_{x=1}^N$ (rather than unitary channels).

If Alice then sends her half of the state to Bob via the channel $\mathcal{N}$, Bob has the state

$$\begin{aligned}\rho_{AB}^{x|\mathcal{N}} &:= (\mathcal{N}_A \otimes \mathcal{I}_B) \circ (\mathcal{E}_{x,A} \otimes \mathcal{I}_B) (|\Phi^+\rangle\langle\Phi^+|_{AB}) \\ &= (\mathcal{I}_A \otimes \mathcal{E}_{x,B}^t) (\mathcal{J}_{AB}^{\mathcal{N}}).\end{aligned}\tag{8.29}$$

From Bob's point of view, he, therefore, has a state from the set $\{\rho_{AB}^{x|\mathcal{N}}\}_{x=1}^N$. In general, not all elements of this set will now have the same rank, as was the case in the proof of Result 8.3.2. Although, due to the fact that $\mathcal{E}^t$ is still a unital channel, and that unital channels are rank non-decreasing, the rank of all states in the set is lower bounded by the Choi-rank, $r_c^{\mathcal{N}}$, of the channel $\mathcal{N}_A$, i.e., $\text{Rank}(\rho_{AB}^{x|\mathcal{N}}) \geq r_c^{\mathcal{N}} \forall x$.

Now, as before, Lemma 8.2.1 is applied to the set of states in Bob possession, and the trace of both sides of Eq. (8.7) is taken. As the rank of each projector onto a state in the set is also lower bounded by $r_c^{\mathcal{N}}$, the trace over the sum of the projectors onto the states in the set is lower bounded by $Nr_c^{\mathcal{N}}$. The right hand side remains unchanged from before. Therefore, $Nr_c^{\mathcal{N}} \leq (N - k)d^2$ still bounds one's ability to succeed in Task 8.3.1 if unital channels are used for the encodings.

Using this same proof technique, it can be seen that the result will still hold whenever a set of rank non-decreasing channels are used for the encoding of the messages. This expands the applicability of Result 8.3.2 further still, and we conjecture that it will generalises beyond rank non-decreasing channels to include encodings via general channels, but the proof is left for future work.

8.5 Conclusions

We give the Choi-rank a novel operational interpretation as the fundamental limit on one's ability to succeed in an entanglement-assisted exclusion task. To derive this result, a necessary condition for conclusive k -state exclusion has been presented, and the notion of weak and strong state exclusion has been introduced. This condition allows the viability of conclusive k -state exclusion to be assessed without the need to first reformulate the set and apply the conditions for 1-state exclusion. Although, by considering $k = 1$, this also adds to the conditions for conclusive 1-state exclusion already present in the literature [64, 66, 189–192]. Whilst it is known that this condition is not sufficient for strong-state exclusion, it would be interesting to know if it is sufficient for weak-state exclusion.

There are several directions in which Result 8.3.2 could be generalised. Firstly, whilst Result 8.3.2 holds for all possible unital-encoding and (general) decoding strategies, it is unknown if it holds for all initial states shared between Alice and Bob. It follows from the definition of the Choi-state that Result 8.3.2 holds for any full-Schmidt-rank state shared between Alice and Bob. And, intuitively, one would imagine that using a less entangled initial state could only reduce one's ability to succeed at the task. This intuition arises from the knowledge that entanglement is a resource for super-dense coding, which is a special case of Task 8.3.1. Understanding this would enable us to determine the underlying resources of Task 8.3.1. Secondly, as stated in the above conjecture, it would be interesting to know if Result 8.3.2 could be generalising to include general (non-unital) encoding strategies.

Understanding how these extensions affect one's ability to succeed in the task will help assess the boundaries of the limitations imposed by a channel's Choi-rank. Moreover, it will allow the significance of this task in quantifying resources to be assessed [67, 186–188, 197–199]

8.6 Additional Material

8.6.1 Proof of Lemma 8.2.1

Here, the following lemma, used in the proof of Lemma 8.2.1, is proved.

Lemma 8.2.2 *Let $\sigma \geq 0$ and $0 \leq Q \leq \mathbb{I}$ be operators such that $\text{tr}[\sigma Q] = 0$, then*

$$\Pi_\sigma \leq \mathbb{I} - Q,\tag{8.30}$$

where Π_σ is the projector onto the support of σ .

Proof. Firstly, note that $\sigma \geq \mu_{\min}(\sigma)\Pi_\sigma$, where $\mu_{\min}(\sigma) > 0$ is the minimal non-zero eigenvalue of σ . From here it can be seen that

$$\begin{aligned}
 0 &= \text{tr}[\sigma Q] \\
 &= \text{tr}[\Pi_\sigma \sigma \Pi_\sigma Q] \\
 &= \text{tr}[\sigma \Pi_\sigma Q \Pi_\sigma] \\
 &\geq \mu_{\min}(\sigma) \text{tr}[\Pi_\sigma \Pi_\sigma Q \Pi_\sigma] \\
 &= \mu_{\min}(\sigma) \text{tr}[\Pi_\sigma Q \Pi_\sigma].
 \end{aligned} \tag{8.31}$$

Next, we note that $\Pi_\sigma Q \Pi_\sigma \geq 0$, as $B^\dagger A B \geq 0$ if $A \geq 0$ and $B \in \mathcal{L}(\mathcal{H})$, and $\Pi_\sigma^\dagger = \Pi_\sigma$. Given this, and the fact that $\mu_{\min}(\sigma) > 0$, it can be seen that it must be the case that $\Pi_\sigma Q \Pi_\sigma = 0$: the only positive semi-definite operator with zero trace is the all zero operator.

From $\Pi_\sigma Q \Pi_\sigma = 0$, it can then be concluded that $\text{Supp}(\sigma) \subseteq \text{Ker}(Q)$, meaning that the support of σ is fully contained within the kernel of Q . Hence, $\Pi_\sigma \leq \Pi_{\text{ker}(Q)} = \mathbb{I} - \Pi_{\text{supp}(Q)}$, where $\Pi_{\text{ker}(Q)}$ and $\Pi_{\text{supp}(Q)}$ are the projectors onto the kernel and support of Q respectively, such that $\Pi_{\text{ker}(Q)} + \Pi_{\text{supp}(Q)} = \mathbb{I}$. Finally, given that $Q \leq \Pi_{\text{supp}(Q)}$, we have that $Q \leq \Pi_{\text{supp}(Q)} \leq \mathbb{I} - \Pi_\sigma$, completing the proof. $\square$

8.6.2 Proof of the Choi-state of a Dephasing Channel

Here, the Choi-state of the dephasing channel is found.

Using the definition of the Choi-state, it can be seen that the Choi-state of a dephasing channel, $\mathcal{D}_p^{\text{ph}}(\rho) := p\rho + (1-p) \sum_{n=0}^{d-1} |n\rangle\langle n| \rho |n\rangle\langle n|$, is given by

$$\mathcal{J}_{AB}^{\mathcal{D}_p^{\text{ph}}} := (\mathcal{D}_p^{\text{ph}}(\cdot) \otimes \mathcal{I}_B) |\Phi_{00}^+\rangle\langle\Phi_{00}^+|_{AB} \tag{8.32}$$

$$= p |\Phi_{00}^+\rangle\langle\Phi_{00}^+|_{AB} + \frac{(1-p)}{d} \sum_{n=0}^{d-1} |nn\rangle\langle nn|_{AB} \tag{8.33}$$

$$= \frac{1}{d} \sum_{n=0}^{d-1} |nn\rangle\langle nn|_{AB} + \frac{p}{d} \sum_{n \neq m} |nn\rangle\langle mm|_{AB}. \tag{8.34}$$

By comparison of matrix elements, one can see that this can be rewritten as

$$\mathcal{J}_{AB}^{\mathcal{D}_p^{\text{ph}}} = \alpha |\Phi_{00}^+\rangle\langle\Phi_{00}^+|_{AB} + \frac{(1-\alpha)}{d-1} \sum_{c=1}^{d-1} |\Phi_{0c}^+\rangle\langle\Phi_{0c}^+|_{AB}, \tag{8.35}$$

such that $\alpha = [p(d-1) + 1]/d$. To see this, consider

$$\begin{aligned}
\sum_{c=1}^{d-1} |\Phi_{0c}^+ \rangle \langle \Phi_{0c}^+|_{AB} &= \sum_{c=1}^{d-1} (\mathbb{I}_A \otimes W_{0,c}) |\Phi_{00}^+ \rangle \langle \Phi_{00}^+|_{AB} (\mathbb{I}_A \otimes W_{0,c}^\dagger) \\
&= \sum_{c=1}^{d-1} (\mathbb{I}_A \otimes \sum_{n=0}^{d-1} \Omega^{cn} |n\rangle \langle n|) |\Phi_{00}^+ \rangle \langle \Phi_{00}^+|_{AB} (\mathbb{I}_A \otimes \sum_{m=0}^{d-1} \Omega^{-cm} |m\rangle \langle m|) \\
&= \sum_{c=1}^{d-1} \sum_{n,m=0}^{d-1} \Omega^{c(n-m)} (\mathbb{I}_A \otimes |n\rangle \langle n|) |\Phi_{00}^+ \rangle \langle \Phi_{00}^+|_{AB} (\mathbb{I}_A \otimes |m\rangle \langle m|) \\
&= d^{-1} \sum_{c=1}^{d-1} \sum_{n,m=0}^{d-1} \sum_{i,j=0}^{d-1} \Omega^{c(n-m)} (\mathbb{I}_A \otimes |n\rangle \langle n|) |ii\rangle \langle jj|_{AB} (\mathbb{I}_A \otimes |m\rangle \langle m|) \\
&= d^{-1} \sum_{n,m=0}^{d-1} \left[\sum_{c=1}^{d-1} \Omega^{c(n-m)} \right] |nn\rangle \langle mm|_{AB}.
\end{aligned} \tag{8.36}$$

It can then be seen, using the sum of a geometric series, that

$$\sum_{c=1}^{d-1} \Omega^{c(n-m)} = \begin{cases} -1 & \text{if } n \neq m \\ (d-1) & \text{if } n = m \end{cases} \tag{8.37}$$

Therefore,

$$\sum_{c=1}^{d-1} |\Phi_{0c}^+ \rangle \langle \Phi_{0c}^+|_{AB} = \frac{d-1}{d} \sum_{n=m}^{d-1} |nn\rangle \langle nn|_{AB} - \frac{1}{d} \sum_{n \neq m} |nn\rangle \langle mm|_{AB}. \tag{8.38}$$

Inputting this into Eq. (8.35) gives

$$\begin{aligned}
\mathcal{J}_{AB}^{\mathcal{D}^{\text{ph}}} &= \frac{\alpha}{d} \sum_{n,m=0}^{d-1} |nn\rangle \langle mm| + \frac{1-\alpha}{d-1} \left[\frac{d-1}{d} \sum_{n=m}^{d-1} |nn\rangle \langle nn|_{AB} - \frac{1}{d} \sum_{n \neq m}^{d-1} |nn\rangle \langle mm|_{AB} \right] \\
&= \left[\frac{\alpha}{d} + \frac{1-\alpha}{d} \right] \sum_{n=m}^{d-1} |nn\rangle \langle nn| + \left[\frac{\alpha}{d} - \frac{1-\alpha}{d(d-1)} \right] \sum_{n \neq m}^{d-1} |nn\rangle \langle mm|_{AB} \\
&= \frac{1}{d} \sum_{n=m} |nn\rangle \langle nn| + \frac{p}{d} \sum_{n \neq m}^{d-1} |nn\rangle \langle mm|_{AB},
\end{aligned} \tag{8.39}$$

if $\alpha = [p(d-1) + 1]/d$, making Eq. (8.34) and Eq. (8.35) equivalent.

8.6.3 Proof that the Transpose of a Unital Channel is a Unital Channel

Consider a quantum channel $\mathcal{E}$ with Kraus decomposition

$$\mathcal{E}(\cdot) = \sum_i K_i(\cdot) K_i^\dagger, \quad \sum_i K_i^\dagger K_i = \mathbb{I}. \tag{8.40}$$

As $\mathcal{E}(\mathbb{I}) = \mathbb{I}$, it then also holds for unital channels that $\sum_i K_i K_i^\dagger = \mathbb{I}$. The Choi-state of this channel is

$$\begin{aligned}
\mathcal{J}_{AB}^{\mathcal{E}} &= (\mathcal{E}_A \otimes \mathcal{I}_B)(|\Phi^+ \rangle \langle \Phi^+|_{AB}) \\
&= \sum_i (K_i \otimes \mathbb{I}) |\Phi^+ \rangle \langle \Phi^+|_{AB} (K_i^\dagger \otimes \mathbb{I}) \\
&= \sum_i (\mathbb{I} \otimes K_i^t) |\Phi^+ \rangle \langle \Phi^+|_{AB} (\mathbb{I} \otimes (K_i^\dagger)^t) \\
&= (\mathcal{I}_A \otimes \mathcal{E}^t) |\Phi^+ \rangle \langle \Phi^+|_{AB},
\end{aligned} \tag{8.41}$$

where we have defined

$$\mathcal{E}^t(\cdot) := \sum_i K_i^t(\cdot)(K_i^\dagger)^t. \quad (8.42)$$

We now prove the following Lemma about $\mathcal{E}^t$.

Lemma 8.6.1. *If $\mathcal{E}$ is a unital channel, then $\mathcal{E}^t$ is a unital channel.*

Proof. If $\mathcal{E}$ is a unital channel with a Kraus decomposition $\mathcal{E}(\cdot) = \sum_i K_i(\cdot)K_i^\dagger$, then the following two conditions hold

$$\sum_i K_i^\dagger K_i = \mathbb{I} \quad \text{and} \quad \sum_i K_i K_i^\dagger = \mathbb{I}. \quad (8.43)$$

Hence, if $\mathcal{E}^t(\cdot) = \sum_i K_i^t(\cdot)(K_i^\dagger)^t$ is a unital channel, then the following two conditions must hold

$$\sum_i (K_i^t)^\dagger K_i^t = \mathbb{I} \quad \sum_i K_i^t (K_i^t)^\dagger = \mathbb{I}. \quad (8.44)$$

We consider each of the equations in Eq. (8.43) in turn. Firstly,

$$\begin{aligned} \mathbb{I} &= \mathbb{I}^t \\ &= \left(\sum_i K_i^\dagger K_i \right)^t \\ &= \sum_i (K_i^\dagger K_i)^t \\ &= \sum_i K_i^t (K_i^\dagger)^t, \\ &= \sum_i K_i^t (K_i^t)^\dagger, \end{aligned} \quad (8.45)$$

where we have used the fact that the transpose and adjoint operations commute. This proves that the first of the conditions in Eq. (8.44) holds. Applying the same proof technique to the second equation in Eq. (8.43) proves that the second equation in Eq. (8.44) holds. Hence, $\mathcal{E}^t$ is a unital channel, completing the proof. $\square$

8.6.4 Proof that Unital Channels are Rank Non-Decreasing

Here, it is shown that the rank of states cannot decrease under unital channels. In what follows, $\text{rank}(\rho)$ denotes the rank of the state ρ , i.e., it is the number of strictly positive eigenvalues that ρ has.

Lemma 8.6.2. *If $\mathcal{E} : \mathcal{H}^d \rightarrow \mathcal{H}^d$ is a unital channel, then $\text{rank}[\mathcal{E}(\rho)] \geq \text{rank}(\rho) \forall \rho$.*

Proof. Firstly, we define $|\text{supp}(\mathbf{a})| = \{\text{number of } i : a_i > 0\}$ where a_i are the components of some n -dimensional vector $\mathbf{a} \in \mathbb{R}^n$. This is the number of strictly positive components of the vector. This was introduced in Ref. [112], where it was noted that given two vectors $\mathbf{a}, \mathbf{b}$ representing probability distributions, such that

$$\begin{aligned} \mathbf{a} &= (p_1, p_2, \dots, p_d) : p_i \in \mathbb{R}^+ \forall i \text{ and } \sum_i p_i = 1 \\ \mathbf{b} &= (q_1, q_2, \dots, q_d) : q_i \in \mathbb{R}^+ \forall i \text{ and } \sum_i q_i = 1, \end{aligned} \quad (8.46)$$

and $p \in [0, 1]$, then

$$|\text{supp}(p\mathbf{a} + (1-p)\mathbf{b})| \geq \max\{|\text{supp}(\mathbf{a})|, |\text{supp}(\mathbf{b})|\}. \quad (8.47)$$

Secondly, we recall the definition of majorisation, detailed in Definition 2.8.1, where a vector $\mathbf{x} \in \mathbb{R}^n$ majorises a vector $\mathbf{y} \in \mathbb{R}^n$, denoted $\mathbf{x} \succ \mathbf{y}$, if

$$\sum_{i=1}^k x_i^\downarrow \geq \sum_{i=1}^k y_i^\downarrow \quad \forall k \in \{1, \dots, n\}, \quad (8.48)$$

where $x_i^\downarrow$ ($y_i^\downarrow$) are the components of the vector $\mathbf{x}$ ($\mathbf{y}$) ordered in decreasing order, such that $x_i^\downarrow \geq x_{i+1}^\downarrow \quad \forall i \in \{1, \dots, n\}$.

Returning to the proof, if there exists a unital channel $\mathcal{E}$ such that $\sigma = \mathcal{E}(\rho)$, then

$$\lambda(\rho) \succ \lambda(\sigma), \quad (8.49)$$

where $\lambda(\rho)$ and $\lambda(\sigma)$ are vectors of the spectrum of ρ and σ , respectively (see [18] or Section 4.3.1.2 for details). This then implies the existence of a doubly stochastic matrix, $\mathbb{D}$, such that [119]

$$\lambda(\sigma) = \mathbb{D}\lambda(\rho) = \sum_i p_i \mathbb{P}_i \lambda(\rho), \quad (8.50)$$

where we have used the fact that a doubly stochastic matrix is a convex combination of permutation matrices [119], such that $\sum_i p_i = 1$, $p_i \geq 0$ and $\mathbb{P}_i$ is a permutation matrix for all i . Employing Eq. (8.47), it can be seen that

$$|\text{supp}(\lambda(\sigma))| = |\text{supp}(\sum_i p_i \mathbb{P}_i \lambda(\rho))| \geq \max_i \{|\text{supp}(\mathbb{P}_i \lambda(\rho))|\} = |\text{supp}(\lambda(\rho))|, \quad (8.51)$$

where we have used the fact that $|\text{supp}(a)|$ is invariant under permutation. To complete the proof, it is noted that $\text{rank}(\sigma) = |\text{supp}(\lambda(\sigma))|$. Hence, Eq. (8.51) gives $\text{rank}(\sigma) \geq \text{rank}(\rho)$, completing the proof. $\square$

Physically, this can be explained by the equal convertibility power of unital channels and noisy operations [18], as discussed in Section 4.3.1.2; unital channels can only output states more or equally as noisy as the input states, and hence, they can only make states more indistinguishable.

CONCLUSION

This thesis aimed to develop our understanding of resource preservation and enhancement for a range of different scenarios and resources. It achieved this by demonstrating the viability of existing tools for studying resource preservation, whilst creating novel ones for investigating resource enhancement. We also investigated the Choi-state, an indispensable tool in the study of quantum dynamics, using the task of quantum state exclusion. Before discussing the takeaways and future directions of each chapter, we will firstly reflect on some more general points concerning this thesis and the field as a whole.

Resource theories have proved to be an excellent framework for quantifying and studying a plethora of different quantum phenomena. Within this thesis, their versatility has been demonstrated: they were used to quantify resource preservability, study the movement of useful properties from one system to another, and to approach the centuries old problem of cooling from a new perspective. Nevertheless, there still remains a significant disconnect between the tools developed via the resource theory community and the practical implementation of quantum information. For example, typically, real hardware is subject to more restrictive physical constraints than those considered in resource theories, where more idealised restrictions are used to allow for analytical or numerical results to be found. There therefore exists an opportunity to apply the now ample resource theory tool-chest to practical systems, with the intention of improving our ability to both process quantum information and to study quantum systems. Magic state distillation — the creation of high fidelity magic states for use in performing universal quantum computation within the stabiliser formalism — does serve as an example of where this has been achieved; the use of catalysis [12, 13] to create methods of quantum state preparation [200] and the use of resource measures to probe the entanglement of many-body quantum systems are further examples [201]. In addition, Chapter 6 partly aimed to achieve this for the process of resource enhancement (see below for further discussion on the success of this). Hence, there does exist research pushing in this direction, however, there still remains plenty of scope for utilising resource theories to aid in the practical implementation of quantum information. Below, we discuss some extensions of the work presented in this thesis in this direction.

Although, the ultimate scope of the quantum resource theories framework remains to be seen, but it has the potential to be vast. The framework shares many attributes with the theory of thermodynamics, a paradigmatic operational theory; both aim to quantify the world by considering what transformations are possible under some set of physical constraints. They do not try to *explain* what the universe *is*, only how it can be used [70]. By adopting this approach, thermodynamics has remained a central pillar of physics for over three centuries, being used to successfully gain insights into systems ranging from black holes to atoms. Given their similar structure and generality, the resource theory framework has the potential for this same broad applicability, with it potentially being utilised all throughout physics, or beyond: not as theories that explain the world, but as a powerful tool for quantifying what is achievable within it.

As a final global observation, we note that the results of Chapter 5 and Chapter 8 both serve as further evidence of the fact that the features of the Choi-state are useful for gaining an insight into the physical properties of the dynamics they represent. In Chapter 5 the eigenvalues of the Choi-state were used to compare the abilities of channels to preserve informational non-equilibrium; in Chapter 8 the rank of the Choi-state was used to bound an agent's ability to gain exclusionary information in a generalisation of super-dense coding. Hence, these mathematical features of the Choi-state are shown to contain highly non-trivial insights into the physical properties of their associated dynamics, even dictating performance in an operational task.

9.1 Future Directions

9.1.1 Chapter 5

In this chapter, resource preservability was investigated in the resource theory of informational non-equilibrium, with necessary and sufficient conditions developed to determine whether one allowed

operation preserves the resource no better than another. This was achieved for qubit channels and n -dimensional Weyl-covariant channels. This progress motivates the following future research directions:

- **Application to more realistic noise models:** Within this work, the resource theory of informational non-equilibrium preservability was used to compare the informational non-equilibrium (purity) preserving abilities of the dephasing and depolarising channels, which are high-level noise models. It would be interesting to see if the framework could be extended to more specific, potentially hardware tailored, noise models. The ultimate aim would be to develop an understanding of how changing certain physical parameters effects the applied noisy channel, and then being able to definitively compare when one type of noise is more ‘noisy’ than another. This could then give an indication of how to tune these physical parameters.
- **Can these result be extended to the resource theory of athermality? i.e., $H \neq 0$:** By applying the framework of resource preservability theories to the resource theory of athermality [17, 19] an understanding of the ability of thermal operations to preserve athermality could be discovered. This could further our knowledge of the dynamical aspects of quantum thermodynamics in the resource theory picture. Moreover, such information could provide a useful insight into how to combat thermal noise in quantum technologies, or inform the design of efficient thermal processes. Lastly, it would be interesting to explore how such tools, in complement to the tools developed in this work, could be utilised in the study of the thermodynamics of communication [26, 130, 202, 203].
- **The relationship between asymmetry and informational non-equilibrium preservability:** Weyl covariant channels played a prominent role in our characterisation of the resource theory of informational non-equilibrium preservability. Namely, it was shown that all qubit noisy operations are equivalent to a Weyl-covariant channel up to a pre-processing and post-processing unitary channel, see Lemma 5.8.1. We then only considered Weyl-covariant channels in the higher dimensional case. Interestingly, these are the allowed operations of the resource theory of asymmetry with respect to the group whose unitary representation are the discrete Weyl-operators (see Definition 2.9.1). Hence, it would be interesting to explore if there exists a relationship between informational non-equilibrium preservability and asymmetry with respect to this group.

9.1.2 Chapter 6

In this chapter, the framework of resource concentration was applied to the resource theories of informational non-equilibrium and unspeakable coherence. In both cases, equalities and bounds on the optimally achievable local resource enhancement were found. In some cases, the dynamics necessary to perform this optimal local resource enhancement was also found. An understanding of the concentration scenario was then used to build an understanding of more general cases of resource enhancement. This work motivates the following possible future research directions:

- **Is the resource concentration framework practically useful:** A motivating factor for developing the resource concentration framework was to bring the notion of resource enhancement away from the asymptotic case and toward the abilities of contemporary quantum hardware. This was done on the assumption that smaller i.e., considering only two states, meant more practically feasible. Whilst an experimental implementation of informational non-equilibrium concentration was proposed in [3], there has not yet been an attempt to try an incorporate the notion of concentration into some practically useful protocol. With these tools having now been developed, it would be pertinent to try and find an application to which they can be applied. Applying the concentration framework to the resource theory of non-stabilizerness [24] could be a good place to start, as the generation of high fidelity magic states is essential for quantum computing.
- **Greedy resource enhancement (concatenation) protocols:** In Section 6.3.4, the notion of a ‘greedy’ resource enhancement, or concatenation, protocol was introduced. Such resource enhancement protocols trade space (number of input states), time (protocol depth) and (potentially) achievable resource enhancement for relative simplicity i.e., each unitary acts on at most two copies. It would be interesting to understand how the achievable local resource enhancement trades off with the

space/time increase, as this would determine if and when these protocols are practically viable i.e., if and when is the simplicity gain worth the increase in time/space and the decrease in the amount of local resource enhancement achieved. Specifically, it would be useful to know if given $n = 2^N$ states, where $N \in \mathbb{N}$, how much local resource enhancement can be achieved using a greedy protocol vs the optimal global operation on all n copies simultaneously. Moreover, it would also be interesting to consider an extension of these protocols where optimal resource enhancement is instead performed on k copies at time, with an understanding of how k trades off with the above quantities being desirable.

- **Concentration of two different resources:** Given two copies of a state, $\rho_A \otimes \rho_B$, the concentration framework studies the relocation of resource from B to A via a resource non-generating unitary. Whilst in this thesis we have only considered an individual resource theory at a time, ρ can be simultaneously resourceful in many different resource theories. It could be insightful to consider the concentration scenario where the dynamics are resource non-generating for multiple different resources; seeing if both resources could be simultaneously enhanced locally, or whether one could be enhanced in the A system whilst the other is enhanced in the B system, could provide a method for investigating the interplay of these resources.

9.1.3 Chapter 7

In this chapter, we considered the problem of cooling a thermal qubit at temperature T_S given access to n other thermal qubits, with varying energy differences, at temperature T_M . We found a set of inequalities that dictated both the optimal amount of cooling possible and informed the dynamics to achieve it. In addition, the notion of machine reducibility was introduced and studied, before the protocol was extended to include cooling under symmetry constraints, and general purity enhancement. This work motivates the following possible future research directions:

- **Black box cooling:** In the cooling protocol presented in this chapter we assumed access to all available information about the state: we knew the temperatures of the system and bath, and the size of the energy differences to perfect precision. Moreover, we knew the eigenbasis of each state. This was also true in the extension of the algorithm to a purity enhancement protocol. Recently, there has been interest in performing work extraction — another thermodynamic task of fundamental importance — from a state without access to prior information about the state, so-called black box work extraction [204]. It would be insightful to apply the same ideas to cooling, and see how far from optimal cooling an agent can get if they don't have all prior information about the state or the machine.
- **Formal link between bound resources and machine reducibility:** In Section 7.5, the notion of machine reducibility was introduced. This occurred when it was possible to perform optimal cooling without having to interact with all qubits in the machine. In Section 6.2.3 and 6.2.6.1, the notion of bound resources were introduced and discussed. These were states for which having additional copies did not necessarily lead to the ability to achieve greater local resource enhancement. Given the links between optimal cooling and purity enhancement, made explicit in Section 7.7, it is not difficult to see the similarities between these two concepts, with both showing that having access to additional states (resources) does not lead to better performance in the respective tasks. Hence, given cooling via unitary dynamics can be seen to be a special case of local purity enhancement, we conjecture that machine reducibility and bound resources will coincide in some scenarios. We leave it to future work to explore this link in detail.

9.1.4 Chapter 8

In this chapter, the Choi-rank was given a novel operational interpretation as the fundamental limit on a newly introduced entanglement-assisted exclusion task. To achieve this, a necessary condition for conclusive k -state exclusion was presented, and the notion of weak and strong state exclusion were introduced. This work motivates the following possible future research directions:

- **Exclusion with some error:** In this work, exclusion is used to quantify two parties' ability to communicate via some channel $\mathcal{N}$ with certainty. It is then shown that the Choi-rank of $\mathcal{N}$ is what limits the certainty that can be achieved. Performing *deterministic* exclusion rather than discrimination at the culmination of the protocol is the method used to ensure that only information that can be gained with certainty is being extracted. However, this relies on there being POVM elements where the probability of them occurring is equal to zero. This can be a strong assumption to make, with any amount of full-rank noise rendering it false. It would therefore be interesting to understand how robust these results are to noise on the input states, or the applied channel, which could prevent deterministic exclusion being possible.
- **Practical use cases of exclusionary information:** Typically, exclusion has been used as a tool for performing proofs in the foundations of physics [64–66]. There is now an effort to determine if there exists practical use cases for quantum state exclusion [205, 206]. Whilst an operational task involving exclusion was detailed in this chapter, the use cases of the task to some practically relevant scenario is not yet clear. It is discussed that the task is a method for recovering certainty in noisy communication, and indeed it does place fundamental bounds on one's ability to do this. However, in reality, communicating classical information is reliable and cheap to do with near certainty. Hence, it still remains to be determined if there exists a quantum information processing task that is both practically useful and utilises the notion of quantum state exclusion.

Bibliography

- [1] B. Stratton, C.-Y. Hsieh, and P. Skrzypczyk, *Unspeakable coherence concentration*, 2025.
- [2] J. Xuereb, B. Stratton, A. Rolandi, J. He, M. Huber, and P. Bakhshinezhad, *Cooling a qubit using n others*, *PRX Quantum* **6**, 040368 (2025).
- [3] C.-Y. Hsieh, B. Stratton, H.-C. Weng, and V. Scarani, *Informational nonequilibrium concentration*, *Phys. Rev. A* **111**, 052423 (2025).
- [4] B. Stratton, C.-Y. Hsieh, and P. Skrzypczyk, *Operational interpretation of the choi rank through exclusion tasks*, *Phys. Rev. A* **110**, L050601 (2024).
- [5] B. Stratton, C.-Y. Hsieh, and P. Skrzypczyk, *Dynamical resource theory of informational nonequilibrium preservability*, *Phys. Rev. Lett.* **132**, 110202 (2024).
- [6] B. Stratton, *An algorithm for estimating α -stabilizer rényi entropies via purity*, 2025.
- [7] C.-Y. Hsieh, B. Stratton, C.-H. Wu, and H.-Y. Ku, *Dynamical resource theory of incompatibility preservability*, *Phys. Rev. A* **111**, 022422 (2025).
- [8] C. E. Shannon, *A mathematical theory of communication*, *The Bell System Technical Journal* **27**, 379–423 (1948).
- [9] M. A. Nielsen and I. L. Chuang, *Quantum computation and quantum information*, 10th anniversary (Cambridge University Press, 2010).
- [10] J. Watrous, *The theory of quantum information* (Cambridge University Press, 2018).
- [11] V. Scarani, S. Iblisdir, N. Gisin, and A. Acín, *Quantum cloning*, *Rev. Mod. Phys.* **77**, 1225–1256 (2005).
- [12] G. Gour, *Quantum resource theories* (Cambridge University Press, 2025).
- [13] E. Chitambar and G. Gour, *Quantum resource theories*, *Rev. Mod. Phys.* **91**, 025001 (2019).
- [14] R. Horodecki, P. Horodecki, M. Horodecki, and K. Horodecki, *Quantum entanglement*, *Rev. Mod. Phys.* **81**, 865–942 (2009).
- [15] T. Baumgratz, M. Cramer, and M. B. Plenio, *Quantifying coherence*, *Phys. Rev. Lett.* **113**, 140401 (2014).
- [16] I. Marvian and R. W. Spekkens, *How to quantify coherence: distinguishing speakable and unspeakable notions*, *Phys. Rev. A* **94**, 052324 (2016).
- [17] M. Horodecki and J. Oppenheim, *Fundamental limitations for quantum and nanoscale thermodynamics*, *Nat. Commun.* **4**, 2059 (2013).
- [18] G. Gour, M. P. Müller, V. Narasimhachar, R. W. Spekkens, and N. Yunger Halpern, *The resource theory of informational nonequilibrium in thermodynamics*, *Phys. Rep.* **583**, 1–58 (2015).
- [19] M. Lostaglio, *An introductory review of the resource theory approach to thermodynamics*, *Rep. Prog. Phys.* **82**, 114001 (2019).
- [20] N. Brunner, D. Cavalcanti, S. Pironio, V. Scarani, and S. Wehner, *Bell nonlocality*, *Rev. Mod. Phys.* **86**, 419–478 (2014).
- [21] P. Skrzypczyk and N. Linden, *Robustness of measurement, discrimination games, and accessible information*, *Phys. Rev. Lett.* **122**, 140403 (2019).
- [22] O. Gühne, E. Haapasalo, T. Kraft, J.-P. Pellonpää, and R. Uola, *Colloquium: incompatible measurements in quantum information science*, *Rev. Mod. Phys.* **95**, 011003 (2023).
- [23] G. Saxena, E. Chitambar, and G. Gour, *Dynamical resource theory of quantum coherence*, *Phys. Rev. Res.* **2**, 023298 (2020).
- [24] V. Veitch, S. A. Hamed Mousavian, D. Gottesman, and J. Emerson, *The resource theory of stabilizer quantum computation*, *New J. Phys.* **16**, 013009 (2014).

- [25] S. Bravyi and A. Kitaev, *Universal quantum computation with ideal clifford gates and noisy ancillas*, *Phys. Rev. A* **71**, 022316 (2005).
- [26] C.-Y. Hsieh, *Resource preservability*, *Quantum* **4**, 244 (2020).
- [27] L. J. Schulman and U. V. Vazirani, *Molecular scale heat engines and scalable quantum computation*, in *Proceedings of the thirty-first annual acm symposium on theory of computing* (1999), pp. 322–329.
- [28] L. J. Schulman, T. Mor, and Y. Weinstein, *Physical limits of heat-bath algorithmic cooling*, *Phys. Rev. Lett.* **94**, 120501 (2005).
- [29] S. Raeisi and M. Mosca, *Asymptotic bound for heat-bath algorithmic cooling*, *Phys. Rev. Lett.* **114**, 100404 (2015).
- [30] L. Bassman Oftelie, A. De Pasquale, and M. Campisi, *Dynamic cooling on contemporary quantum computers*, *PRX Quantum* **5**, 030309 (2024).
- [31] Á. M. Alhambra, M. Lostaglio, and C. Perry, *Heat-Bath Algorithmic Cooling with optimal thermalization strategies*, *Quantum* **3**, 188 (2019).
- [32] F. Clivaz, R. Silva, G. Haack, J. B. Brask, N. Brunner, and M. Huber, *Unifying paradigms of quantum refrigeration: fundamental limits of cooling and associated work costs*, *Phys. Rev. E* **100**, 042130 (2019).
- [33] F. Clivaz, R. Silva, G. Haack, J. B. Brask, N. Brunner, and M. Huber, *Unifying paradigms of quantum refrigeration: a universal and attainable bound on cooling*, *Phys. Rev. Lett.* **123**, 170605 (2019).
- [34] A. E. Allahverdyan, K. V. Hovhannisyan, D. Janzing, and G. Mahler, *Thermodynamic limits of dynamic cooling*, *Phys. Rev. E* **84**, 041109 (2011).
- [35] P. Taranto, F. Bakhshinezhad, A. Bluhm, R. Silva, N. Friis, M. P. Lock, G. Vitagliano, F. C. Binder, T. Debarba, E. Schwarzhans, F. Clivaz, and M. Huber, *Landauer versus nernst: what is the true cost of cooling a quantum system?*, *PRX Quantum* **4**, 010332 (2023).
- [36] P. Taranto, P. Lipka-Bartosik, N. A. Rodríguez-Briones, M. Perarnau-Llobet, N. Friis, M. Huber, and P. Bakhshinezhad, *Efficiently cooling quantum systems with finite resources: insights from thermodynamic geometry*, *Phys. Rev. Lett.* **134**, 070401 (2025).
- [37] N. Linden, S. Popescu, and P. Skrzypczyk, *How small can thermal machines be? the smallest possible refrigerator*, *Phys. Rev. Lett.* **105**, 130401 (2010).
- [38] R. Silva, G. Manzano, P. Skrzypczyk, and N. Brunner, *Performance of autonomous quantum thermal machines: hilbert space dimension as a thermodynamical resource*, *Phys. Rev. E* **94**, 032120 (2016).
- [39] M. A. Nielsen and I. L. Chuang, *Quantum computation and quantum information: 10th anniversary edition* (Cambridge University Press, 2010).
- [40] G. Chiribella, G. M. D’Ariano, and P. Perinotti, *Transforming quantum operations: quantum supermaps*, *EPL* **83**, 30004 (2008).
- [41] R. A. Horn and C. R. Johnson, *Matrix analysis* (Cambridge University Press, 1985).
- [42] F. Riesz, *Sur les opérations fonctionnelles linéaires*, *Comptes Rendus Hebdomadaires des Séances de l’Académie des Sciences* **149**, 974–977 (1909).
- [43] S. Kullback and R. A. Leibler, *On information and sufficiency*, *Ann. Math. Stat.* **22**, 79–86 (1951).
- [44] A. Rényi, *On measures of information and entropy*, in *Proceedings of the 4th berkeley symposium on mathematical statistics and probability*, volume 1: contributions to the theory of statistics, edited by J. Neyman (1961), pp. 547–561.
- [45] M. Tomamichel, *Quantum information processing with finite resources* (Springer International Publishing, 2016).
- [46] H. Umegaki, *Conditional expectation in an operator algebra. iv. entropy and information*, *Kodai Math. Semin. Rep.* **14**, 59–85 (1962).
- [47] N. Datta, *Min- and max-relative entropies and a new entanglement monotone*, *IEEE Trans. Inf. Theory* **55**, 2816–2826 (2009).

- [48] R. Takagi and B. Regula, *General resource theories in quantum mechanics and beyond: operational characterization via discrimination tasks*, *Phys. Rev. X* **9**, 031053 (2019).
- [49] C.-Y. Hsieh, G. N. M. Tabia, Y.-C. Yin, and Y.-C. Liang, *Resource marginal problems*, *Quantum* **8**, 1353 (2024).
- [50] C.-Y. Hsieh, *Communication, dynamical resource theory, and thermodynamics*, *PRX Quantum* **2**, 020318 (2021).
- [51] R. Takagi, K. Wang, and M. Hayashi, *Application of the resource theory of channels to communication scenarios*, *Phys. Rev. Lett.* **124**, 120502 (2020).
- [52] A. W. Marshall, I. Olkin, and B. C. Arnold, *Inequalities: theory of majorization and its applications*, Second, Vol. 143 (Springer, 2011).
- [53] M. A. Nielsen, *Conditions for a class of entanglement transformations*, *Phys. Rev. Lett.* **83**, 436–439 (1999).
- [54] G. H. Hardy, J. E. Littlewood, and G. Pólya, *Inequalities* (Cambridge University Press, 1952).
- [55] A. Kleiner, B. Moldovanu, and P. Strack, *Extreme points and majorization: economic applications*, *Econometrica* **89**, 1557–1593 (2021).
- [56] P. J. Boland, *Majority systems and the condorcet jury theorem*, *J. R. Stat. Soc., Ser. D* **38**, 181–189 (1989).
- [57] W. K. Wootters, *A wigner-function formulation of finite-state quantum mechanics*, *Ann. Phys.* **176**, 1–21 (1987).
- [58] M. M. Wolf, *Quantum channels and operations-guided tour* (2012).
- [59] J. Bae and L.-C. Kwek, *Quantum state discrimination and its applications*, *J. Phys. A Math. Theor.* **48**, 083001 (2015).
- [60] C. W. Helstrom, *Quantum detection and estimation theory*, *J. Stat. Phys.* **1**, 231–252 (1969).
- [61] P. Skrzypczyk and D. Cavalcanti, *Semidefinite programming in quantum information science*, 2053–2563 (IOP Publishing, 2023).
- [62] A. Chefles, *Unambiguous discrimination between linearly independent quantum states*, *Phys. Lett. A* **239**, 339–347 (1998).
- [63] T. Rudolph, R. W. Spekkens, and P. S. Turner, *Unambiguous discrimination of mixed states*, *Phys. Rev. A* **68**, 010301 (2003).
- [64] C. M. Caves, C. A. Fuchs, and R. Schack, *Conditions for compatibility of quantum-state assignments*, *Phys. Rev. A* **66**, 062111 (2002).
- [65] M. F. Pusey, J. Barrett, and T. Rudolph, *On the reality of the quantum state*, *Nat. Phys.* **8**, 475–478 (2012).
- [66] S. Bandyopadhyay, R. Jain, J. Oppenheim, and C. Perry, *Conclusive exclusion of quantum states*, *Phys. Rev. A* **89**, 022336 (2014).
- [67] R. Uola, T. Bullock, T. Kraft, J.-P. Pellonpää, and N. Brunner, *All quantum resources provide an advantage in exclusion tasks*, *Phys. Rev. Lett.* **125**, 110402 (2020).
- [68] R. Takagi, B. Regula, K. Bu, Z.-W. Liu, and G. Adesso, *Operational advantage of quantum resources in subchannel discrimination*, *Phys. Rev. Lett.* **122** (2019).
- [69] C. H. Bennett and S. J. Wiesner, *Communication via one- and two-particle operators on einstein-podolsky-rosen states*, *Phys. Rev. Lett.* **69**, 2881–2884 (1992).
- [70] J. Goold, M. Huber, A. Riera, L. del Rio, and P. Skrzypczyk, *The role of quantum information in thermodynamics—a topical review*, *J. Phys. A Math* **49**, 143001 (2016).
- [71] C. Weedbrook, S. Pirandola, R. García-Patrón, N. J. Cerf, T. C. Ralph, J. H. Shapiro, and S. Lloyd, *Gaussian quantum information*, *Rev. Mod. Phys.* **84**, 621–669 (2012).
- [72] C. Palazuelos, *Superactivation of quantum nonlocality*, *Phys. Rev. Lett.* **109**, 190401 (2012).

- [73] H. Tajima and R. Takagi, *Gibbs-preserving operations requiring infinite amount of quantum coherence*, *Phys. Rev. Lett.* **134**, 170201 (2025).
- [74] H. Tajima, K. Yamaguchi, R. Takagi, and Y. Kuramochi, *Universal tradeoff relations between resource cost and irreversibility of channels: general-resource wigner-araki-yanase theorems and beyond*, 2025.
- [75] P. Faist, J. Oppenheim, and R. Renner, *Gibbs-preserving maps outperform thermal operations in the quantum regime*, *New J. Phys.* **17**, 043003 (2015).
- [76] F. Brandão, M. Horodecki, N. Ng, J. Oppenheim, and S. Wehner, *The second laws of quantum thermodynamics*, *PNAS* **112**, 3275–3279 (2015).
- [77] C. H. Bennett, H. J. Bernstein, S. Popescu, and B. Schumacher, *Concentrating partial entanglement by local operations*, *Phys. Rev. A* **53**, 2046–2052 (1996).
- [78] A. Winter and D. Yang, *Operational resource theory of coherence*, *Phys. Rev. Lett.* **116**, 120404 (2016).
- [79] V. Giovannetti, S. Lloyd, and L. Maccone, *Quantum metrology*, *Phys. Rev. Lett.* **96**, 010401 (2006).
- [80] J. Chen, *Review on quantum communication and quantum computation*, *J. Phys.: Conf. Ser.* **1865**, 022008 (2021).
- [81] I. Marvian and R. W. Spekkens, *Extending noether’s theorem by quantifying the asymmetry of quantum states*, *Nat. Commun.* **5**, 3821 (2014).
- [82] M. Lostaglio, D. Jennings, and T. Rudolph, *Description of quantum coherence in thermodynamic processes requires constraints beyond free energy*, *Nat. Comms.* **6**, 6383 (2015).
- [83] T. V. Kondra, R. Ganardi, and A. Streltsov, *Coherence manipulation in asymmetry and thermodynamics*, *Phys. Rev. Lett.* **132**, 200201 (2024).
- [84] M. Lostaglio, K. Korzekwa, D. Jennings, and T. Rudolph, *Quantum coherence, time-translation symmetry, and thermodynamics*, *Phys. Rev. X* **5**, 021001 (2015).
- [85] N. Shiraishi and R. Takagi, *Arbitrary amplification of quantum coherence in asymptotic and catalytic transformation*, *Phys. Rev. Lett.* **132**, 180202 (2024).
- [86] I. Marvian, *Coherence distillation machines are impossible in quantum thermodynamics*, *Nat. Commun.* **11**, 25 (2020).
- [87] M. Lostaglio and M. P. Müller, *Coherence and asymmetry cannot be broadcast*, *Phys. Rev. Lett.* **123**, 020403 (2019).
- [88] I. Marvian and R. W. Spekkens, *No-broadcasting theorem for quantum asymmetry and coherence and a trade-off relation for approximate broadcasting*, *Phys. Rev. Lett.* **123**, 020404 (2019).
- [89] J. Åberg, *Catalytic coherence*, *Phys. Rev. Lett.* **113**, 150402 (2014).
- [90] F. Ding, X. Hu, and H. Fan, *Amplifying asymmetry with correlating catalysts*, *Phys. Rev. A* **103**, 022403 (2021).
- [91] R. Takagi and N. Shiraishi, *Correlation in catalysts enables arbitrary manipulation of quantum coherence*, *Phys. Rev. Lett.* **128**, 240501 (2022).
- [92] S. D. Bartlett, T. Rudolph, and R. W. Spekkens, *Reference frames, superselection rules, and quantum information*, *Rev. Mod. Phys.* **79**, 555–609 (2007).
- [93] I. Marvian, R. W. Spekkens, and P. Zanardi, *Quantum speed limits, coherence, and asymmetry*, *Phys. Rev. A* **93**, 052331 (2016).
- [94] G. Gour and R. W. Spekkens, *The resource theory of quantum reference frames: manipulations and monotones*, *New J. Phys.* **10**, 033023 (2008).
- [95] G. Gour, I. Marvian, and R. W. Spekkens, *Measuring the quality of a quantum reference frame: the relative entropy of frameness*, *Phys. Rev. A* **80**, 012307 (2009).
- [96] I. Marvian and R. W. Spekkens, *The theory of manipulations of pure state asymmetry: i. basic tools, equivalence classes and single copy transformations*, *New J. Phys.* **15**, 033001 (2013).
- [97] I. Marvian and R. W. Spekkens, *Asymmetry properties of pure quantum states*, *Phys. Rev. A* **90**, 014102 (2014).

- [98] I. Marvian and R. W. Spekkens, *Modes of asymmetry: the application of harmonic analysis to symmetric quantum dynamics and quantum reference frames*, *Phys. Rev. A* **90**, 062110 (2014).
- [99] M. Skotiniotis and G. Gour, *Alignment of reference frames and an operational interpretation for the g -asymmetry*, *New J. Phys.* **14**, 073022 (2012).
- [100] G. Gour and C. M. Scandolo, *Dynamical resources*, 2020.
- [101] G. Gour and C. M. Scandolo, *Dynamical entanglement*, *Phys. Rev. Lett.* **125**, 180505 (2020).
- [102] K. Ji and E. Chitambar, *Entropic and operational characterizations of dynamic quantum resources*, *IEEE Trans. Inf. Theory* **71**, 7753–7787 (2025).
- [103] Z.-W. Liu and A. Winter, *Resource theories of quantum channels and the universal role of resource erasure*, 2019.
- [104] Y. Liu and X. Yuan, *Operational resource theory of quantum channels*, *Phys. Rev. Res.* **2**, 012035(R) (2020).
- [105] T. Theurer, D. Egloff, L. Zhang, and M. B. Plenio, *Quantifying operations with an application to coherence*, *Phys. Rev. Lett.* **122**, 190405 (2019).
- [106] D. Rosset, F. Buscemi, and Y.-C. Liang, *Resource theory of quantum memories and their faithful verification with minimal assumptions*, *Phys. Rev. X* **8**, 021033 (2018).
- [107] H.-J. Kim, S. Lee, L. Lami, and M. B. Plenio, *One-shot manipulation of entanglement for quantum channels*, *IEEE Trans. Inf. Theory* **67**, 5339–5351 (2021).
- [108] C. Sparaciari, L. del Rio, C. M. Scandolo, P. Faist, and J. Oppenheim, *The first law of general quantum resource theories*, *Quantum* **4**, 259 (2020).
- [109] M. Horodecki, P. Horodecki, and J. Oppenheim, *Reversible transformations from pure to mixed states and the unique measure of information*, *Phys. Rev. A* **67**, 062104 (2003).
- [110] A. Streltsov, H. Kampermann, S. Wölk, M. Gessner, and D. Bruß, *Maximal coherence and the resource theory of purity*, *New J. Phys.* **20**, 053058 (2018).
- [111] M. P. Woods and M. Horodecki, *Autonomous quantum devices: when are they realizable without additional thermodynamic costs?*, *Phys. Rev. X* **13**, 011016 (2023).
- [112] M. Lostaglio, Á. M. Alhambra, and C. Perry, *Elementary thermal operations*, *Quantum* **2**, 52 (2018).
- [113] C. Perry, P. Ćwikliński, J. Anders, M. Horodecki, and J. Oppenheim, *A sufficient set of experimentally implementable thermal operations for small systems*, *Phys. Rev. X* **8**, 041049 (2018).
- [114] P. Faist, M. Berta, and F. Brandão, *Thermodynamic capacity of quantum processes*, *Phys. Rev. Lett.* **122**, 200601 (2019).
- [115] A. Serafini, M. Lostaglio, S. Longden, U. Shackerley-Bennett, C.-Y. Hsieh, and G. Adesso, *Gaussian thermal operations and the limits of algorithmic cooling*, *Phys. Rev. Lett.* **124**, 010602 (2020).
- [116] V. Narasimhachar, S. Assad, F. C. Binder, J. Thompson, B. Yadin, and M. Gu, *Thermodynamic resources in continuous-variable quantum systems*, *npj Quantum Inf.* **7** (2021).
- [117] K. Heshami, D. G. England, P. C. Humphreys, P. J. Bustard, V. M. Acosta, J. Nunn, and B. J. Sussman, *Quantum memories: emerging applications and recent advances*, *J. Mod. Opt.* **63**, 2005–2028 (2016).
- [118] I. Bengtsson, Å. Ericsson, M. Kuś, W. Tadej, and K. Życzkowski, *Birkhoff's polytope andunistochastic matrices, $n = 3$ and $n = 4$* , *Commun. Math. Phys.* **259**, 307–324 (2005).
- [119] R. B. Bapat and T. E. S. Raghavan, *Doubly stochastic matrices*, in *Nonnegative matrices and applications*, Encyclopedia of Mathematics and its Applications (Cambridge University Press, 1997), pp. 59–114.
- [120] M. Hohenwarter, M. Borchers, G. Ancsin, B. Bencze, M. Blossier, A. Delobelle, C. Denizet, J. Éliás, Á. Fekete, L. Gál, Z. Konečný, Z. Kovács, S. Lizelfelner, B. Parisse, and G. Sturr, *GeoGebra 4.4*, <http://www.geogebra.org>, (Dec. 2013)
- [121] C. H. Bennett, P. W. Shor, J. A. Smolin, and A. V. Thapliyal, *Entanglement-assisted classical capacity of noisy quantum channels*, *Phys. Rev. Lett.* **83**, 3081–3084 (1999).

- [122] S. Lloyd, *Capacity of the noisy quantum channel*, *Phys. Rev. A* **55**, 1613–1622 (1997).
- [123] I. Devetak, *The private classical capacity and quantum capacity of a quantum channel*, *IEEE Trans. Inf. Theory* **51**, 44–55 (2005).
- [124] L. Gyongyosi, S. Imre, and H. V. Nguyen, *A survey on quantum channel capacities*, *IEEE Commun. Surv. Tutor.* **20**, 1149–1205 (2018).
- [125] A. S. Holevo, 2002.
- [126] G. G. Amosov, A. S. Holevo, and R. F. Werner, 2000.
- [127] J. Cortese, *Holevo-schumacher-westmoreland channel capacity for a class of qudit unital channels*, *Phys. Rev. A* **69**, 022302 (2004).
- [128] K. Siudzińska, *Classical capacity of generalized pauli channels*, *J. Phys. A: Math. Theor.* **53**, 445301 (2020).
- [129] R. Takagi, K. Wang, and M. Hayashi, *Application of the resource theory of channels to communication scenarios*, *Phys. Rev. Lett.* **124**, 120502 (2020).
- [130] C.-Y. Hsieh, *Communication, dynamical resource theory, and thermodynamics*, *PRX Quantum* **2**, 020318 (2021).
- [131] R. Horodecki and M. Horodecki, *Information-theoretic aspects of inseparability of mixed states*, *Phys. Rev. A* **54**, 1838–1843 (1996).
- [132] C.-K. Li and M.-D. Choi, *On unital qubit channels*, 2023.
- [133] F. Dolde, I. Jakobi, B. Naydenov, N. Zhao, S. Pezzagna, C. Trautmann, J. Meijer, P. Neumann, F. Jelezko, and J. Wrachtrup, *Room-temperature entanglement between single defect spins in diamond*, *Nat. Phys.* **9**, 139–143 (2013).
- [134] F. Dolde, V. Bergholm, Y. Wang, et al., *High-fidelity spin entanglement using optimal control*, *Nat. Commun.* **5**, 3371 (2014).
- [135] C. E. Bradley, J. Randall, M. H. Abobeih, R. C. Berrevoets, M. J. Degen, M. A. Bakker, M. Markham, D. J. Twitchen, and T. H. Taminiau, *A ten-qubit solid-state spin register with quantum memory up to one minute*, *Phys. Rev. X* **9**, 031045 (2019).
- [136] C. H. Bennett, G. Brassard, C. Crépeau, R. Jozsa, A. Peres, and W. K. Wootters, *Teleporting an unknown quantum state via dual classical and einstein-podolsky-rosen channels*, *Phys. Rev. Lett.* **70**, 1895–1899 (1993).
- [137] M. Horodecki, P. Horodecki, and R. Horodecki, *General teleportation channel, singlet fraction, and quasidistillation*, *Phys. Rev. A* **60**, 1888–1898 (1999).
- [138] M. Horodecki, P. Horodecki, and R. Horodecki, *Mixed-state entanglement and distillation: is there a “bound” entanglement in nature?*, *Phys. Rev. Lett.* **80**, 5239–5242 (1998).
- [139] E. T. Campbell and D. E. Browne, *Bound states for magic state distillation in fault-tolerant quantum computation*, *Phys. Rev. Lett.* **104**, 030503 (2010).
- [140] K. Fan, *Maximum properties and inequalities for the eigenvalues of completely continuous operators*, *PNAS* **37**, 760 (1951).
- [141] L. Szilard, *Über die entropieverminderung in einem thermodynamischen system bei eingriffen intelligenter wesen*, *Zeitschrift für Physik* **53**, 840 (1929).
- [142] T. Apostol, *Mathematical analysis: A modern approach to advanced calculus* (Pearson, 1974).
- [143] J. Von Neumann, *Some matrix-inequalities and metrization of matric space* (1937).
- [144] D. Reeb and M. M. Wolf, *An improved landauer principle with finite-size corrections*, *New J. Phys.* **16**, 103011 (2014).
- [145] H. Wilming and R. Gallego, *Third law of thermodynamics as a single inequality*, *Phys. Rev. X* **7**, 041033 (2017).
- [146] L. Masanes and J. Oppenheim, *A general derivation and quantification of the third law of thermodynamics*, *Nat. Commun.* **8**, 14538 (2017).

- [147] N. A. Rodríguez-Briones, E. Martín-Martínez, A. Kempf, and R. Laflamme, *Correlation-enhanced algorithmic cooling*, *Phys. Rev. Lett.* **119**, 050502 (2017).
- [148] J. Lin, N. A. Rodríguez-Briones, E. Martín-Martínez, and R. Laflamme, *Thermodynamic analysis of algorithmic cooling protocols: efficiency metrics and improved designs*, *Phys. Rev. A* **110**, 022215 (2024).
- [149] A. Levy and R. Kosloff, *Quantum absorption refrigerator*, *Phys. Rev. Lett.* **108**, 070604 (2012).
- [150] M. T. Mitchison, *Quantum thermal absorption machines: refrigerators, engines and clocks*, *Contemp. Phys.* **60**, 164–187 (2019).
- [151] V. Shende, A. Prasad, I. Markov, and J. Hayes, *Synthesis of reversible logic circuits*, *IEEE TCAD* **22**, 710–722 (2003).
- [152] A. Barenco, C. H. Bennett, R. Cleve, D. P. DiVincenzo, N. Margolus, P. Shor, T. Sleator, J. A. Smolin, and H. Weinfurter, *Elementary gates for quantum computation*, *Phys. Rev. A* **52**, 3457–3467 (1995).
- [153] N. Brunner, N. Linden, S. Popescu, and P. Skrzypczyk, *Virtual qubits, virtual temperatures, and the foundations of thermodynamics*, *Phys. Rev. E* **85**, 051117 (2012).
- [154] E. Noether, *Invariante variationsprobleme*, *Nachr. Ges. Wiss. Göttingen, Math.-Phys. Kl.* **1918**, 235–257 (1918).
- [155] R. Silva, P. Bakhshinezhad, and F. Clivaz, *Optimal unitary trajectories under commuting target and cost observables; applications to cooling*, 2024.
- [156] S. Majidy, W. F. Braasch, A. Lasek, T. Upadhyaya, A. Kalev, and N. Yunger Halpern, *Noncommuting conserved charges in quantum thermodynamics and beyond*, *Nat. Rev. Phys.* **5**, 689–698 (2023).
- [157] S. Aaronson, D. Grier, and L. Schaeffer, *The classification of reversible bit operations*, 2015.
- [158] W. F. Stinespring, *Positive functions on c^* -algebras*, *AMS* **6**, 211–216 (1955).
- [159] A. Jamiołkowski, *Linear transformations which preserve trace and positive semidefiniteness of operators*, *Rep. Math. Phys.* **3**, 275–278 (1972).
- [160] M.-D. Choi, *Completely positive linear maps on complex matrices*, *Linear Algebr. Appl.* **10**, 285 (1975).
- [161] E. M. Rains, *A semidefinite program for distillable entanglement*, 2001.
- [162] X. Wang, M. M. Wilde, and Y. Su, *Quantifying the magic of quantum channels*, *New J. Phys.* **21**, 103002 (2019).
- [163] M. Horodecki, P. W. Shor, and M. B. Ruskai, *Entanglement breaking channels*, *Rev. Math. Phys.* **15**, 629 (2003).
- [164] H.-Y. Ku, J. Kadlec, A. Cernoch, M. T. Quintino, W. Zhou, K. Lemr, N. Lambert, A. Miranowicz, S.-L. Chen, F. Nori, and Y.-N. Chen, *Quantifying quantumness of channels without entanglement*, *PRX Quantum* **3**, 020338 (2022).
- [165] M. Piani, M. Horodecki, P. Horodecki, and R. Horodecki, *Properties of quantum nonsignaling boxes*, *Phys. Rev. A* **74**, 012305 (2006).
- [166] C.-Y. Hsieh, M. Lostaglio, and A. Acín, *Quantum channel marginal problem*, *Phys. Rev. Res.* **4**, 013249 (2022).
- [167] X. Wang, W. Xie, and R. Duan, *Semidefinite programming strong converse bounds for classical capacity*, *IEEE Trans. Inf. Theory* **64**, 640 (2018).
- [168] M. J. Hoban and A. B. Sainz, *A channel-based framework for steering, non-locality and beyond*, *New J. Phys.* **20**, 053048 (2018).
- [169] D. Beckman, D. Gottesman, M. A. Nielsen, and J. Preskill, *Causal and localizable quantum operations*, *Phys. Rev. A* **64**, 052309 (2001).
- [170] T. Eggeling, D. Schlingemann, and R. F. Werner, *Semicausal operations are semilocalizable*, *EPL* **57**, 782 (2002).
- [171] R. Duan and A. Winter, *No-signalling-assisted zero-error capacity of quantum channels and an information theoretic interpretation of the lovász number*, *IEEE Trans. Inf. Theory* **62**, 891 (2016).

- [172] R. Takagi and B. Regula, *General resource theories in quantum mechanics and beyond: operational characterization via discrimination tasks*, *Phys. Rev. X* **9**, 031053 (2019).
- [173] E. Zanoni and C. M. Scandolo, 2024.
- [174] G. Gour and M. M. Wilde, *Entropy of a quantum channel*, *Phys. Rev. Res.* **3**, 023096 (2021).
- [175] M. M. Wilde, *Entanglement cost and quantum channel simulation*, *Phys. Rev. A* **98**, 042338 (2018).
- [176] D. Rosset, F. Buscemi, and Y.-C. Liang, *Resource theory of quantum memories and their faithful verification with minimal assumptions*, *Phys. Rev. X* **8**, 021033 (2018).
- [177] J. R. Seddon and E. T. Campbell, *Quantifying magic for multi-qubit operations*, *Proc. R. Soc. A* **475**, 20190251 (2019).
- [178] G. D. Berk, A. J. P. Garner, B. Yadin, K. Modi, and F. A. Pollock, *Resource theories of multi-time processes: a window into quantum non-markovianity*, *Quantum* **5**, 435 (2021).
- [179] S. Bäuml, S. Das, X. Wang, and M. M. Wilde, 2019.
- [180] E. Haapasalo, T. Kraft, N. Miklin, and R. Uola, *Quantum marginal problem and incompatibility*, *Quantum* **5**, 476 (2021).
- [181] Y. Liu and X. Yuan, *Operational resource theory of quantum channels*, *Phys. Rev. Res.* **2**, 012035 (2020).
- [182] C.-Y. Hsieh, B. Stratton, C.-H. Wu, and H.-Y. Ku, 2024.
- [183] F. Buscemi and N. Datta, *The quantum capacity of channels with arbitrarily correlated noise*, *IEEE Trans. Inf. Theory* **56**, 1447 (2010).
- [184] M. Girard, D. Leung, J. Levick, C.-K. Li, V. Paulsen, Y. T. Poon, and J. Watrous, *On the mixed-unitary rank of quantum channels*, *Commun. Math. Phys.* **394**, 919 (2022).
- [185] S. Singh and N. Datta, *Detecting positive quantum capacities of quantum channels*, *npj Quantum Inf.* **8**, 50 (2022).
- [186] A. F. Ducuara and P. Skrzypczyk, *Operational interpretation of weight-based resource quantifiers in convex quantum resource theories*, *Phys. Rev. Lett.* **125**, 110401 (2020).
- [187] C.-Y. Hsieh, R. Uola, and P. Skrzypczyk, 2023.
- [188] M. Ye, Y. Li, and Z. Li, *Operational characterization of weight-based resource quantifiers via exclusion tasks in general probabilistic theories*, *Quantum Inf. Process.* **20**, 317 (2021).
- [189] T. Heinosaari and O. Kerppo, *Antidistinguishability of pure quantum states*, *J. Phys. A: Math. Theor.* **51**, 365303 (2018).
- [190] H. K. Mishra, M. Nussbaum, and M. M. Wilde, *On the optimal error exponents for classical and quantum antidistinguishability*, *Lett. Math. Phys.* **114**, 76 (2024).
- [191] N. Johnston, V. Russo, and J. Sikora, *Tight bounds for antidistinguishability and circulant sets of pure quantum states*, *Quantum* **9**, 1622 (2025).
- [192] G. C. Knee, *Towards optimal experimental tests on the reality of the quantum state*, *New J. Phys.* **19**, 023004 (2017).
- [193] V. Russo and J. Sikora, *Inner products of pure states and their antidistinguishability*, *Phys. Rev. A* **107**, L030202 (2023).
- [194] V. Havlíček and J. Barrett, *Simple communication complexity separation from quantum state antidistinguishability*, *Phys. Rev. Res.* **2**, 013326 (2020).
- [195] A. Molina, *Povms are equivalent to projections for perfect state exclusion of three pure states in three dimensions*, *Quantum* **3**, 117 (2019).
- [196] C.-Y. Hsieh, *Communication, dynamical resource theory, and thermodynamics*, *PRX Quantum* **2**, 020318 (2021).
- [197] A. F. Ducuara and P. Skrzypczyk, *Characterization of quantum betting tasks in terms of arimoto mutual information*, *PRX Quantum* **3**, 020366 (2022).
- [198] A. F. Ducuara and P. Skrzypczyk, 2023.

- [199] A. F. Ducuara, P. Skrzypczyk, F. Buscemi, P. Sidajaya, and V. Scarani, *Maxwell's demon walks into wall street: stochastic thermodynamics meets expected utility theory*, *Phys. Rev. Lett.* **131**, 197103 (2023).
- [200] A. d. O. Junior, M. Perarnau-Llobet, N. Brunner, and P. Lipka-Bartosik, *Quantum catalysis in cavity quantum electrodynamics*, *Phys. Rev. Res.* **6**, 023127 (2024).
- [201] M. Cramer, M. B. Plenio, and H. Wunderlich, *Measuring entanglement in condensed matter systems*, *Phys. Rev. Lett.* **106**, 020401 (2011).
- [202] C.-Y. Hsieh, *Dynamical landauer principle: quantifying information transmission by thermodynamics*, *Phys. Rev. Lett.* **134**, 050404 (2025).
- [203] C.-Y. Hsieh, *Dynamical landauer principle: thermodynamic criteria of transmitting classical information*, *Phys. Rev. A* **111**, 022207 (2025).
- [204] K. Watanabe and R. Takagi, *Black box work extraction and composite hypothesis testing*, *Phys. Rev. Lett.* **133**, 250401 (2024).
- [205] J. Bae, K. Flatt, T. Heinosaari, O. Kerppo, K. Mohan, A. Muñoz-Moller, and A. Rai, *Random exclusion codes: quantum advantages of single-shot communication*, 2025.
- [206] C.-Y. Hsieh, R. Uola, and P. Skrzypczyk, *Quantum complementarity: a novel resource for unambiguous exclusion and encryption*, 2023.